

Service Manual

LG-D337

Table Of Contents

1. INTRODUCTION.....	3	4. DOWNLOAD.....	112
1.1 Purpose	3	5. BLOCK DIAGRAM.....	129
1.2 Regulatory Information	3	6. CIRCUIT DIAGRAM	141
2. PERFORMANCE.....	5	7. BGA PIN MAP	158
2.1 Product Name.....	5	8. PCB LAYOUT	178
2.2 Supporting Standard.....	5	9. CALIBRATION.....	182
2.3 Main Parts & Solution	6	10. HIDDEN MENU.....	200
2.4 HW Features	6	11. DISASSEMBLE GUIDE.....	235
2.5 SW Features	7	12. EXPLODED VIEW & REPLACEMENT PART LIST	242
2.6 H/W SPEC.....	12		
3. TROUBLE SHOOTING	21		
3.1 WCDMA Rx Part.....	21		
3.2 WCDMA Tx Part	33		
3.3 GSM Rx Part	49		
3.4 GSM Tx Part Troubleshooting.....	59		
3.5. BT/WiFi.....	67		
3.6. GPS.....	69		
3.7. FM Radio	72		
3.8 DTV.....	74		
3.9 Power TS.....	76		
3.10. Charging TS	80		
3.11. USB TS.....	82		
3.12.USB TS.....	83		
3.13. Accelerometer & Compass Sensor on/off TS	99		
3.14. Proximity Sensor on/off TS.....	100		
3.15. Touch Sensor Trouble Shooting.....	102		
3.16. Vibrator Trouble Shooting.....	104		
3.17. 8M Main Camera TS	106		
3.18. VT Camera TS	108		
3.19. Main LCD TS	110		



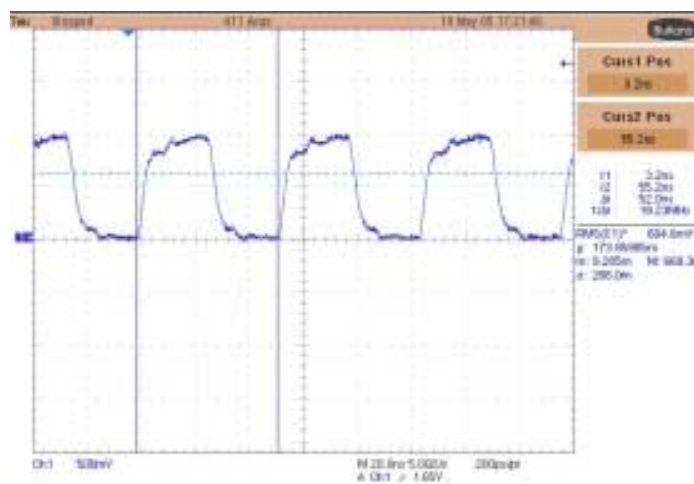
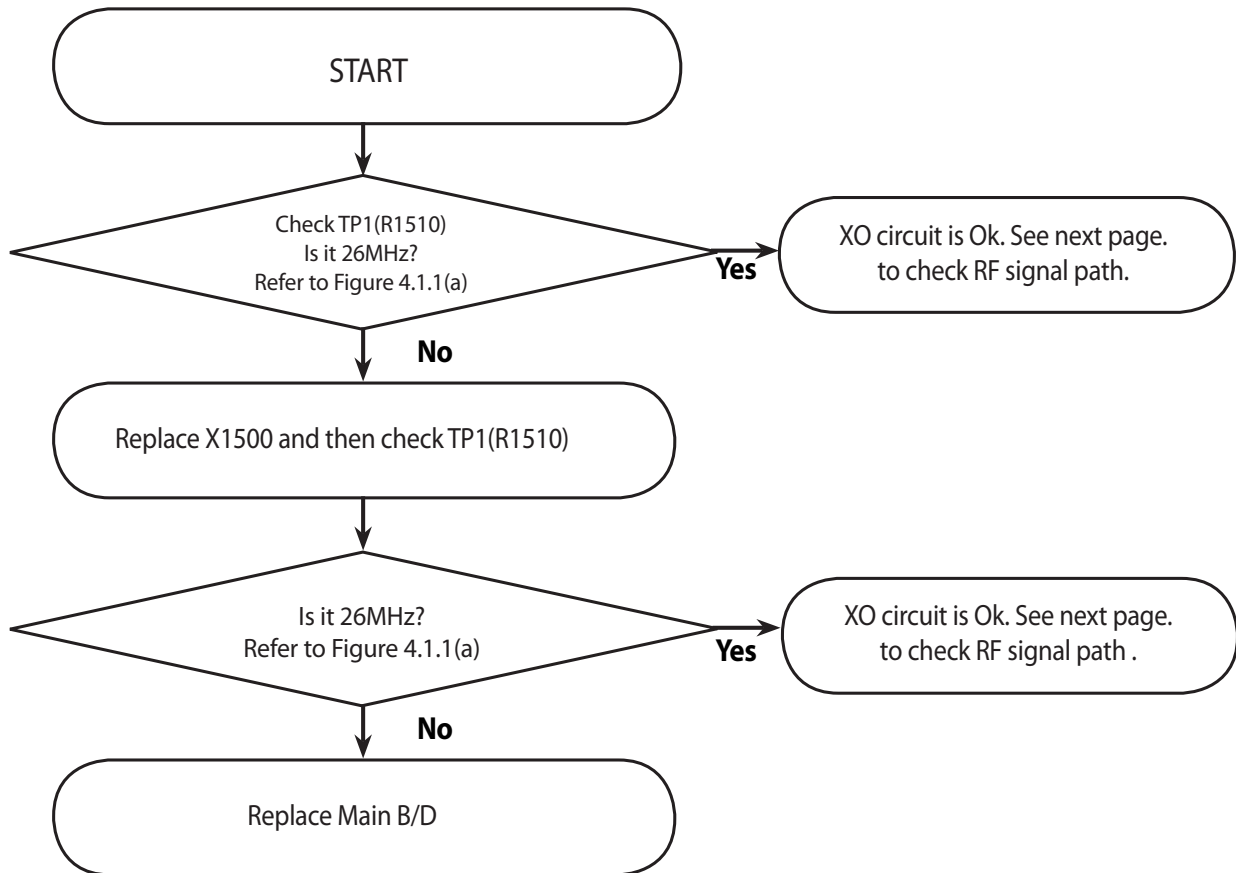
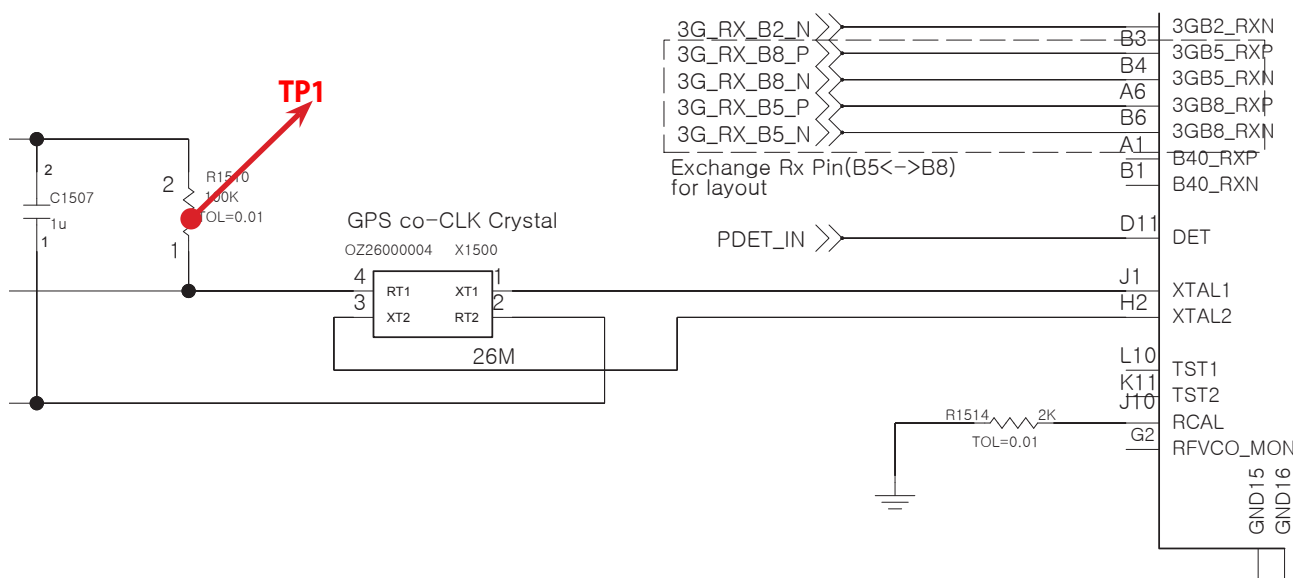
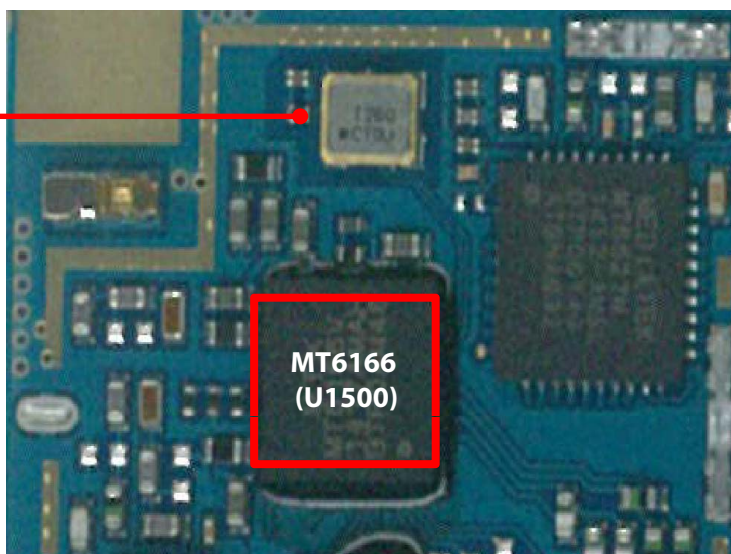


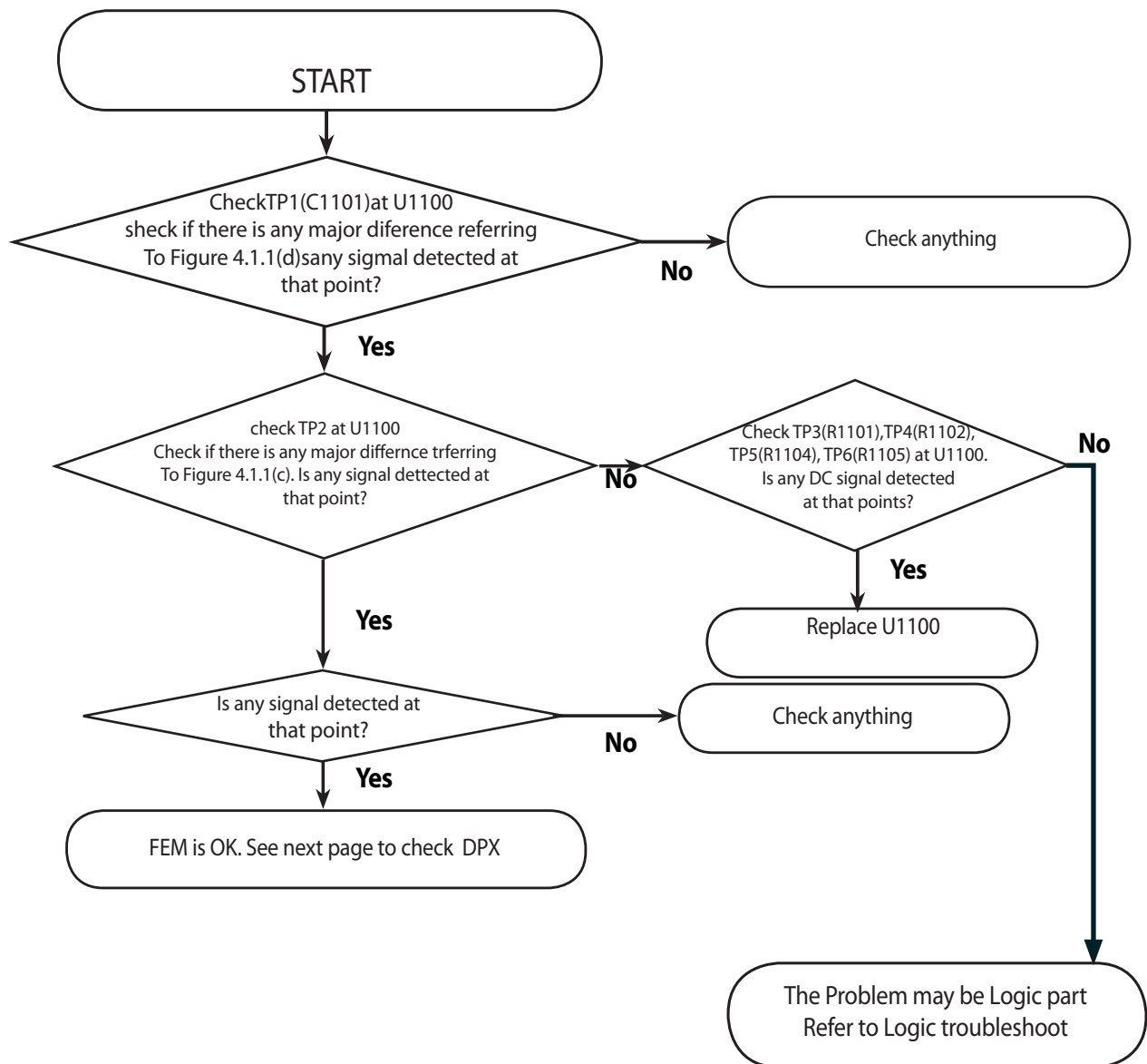
Figure 4.1.1 (a)

3.1.1.2 Checking MT6166 XO circuit

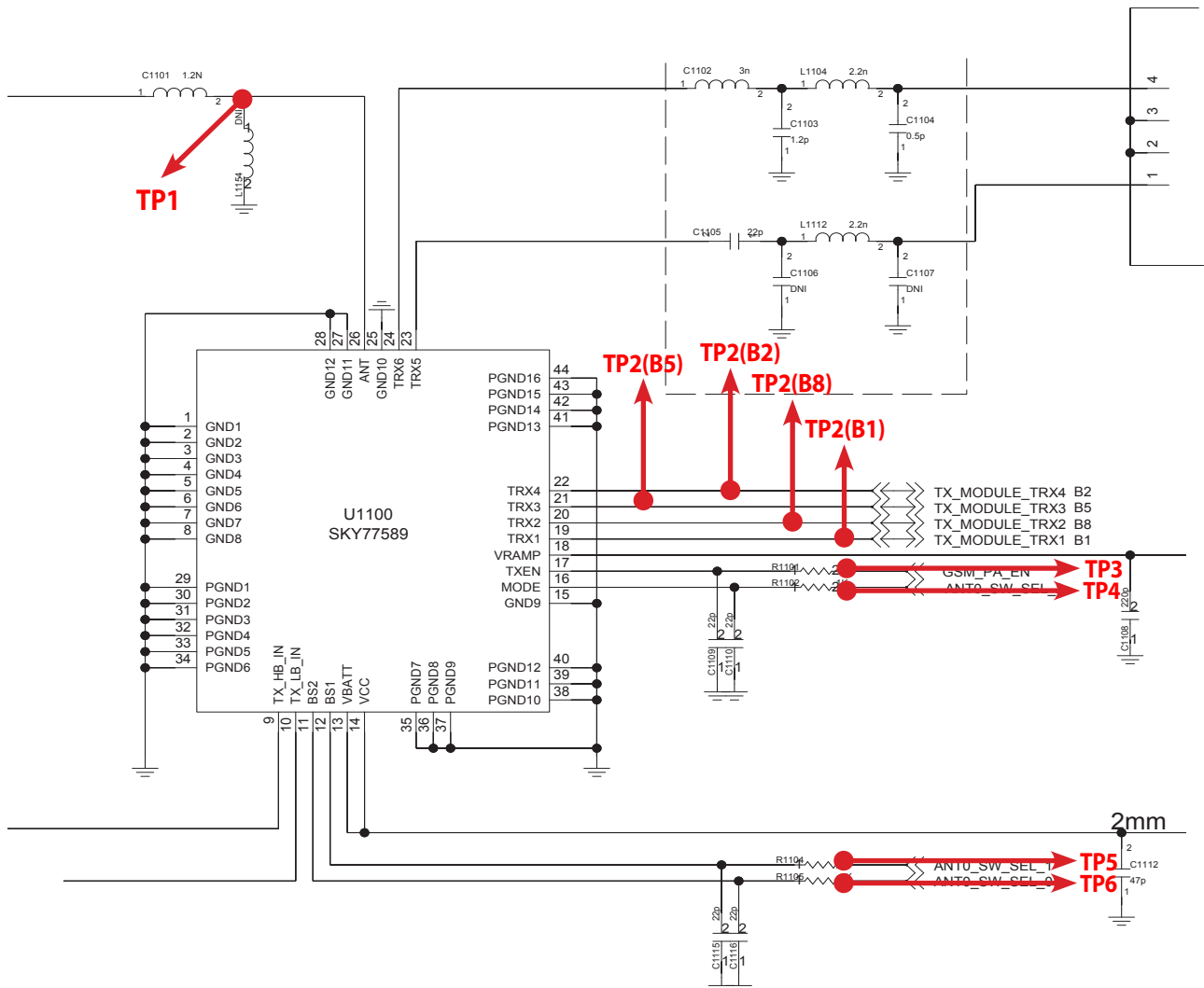
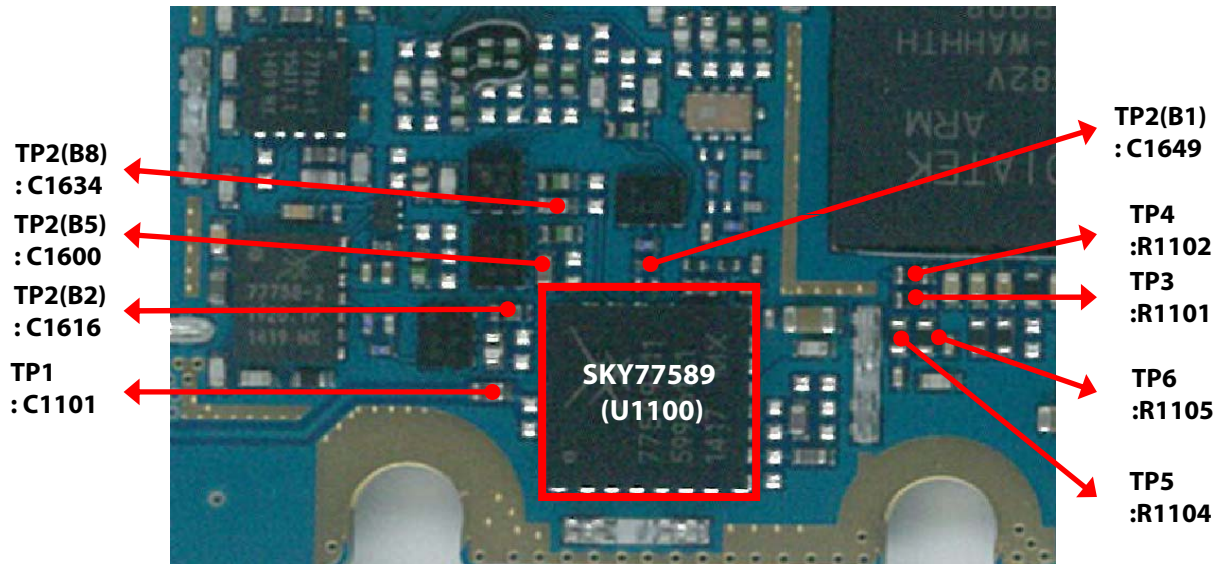
TP1 : R1510
(MAIN_CLK_26MHz)



3.1.1.3 Checking RF signal path (FEM)



4. DOWNLOAD



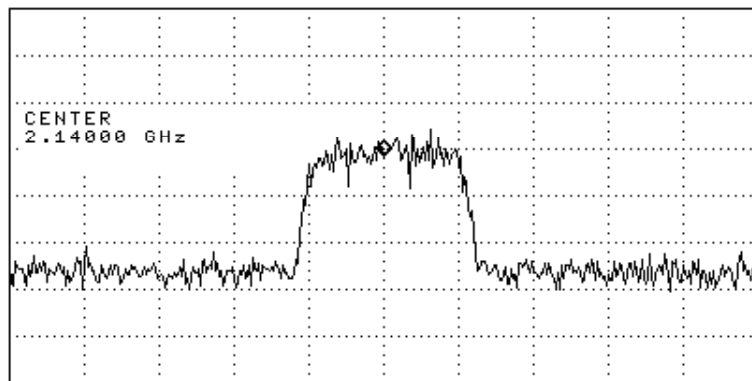


Figure 4.1.1 (b)

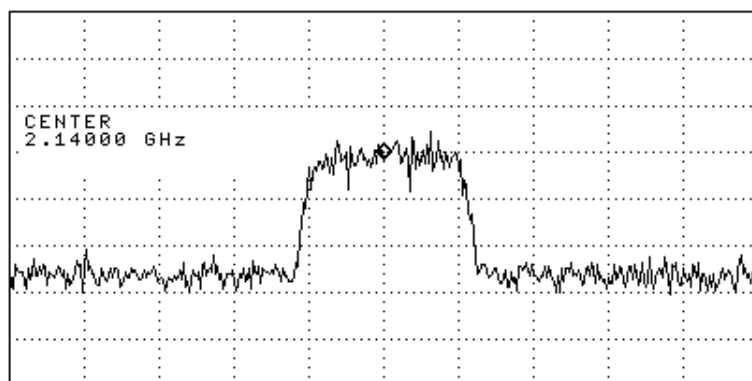
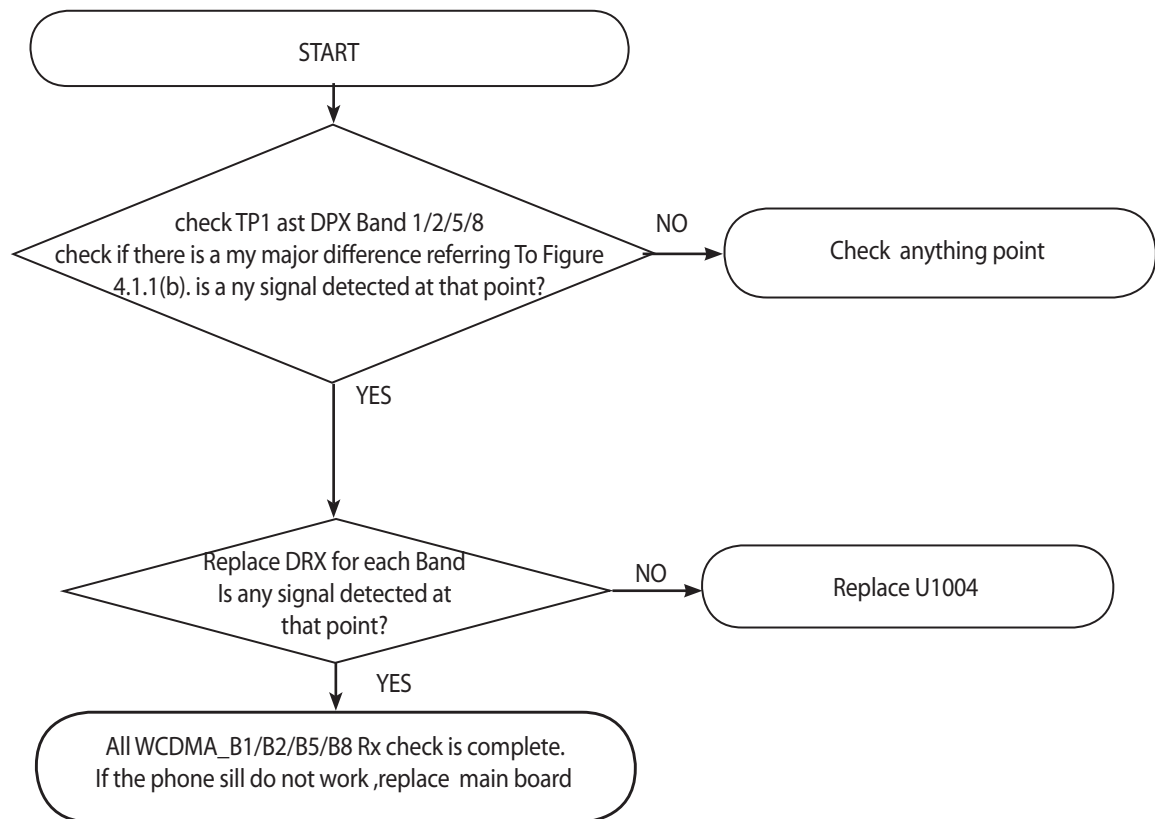
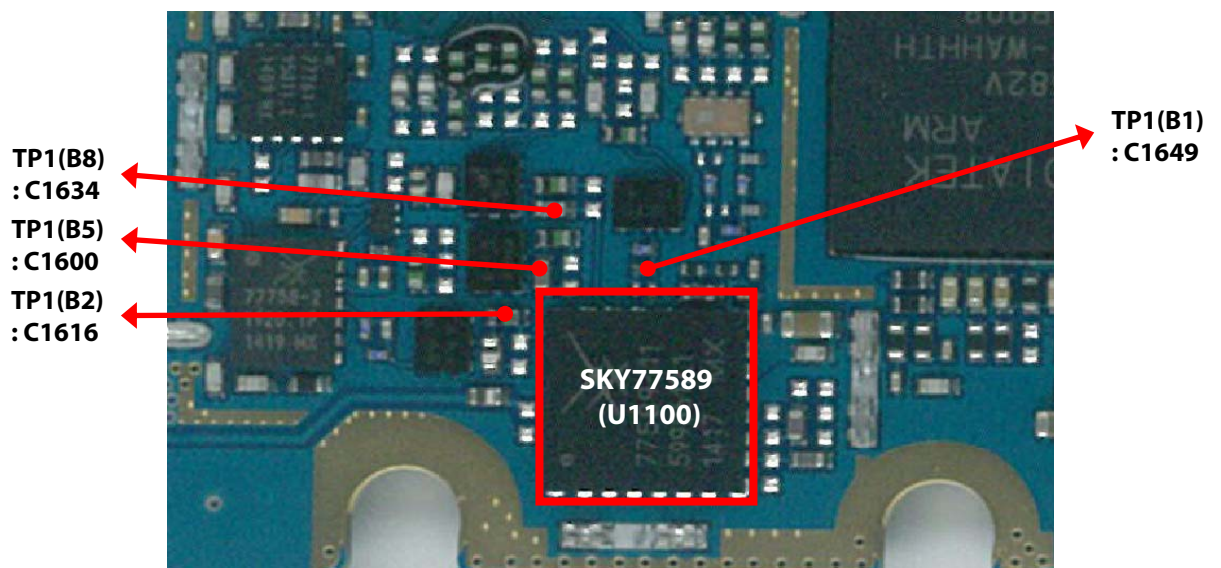


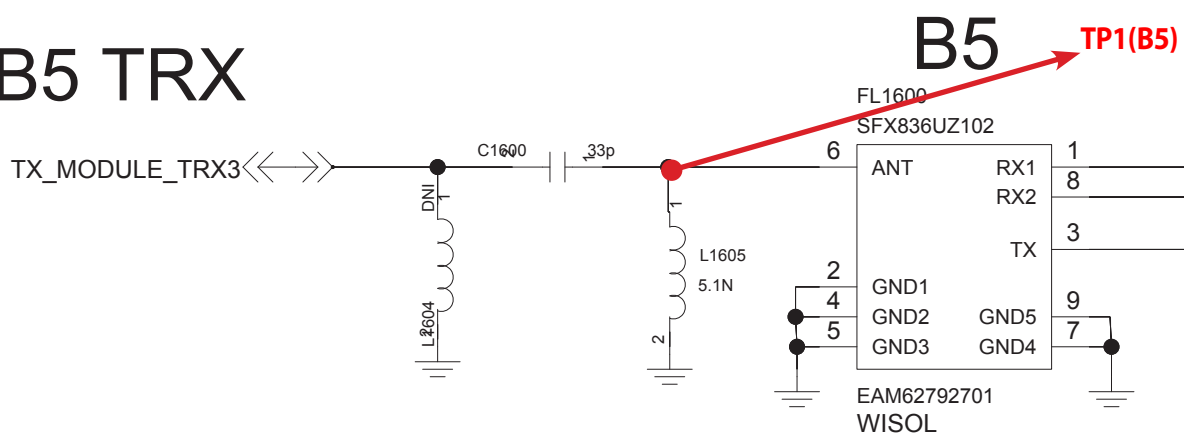
Figure 4.1.1 (c)



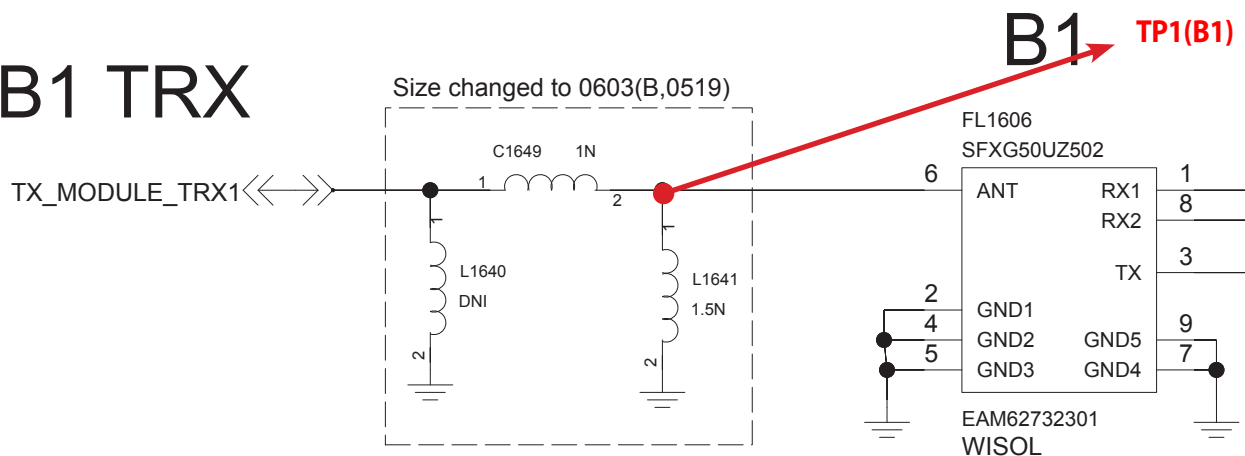
3.1.1.4 Checking RF signal path (FEM)



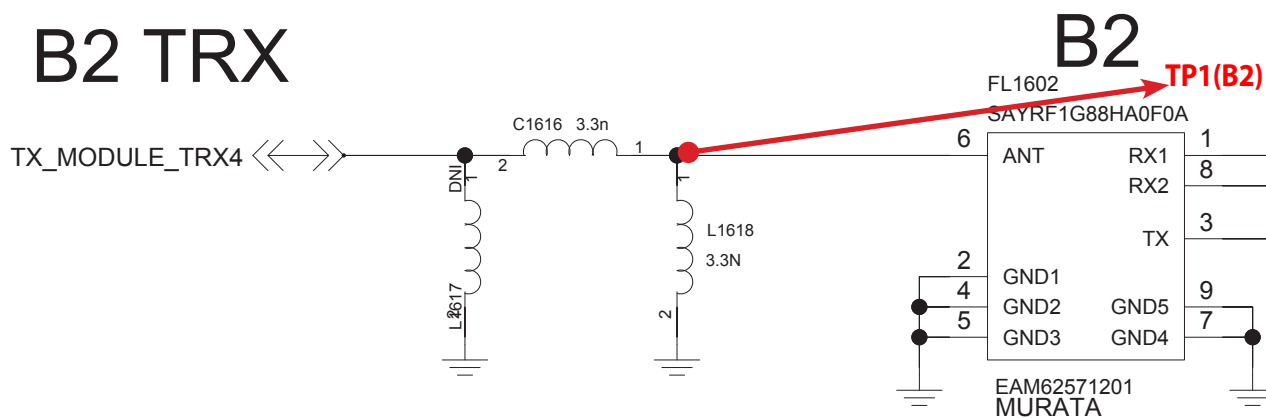
B5 TRX



B1 TRX



B2 TRX



B8 TRX

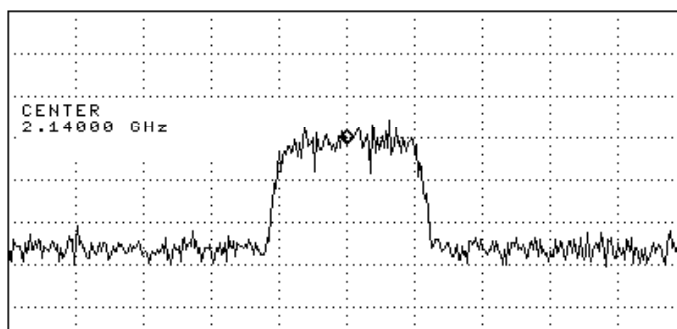
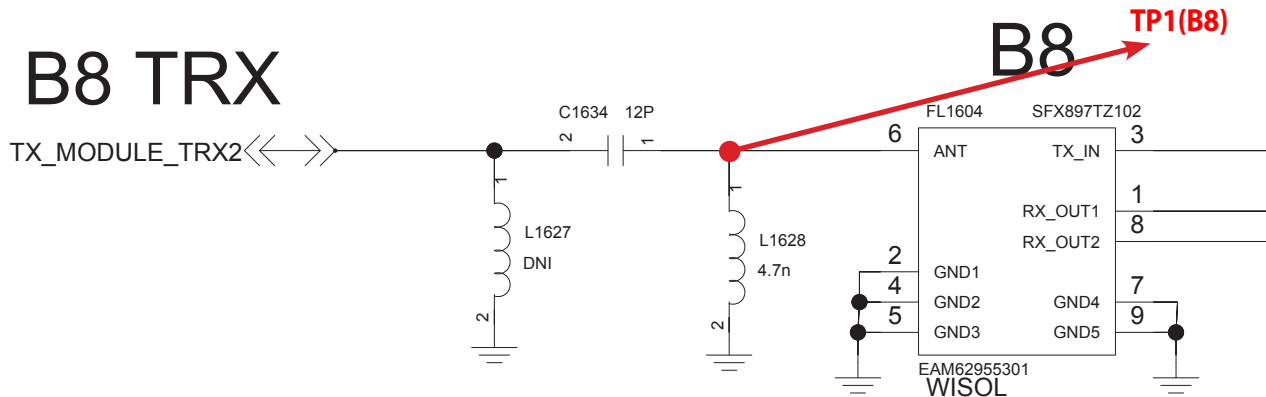


Figure 4.1.1 (b)

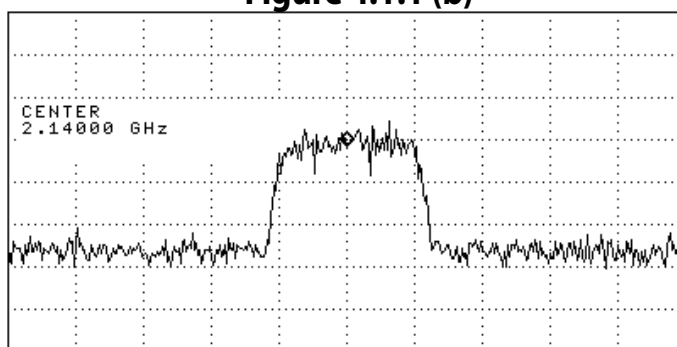
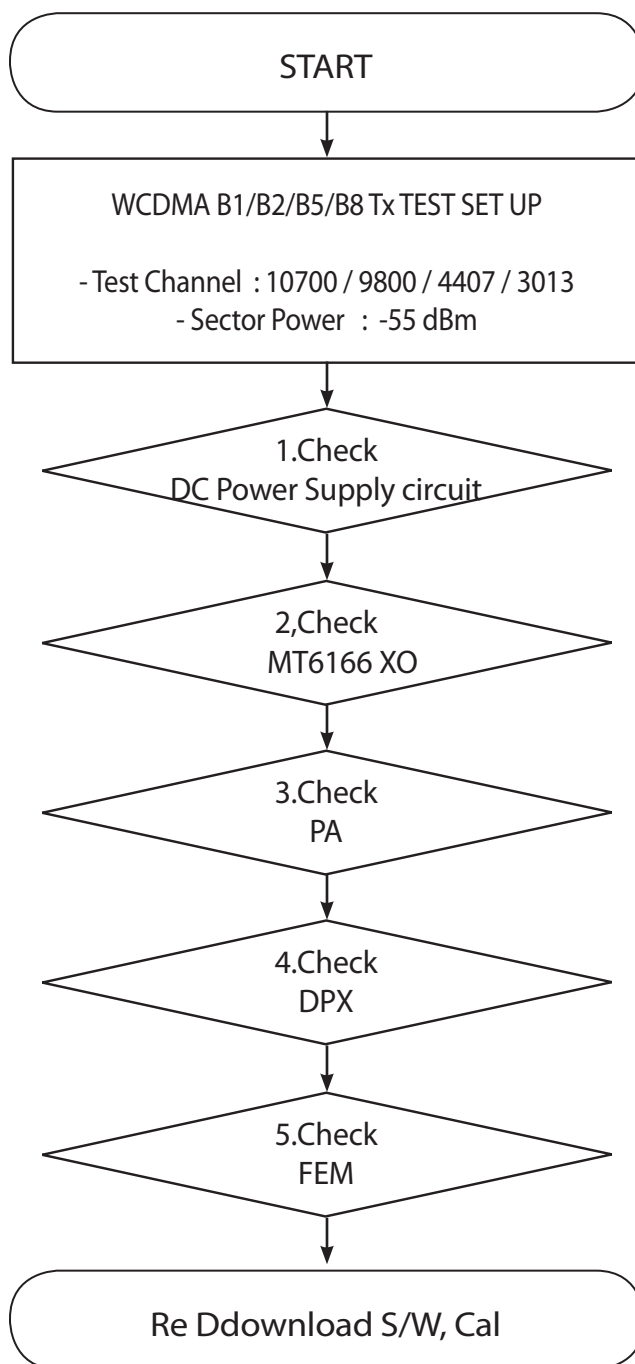


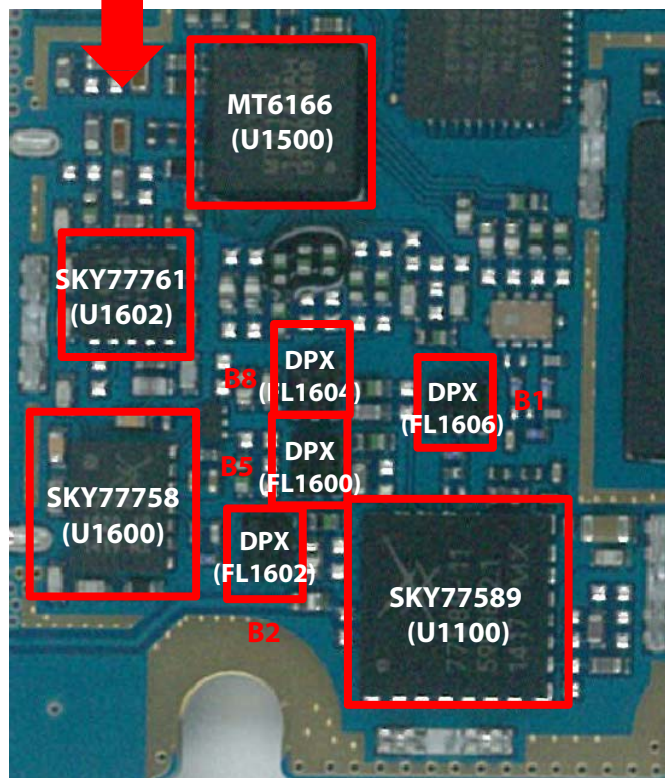
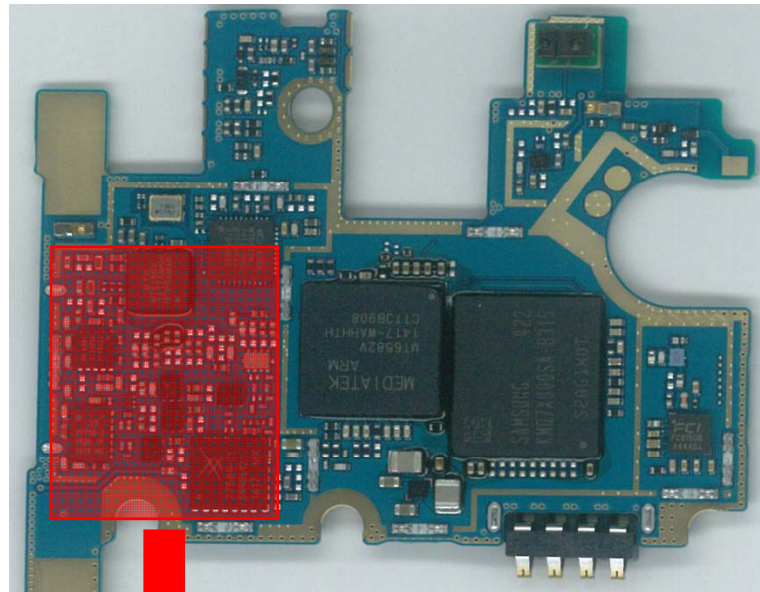
Figure 4.1.1 (c)

3.2 WCDMA Tx Part

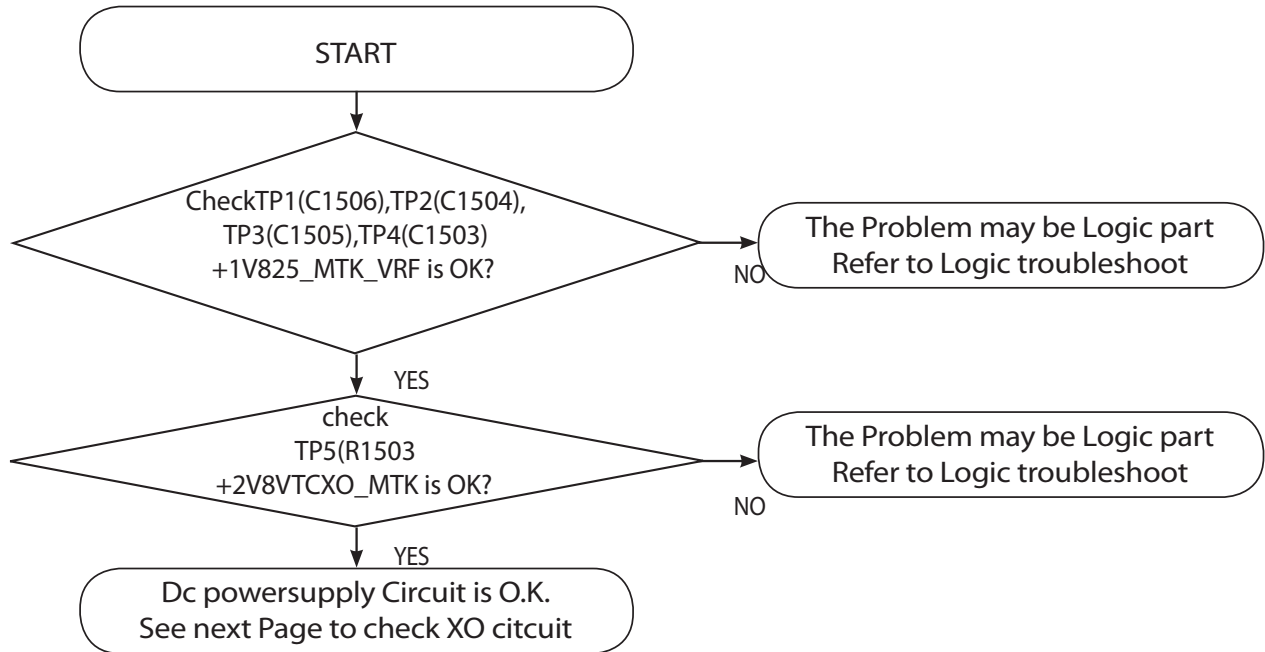
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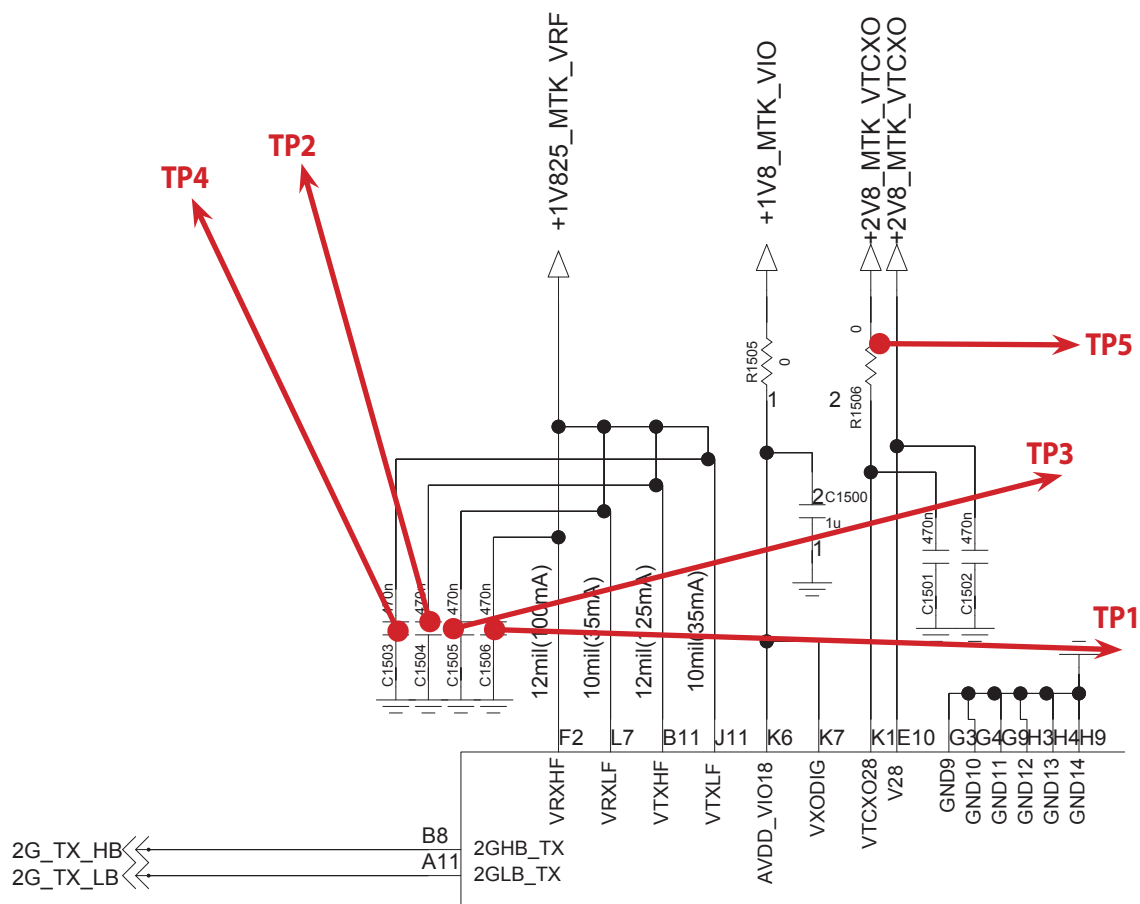
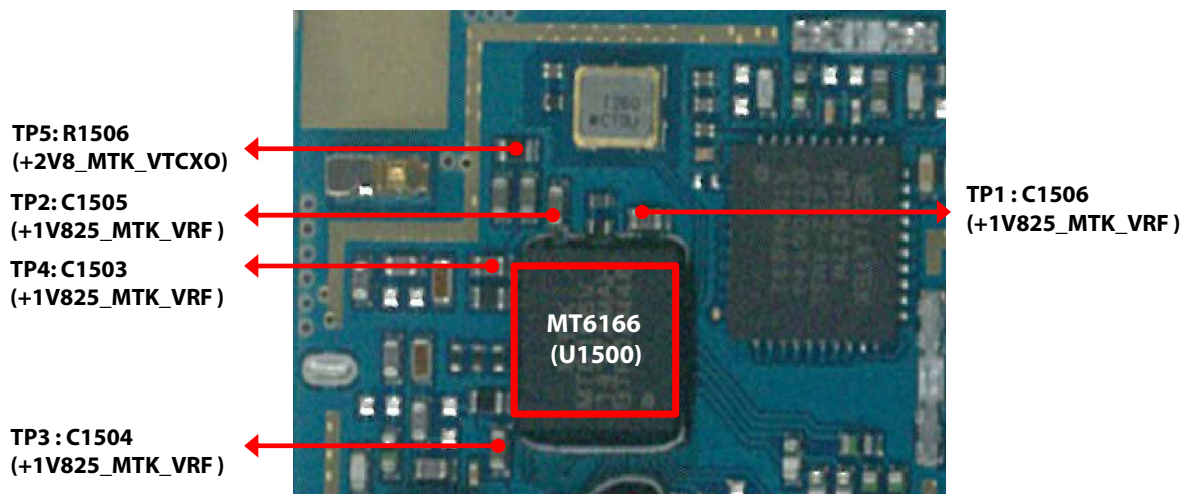


There are no Test Points on the IQ Data Line. So It is impossible to check the IQ Data Wave form



3.2.1.1 Checking DC Power supply circuit





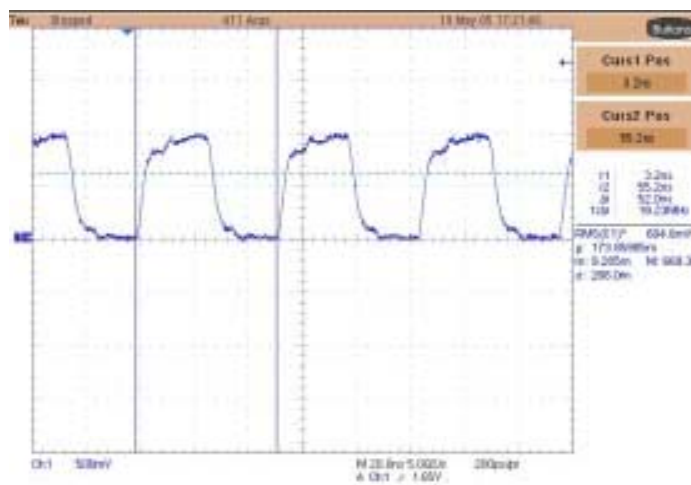
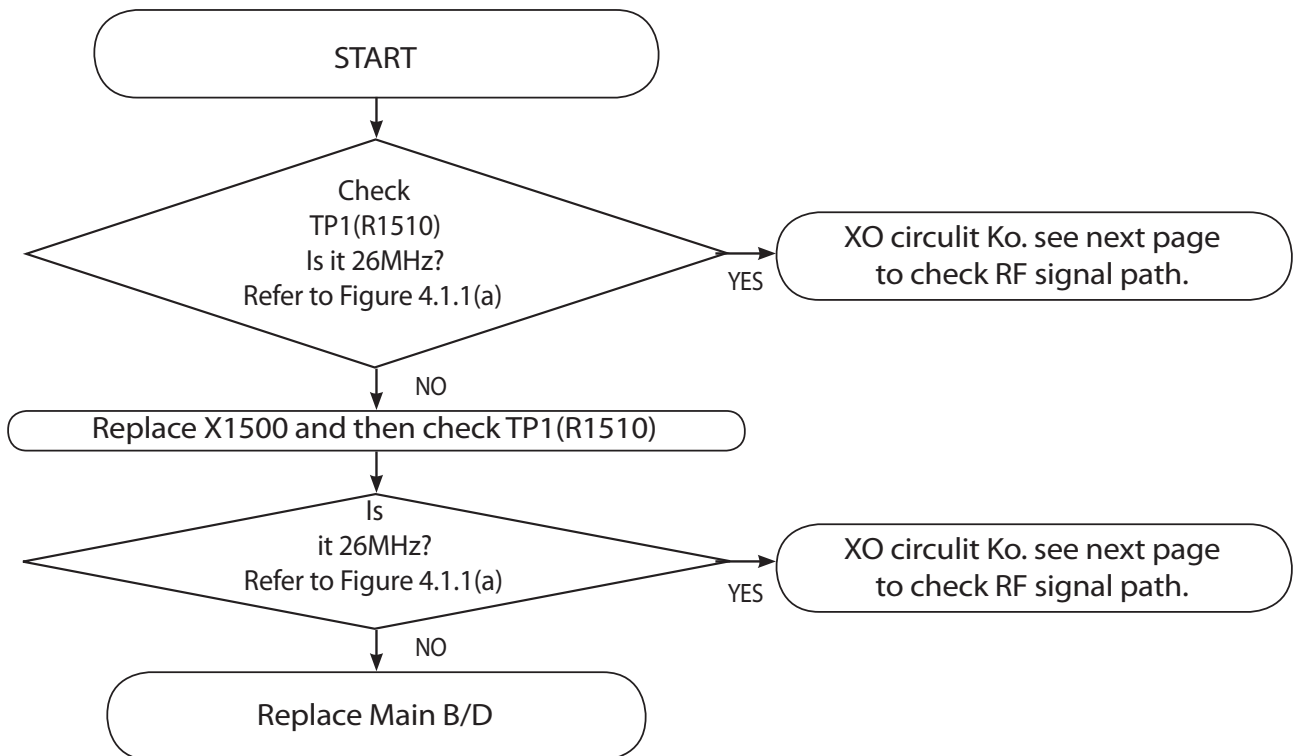
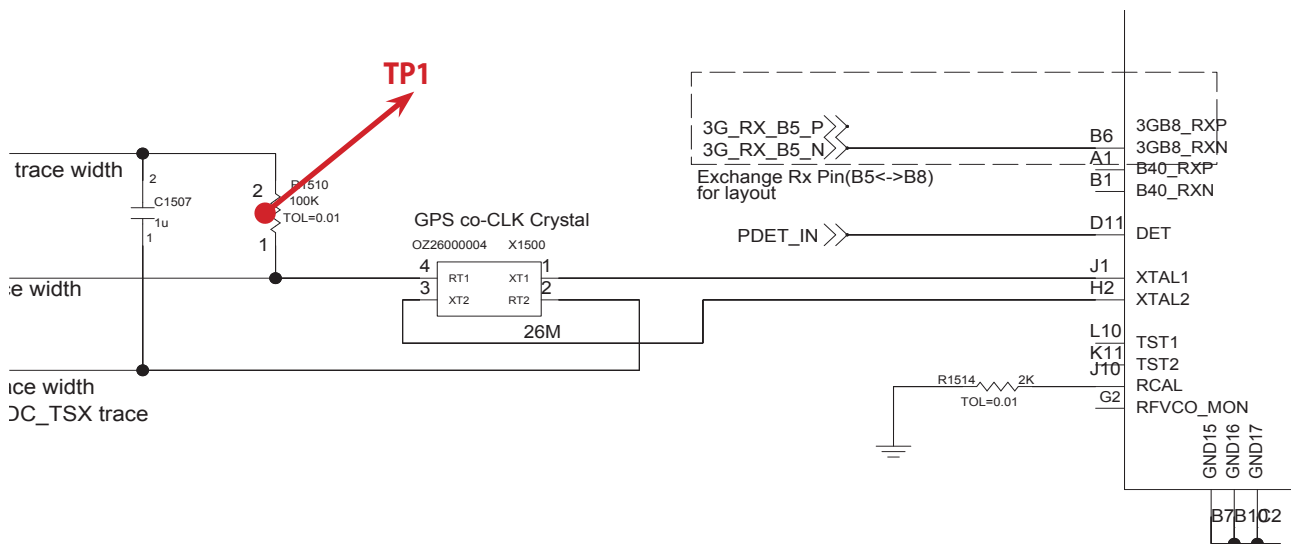
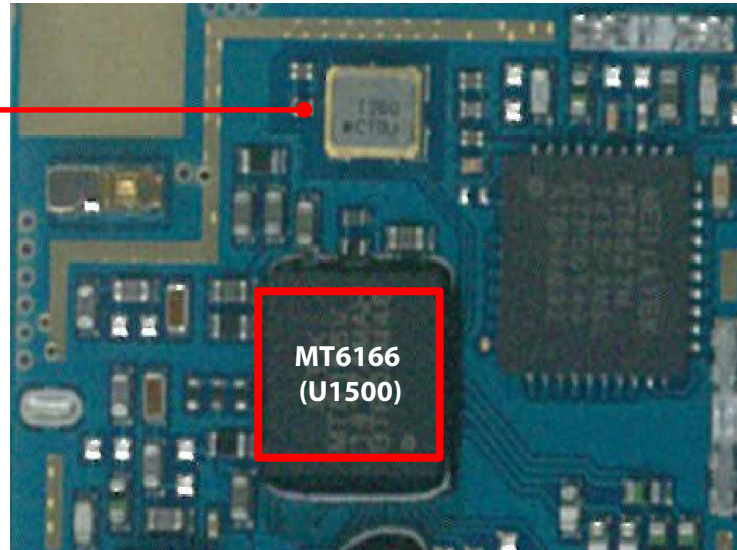
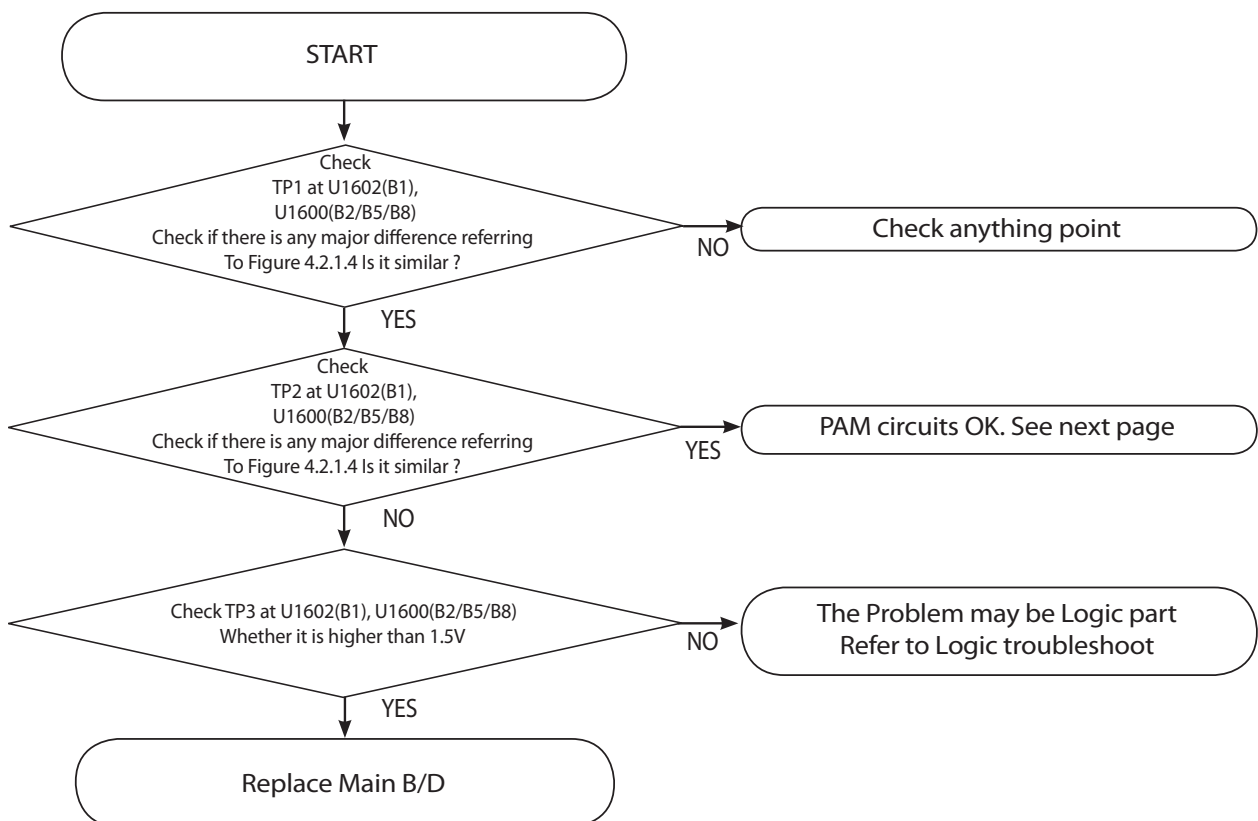


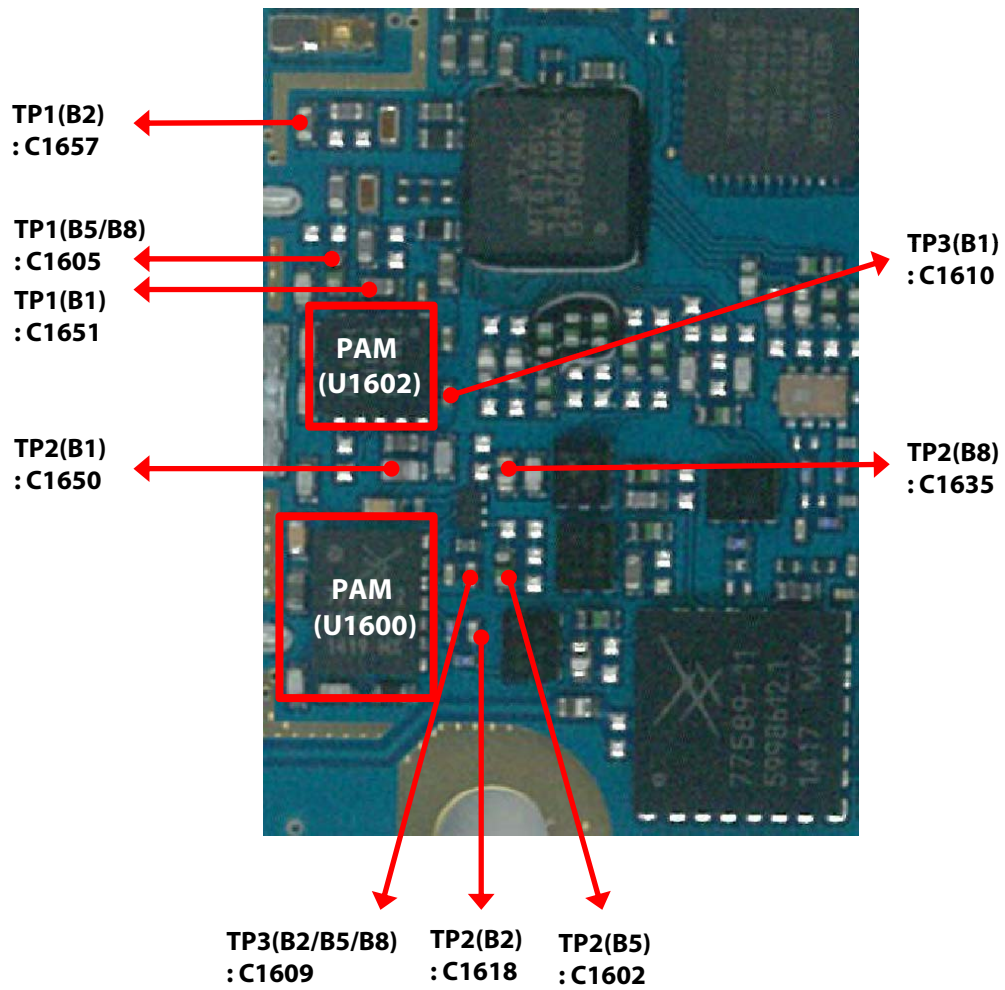
Figure 4.1.1 (a)

3.2.1.2 Checking MT6166 XO circuit

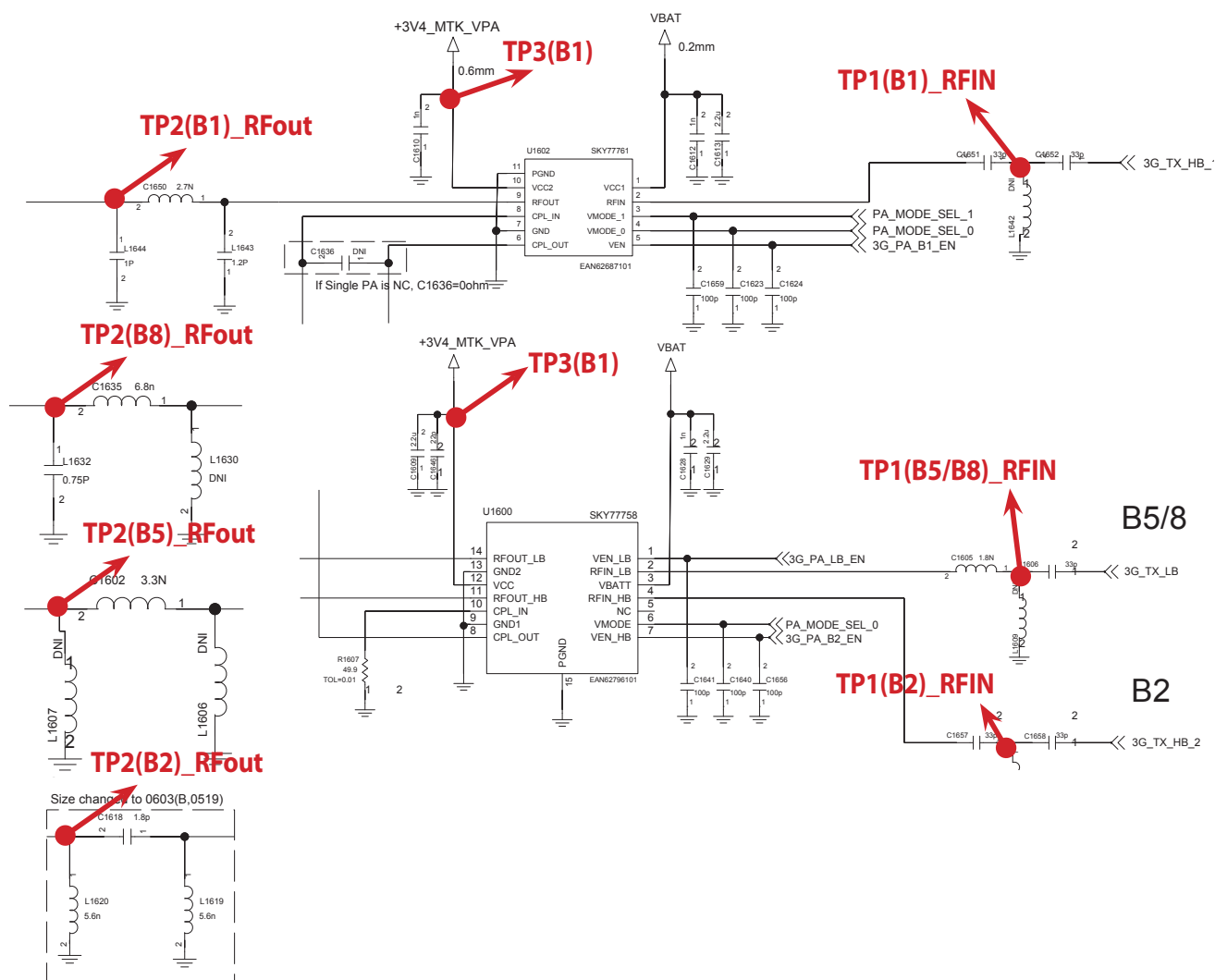
TP1 : R1510
(MAIN_CLK_26MHz)

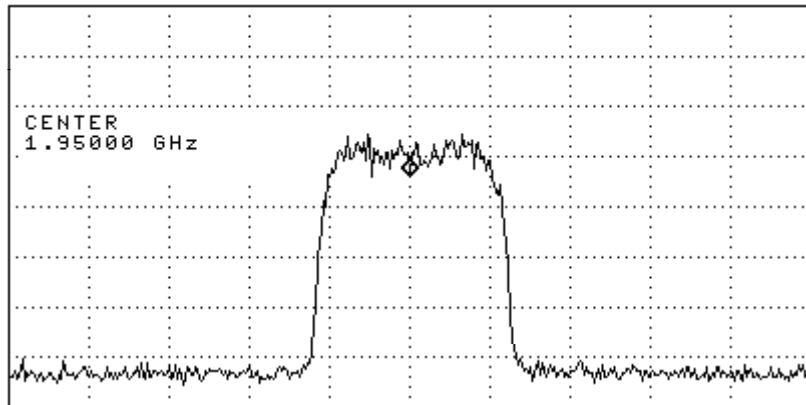


3.2.1.3 Checking B1/B2/B5/B8 TX PA

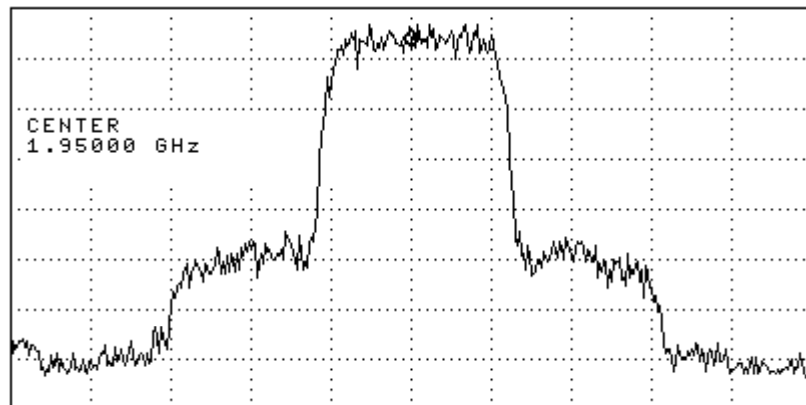


4. DOWNLOAD



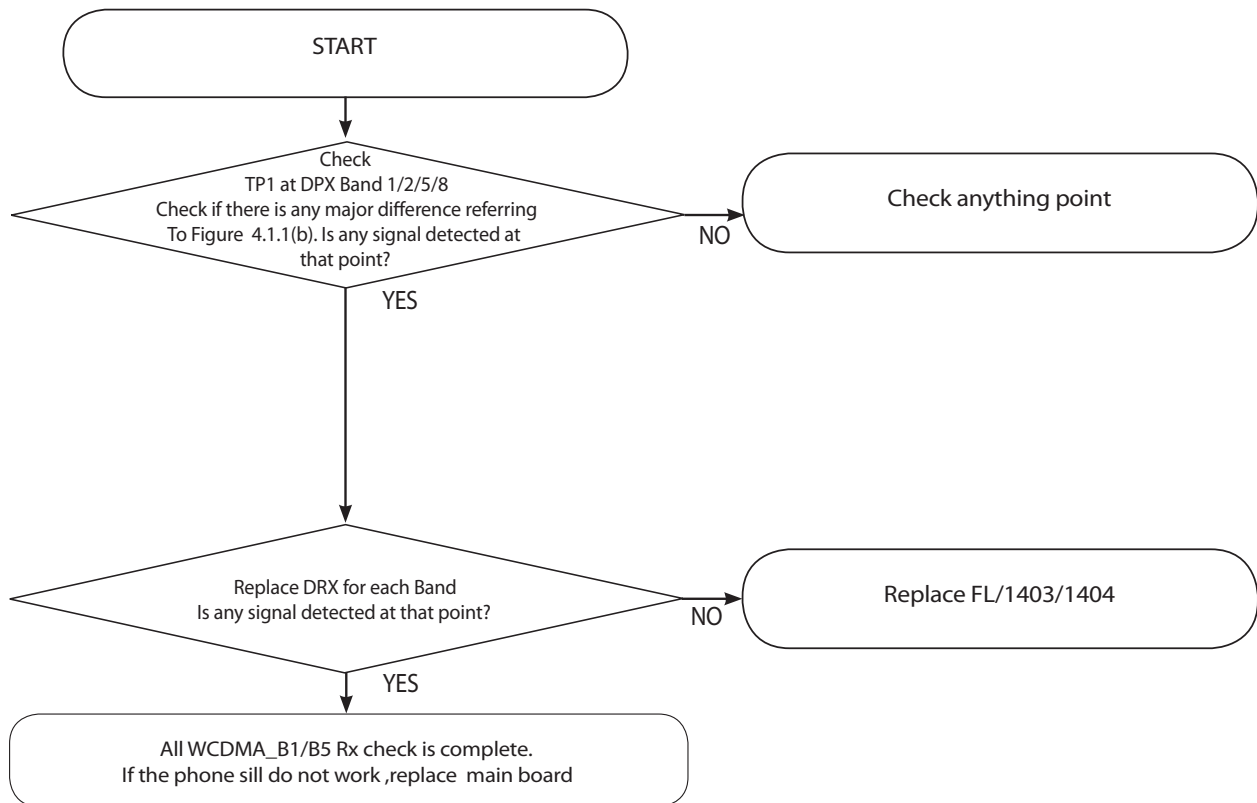


TP1_RF IN

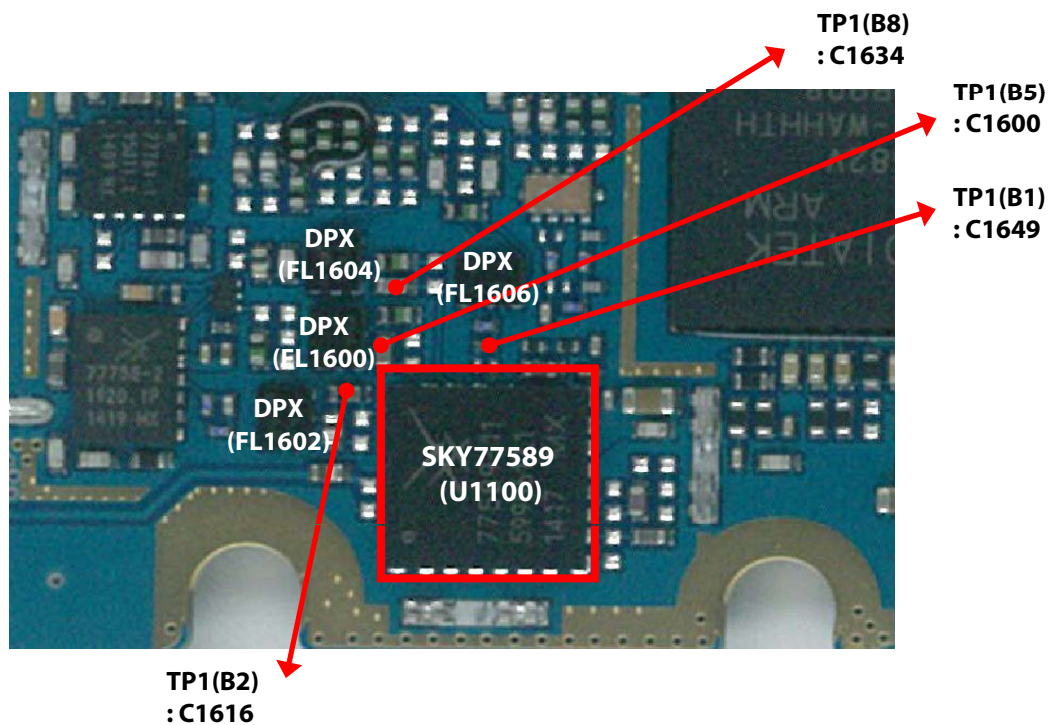


TP2_RF out

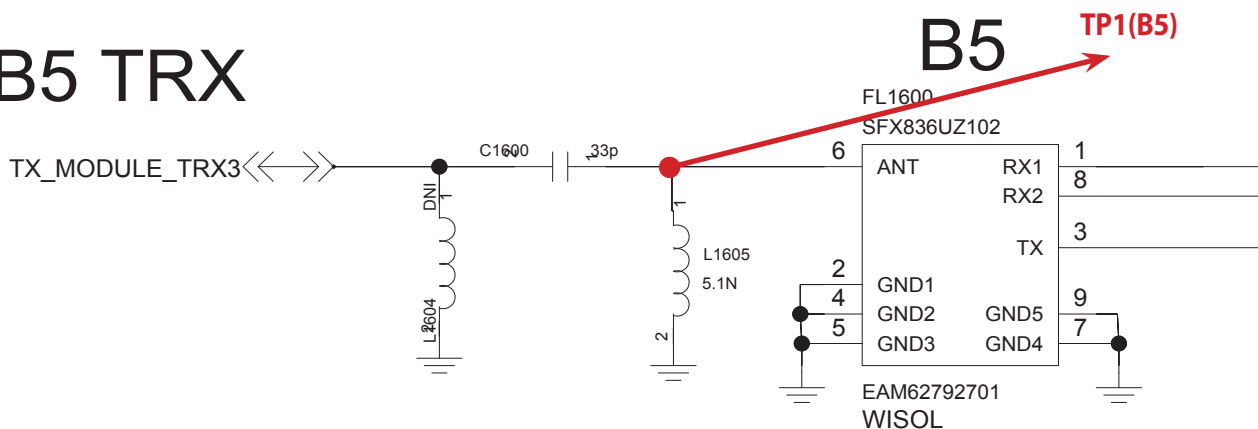
3.2.1.4 Checking RF signal path (DPX)



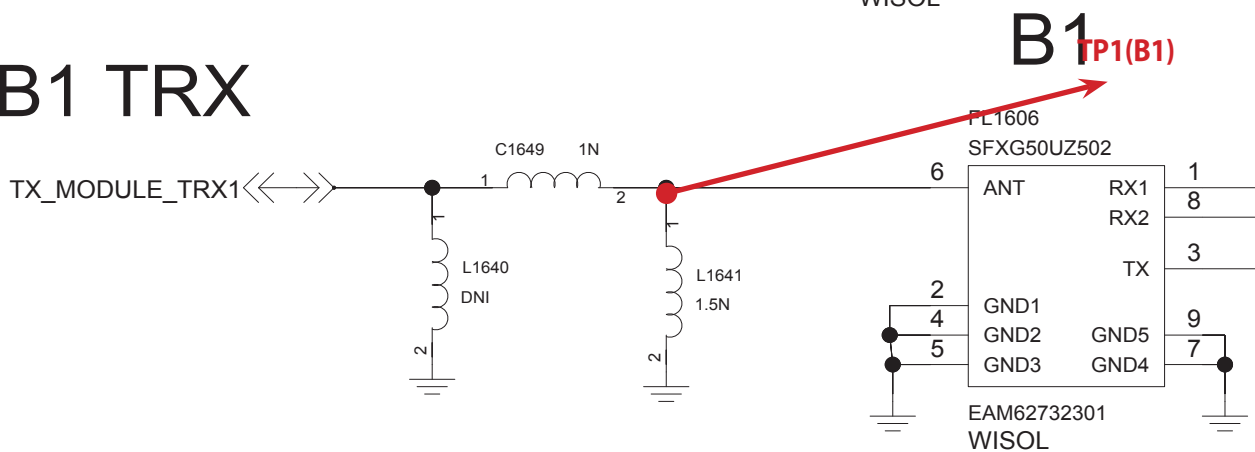
3.2.1.4 Checking RF signal path (DPX)

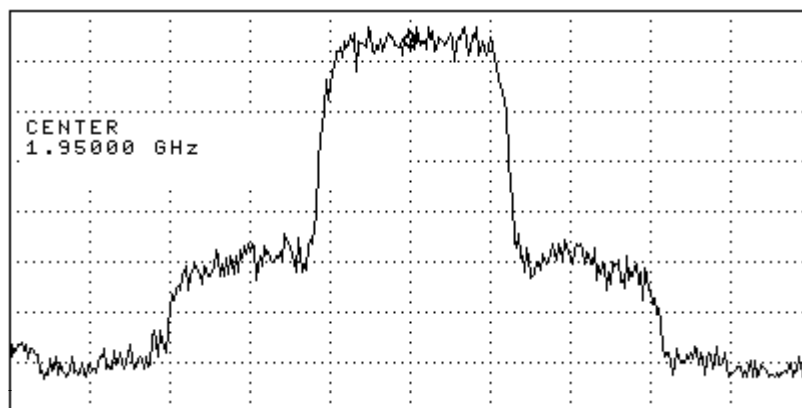
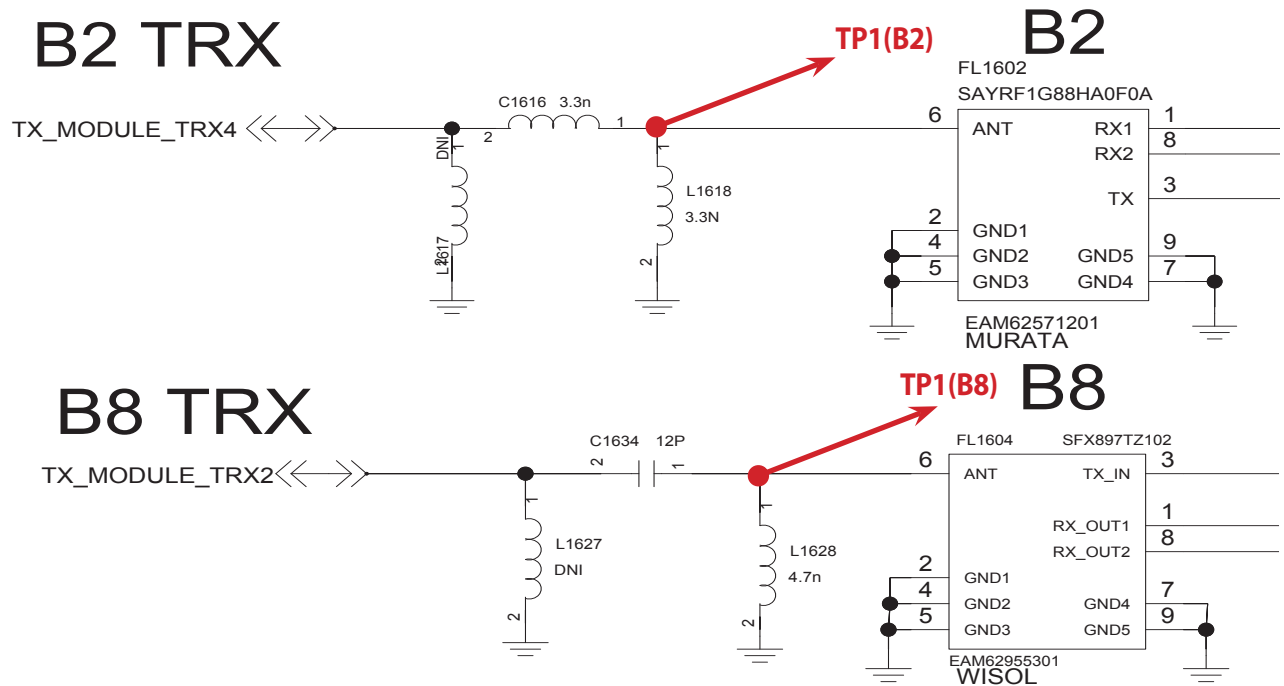


B5 TRX



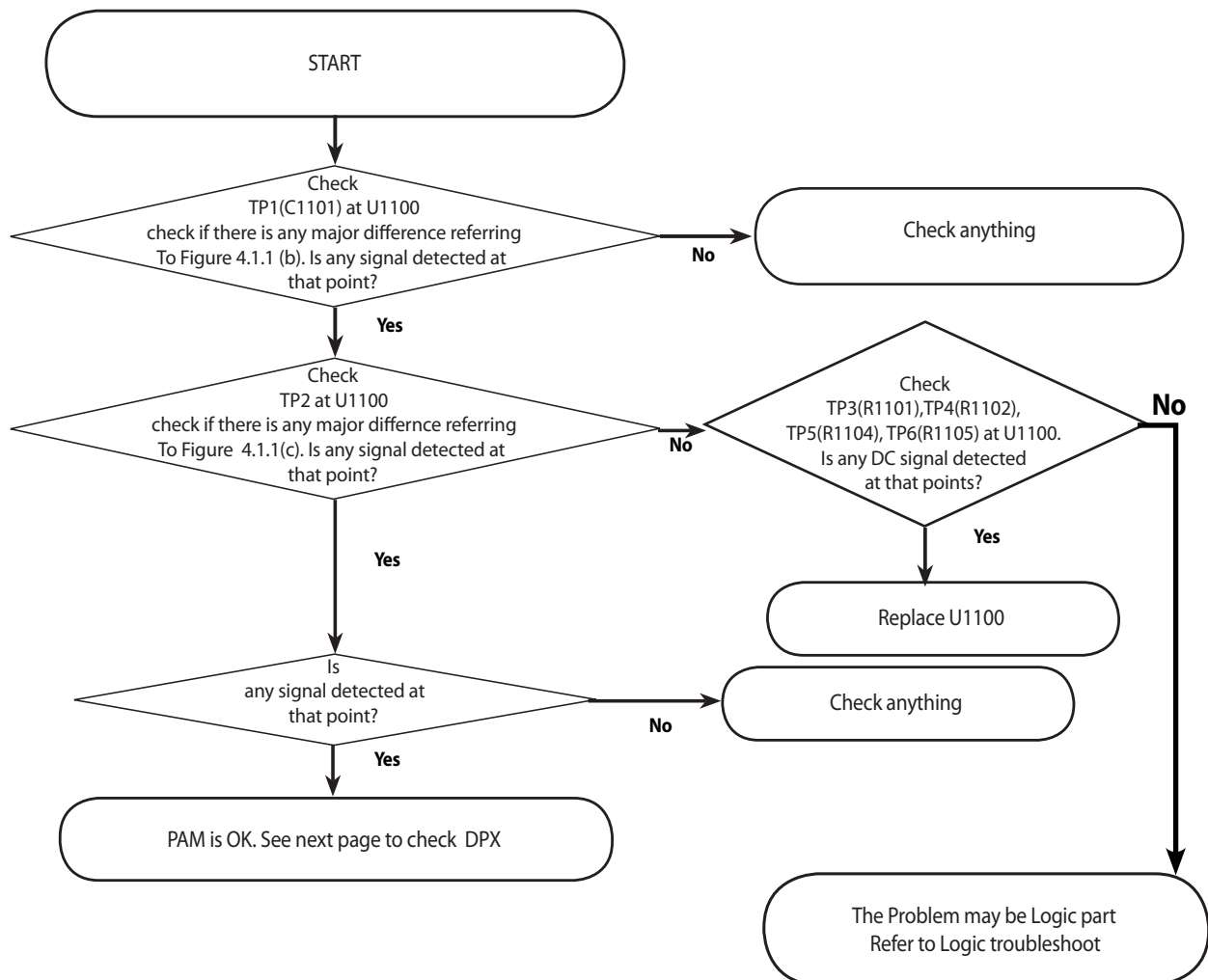
B1 TRX

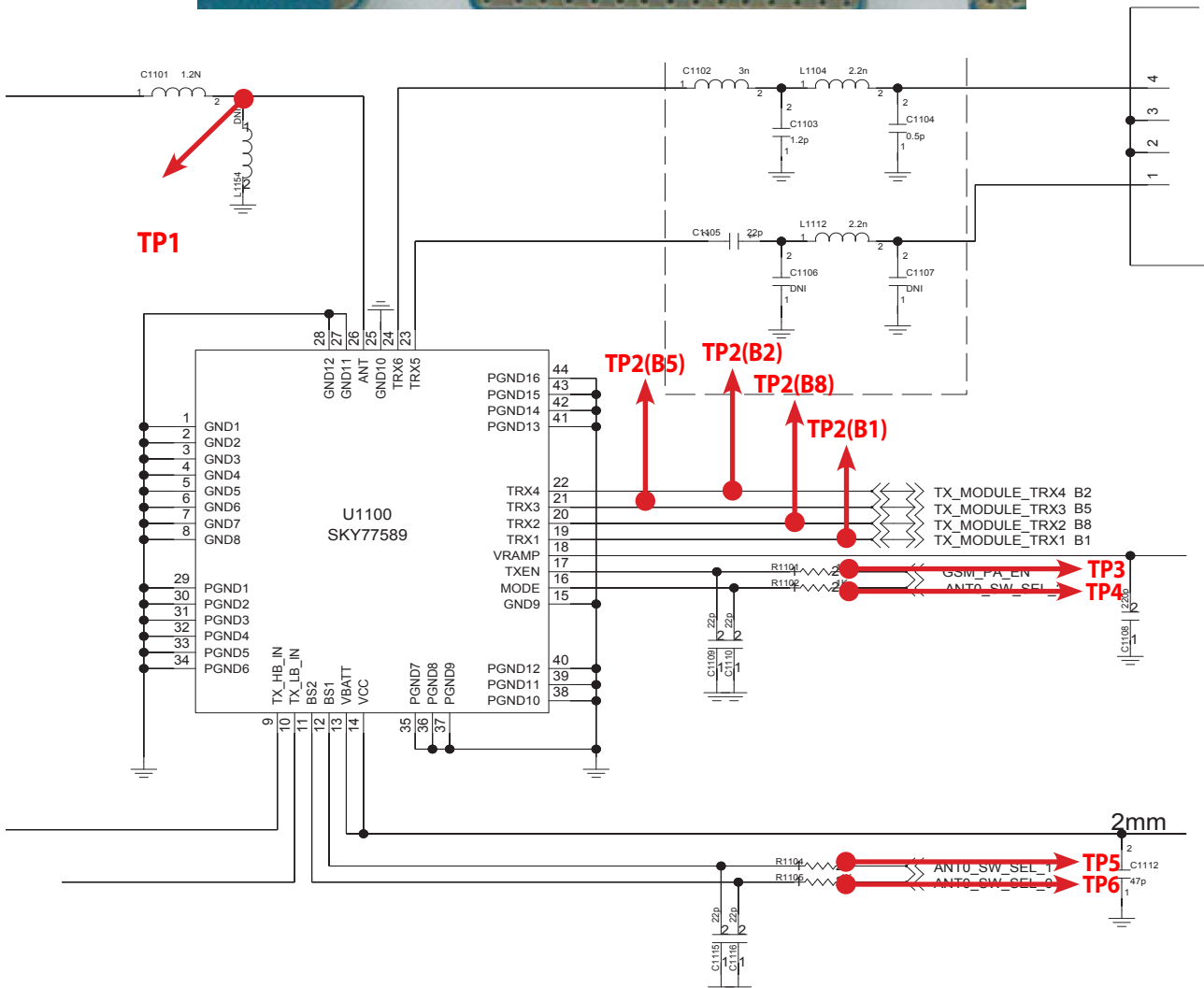
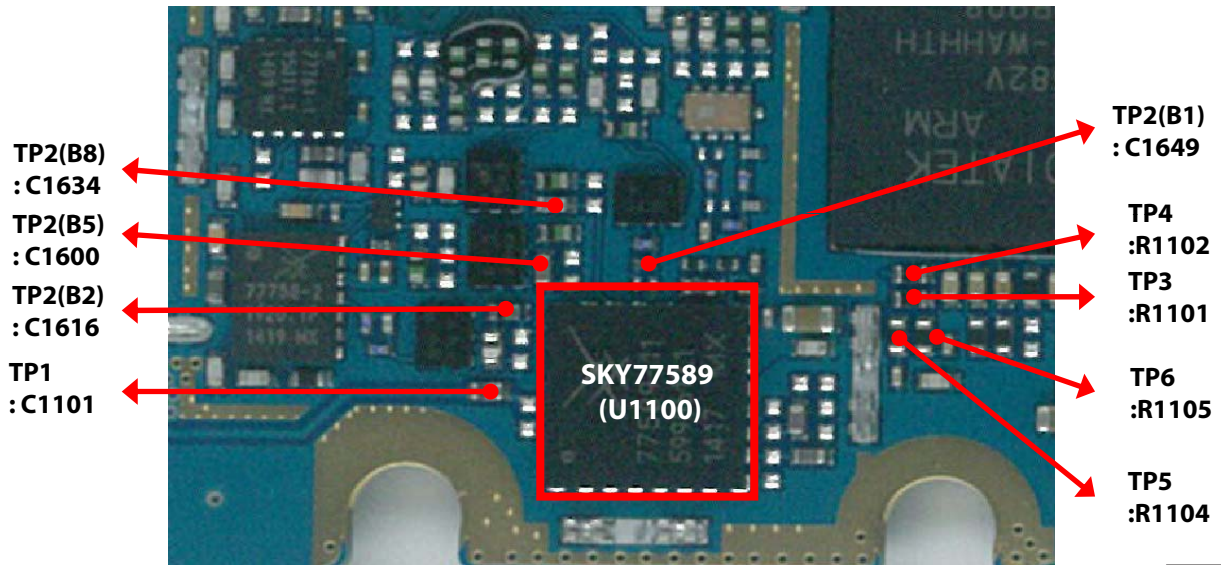




Figuer4.1.1(b)

3.2.1.5 Checking RF signal path (FEM)





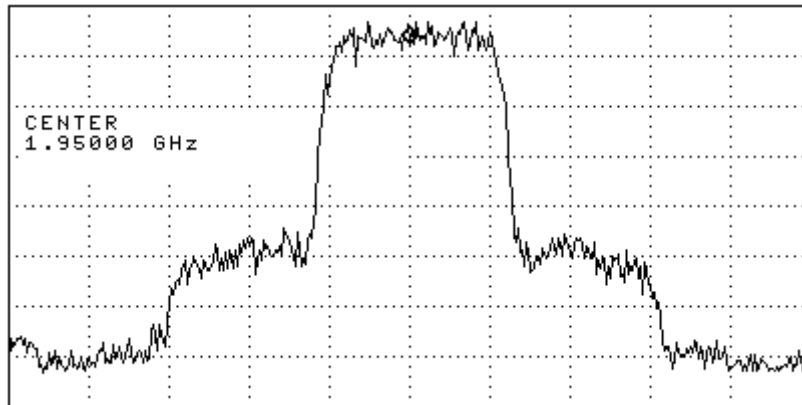
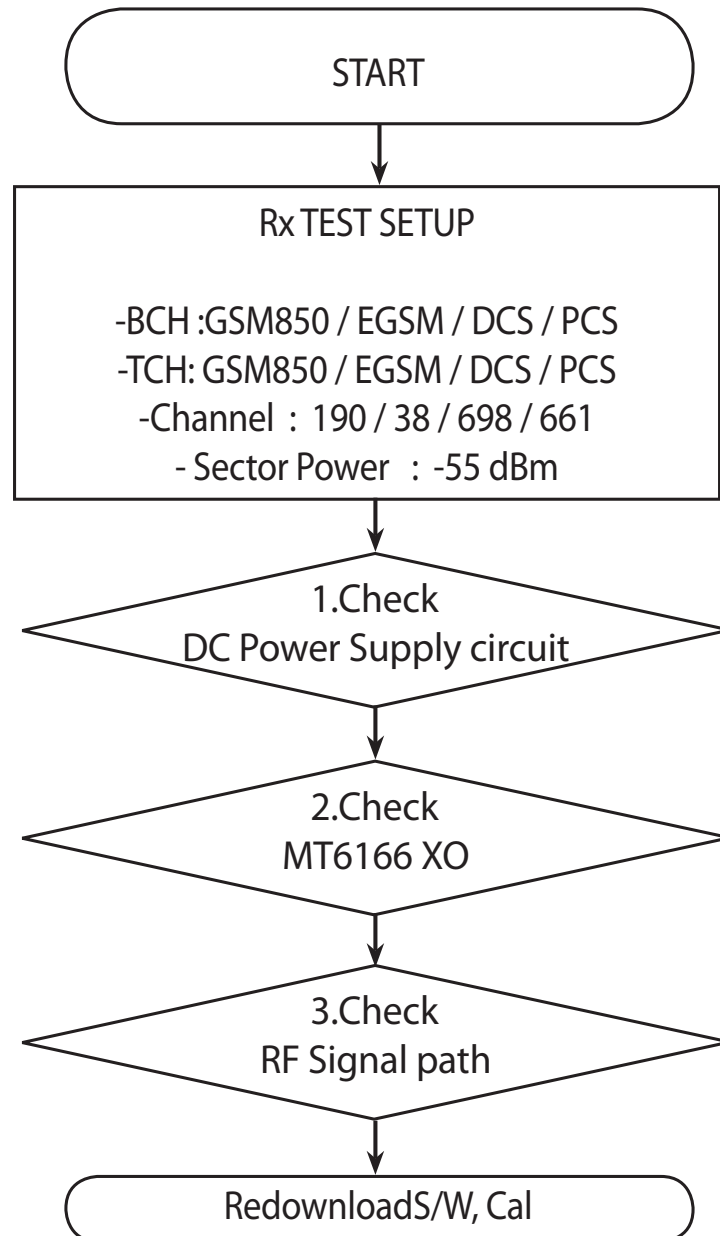


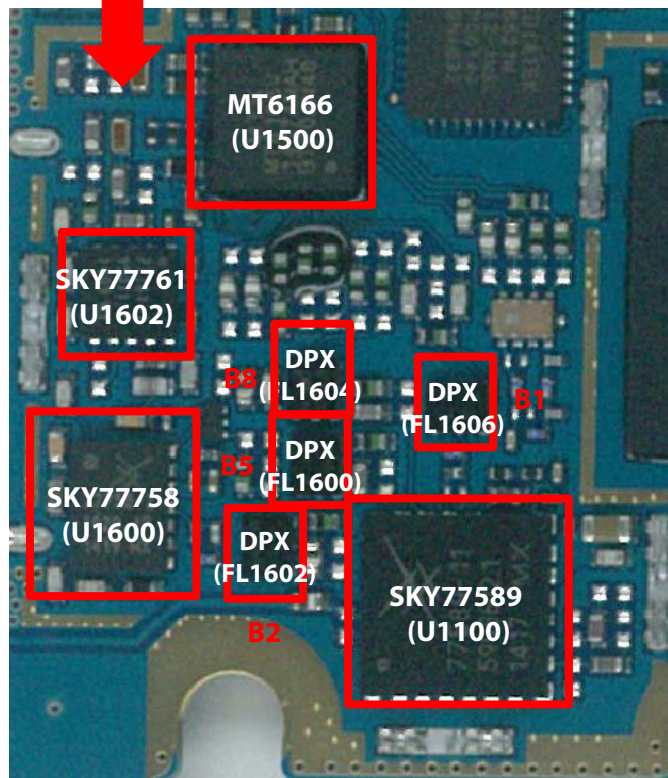
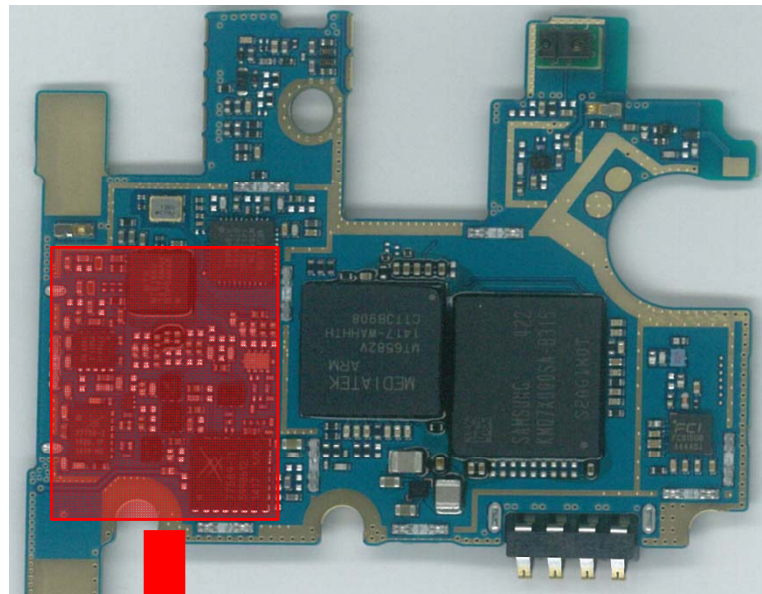
Figure 4.1.1(b)

3.3 GSM Rx Part

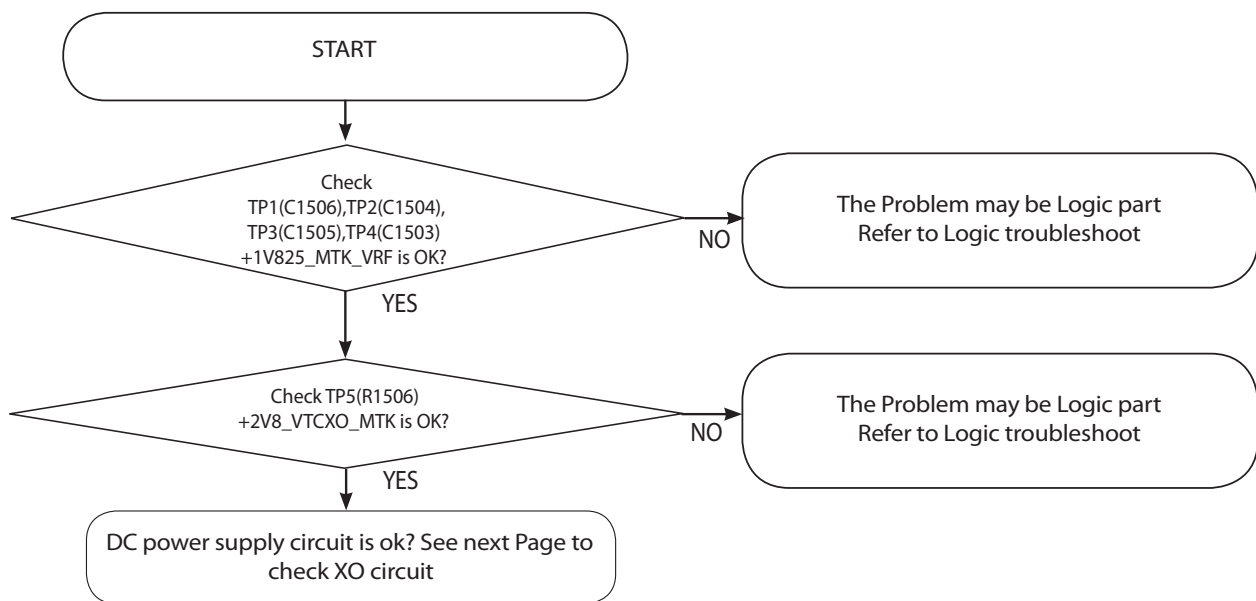
3.3.1 GSM850/900/DCS/PCS Rx



there are no Test Points on the IQ Data Line. So It is impossible to check the IQ Data Wave form



3.3.1.1 Checking DC Power supply circuit



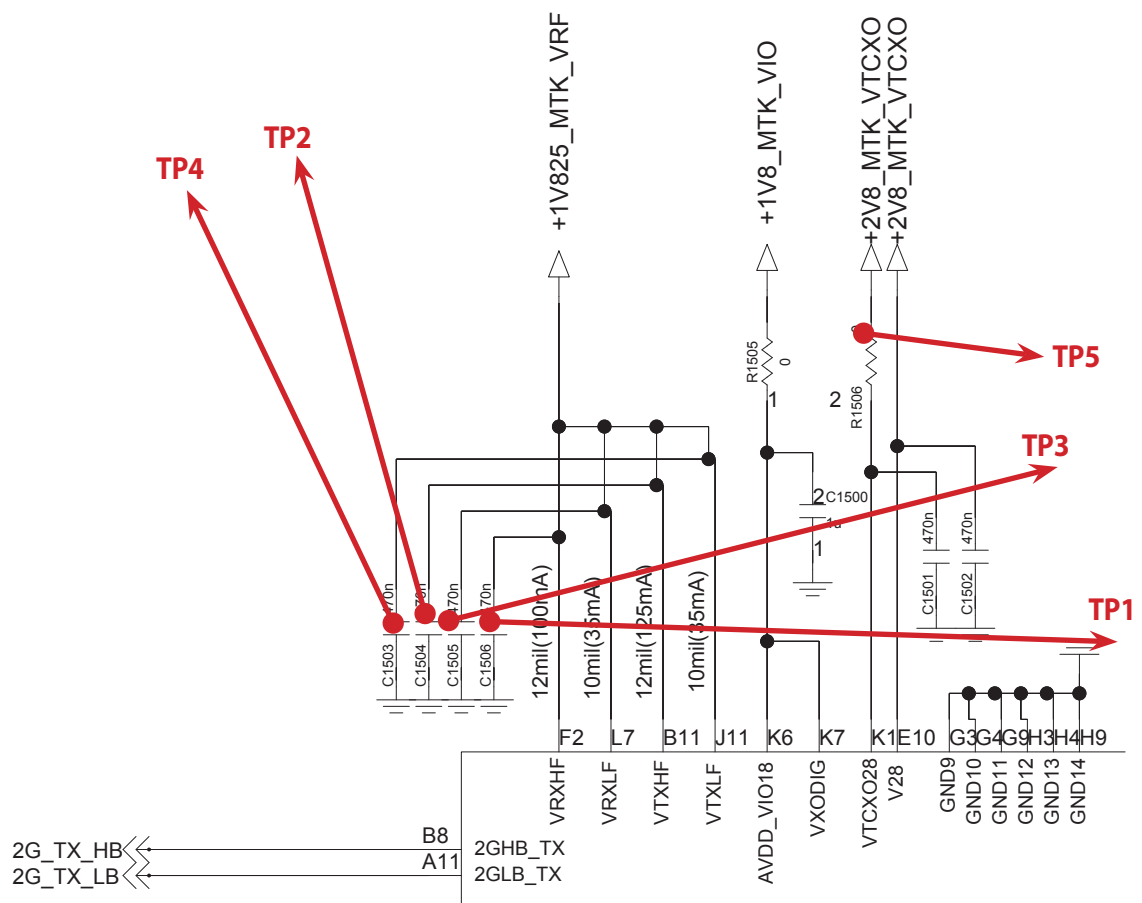
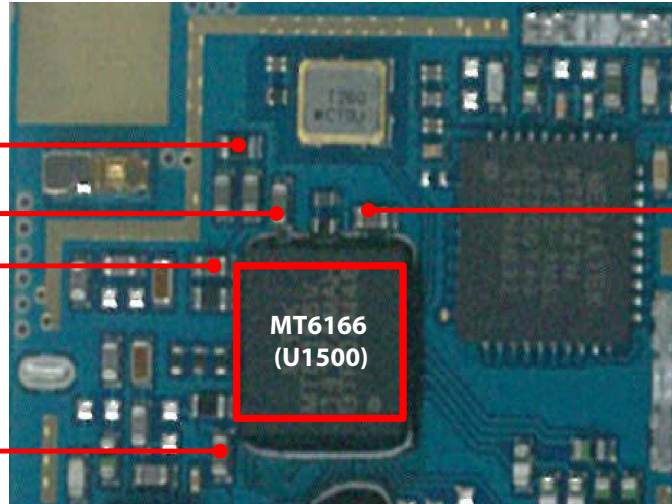
TP5: R1506
(+2V8_MTK_VTCXO)

TP2: C1505
(+1V825_MTK_VRF)

TP4: C1503
(+1V825_MTK_VRF)

TP3: C1504
(+1V825_MTK_VRF)

TP1: C1506
(+1V825_MTK_VRF)



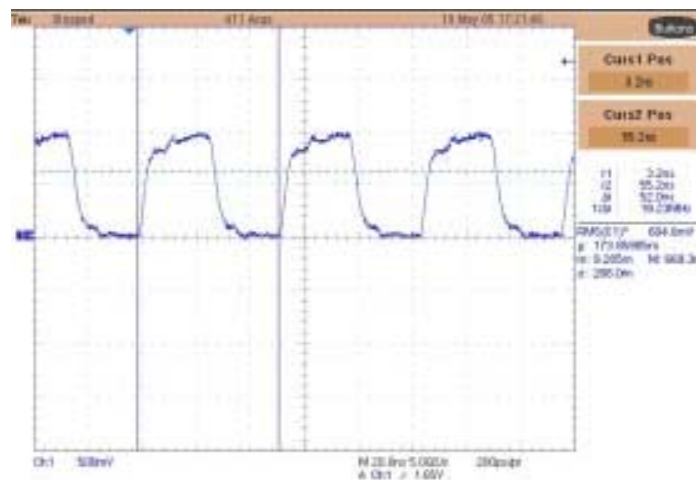
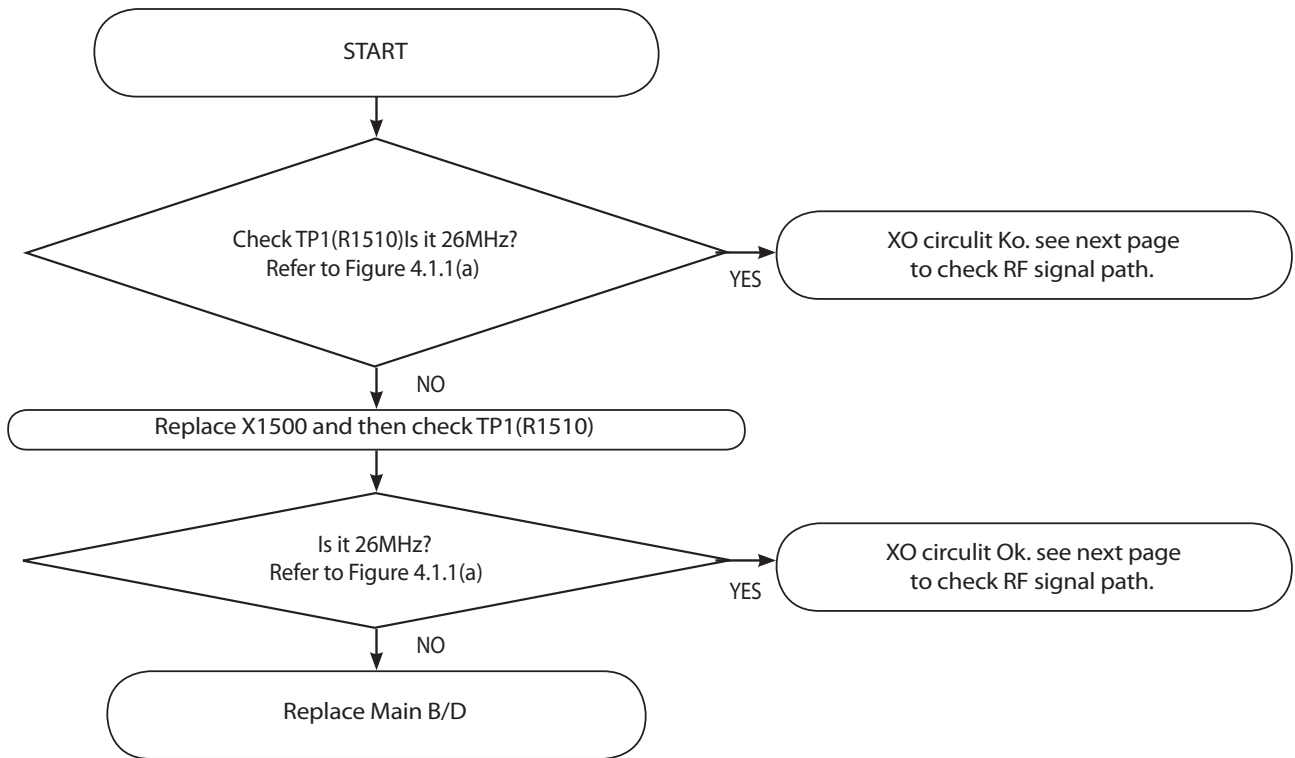
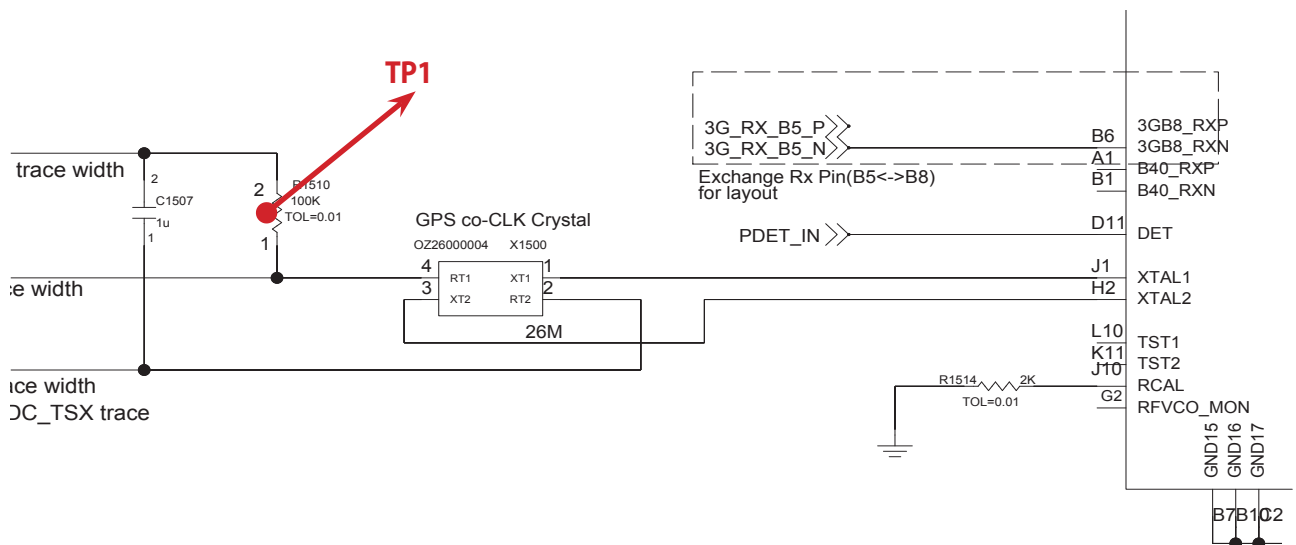
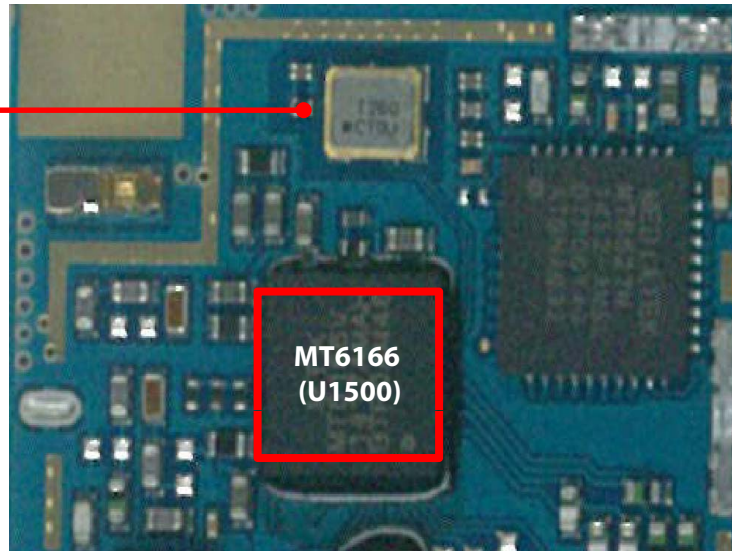


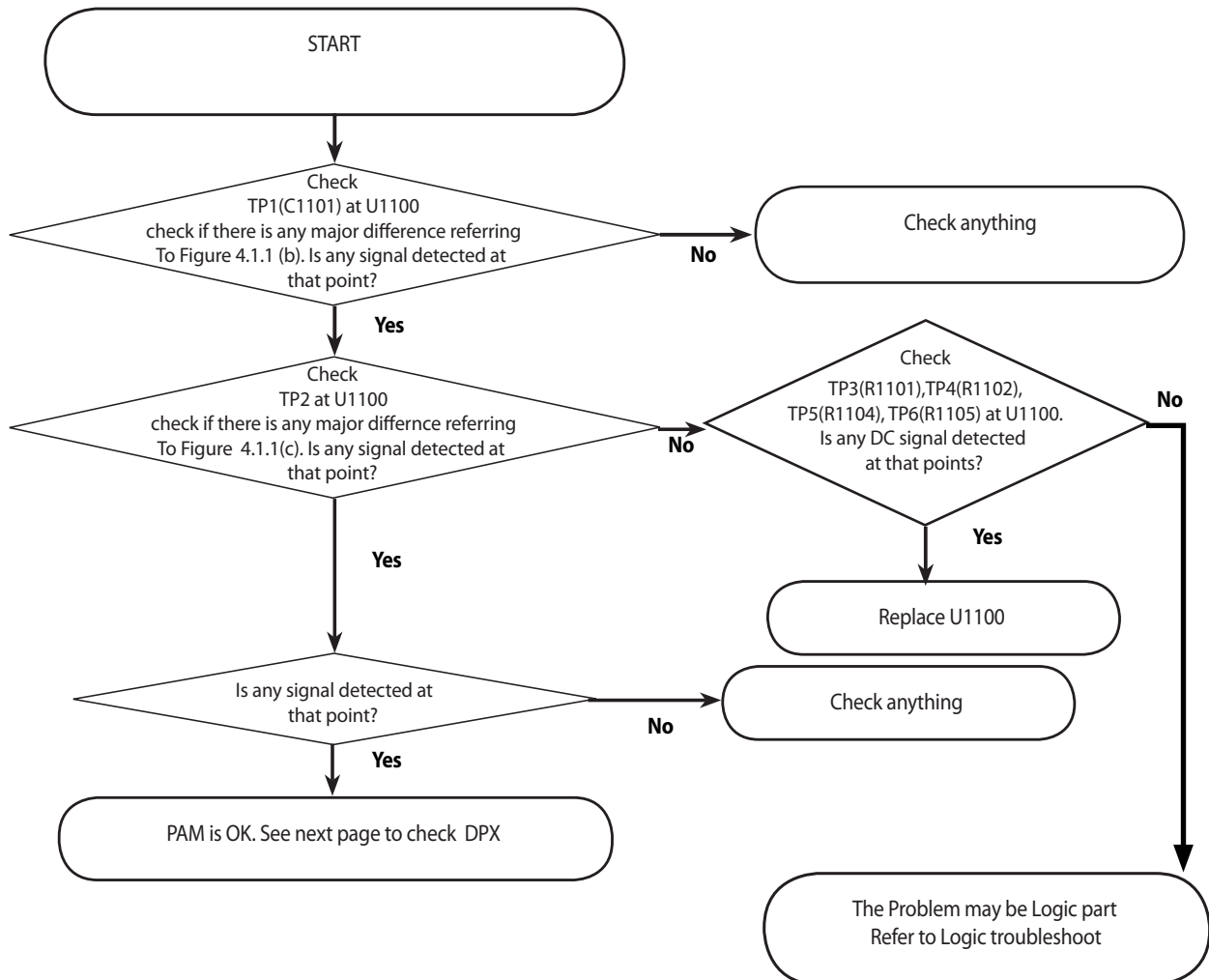
Figure 4.1.1(a)

3.3.1.2 Checking MT6166 XO circuit

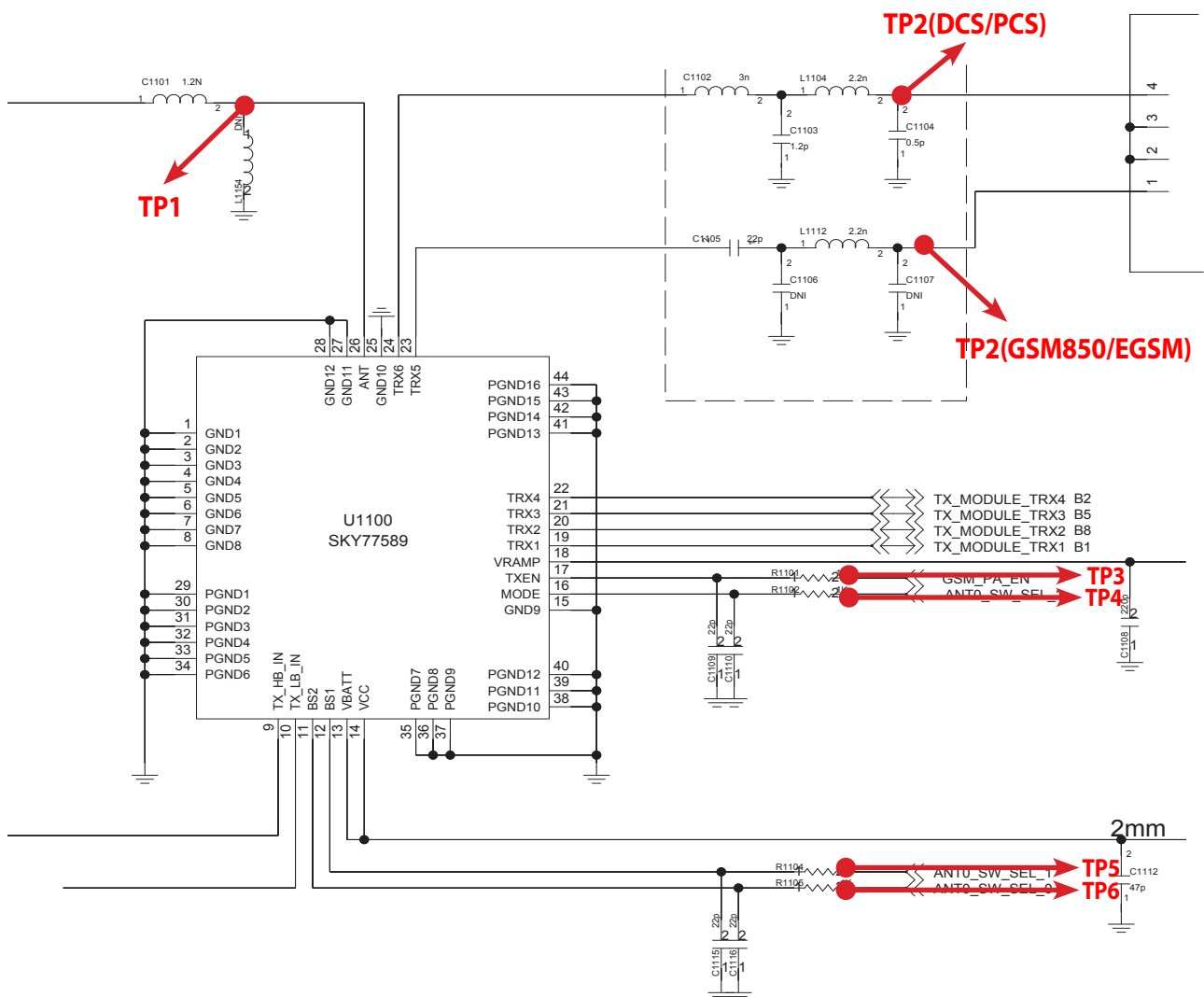
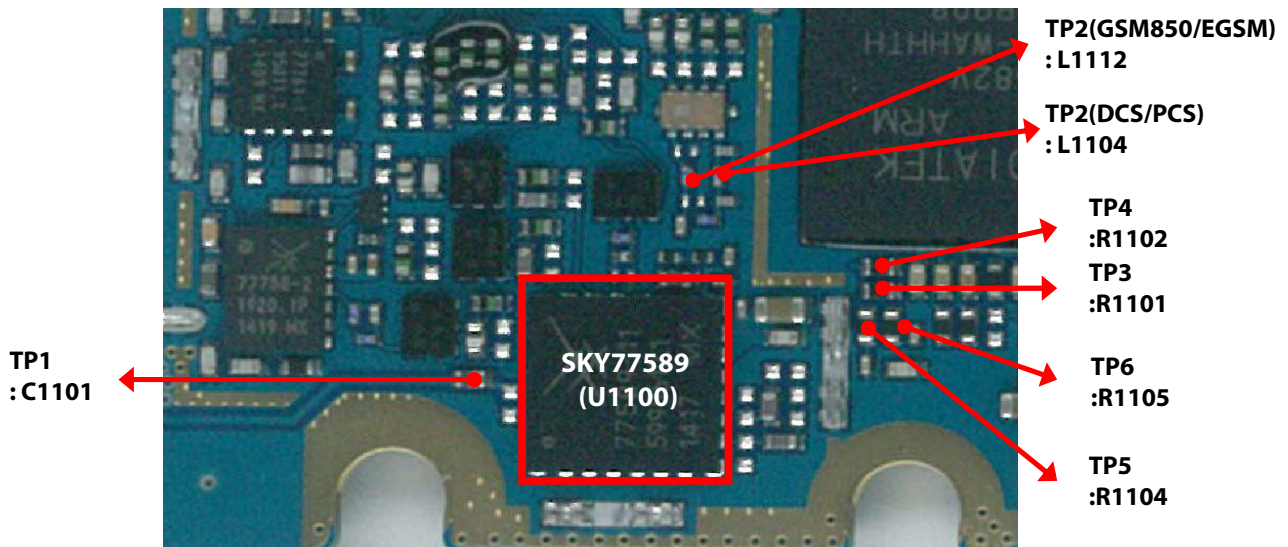
TP1 : R1510
(MAIN_CLK_26MHz)



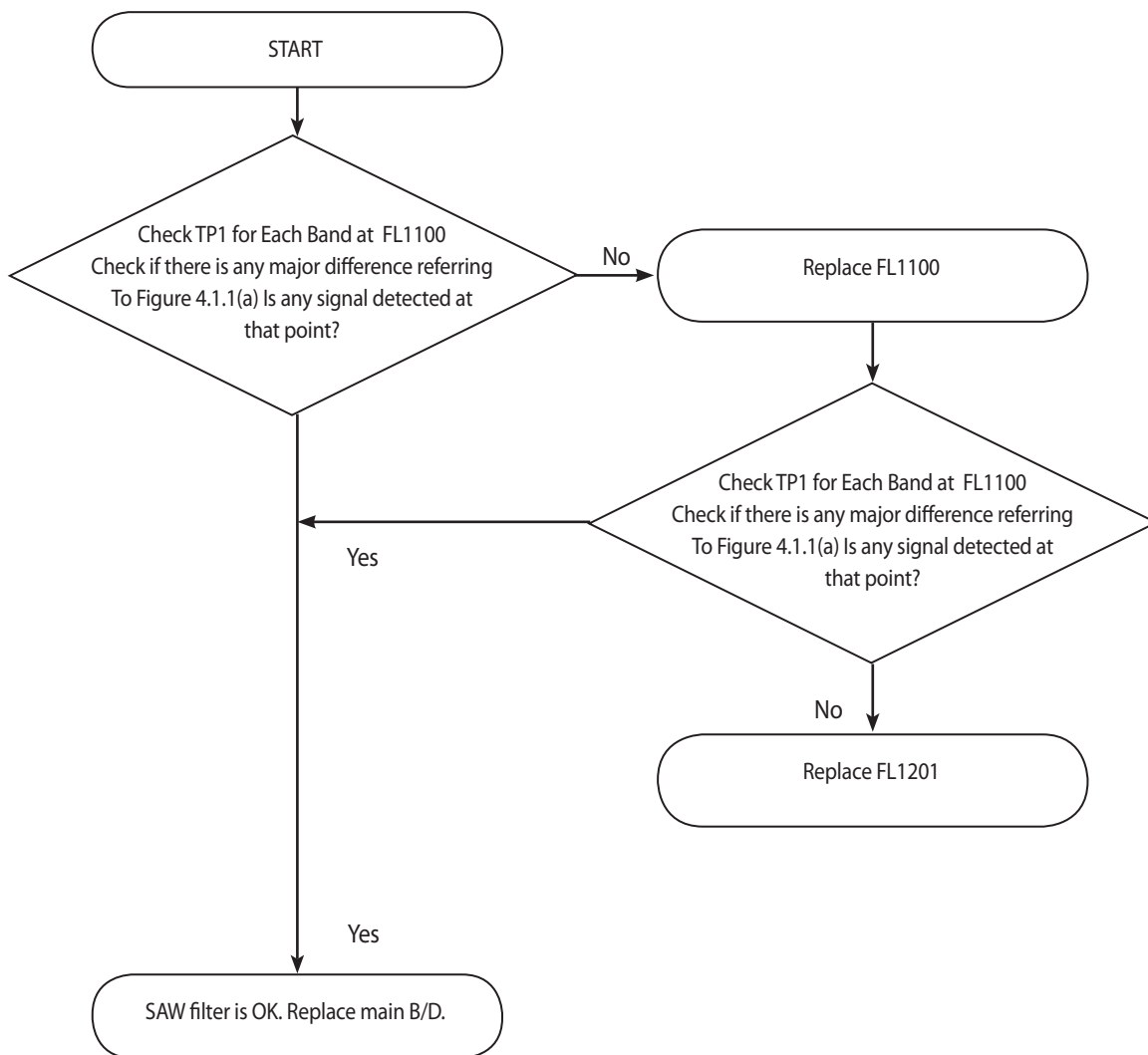
3.3.1.3 Checking RF signal path (FEM)



4. DOWNLOAD



3.3.1.4 Checking RF Flow(SAW filter)



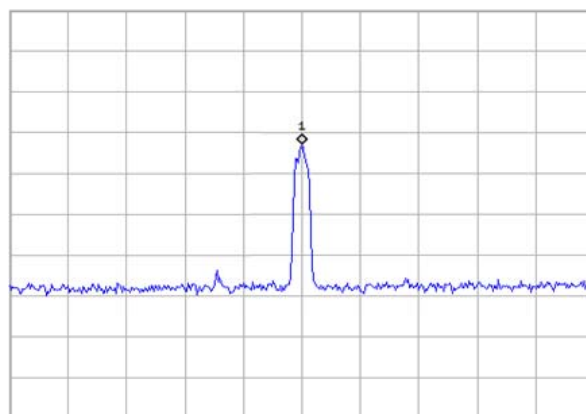
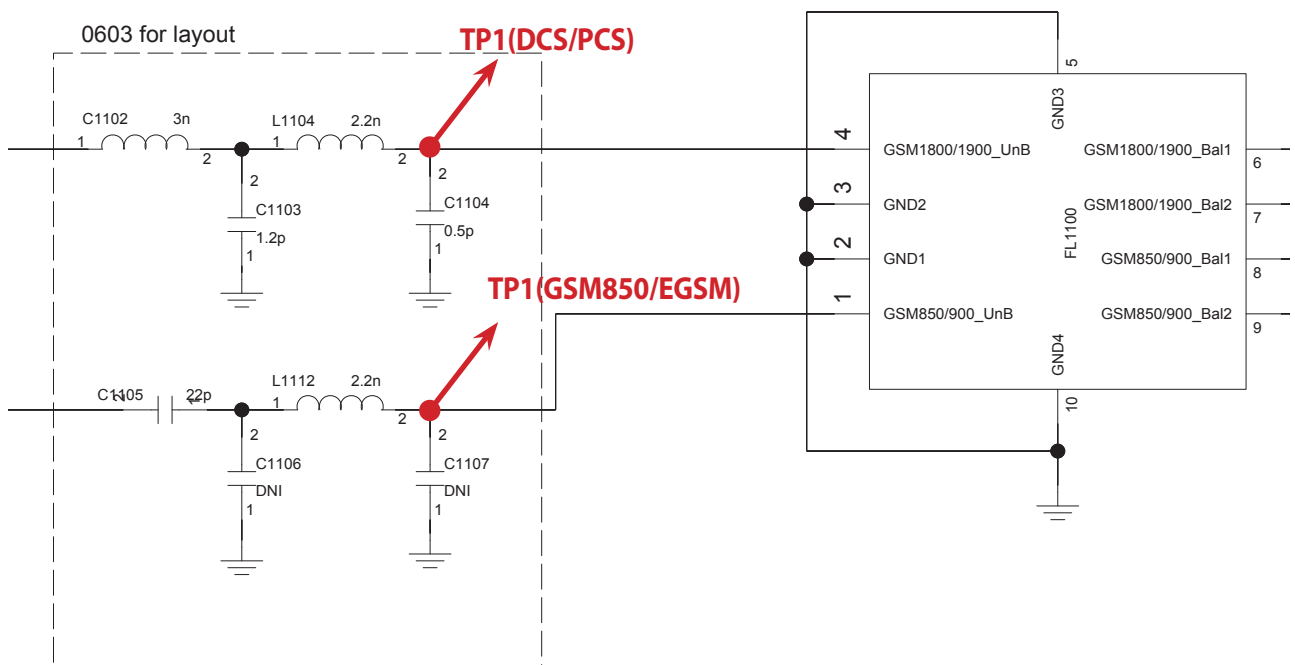
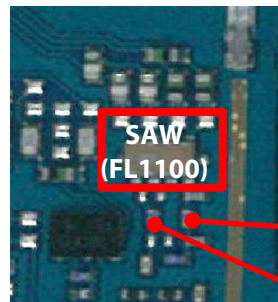
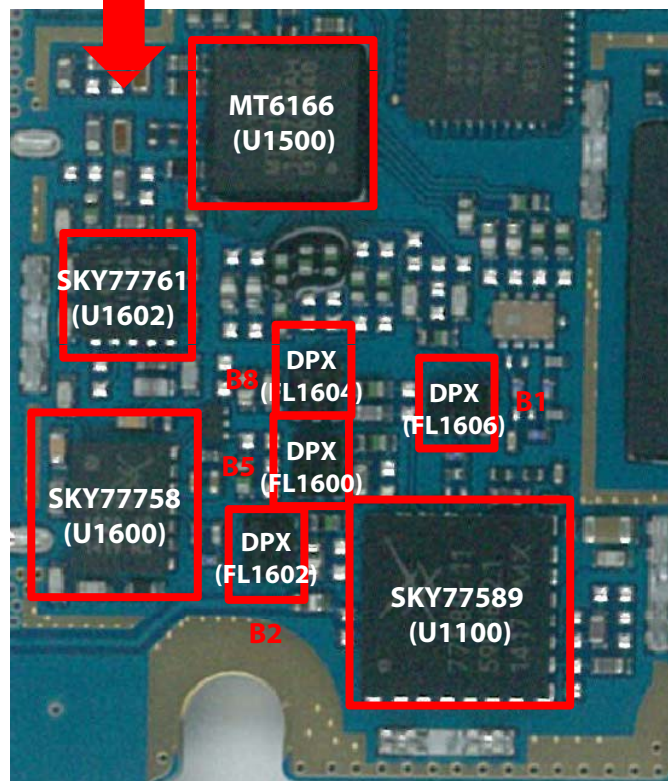
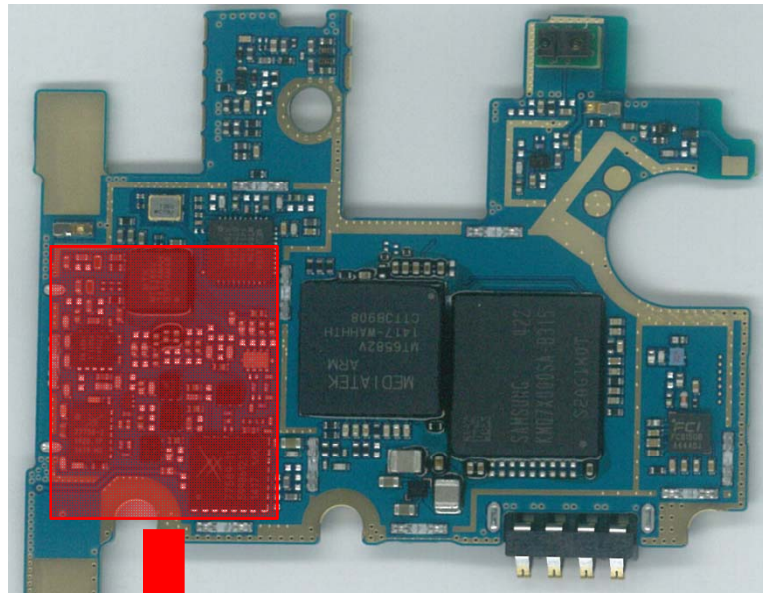


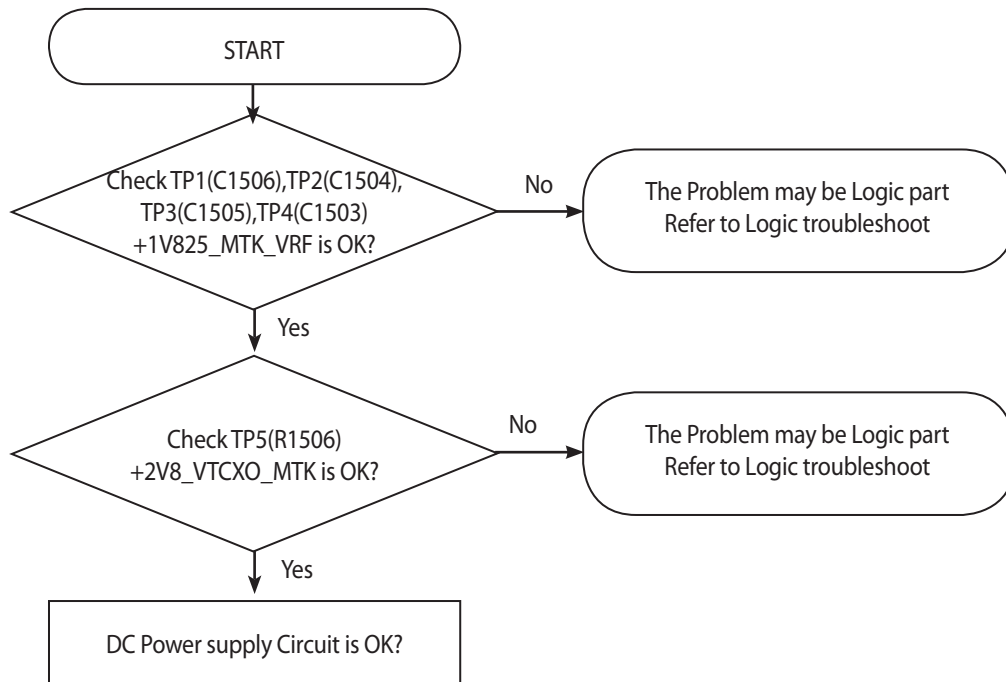
Figure 4.1.1 (a)

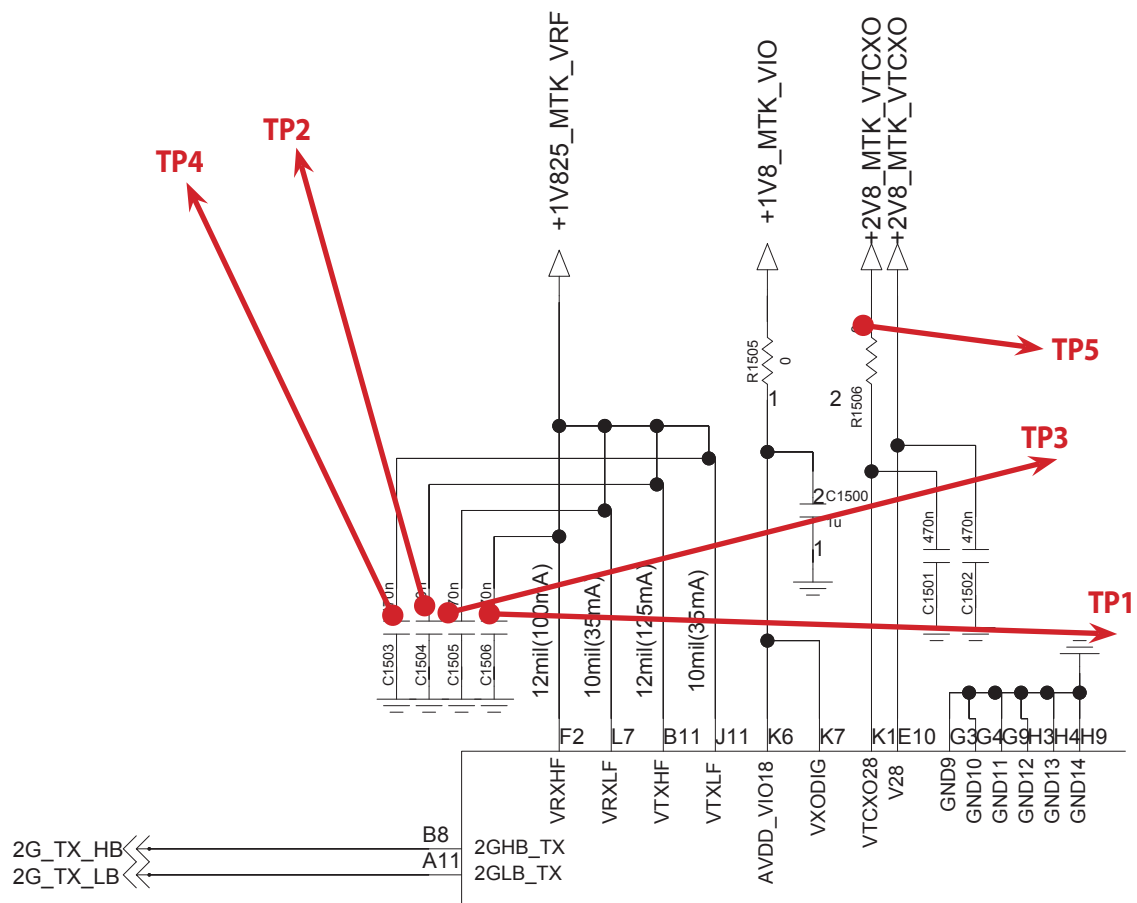
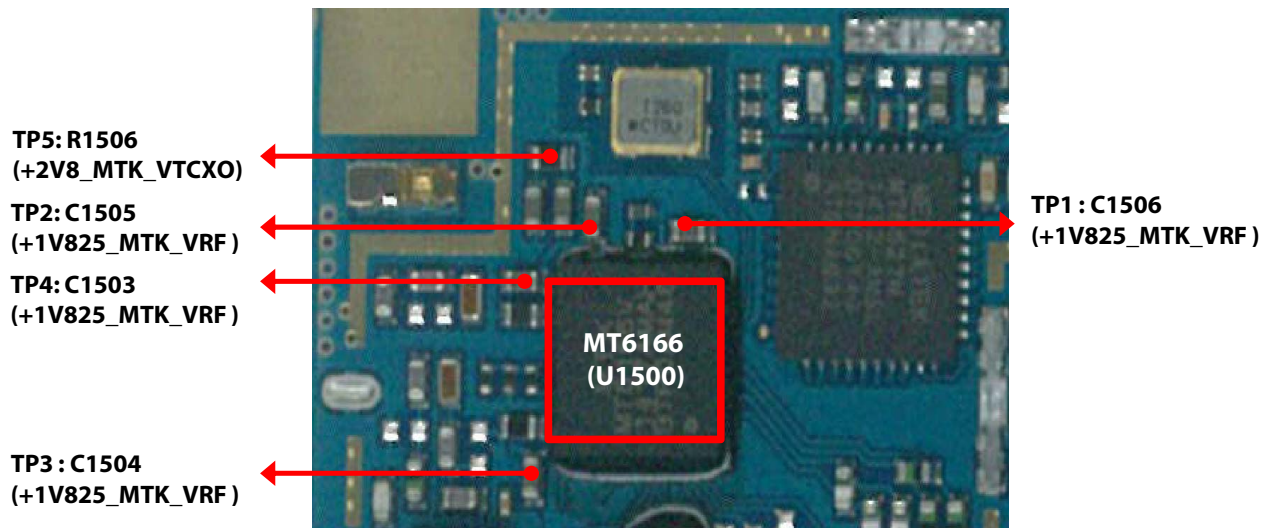
3.4 GSM Tx Part Troubleshooting

3.4.1 GSM850/90/DCS/PCS Tx



3.4.1.1 Checking DC Power supply circuit





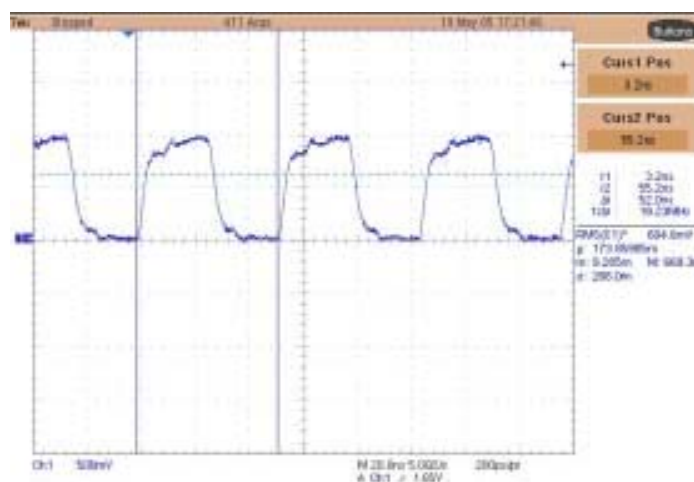
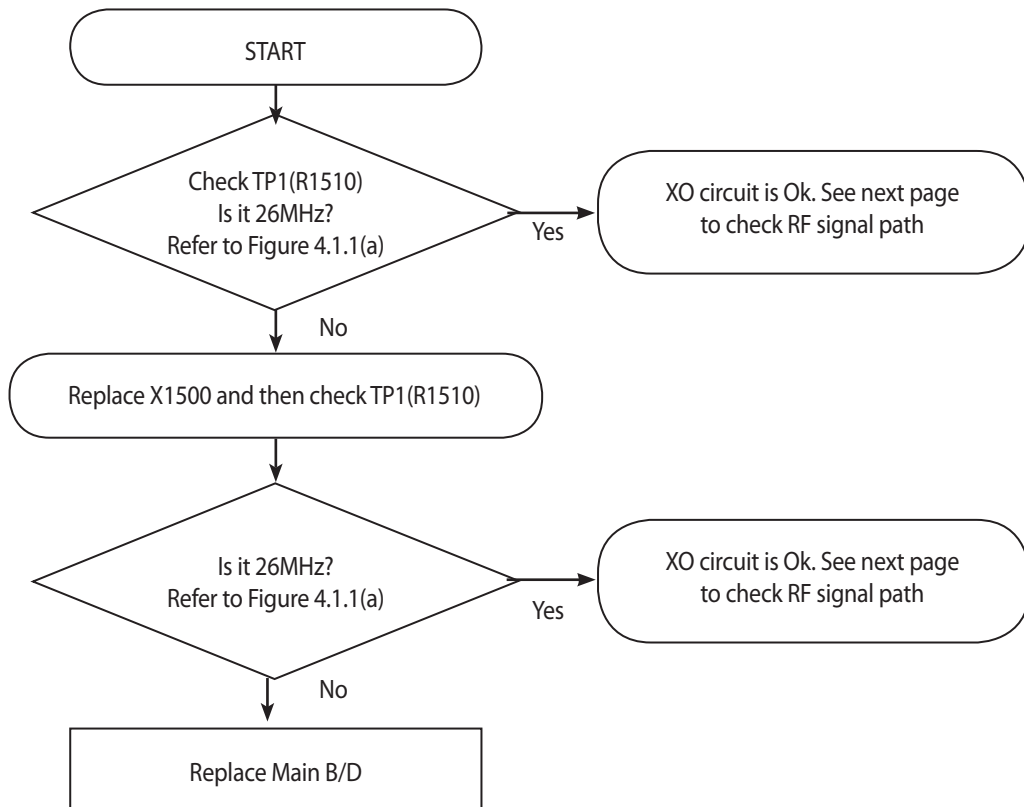
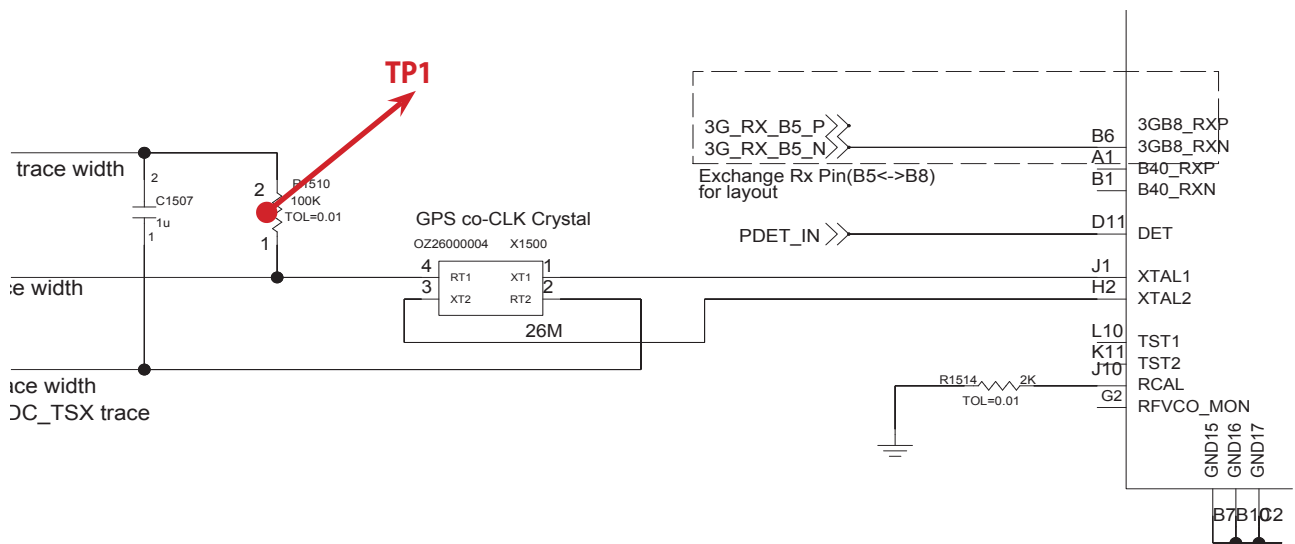
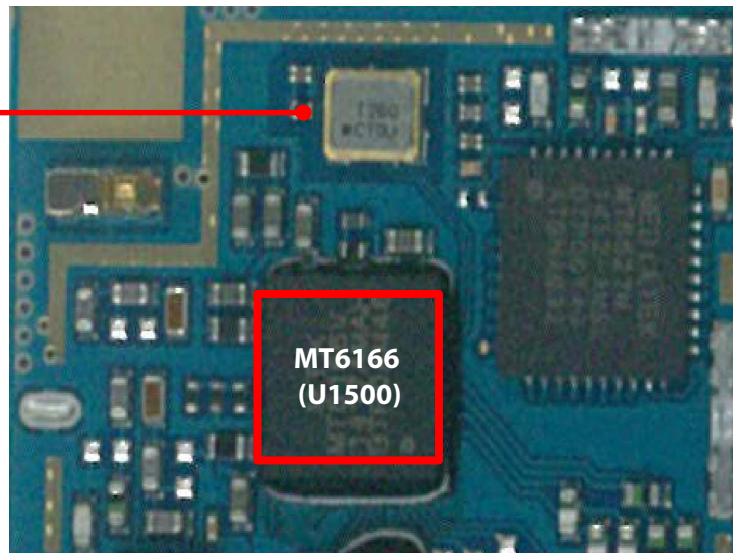


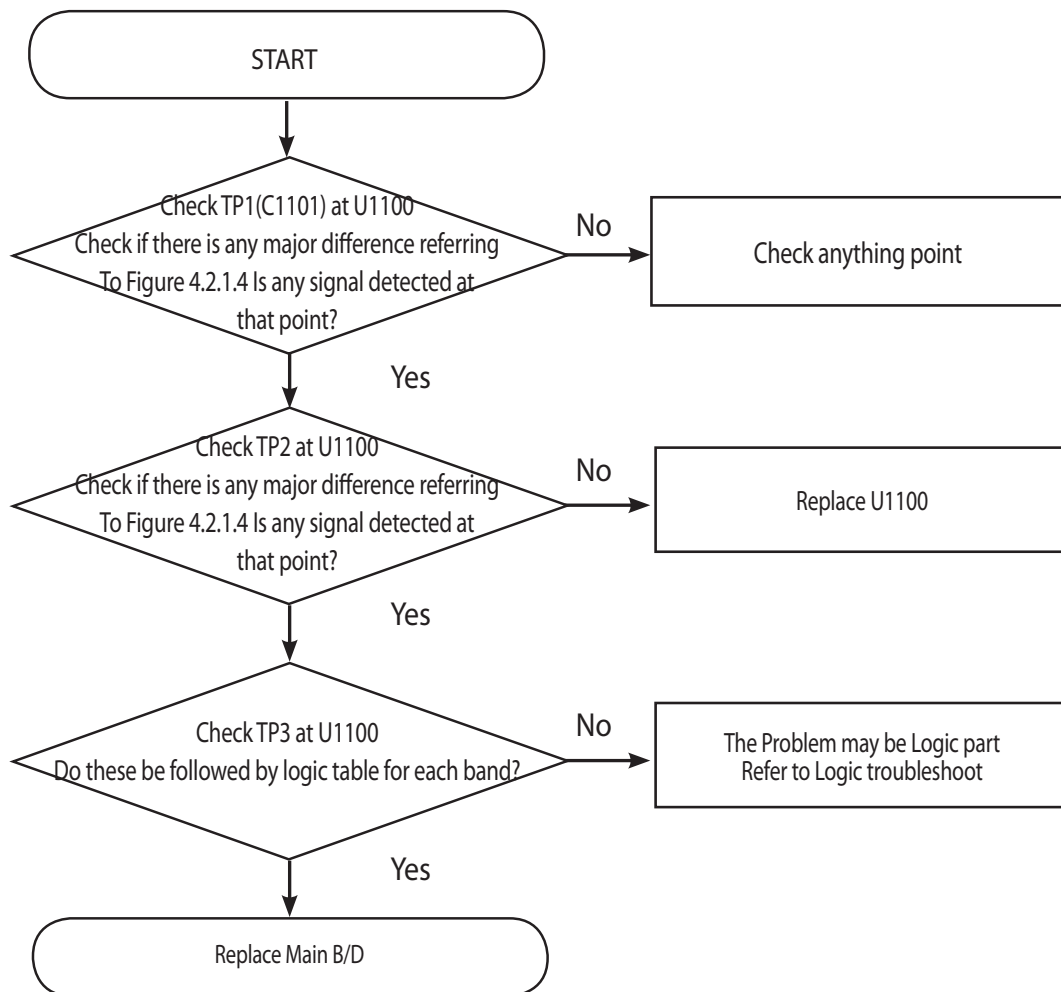
Figure4.1.1(a)

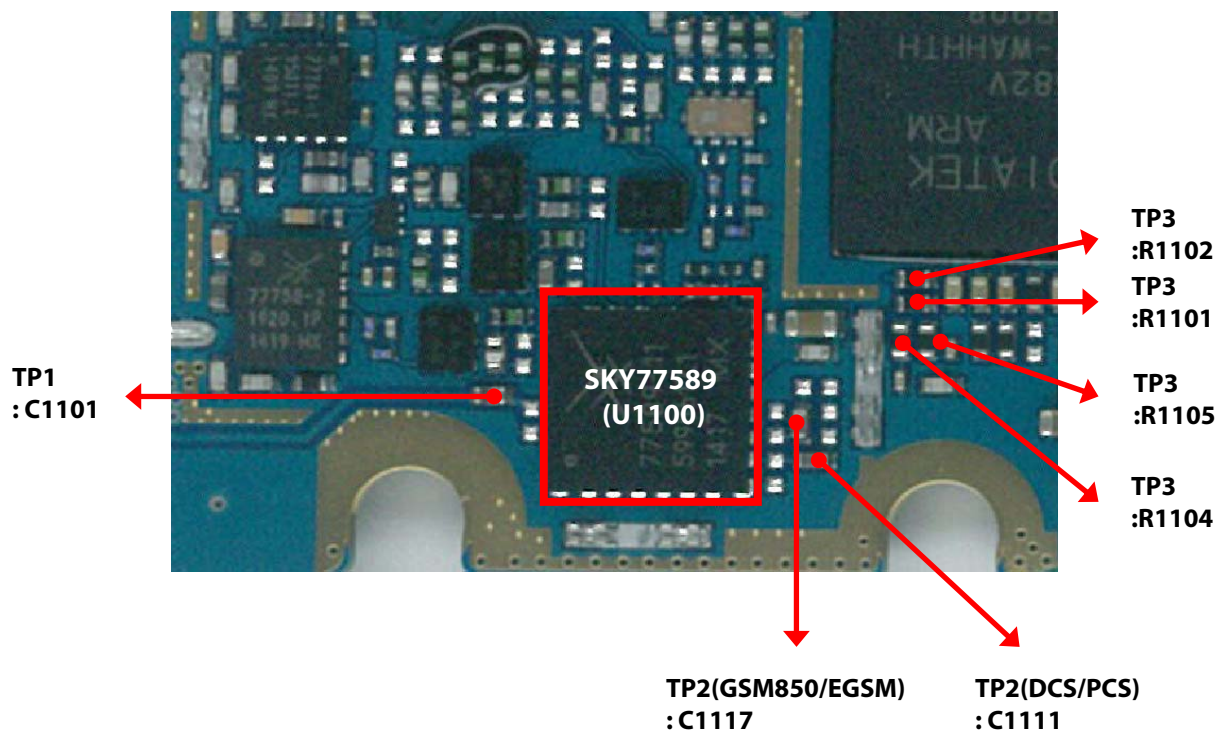
3.4.1.2 Checking MT6166 XO circuit

**TP1 : R1510
(MAIN_CLK_26MHz)**

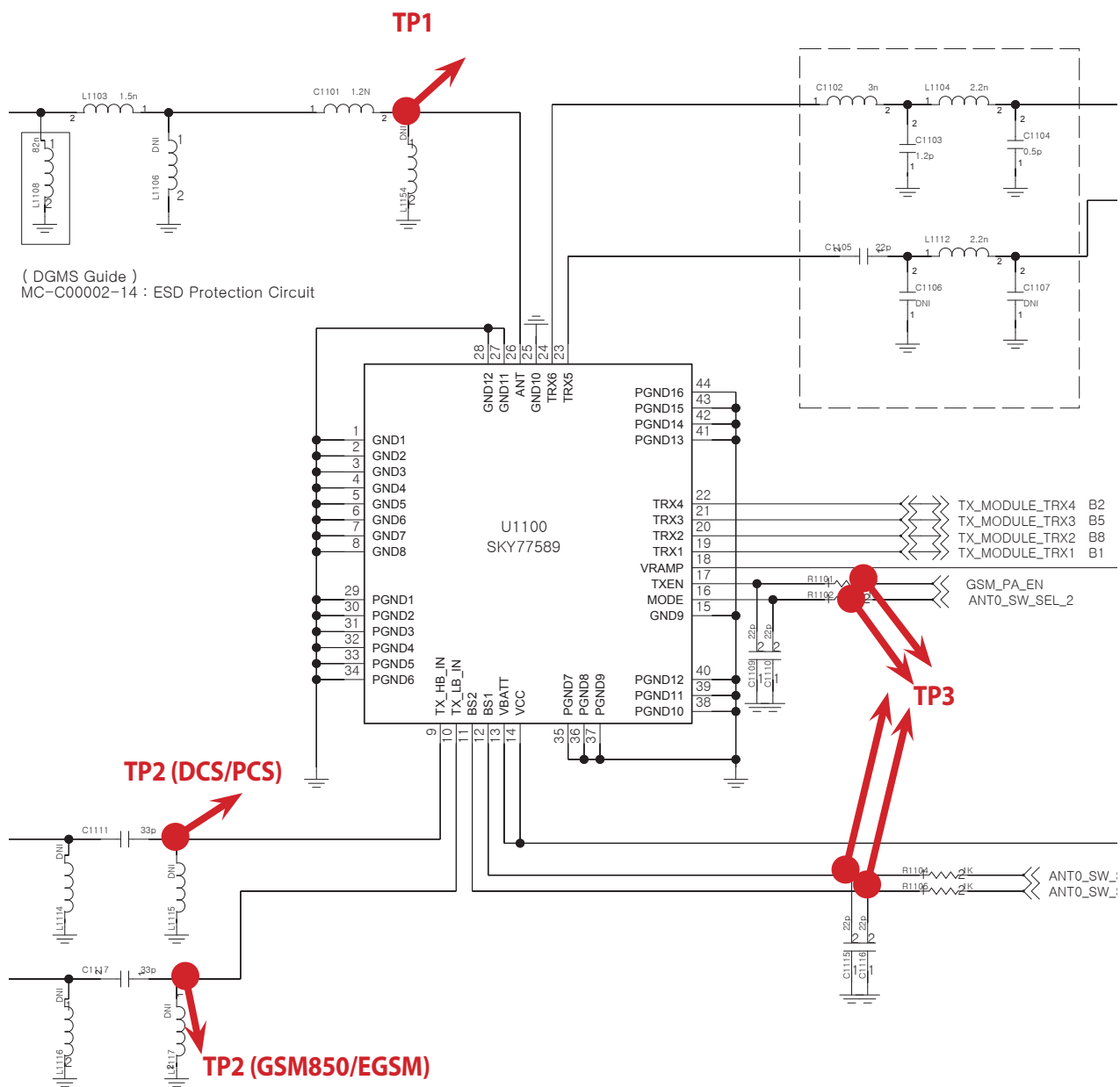


3.4.1.3 Checking RF signal path (FEM)





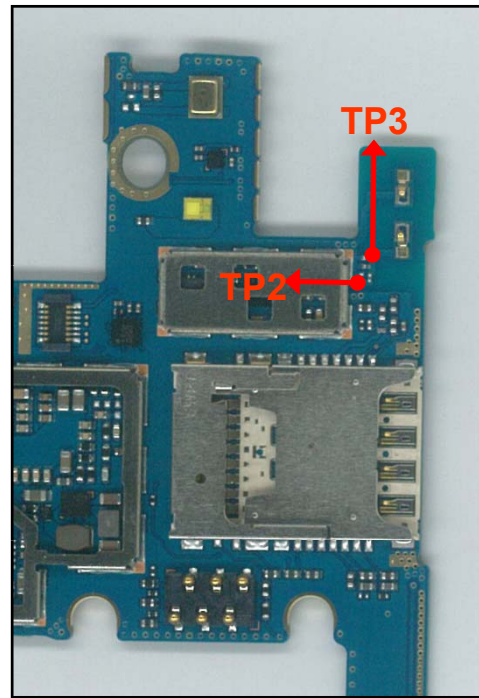
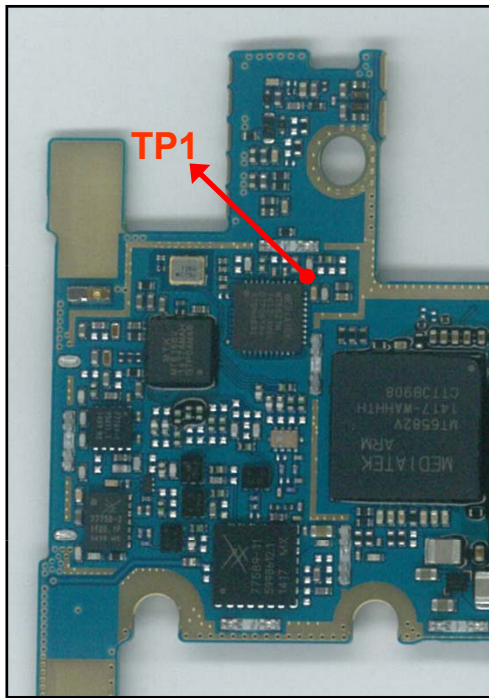
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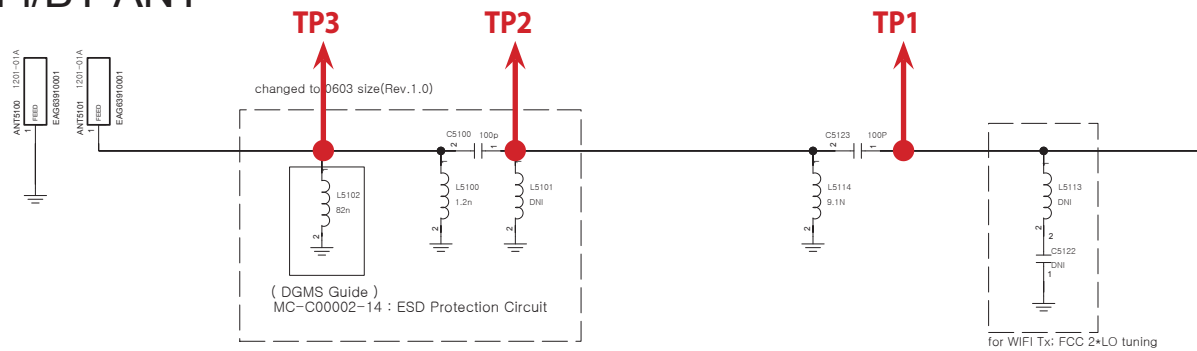
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		GSM PA_EN	ANT_SEL2	ANT_SEL1	ANT_SEL0
Stanby	Stanby	0	0	0	0
LB_TX	G_LB_TX	1	0	0	1
HB_TX	G_HB_TX	1	0	1	1
TRX1	B1	0	1	0	0
TRX2	B8	0	1	1	0
TRX3	B5	0	1	0	1
TRX4	B2	0	1	1	1
TRX5	G_LB_Rx	0	0	1	0
TRX6	G_HB_Rx	0	0	0	1

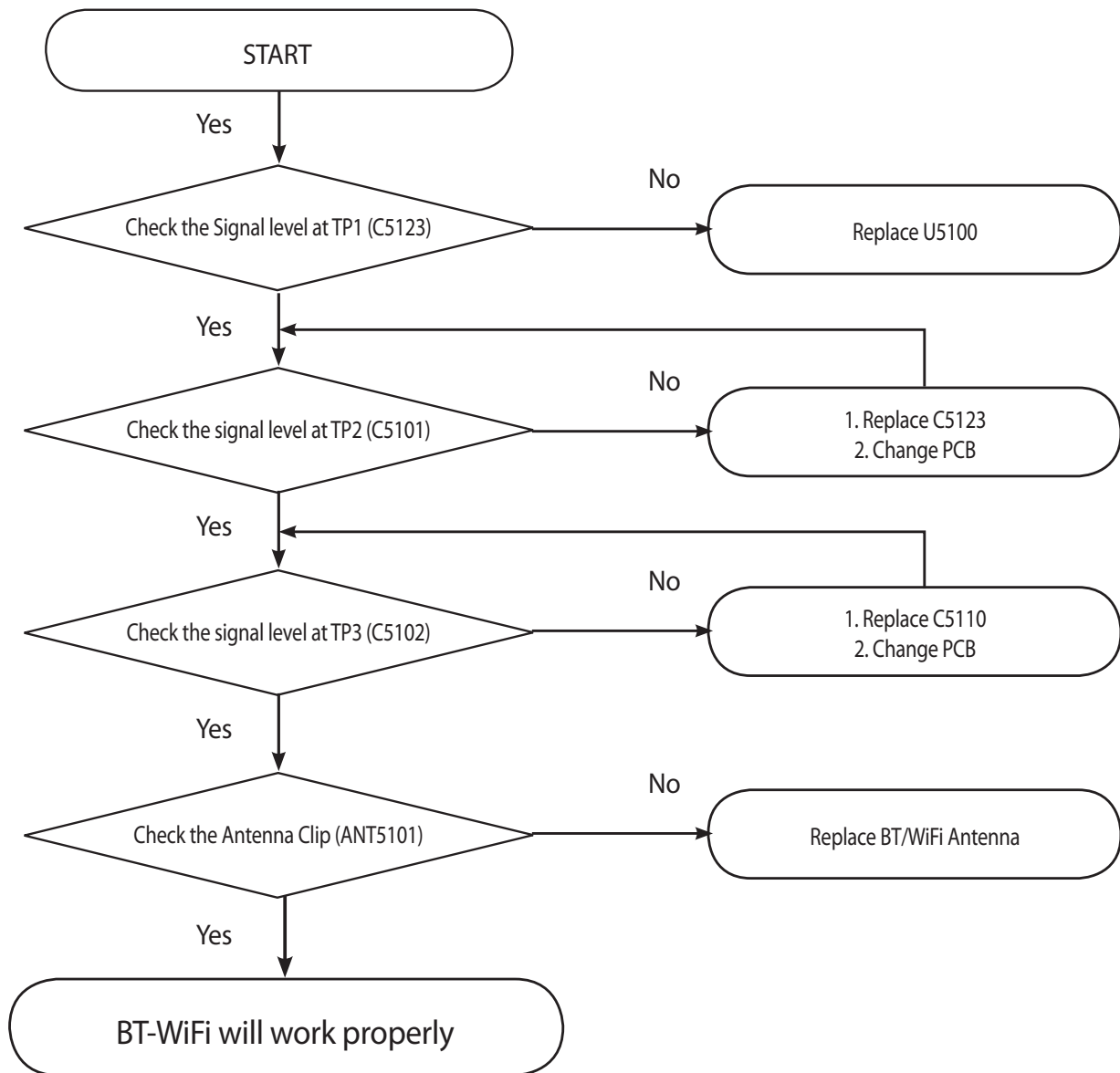
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3.5. BT/WiFi

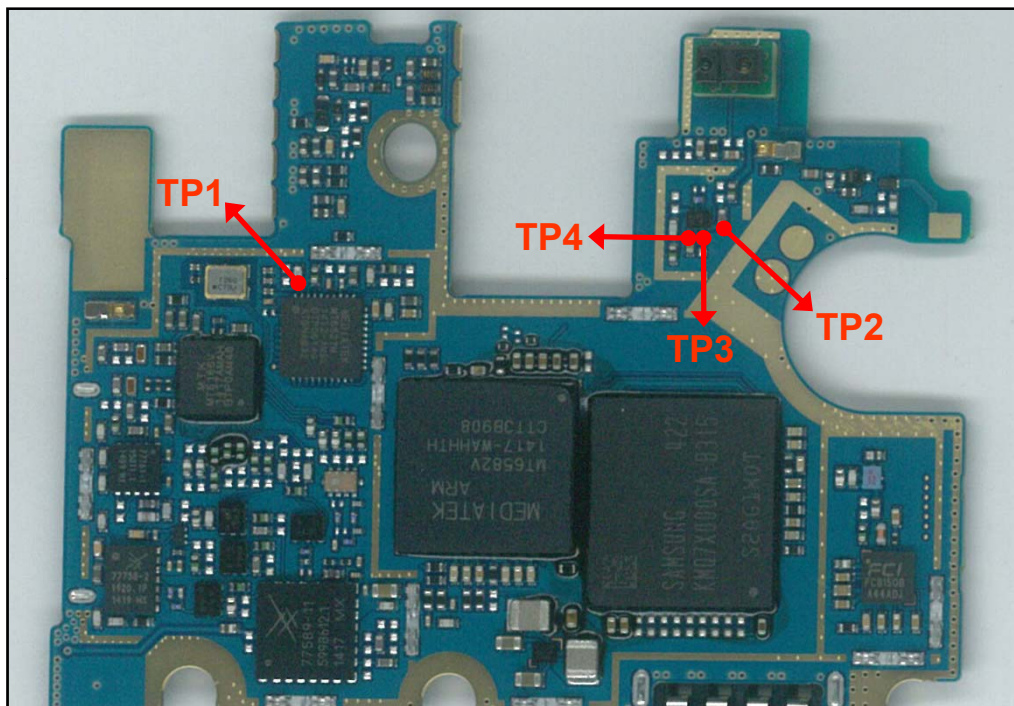


WiFi/BT ANT

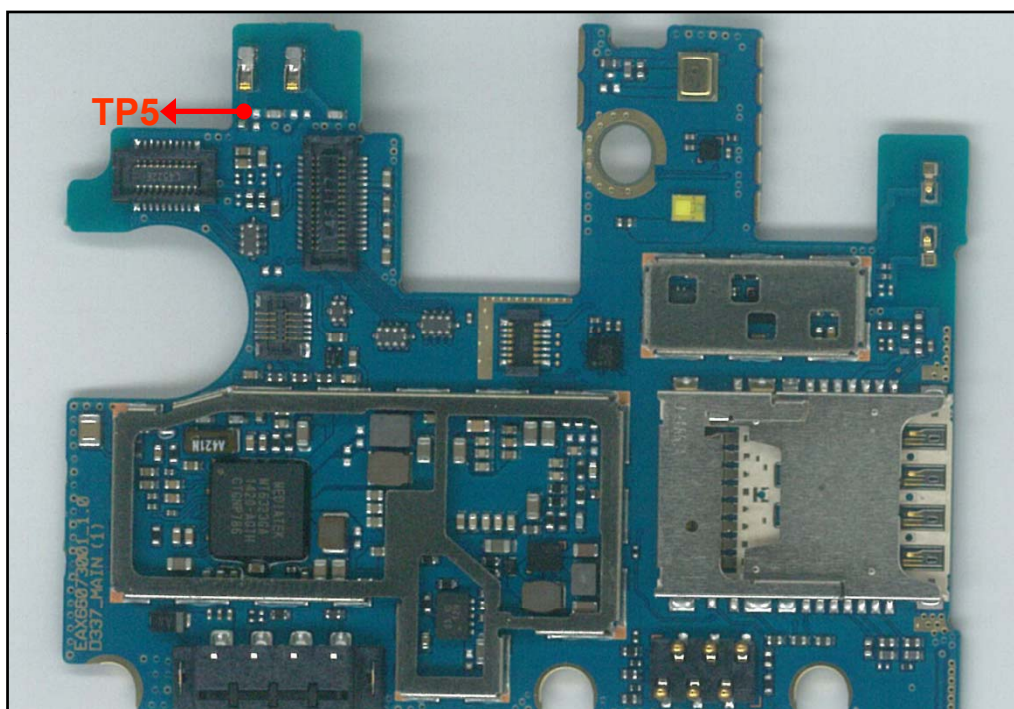




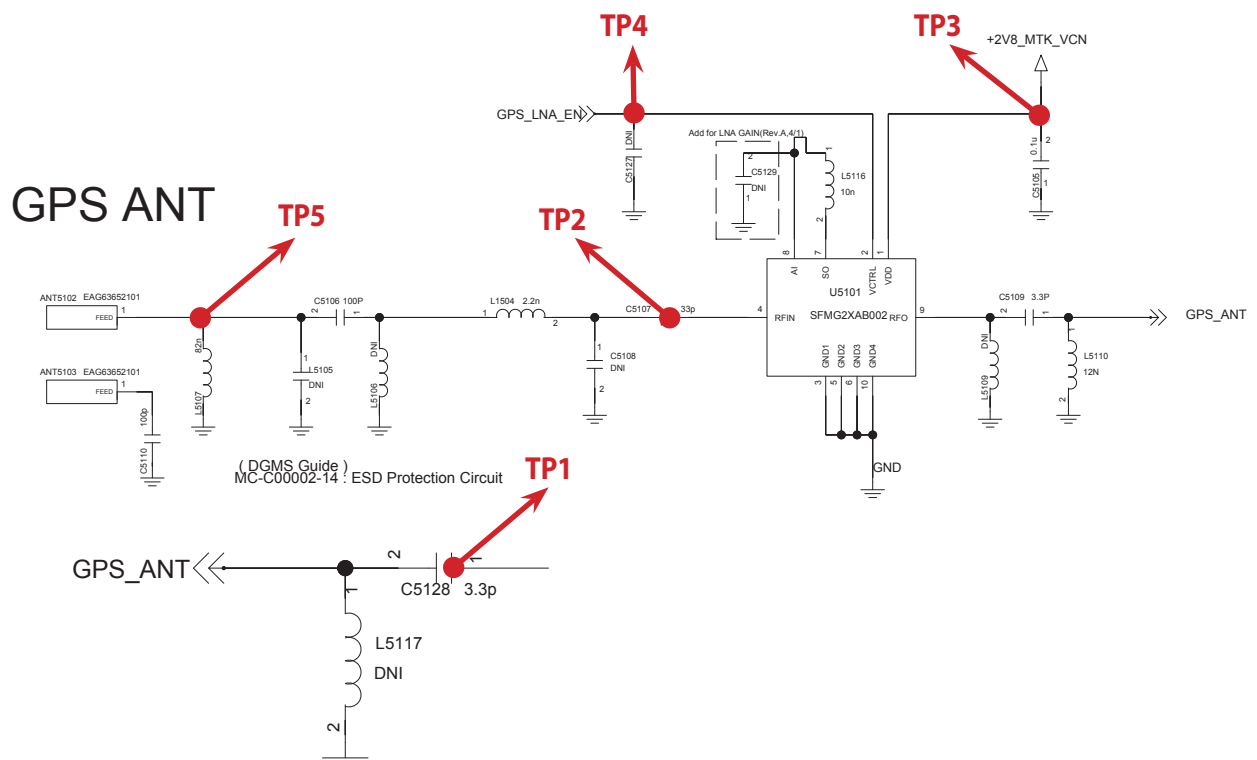
3.6. GPS

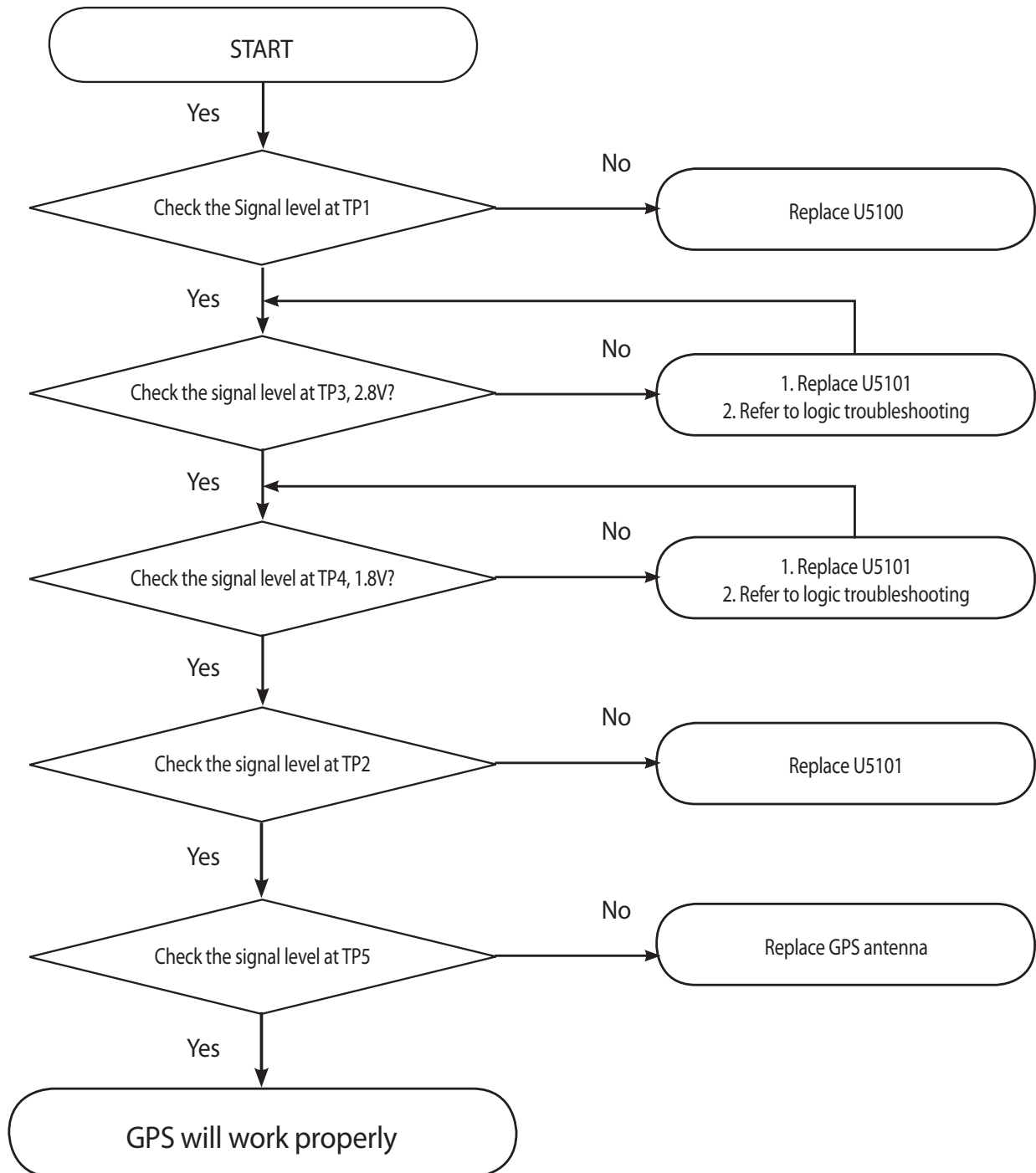


<Top side>

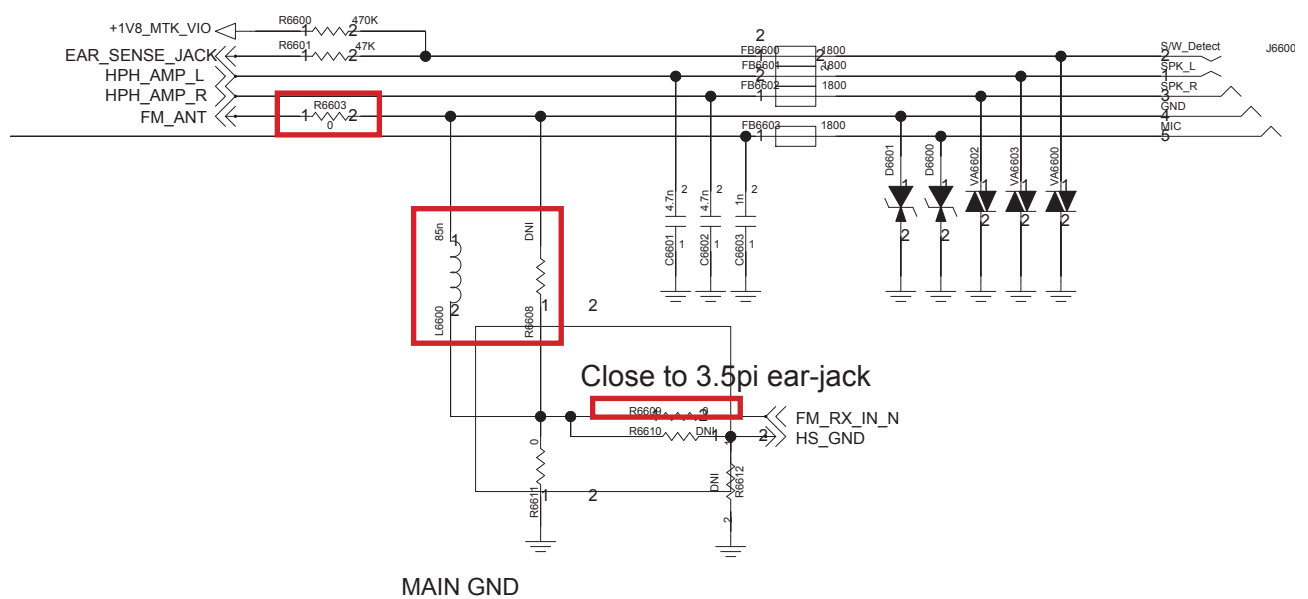
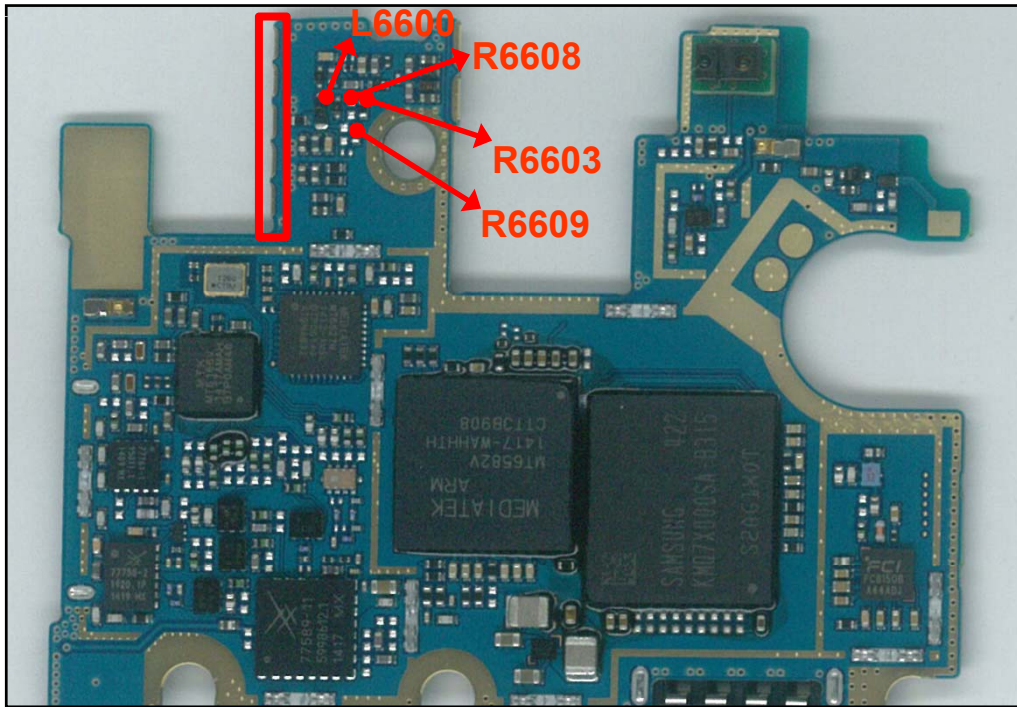


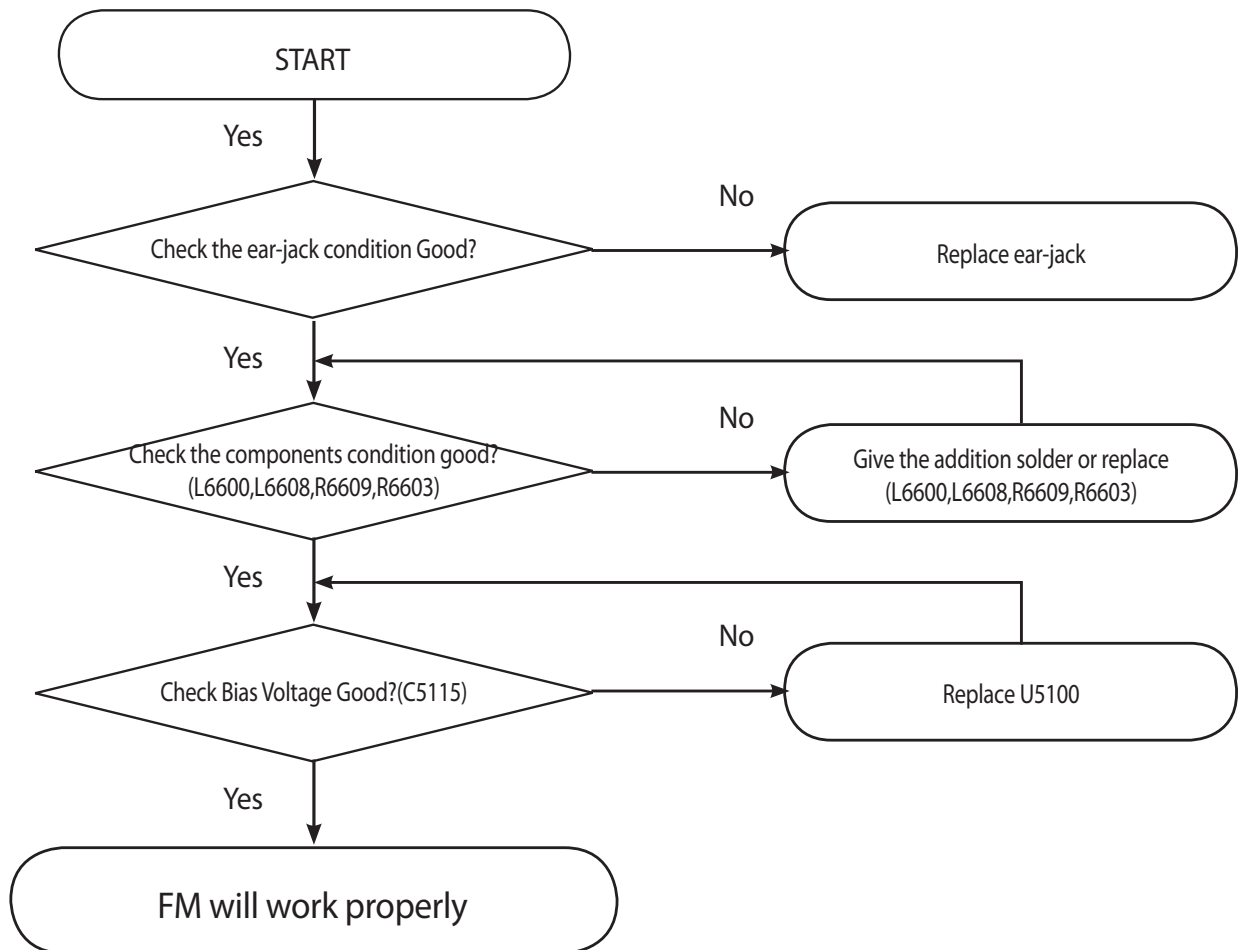
<Bottom side>



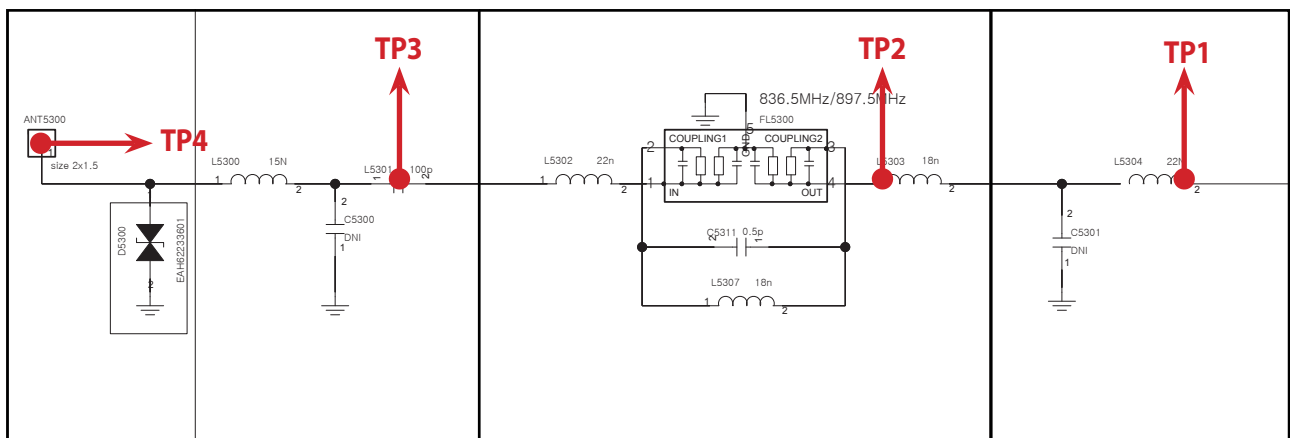
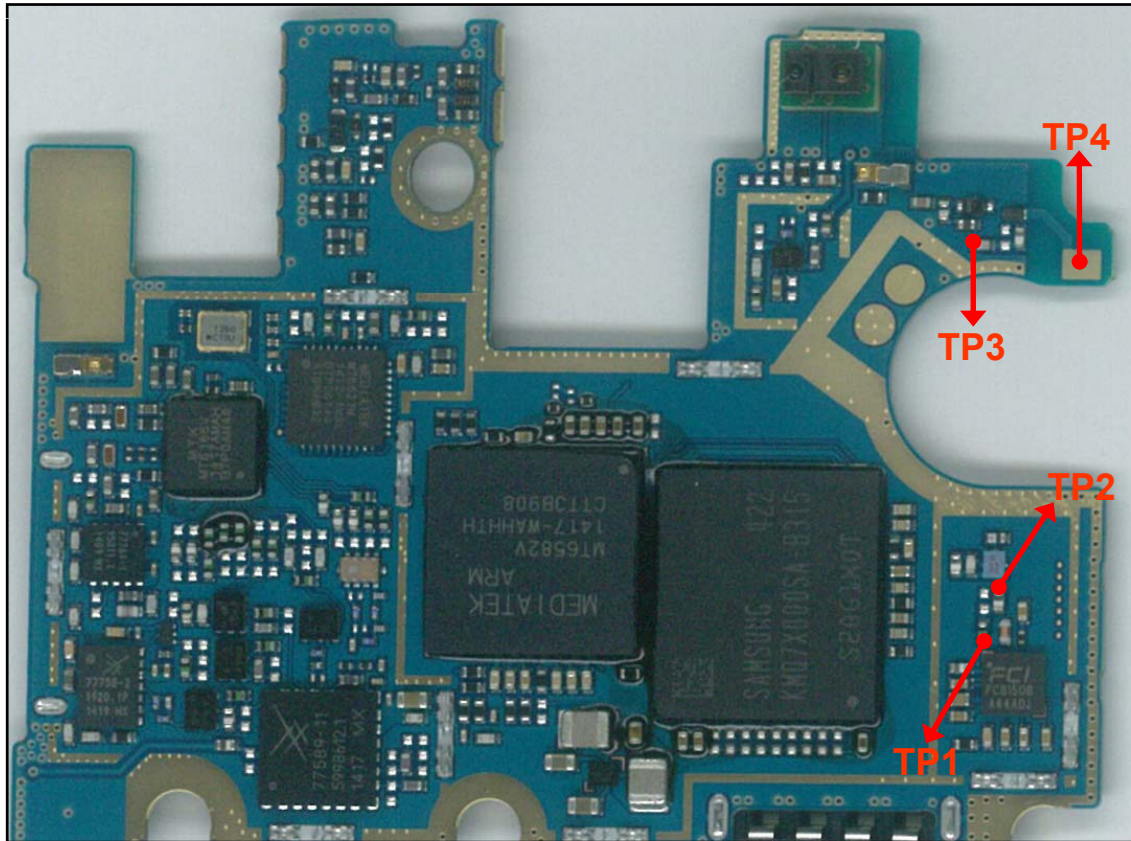


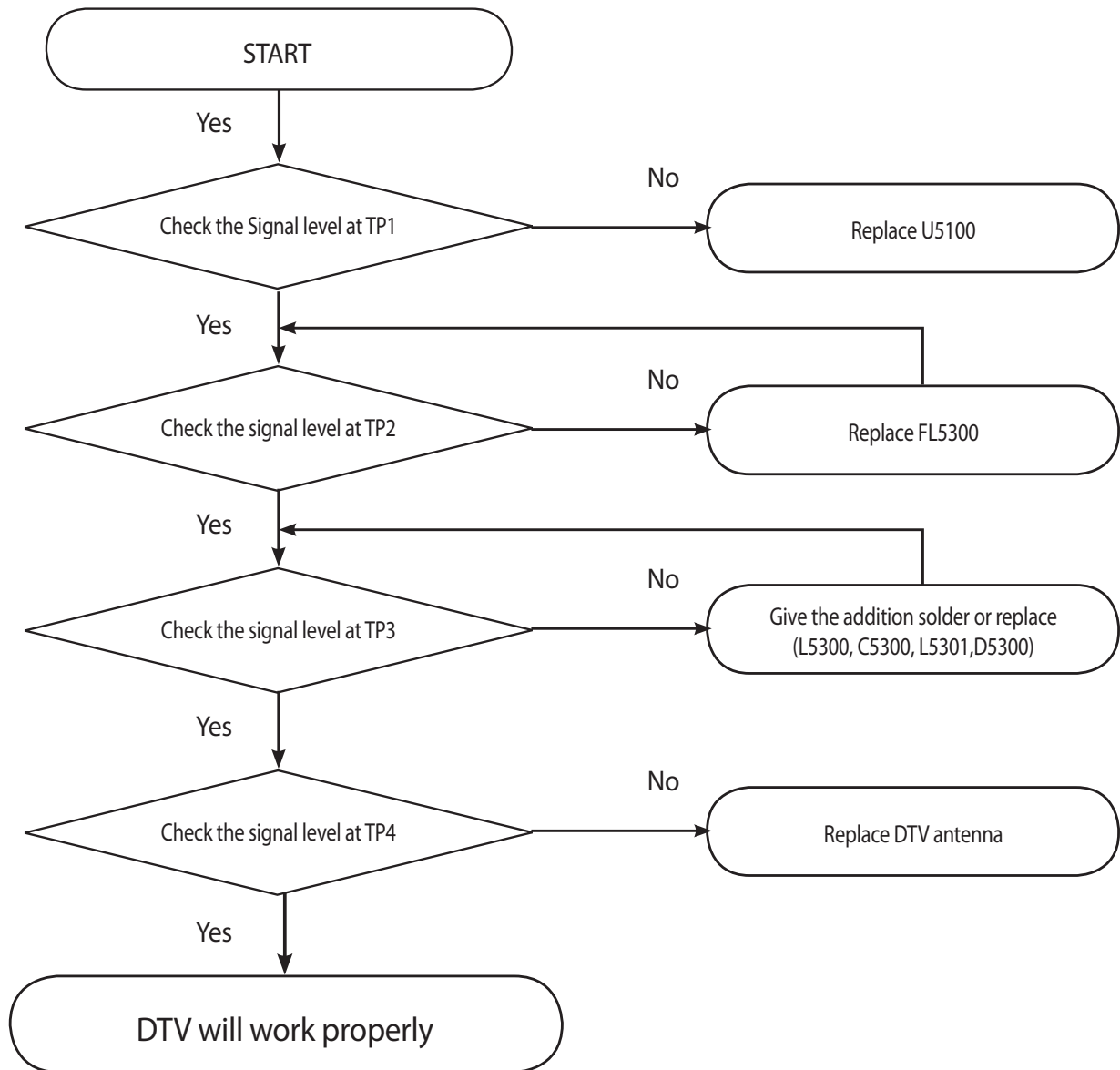
3.7. FM Radio





3.8. DTV

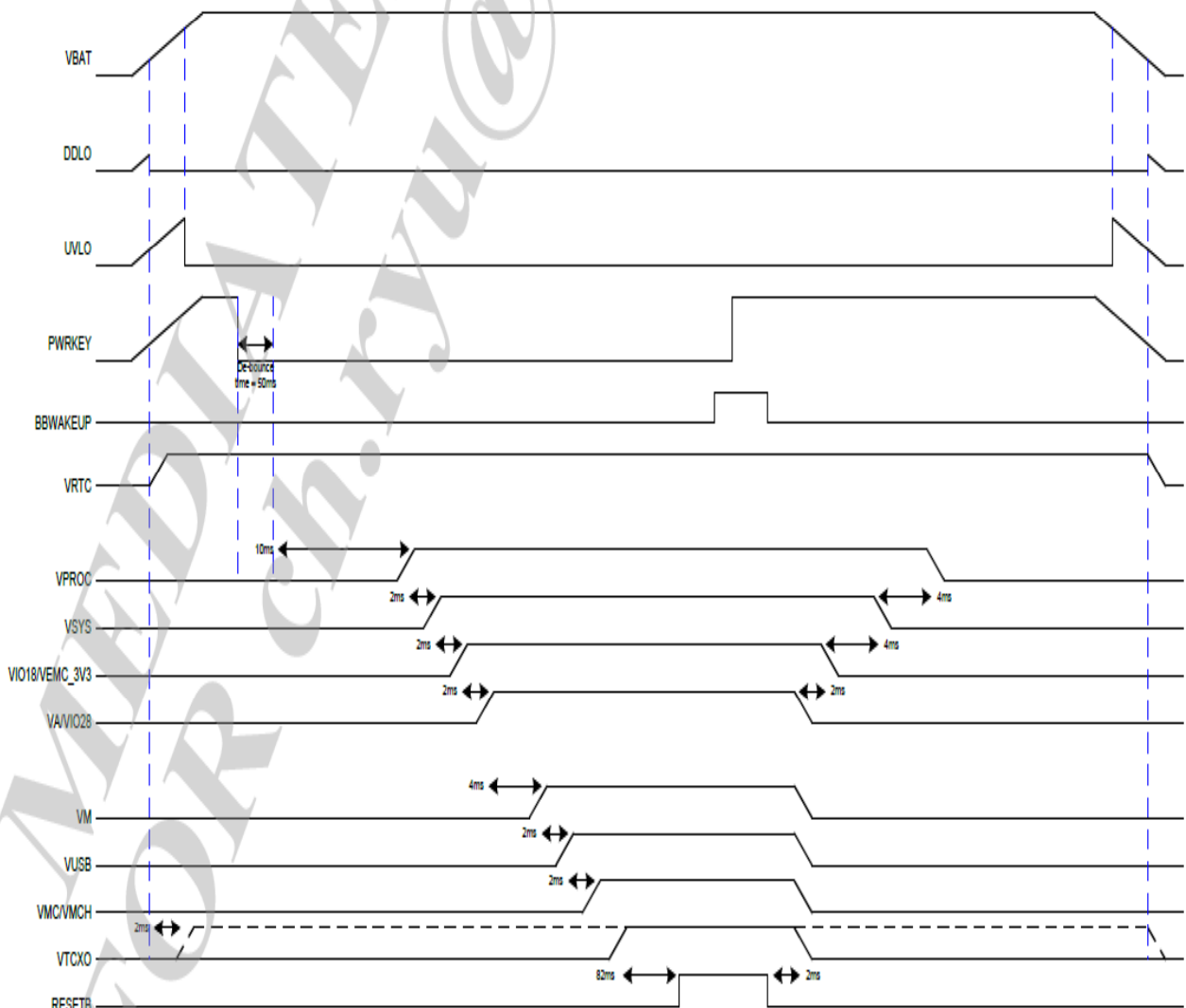


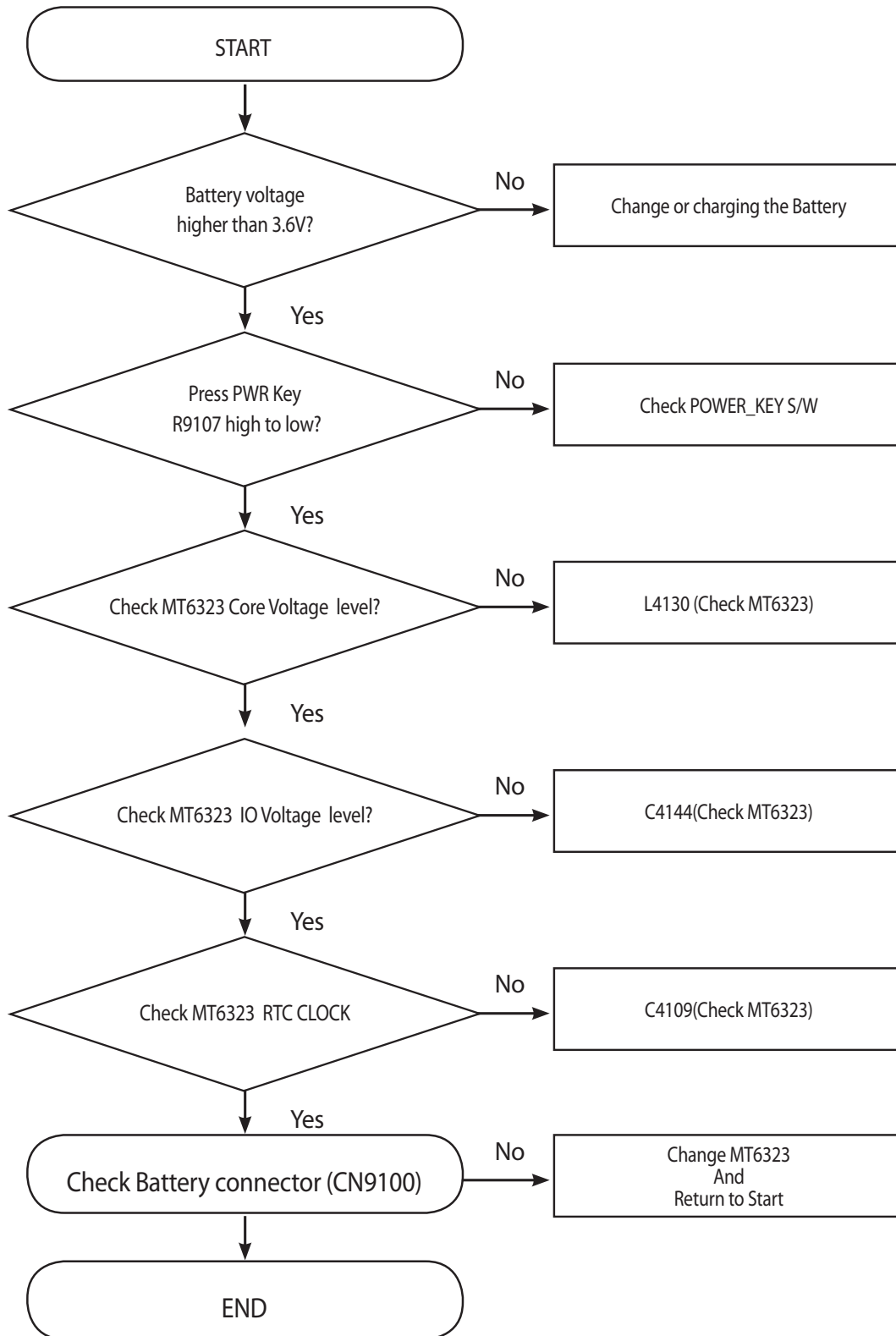


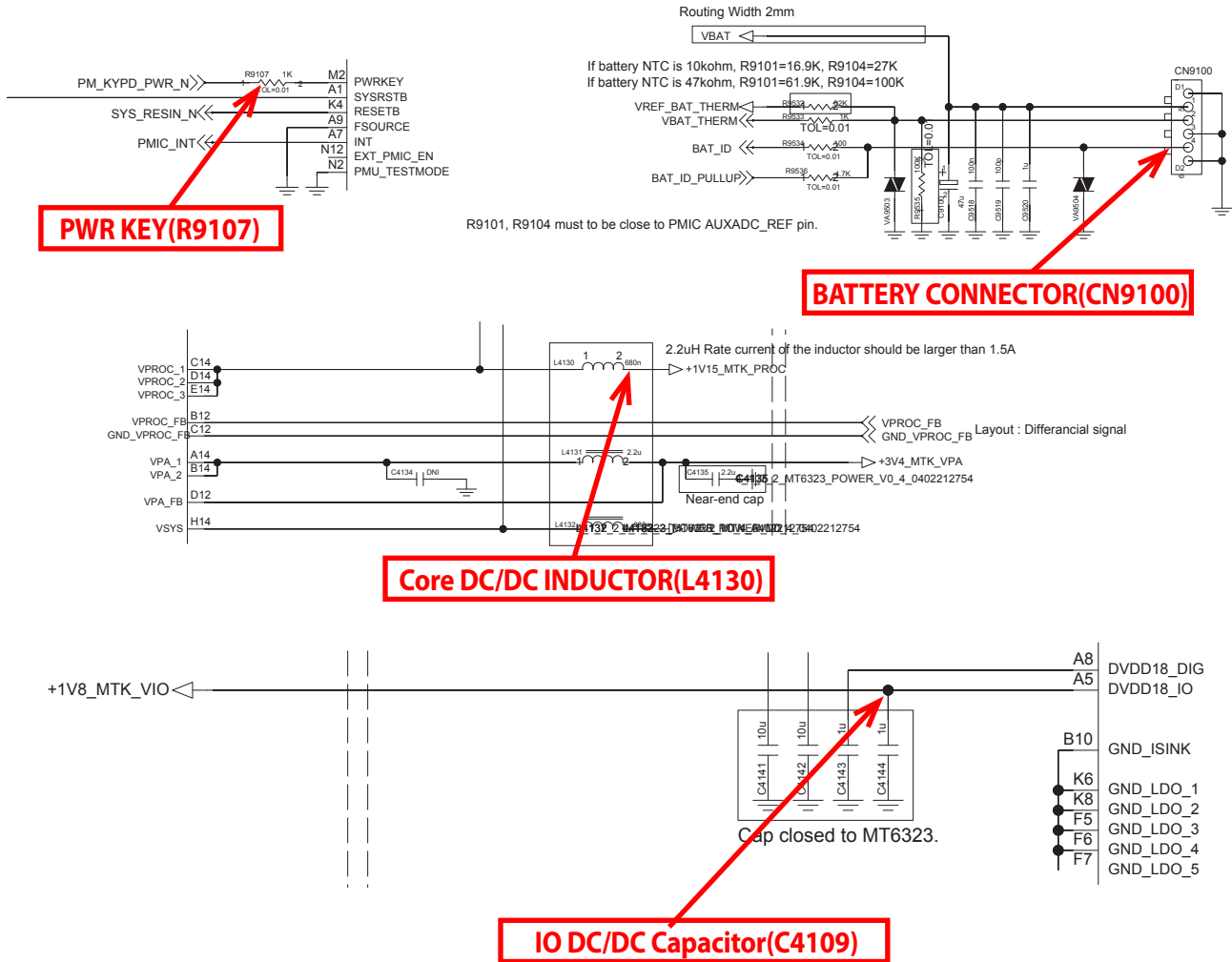
3.9. Power TS

The main power source of D337 is provided by 1 chip which is MT6323
Power ON sequence of D337 is

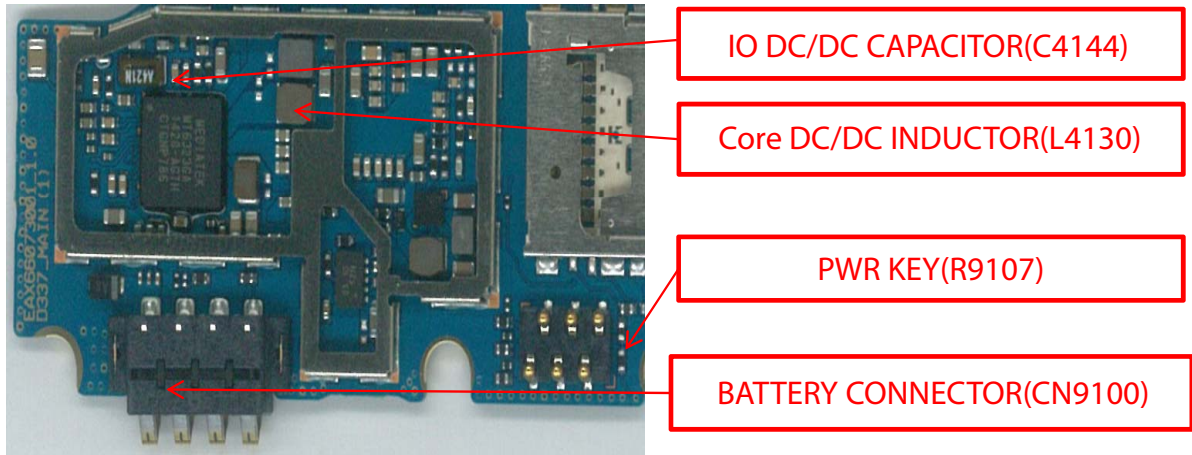
Power on/off sequence







4. DOWNLOAD



3.10. Charging TS

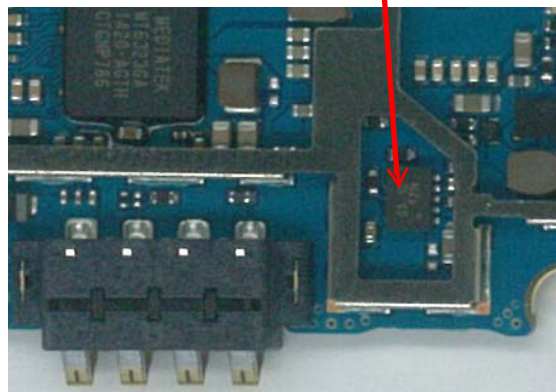
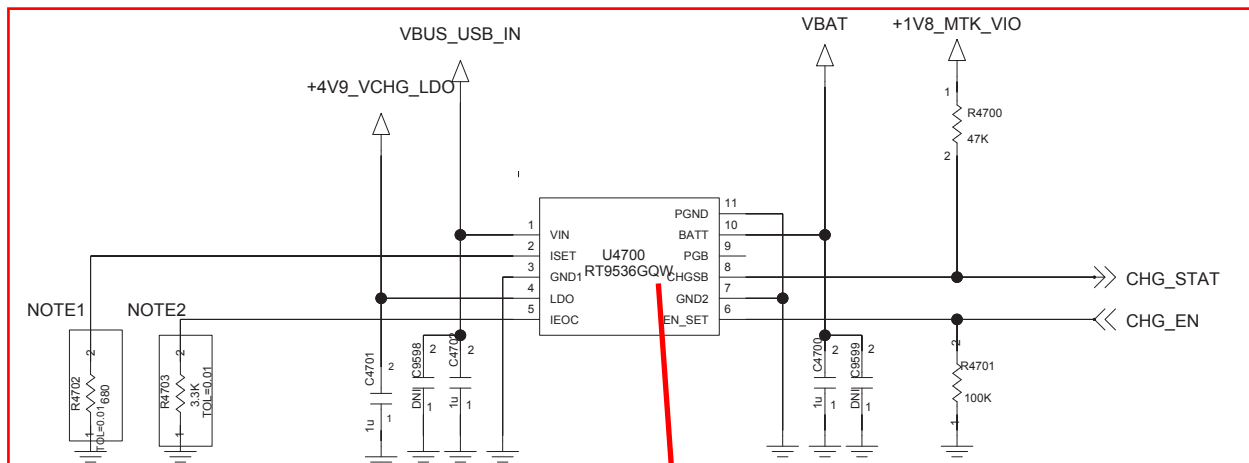
D337 uses Charger IC (linear Charger) for charging control. The Charger IC circuit includes the charging pass transistor, detects the VDD and VBAT voltages, and its control driver. **The Charger IC** maximum charging current is set around 820mA.

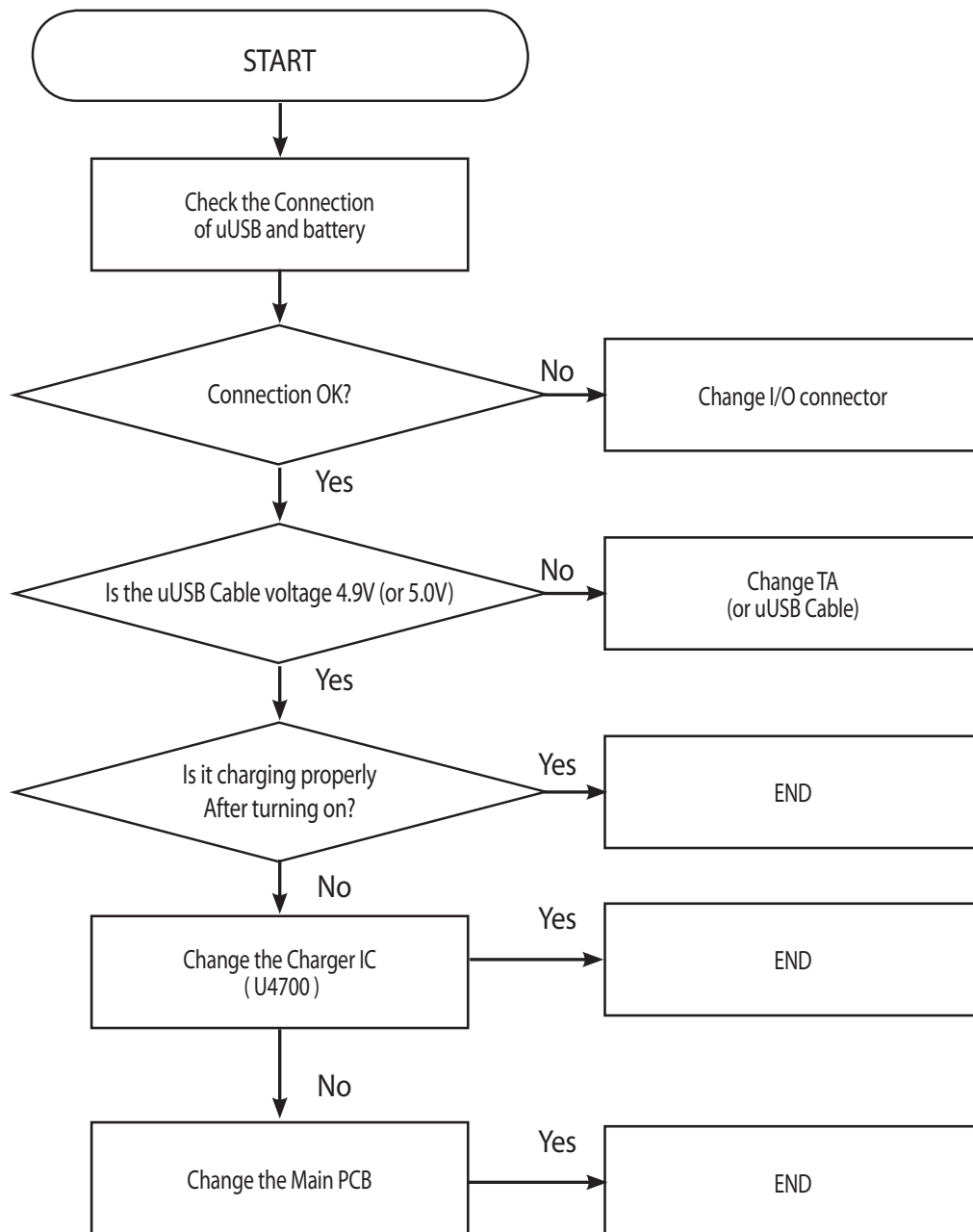
Charging Procedure

- Supply the VCHG use a TA or u-USB cable
- **The Charger IC** control the charging current
- Charging current flows into the main battery

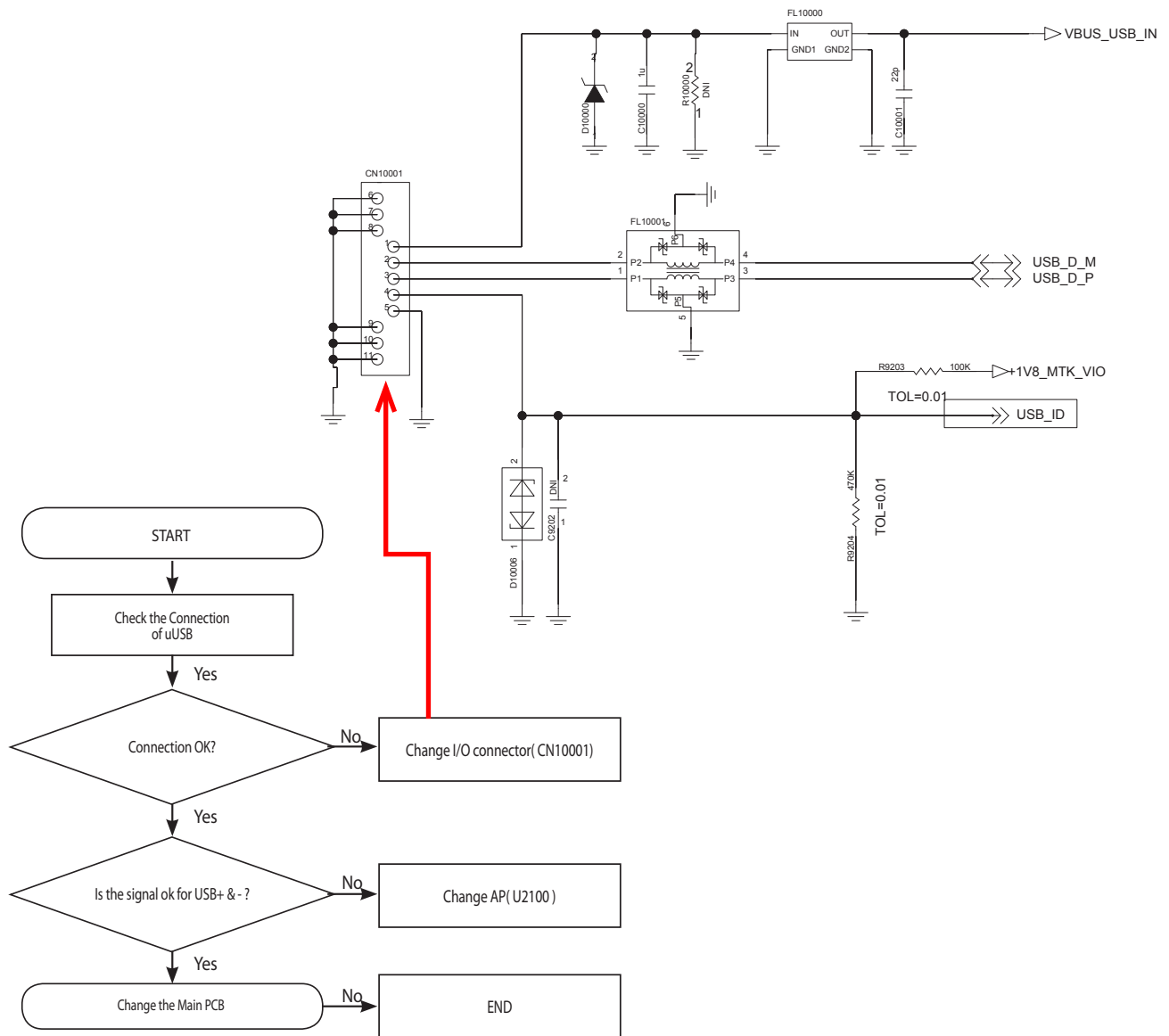
Troubleshooting Check Point

- Connection of the TA or u-USB cable





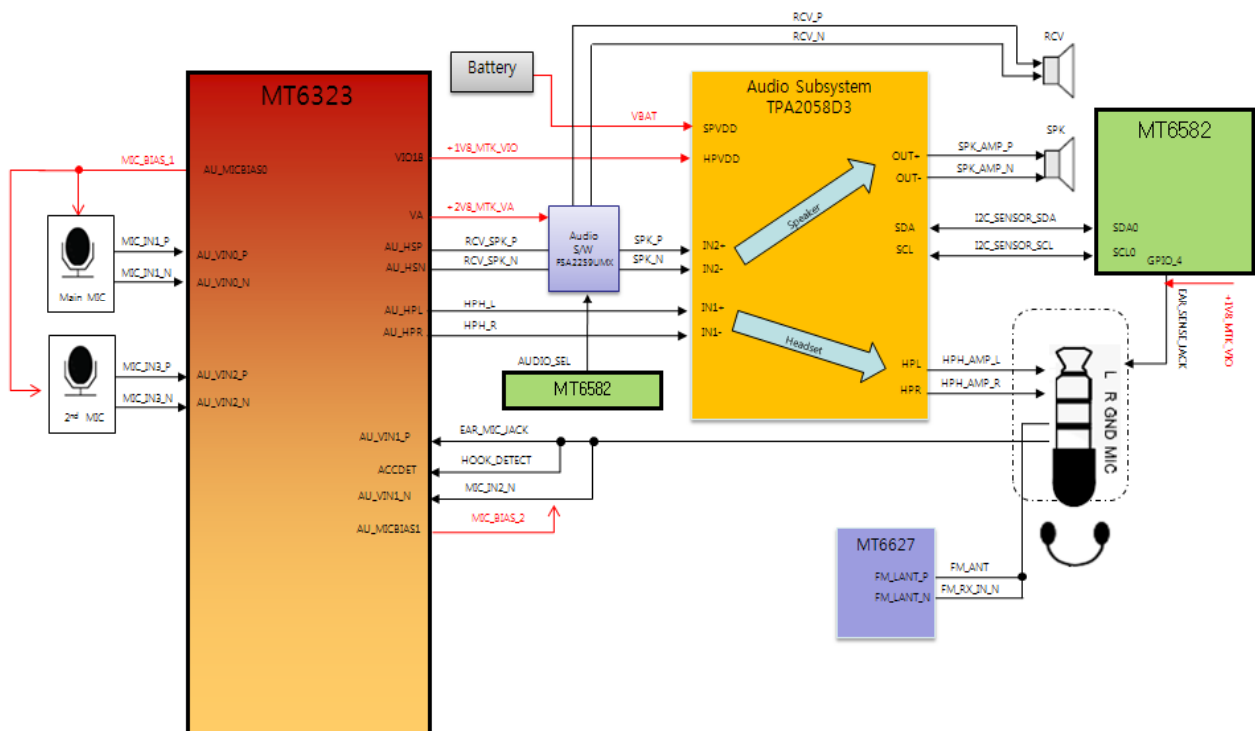
3.11. USB TS



3.12.USB TS

3.12.1. Overview

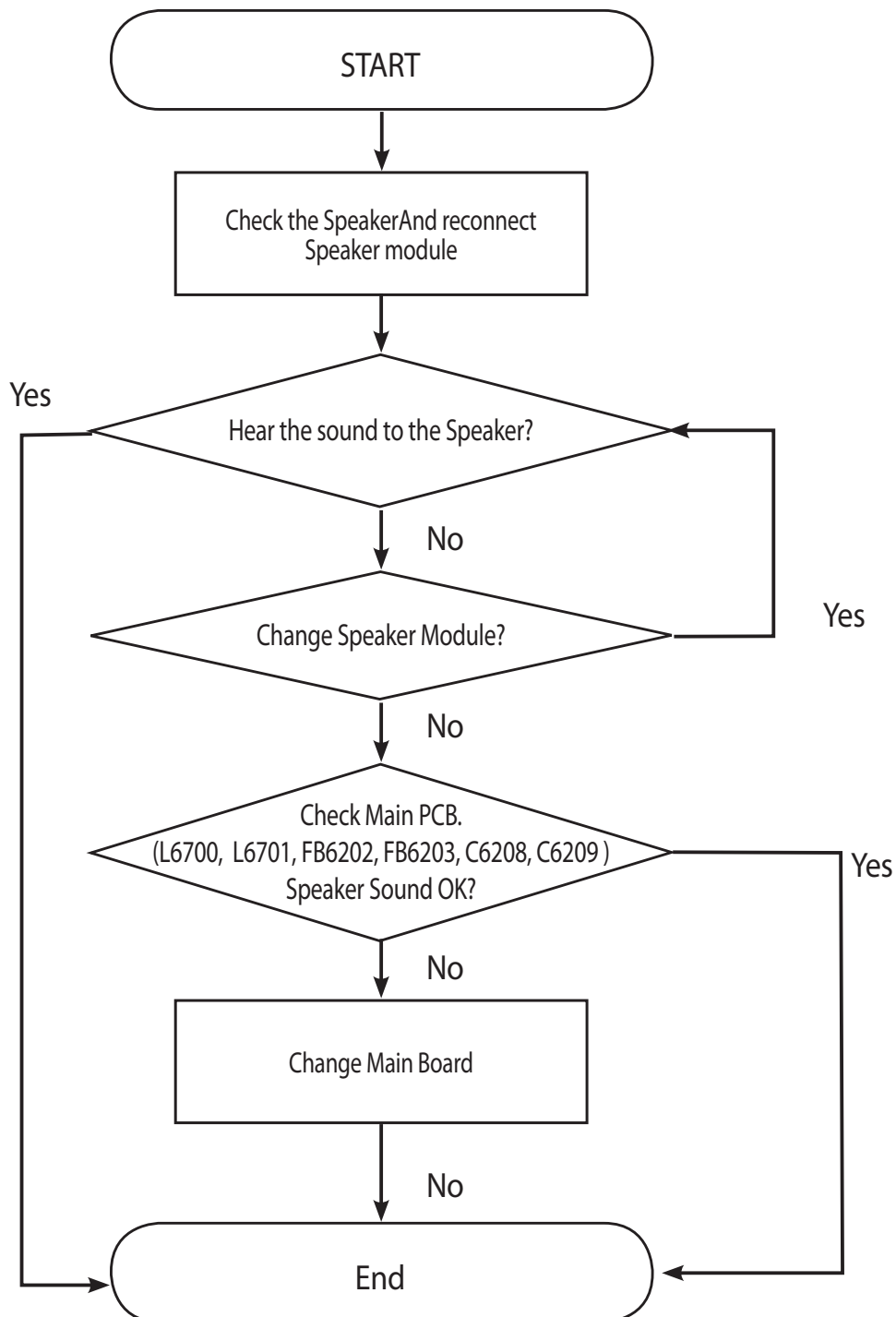
Audio Block Diagram



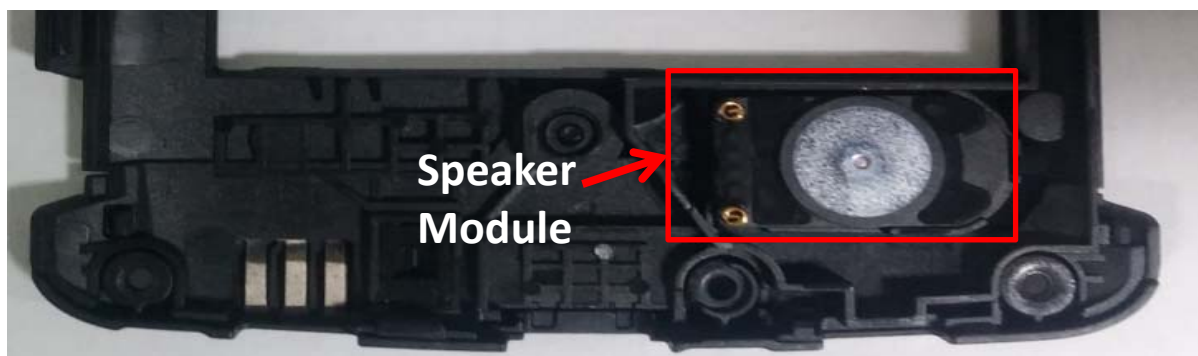
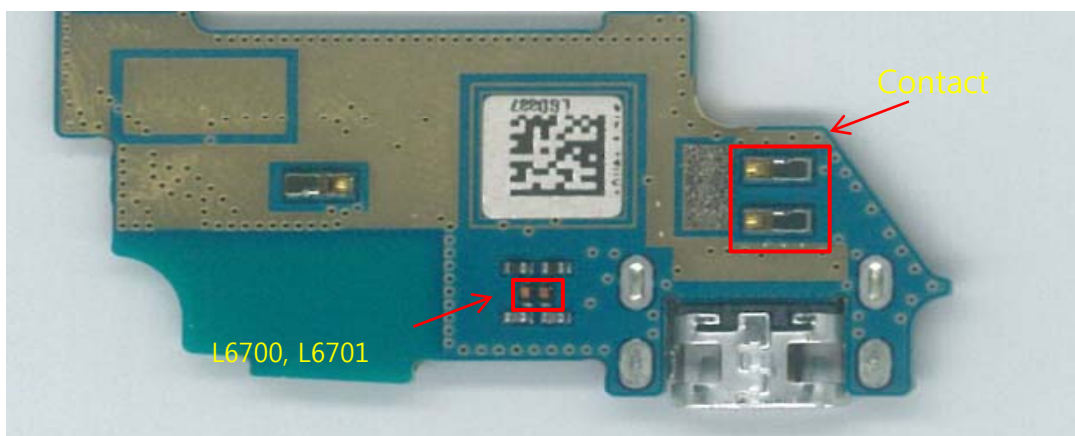
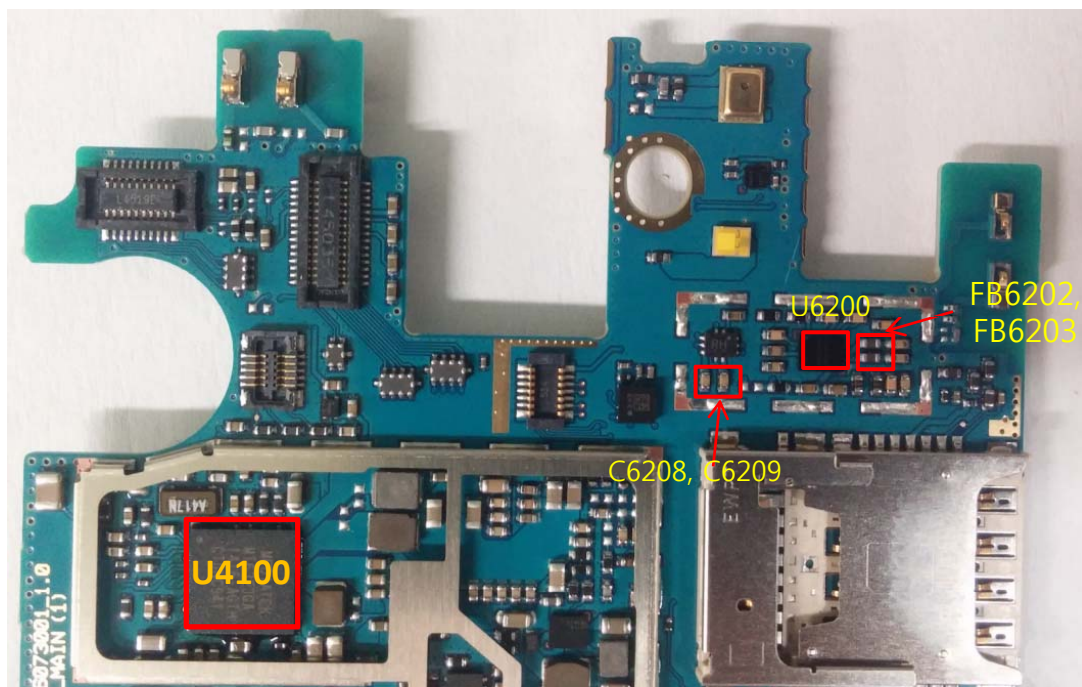
3.12.2. Speaker Trouble Shooting

It's trouble shooting guide for no sound case.

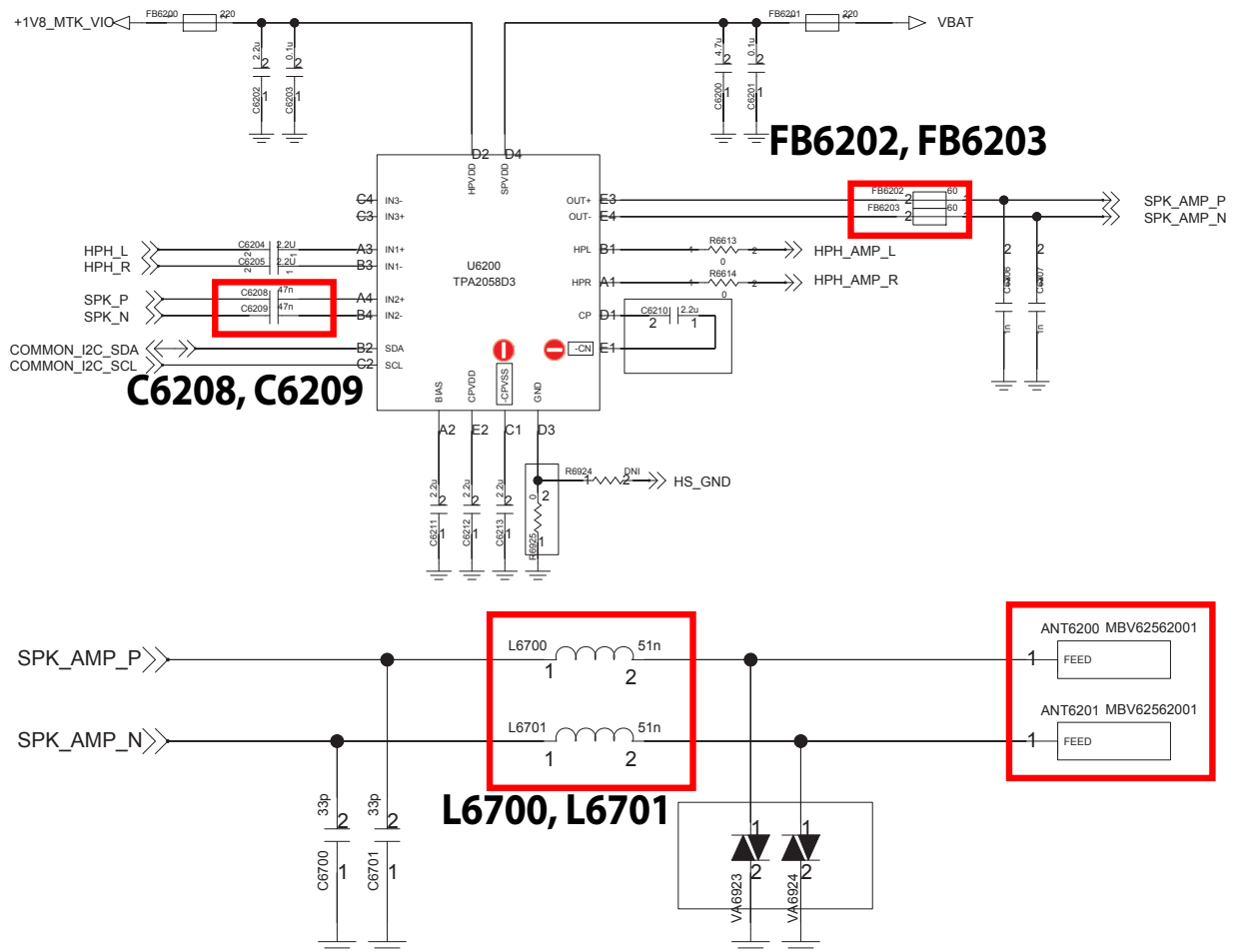
Speaker control signals are generate by MT6323(U4100)If low level volume or breaking the sound, change speaker module



4. DOWNLOAD



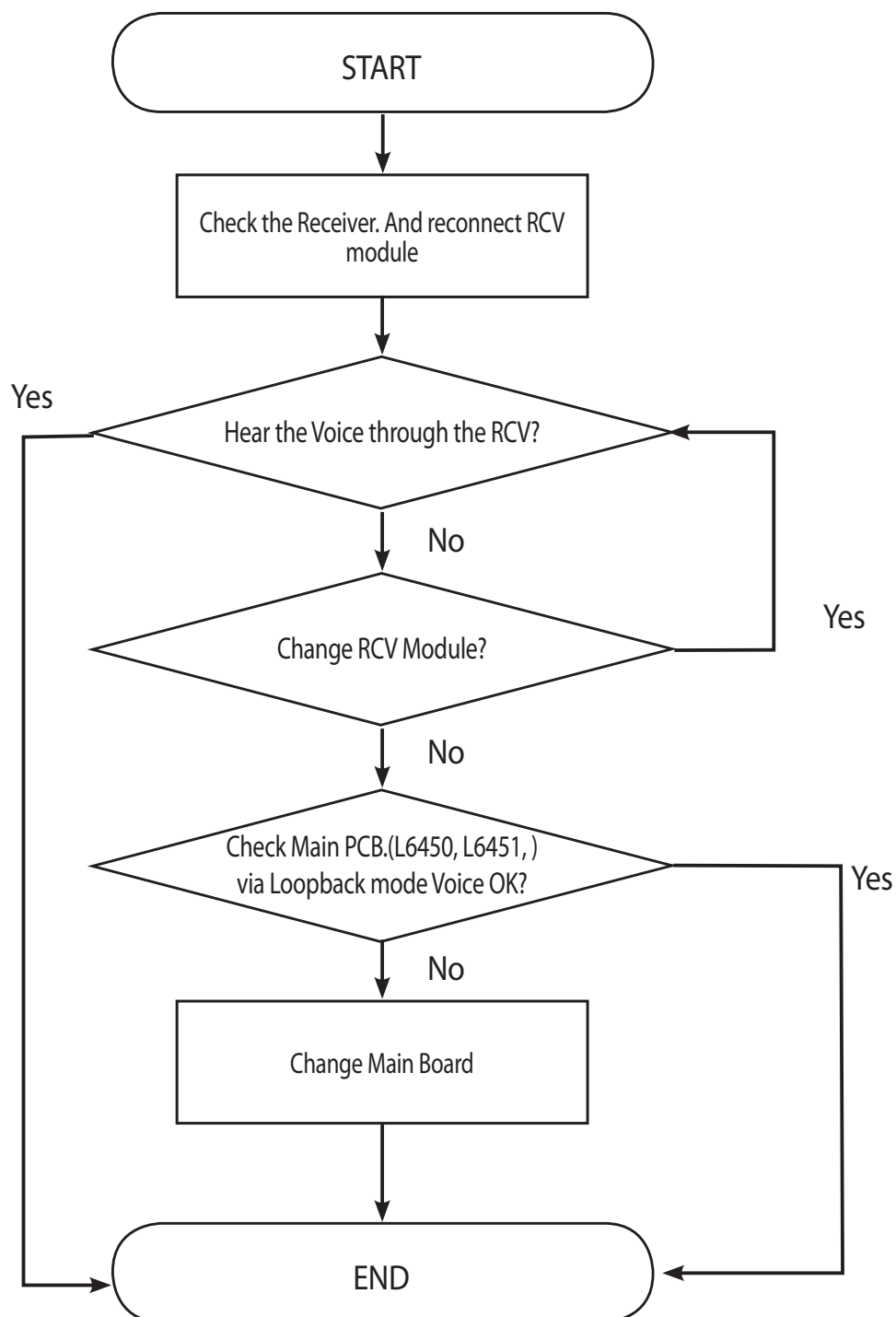
Main PCB



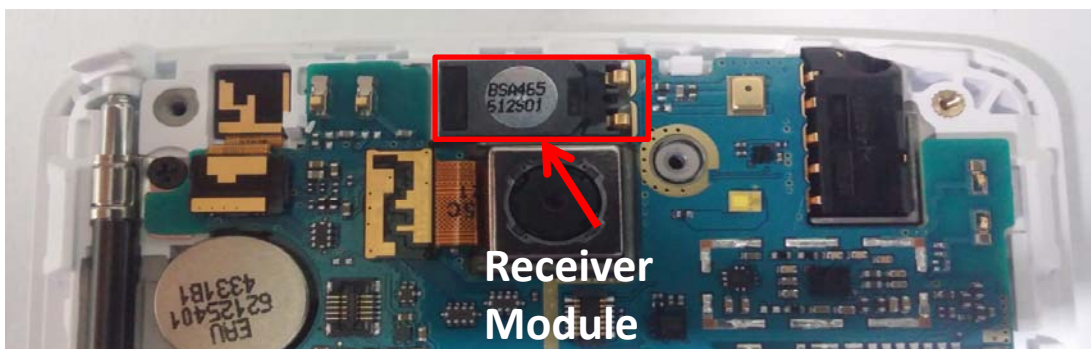
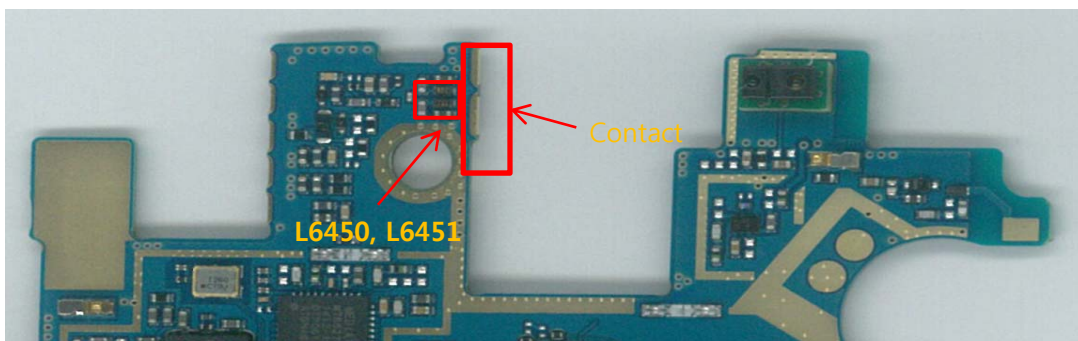
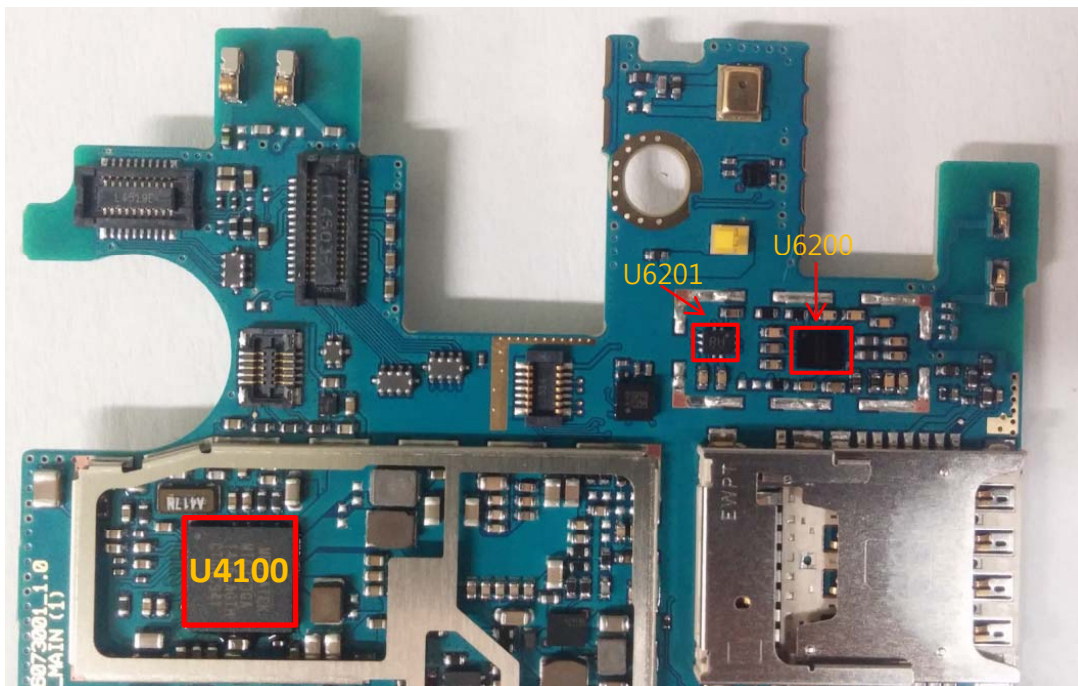
3.12.3. Receiver Trouble Shooting

It's trouble shooting guide for no sound case.

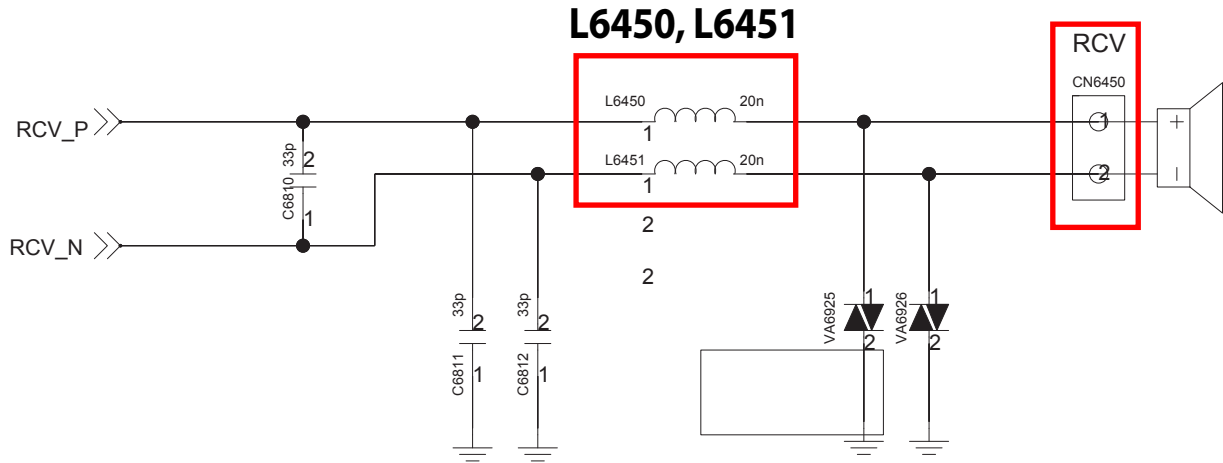
Receiver control signals are generate by MT6323(U4100)If low level volume or breaking the voice, change RCV module



4. DOWNLOAD

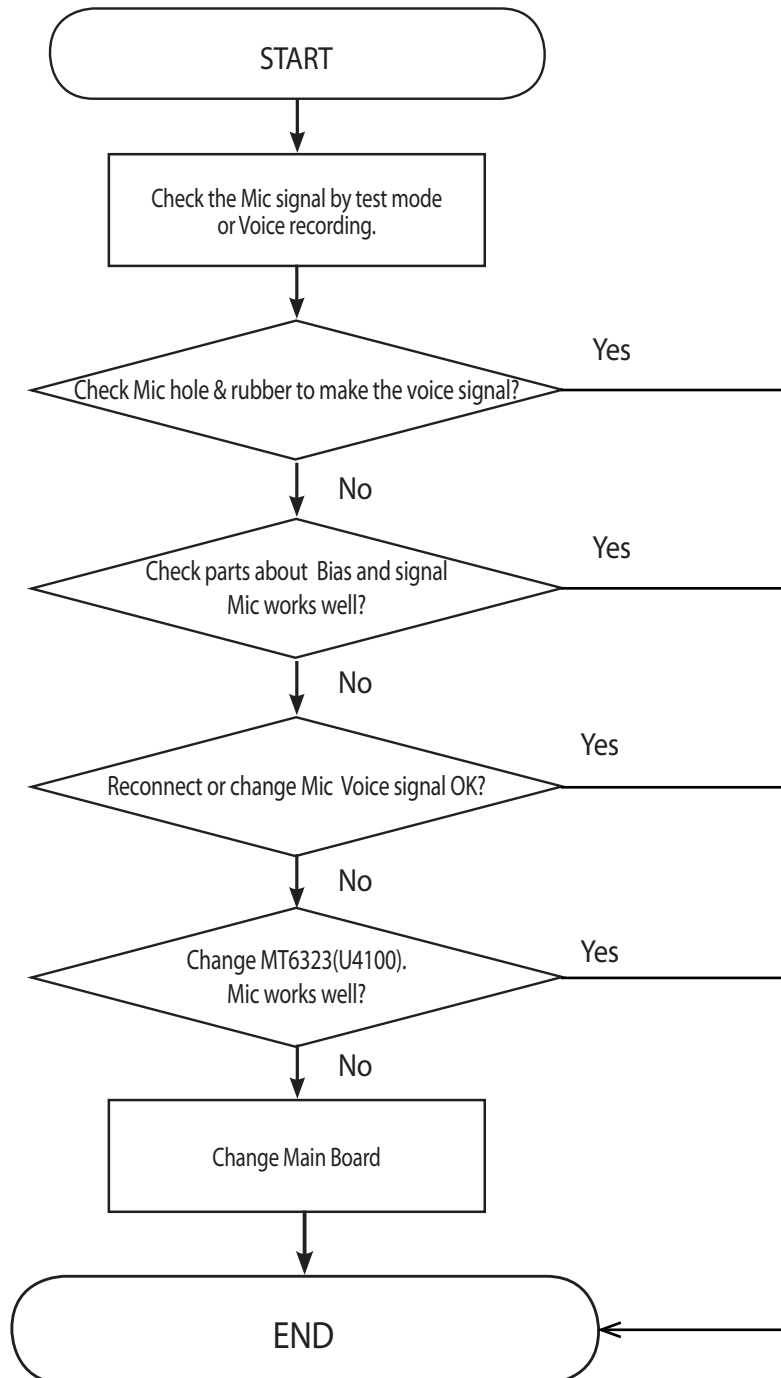


Main PCB

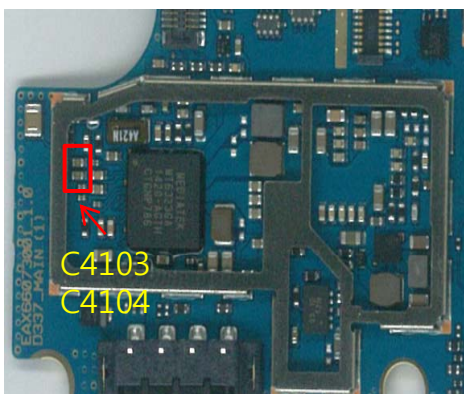
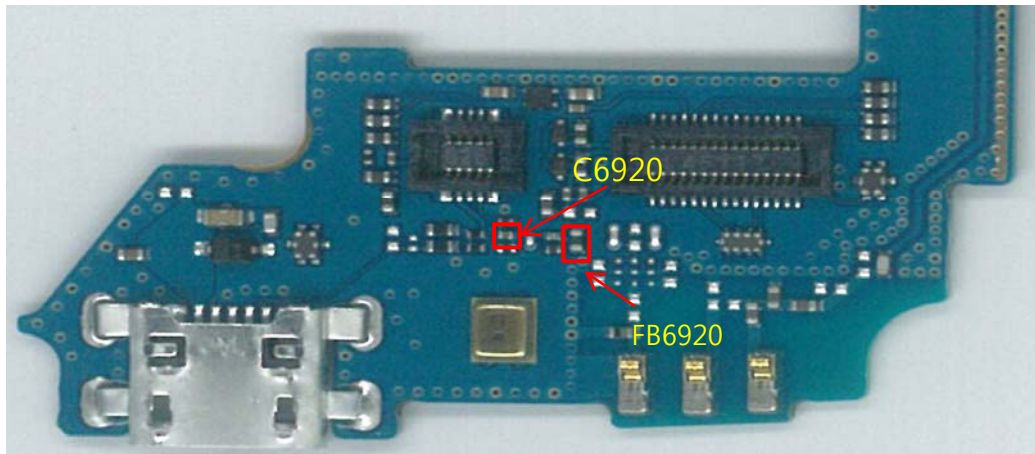


3.12.4. Main Mic Trouble Shooting

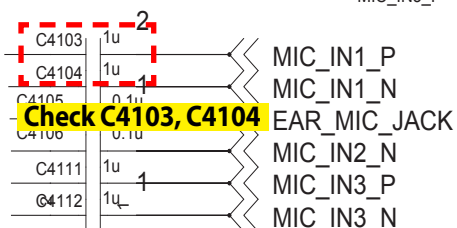
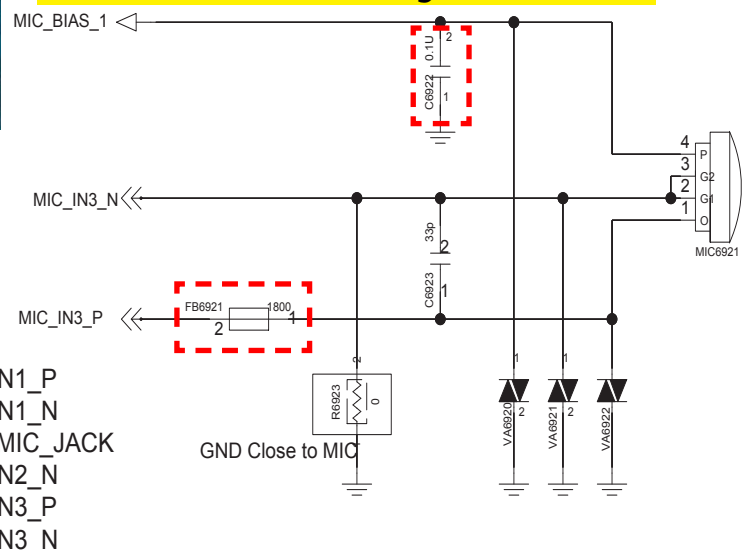
It's operating voice call(except speakerphone), voice recording, camcorder recording



Check Main Mic hole.



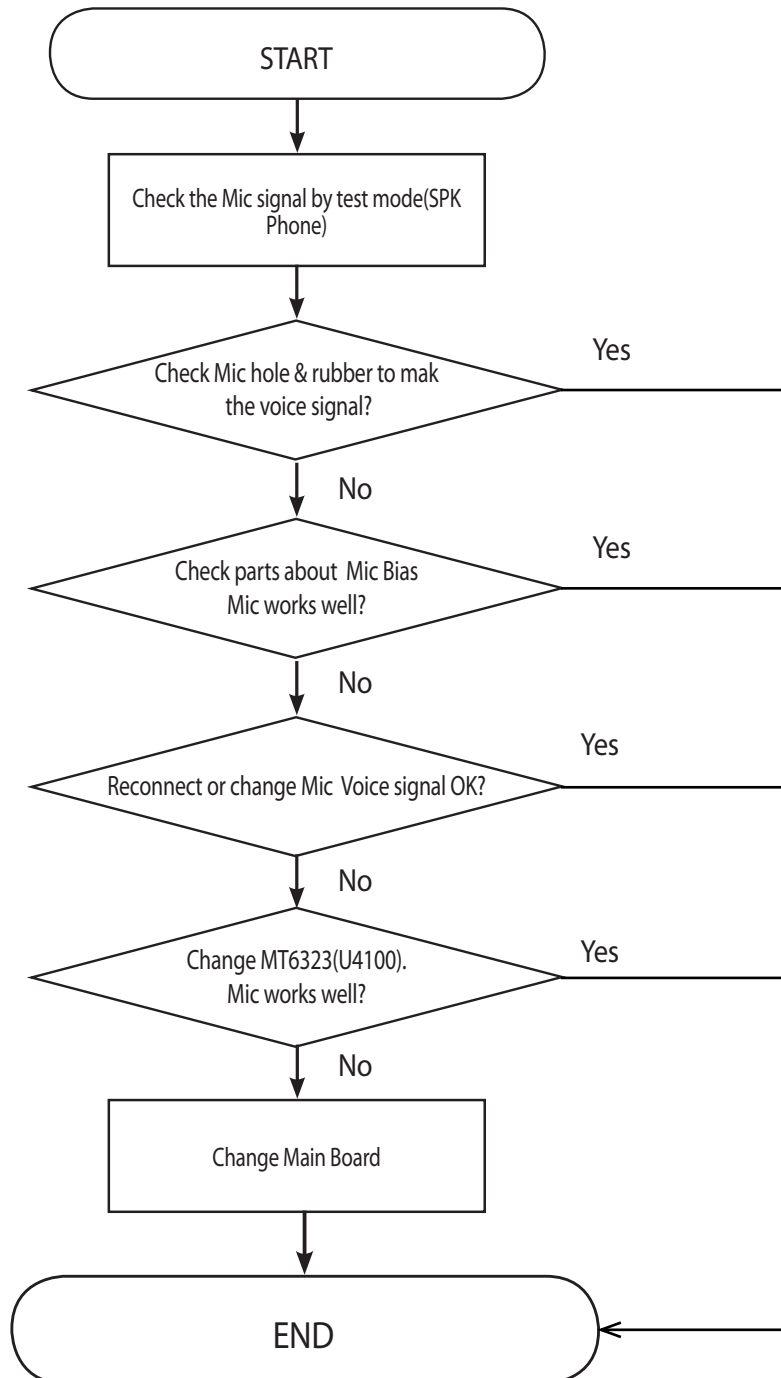
Check FB6920, VMIC voltage level (C6920)



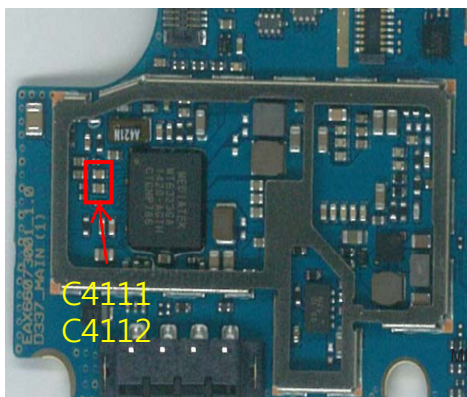
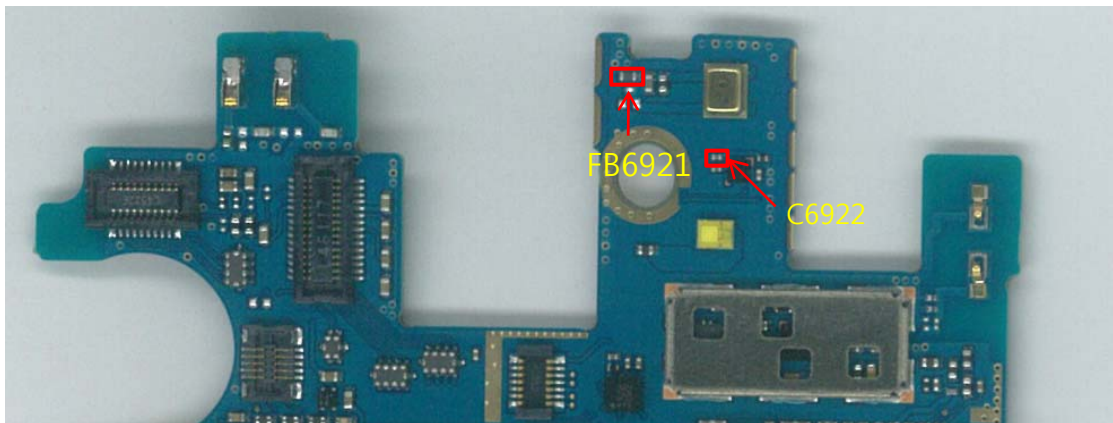
Check C4103, C4104

3.12.5. Sub Mic Trouble Shooting

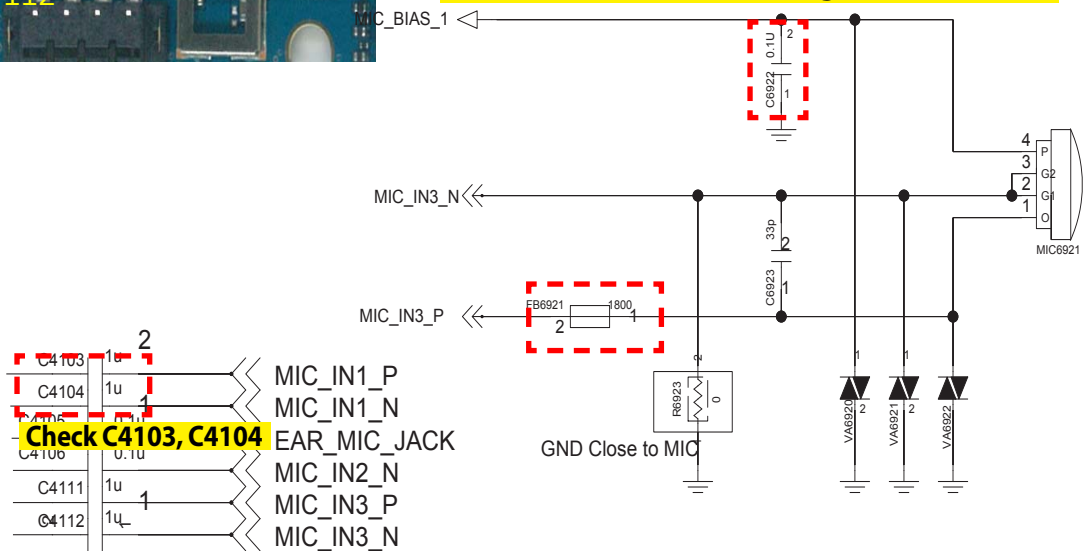
It's operating Speakerphone call



Check Main Mic hole.



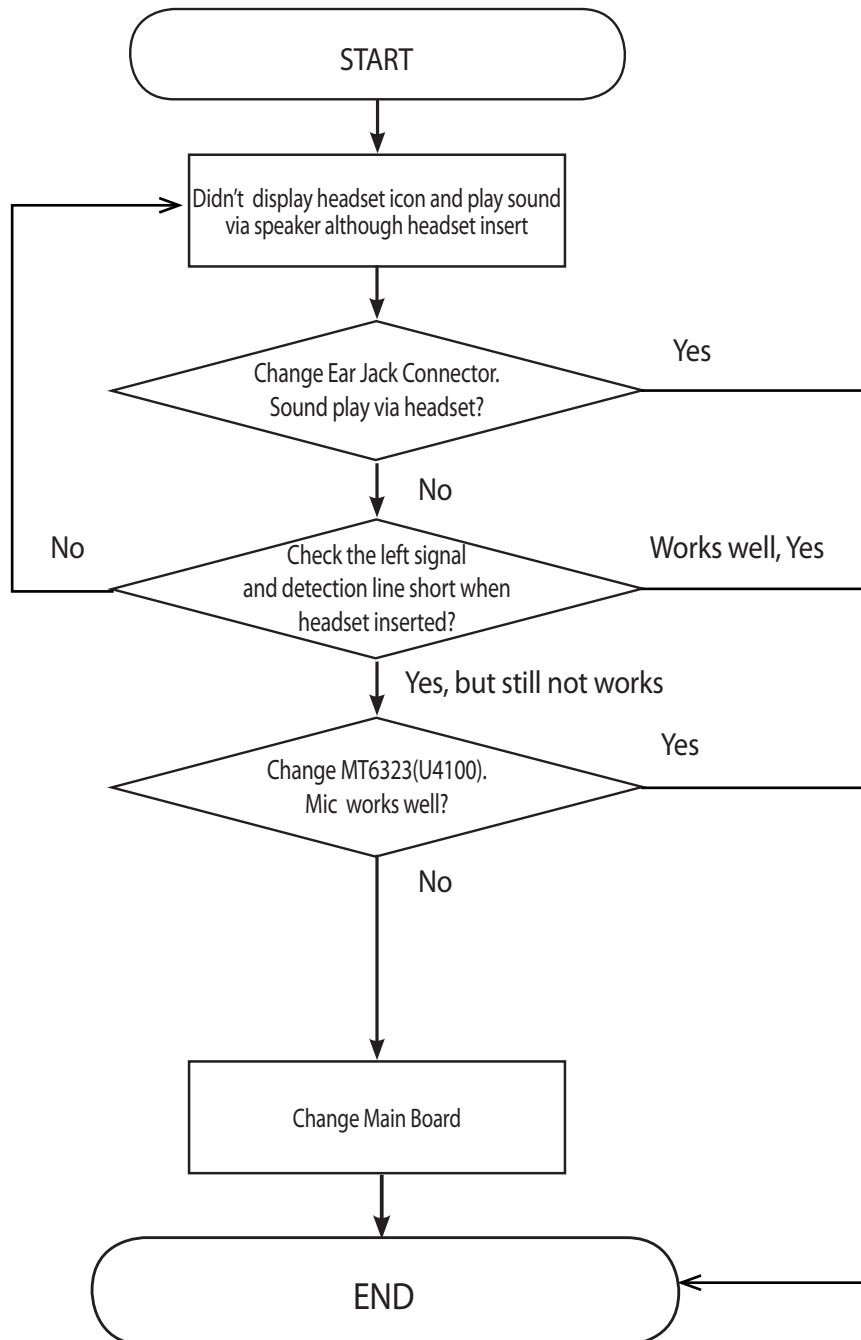
Check FB6921, VMIC voltage level (C6922)

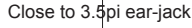


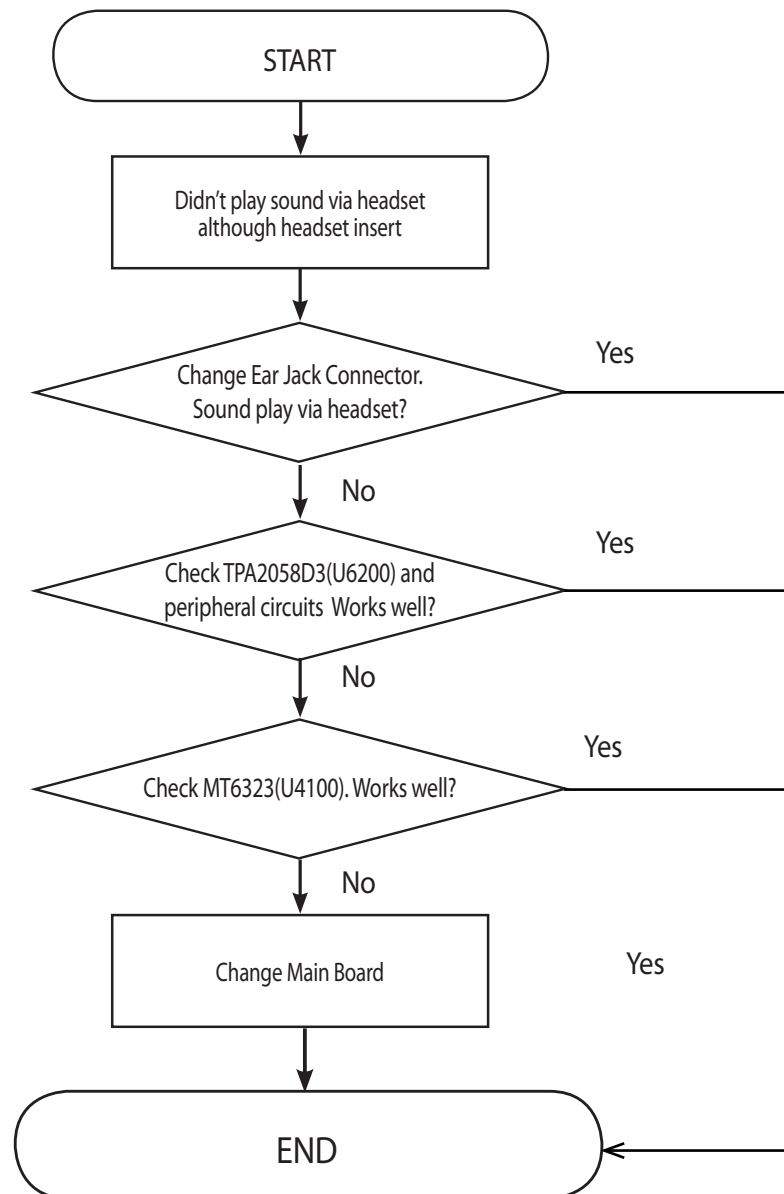
3.12.6. Ear-Mic Trouble Shooting

Ear MIC control signals are generate by MT6323(U4100)

Case A. Disable detected headset insert.

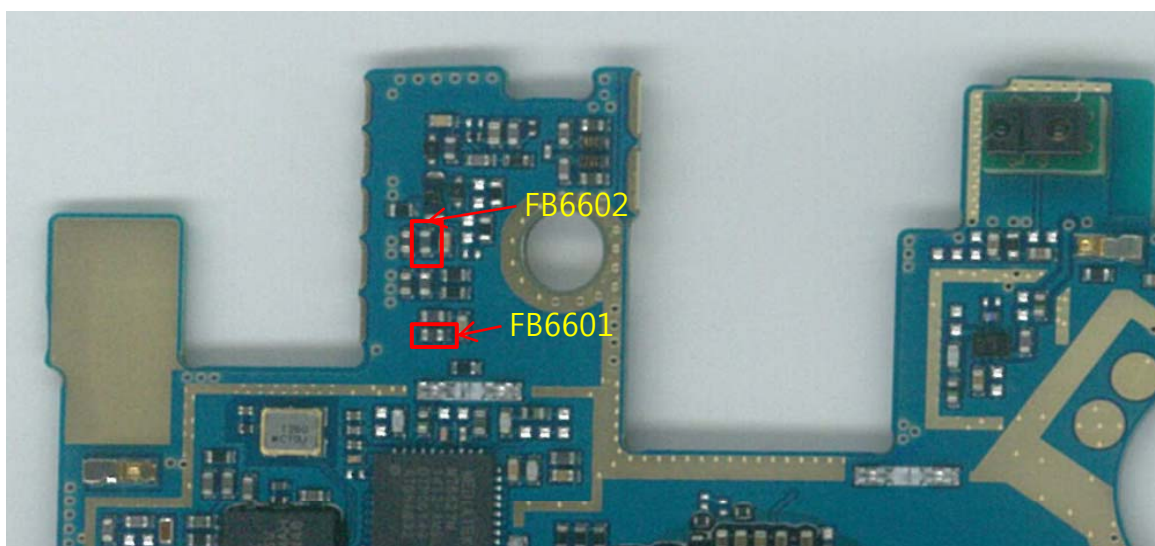
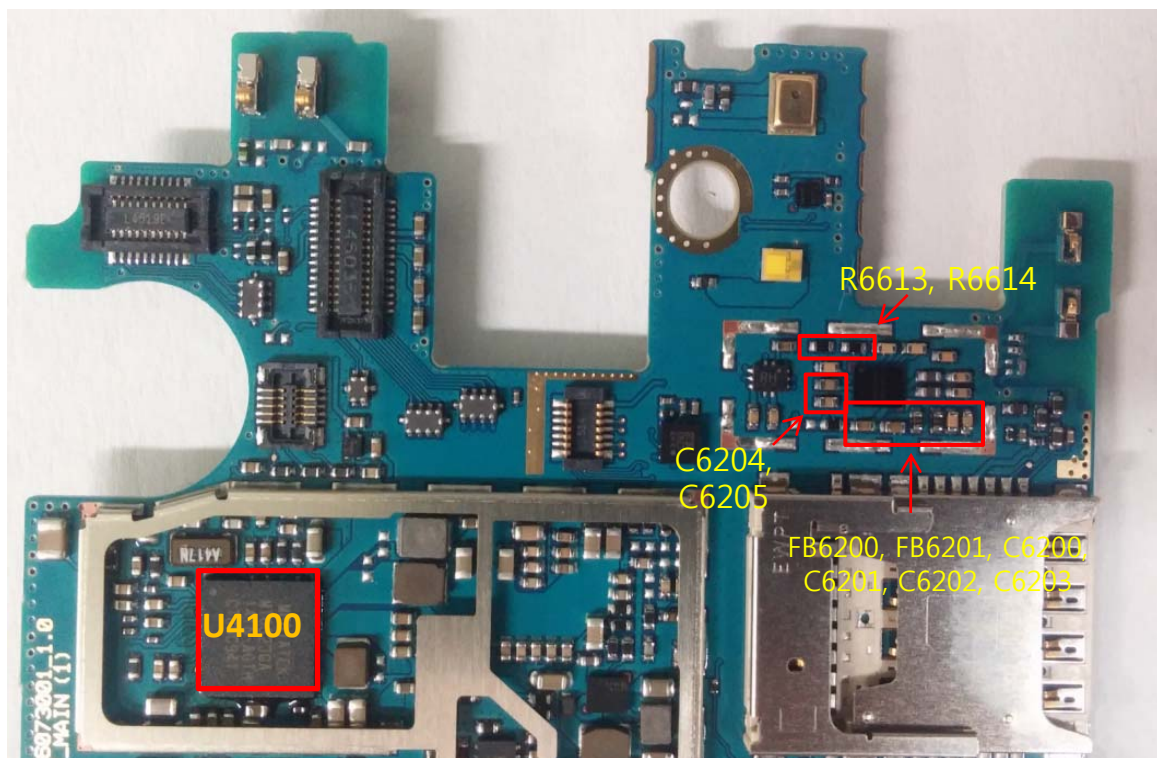




Case B. No Sound from Headset

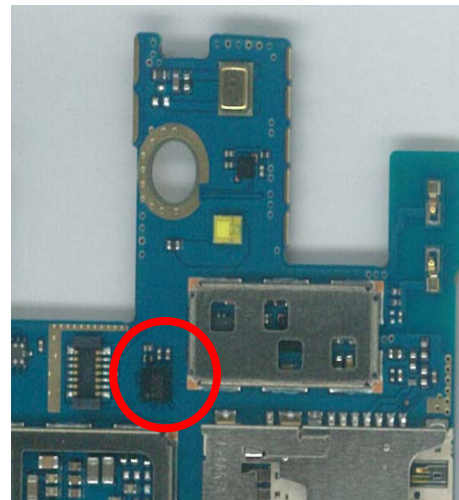
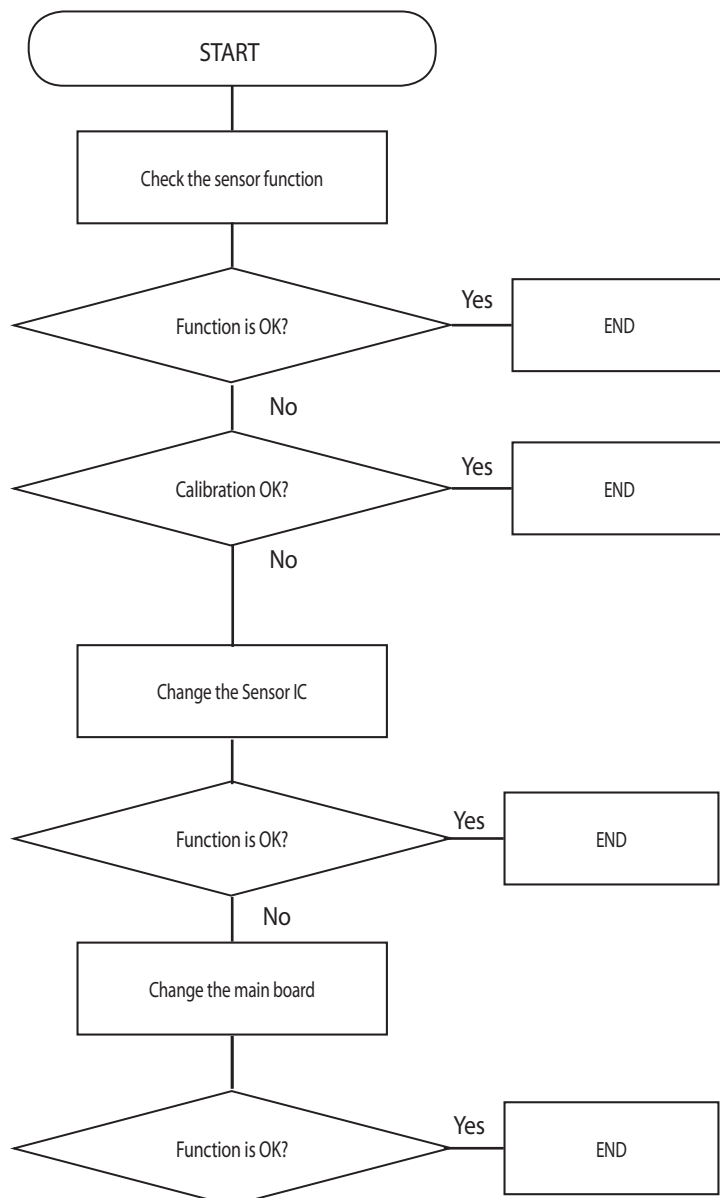


In case of you can't heard anything, check C6204, C6205, R6613, R6614, FB6601, FB6602.

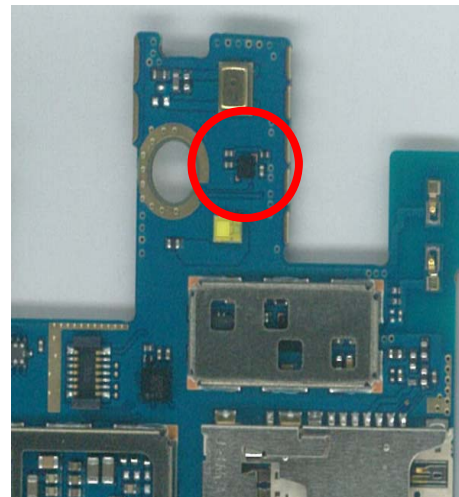


3.13. Accelerometer & Compass Sensor on/off TS

The **Accl. & compass** sensors are calibrated by using SW algorithm.



Accelerometer



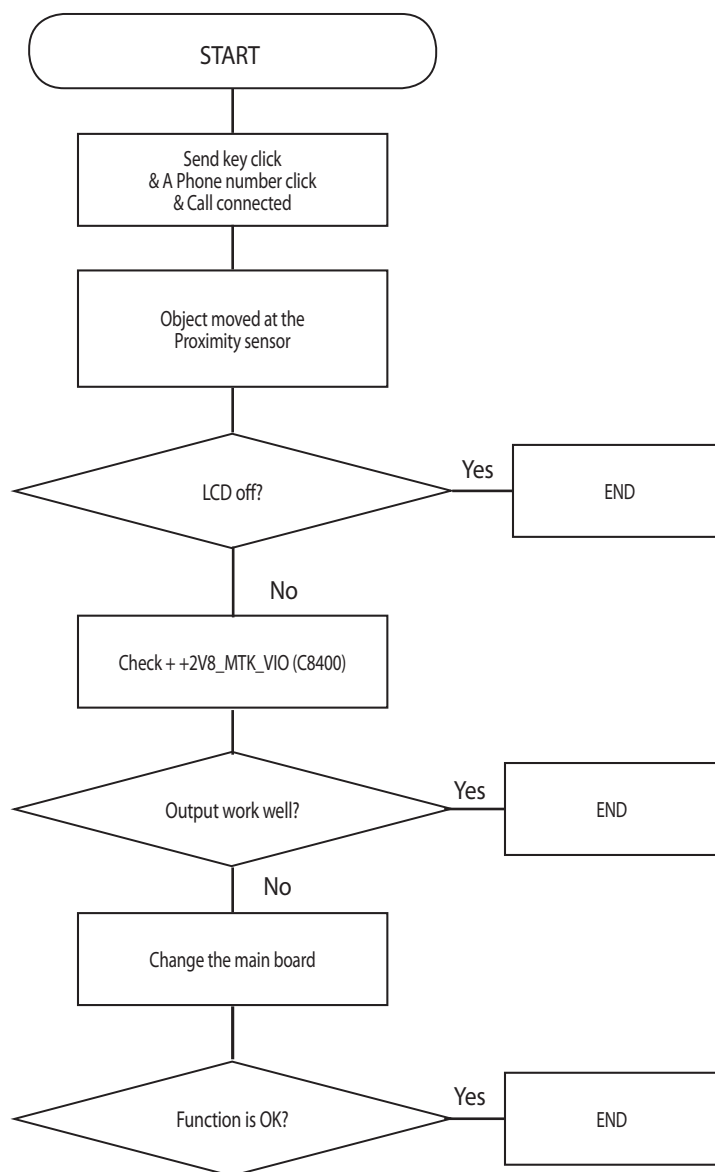
Compass sensor

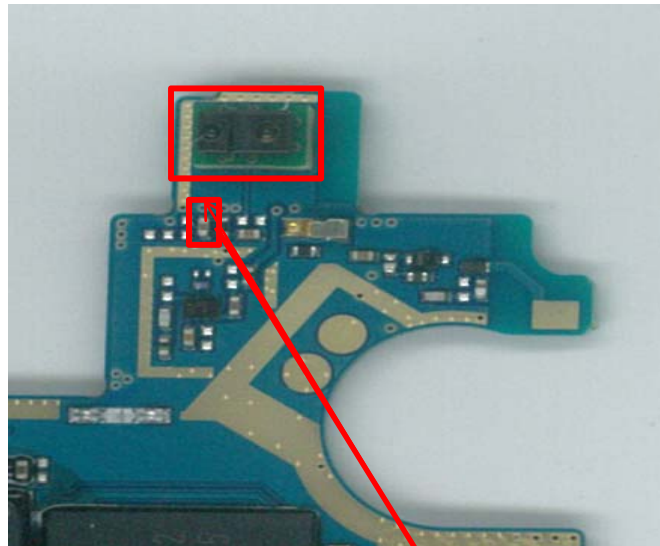
3.14. Proximity Sensor on/off TS

Proximity Sensor is worked as below:

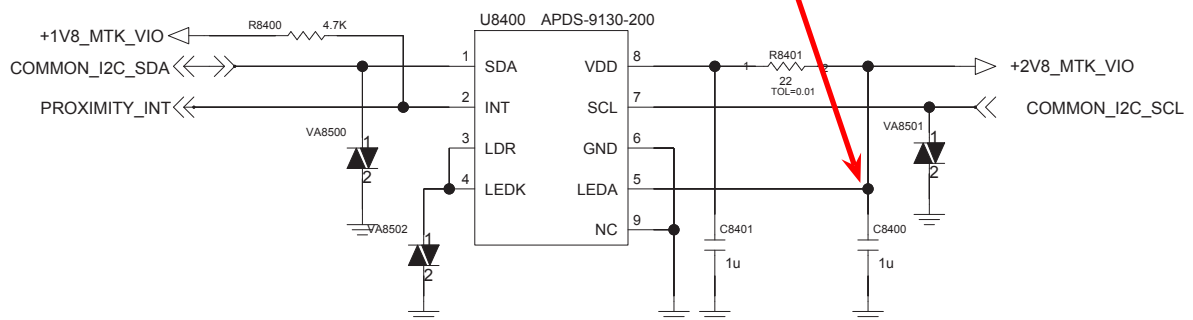
Send Key click -> Phone number click -> Call connected -> Object moved at the sensor

-> Control the screen's on/off operation automatically

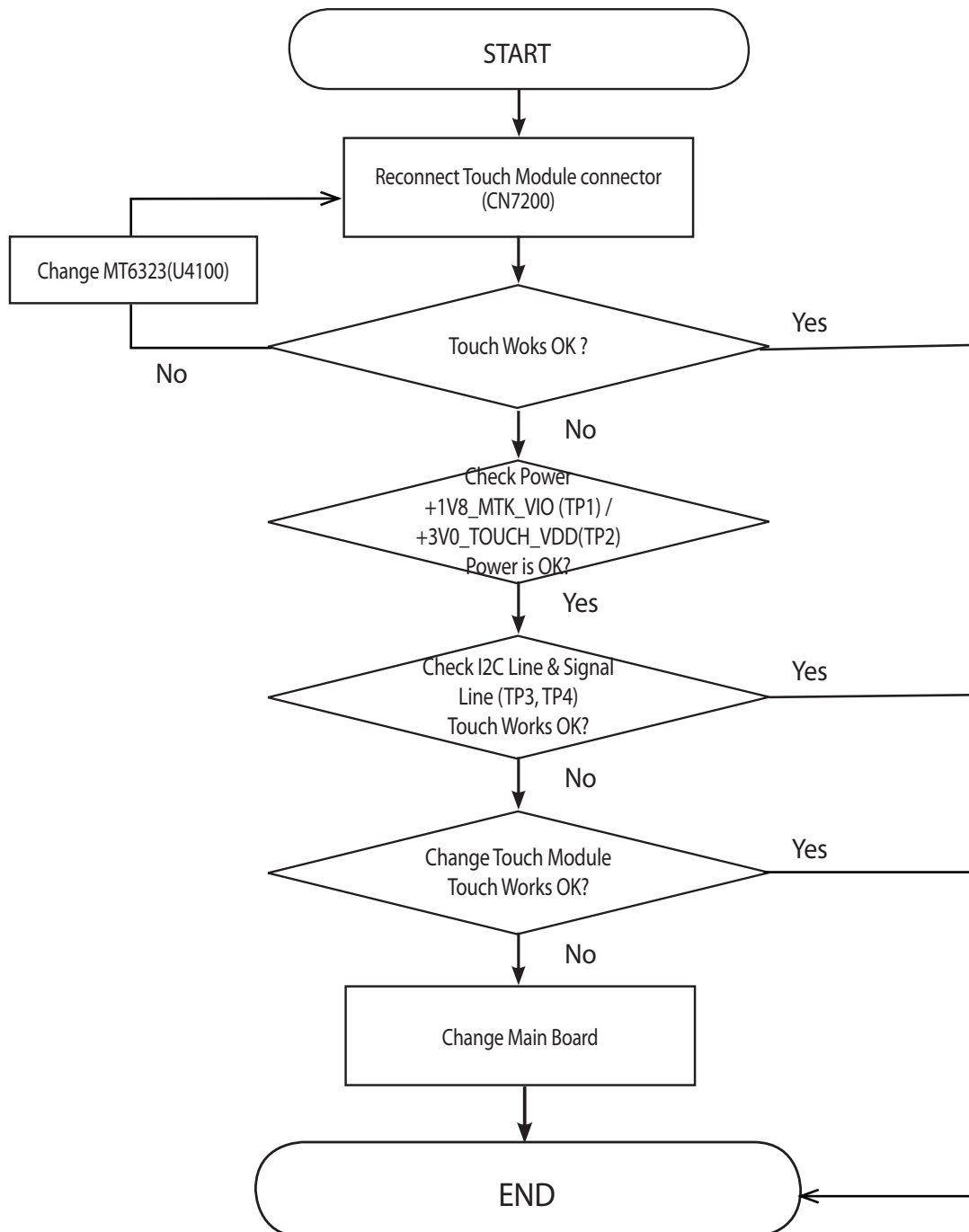


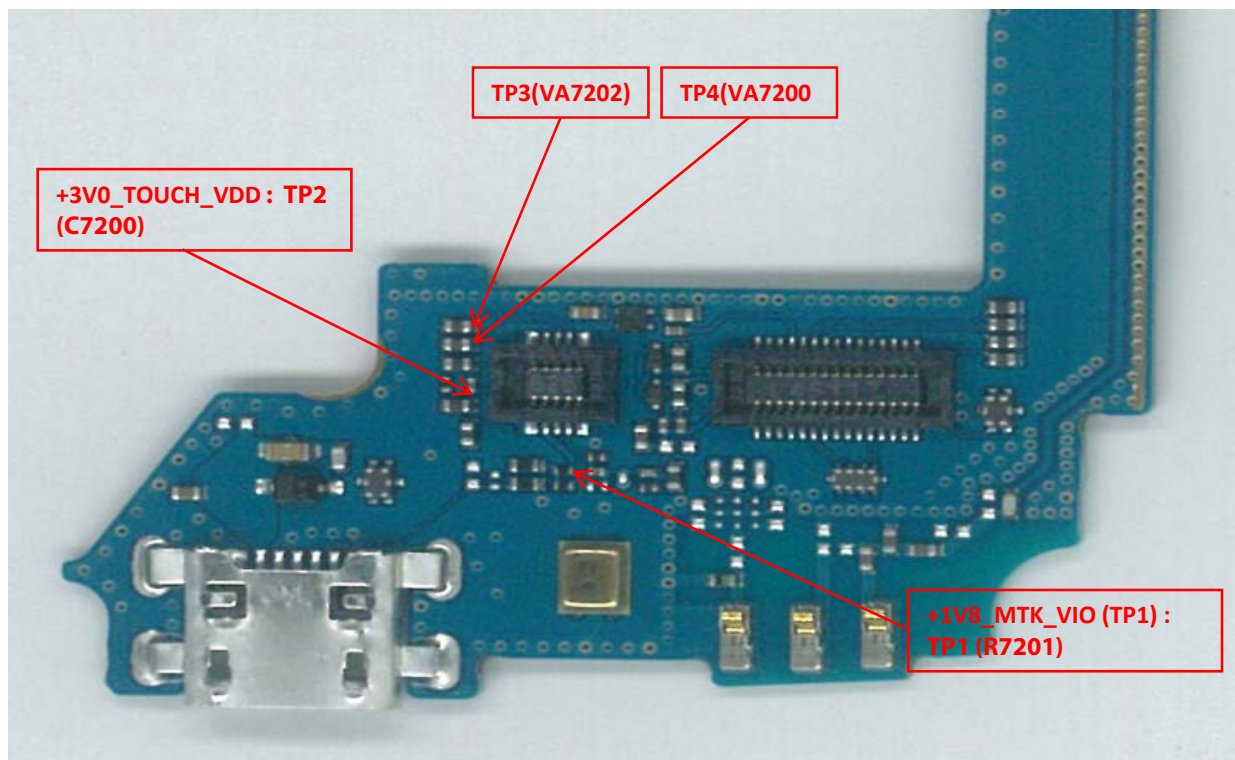
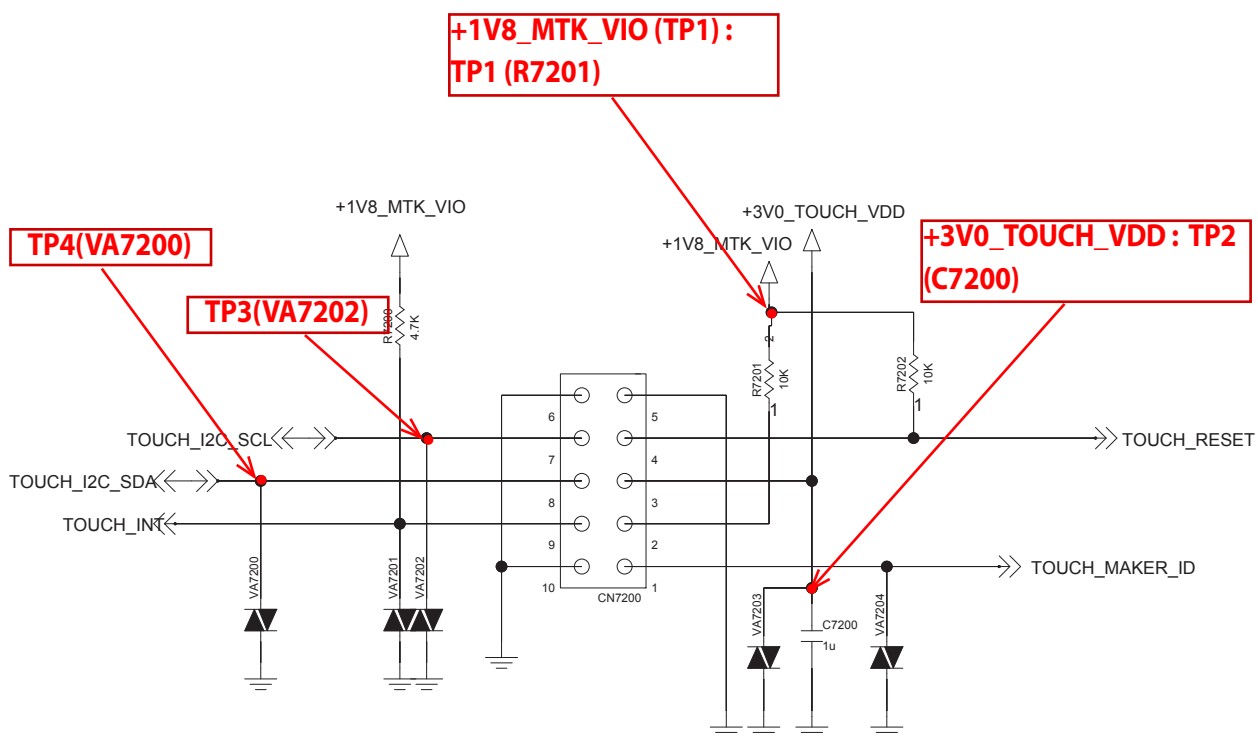


Sensor Power: C8400



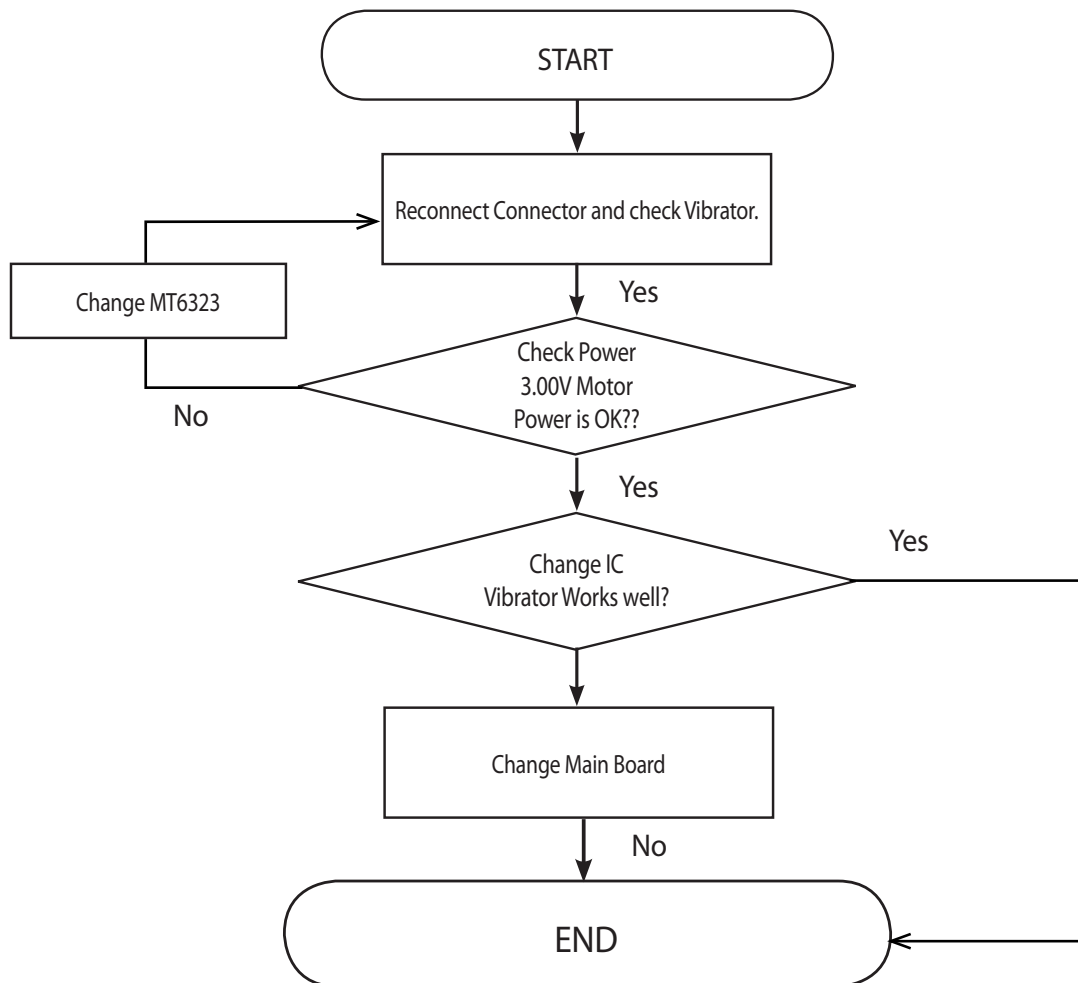
3.15. Touch Sensor Trouble Shooting

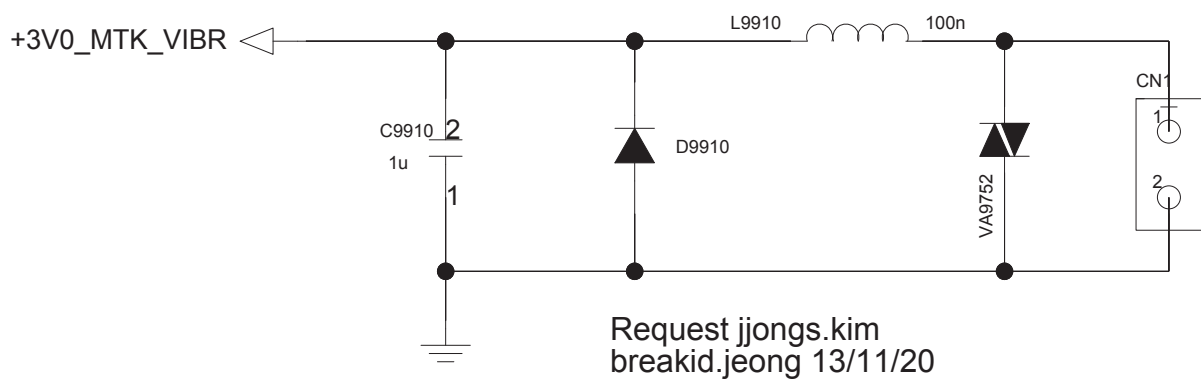
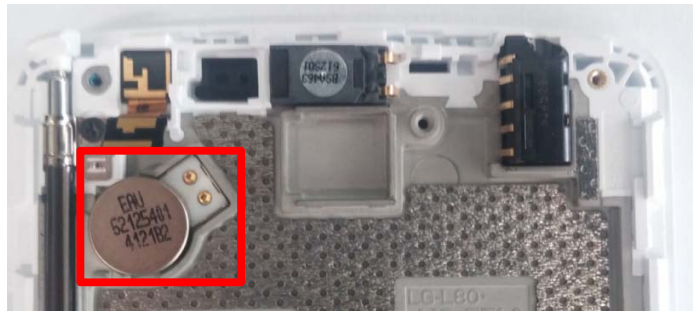
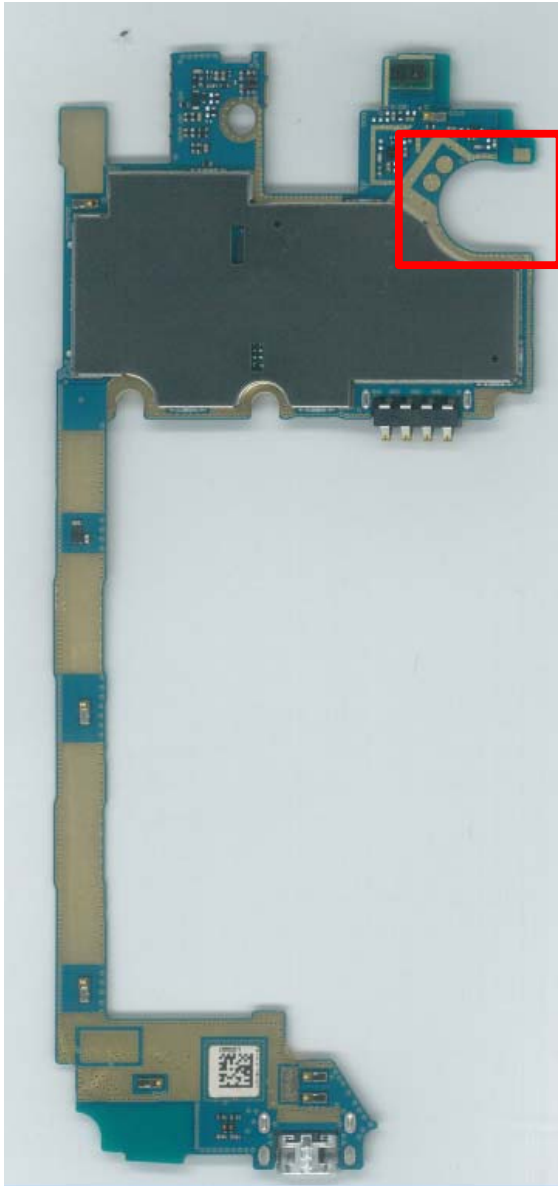




3.16. Vibrator Trouble Shooting

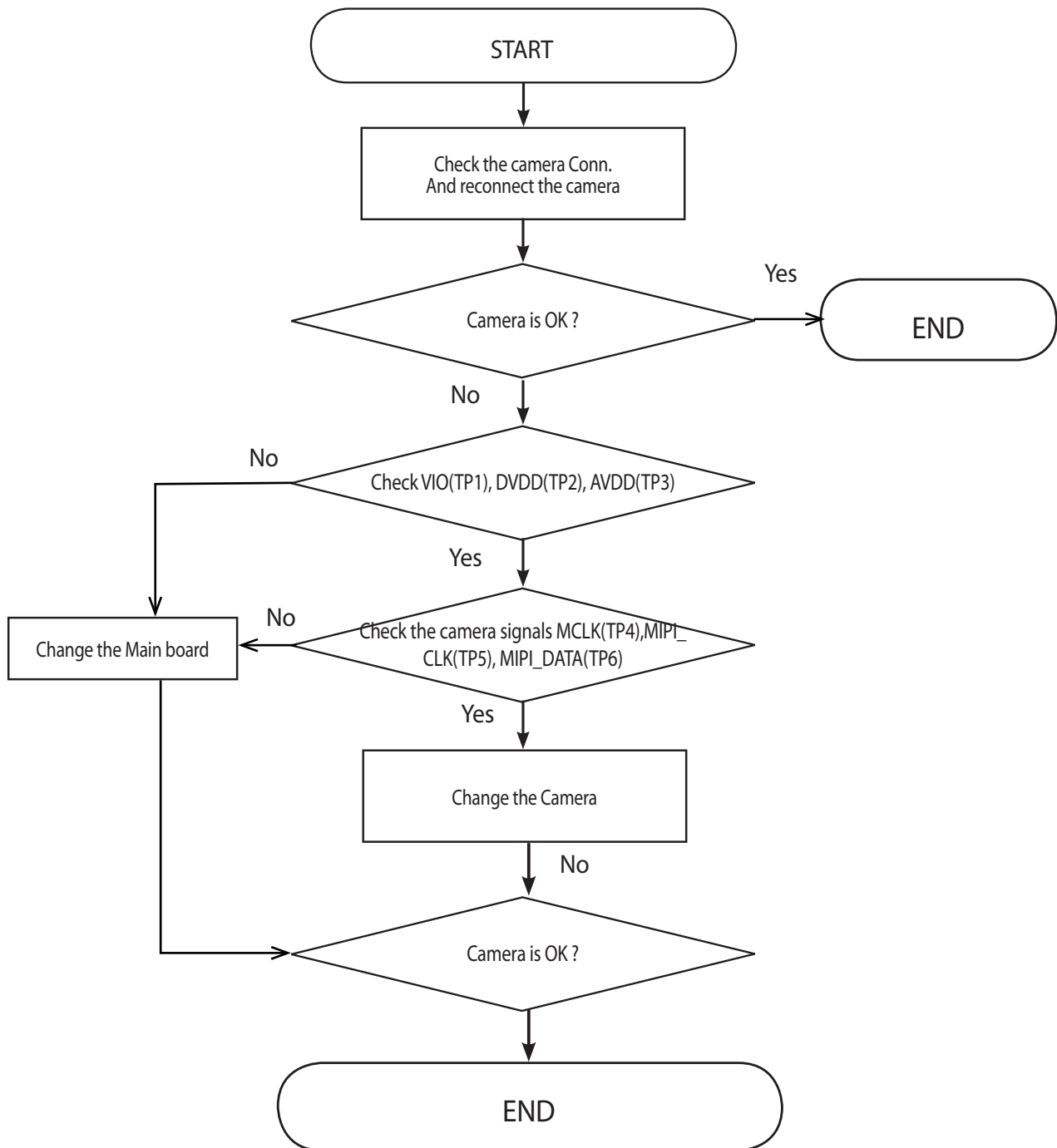
Motor Type is small cylinder.



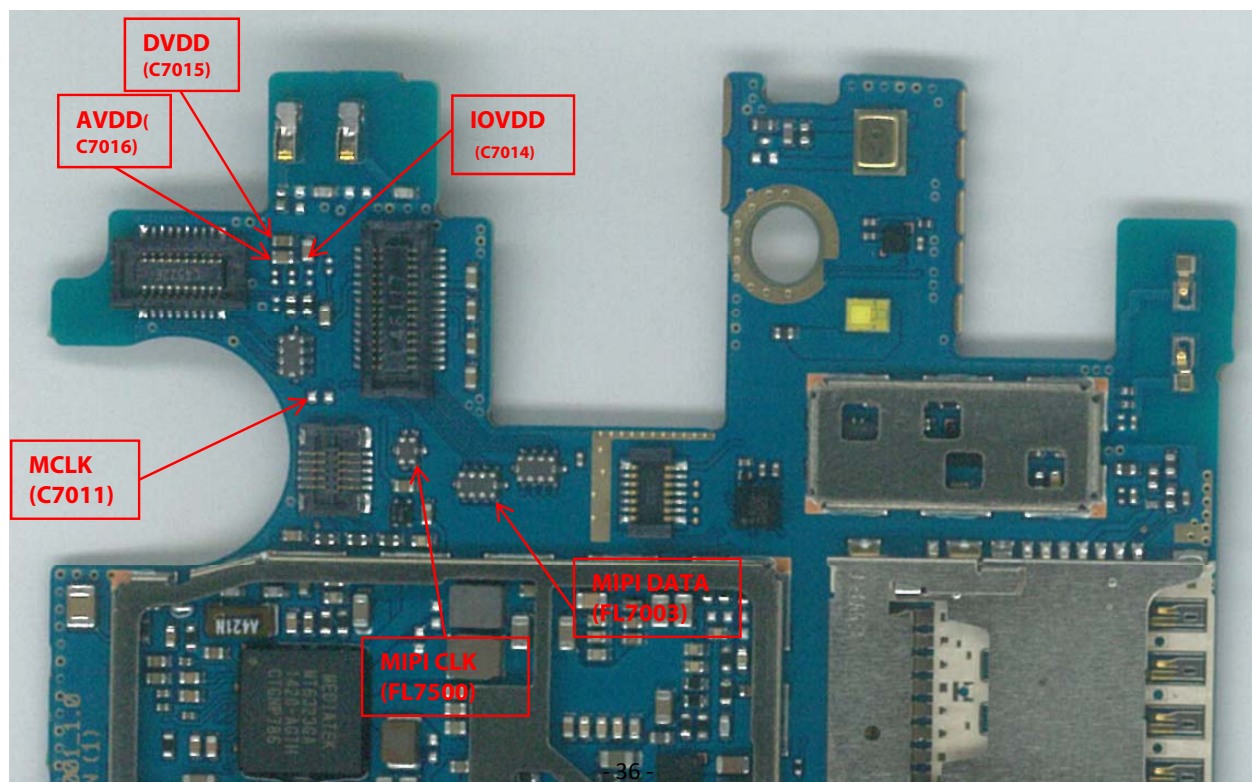
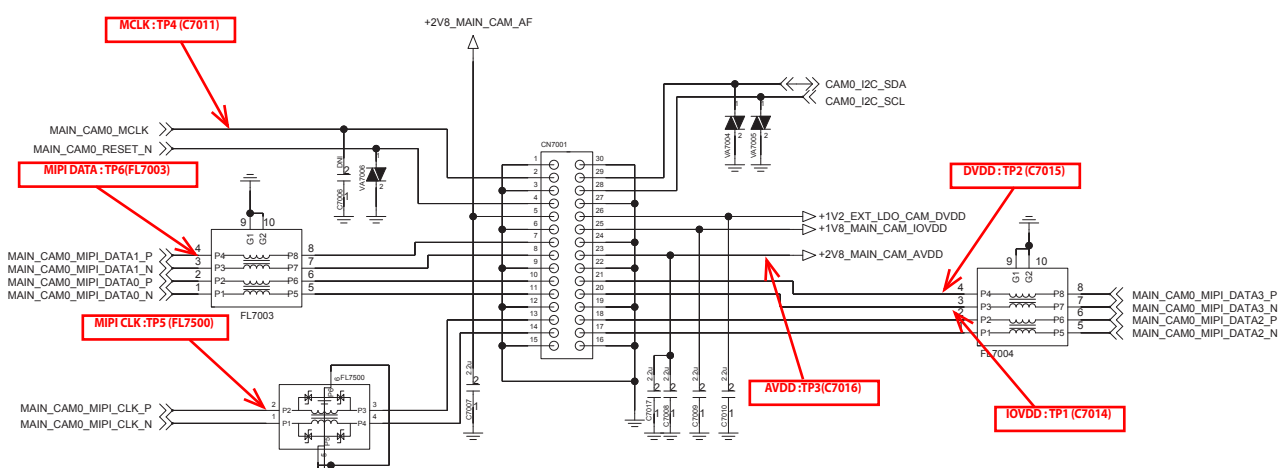


3.17. 8M Main Camera TS

8M camera control signals are generated by **MT6582 (U2100 : Main Chipset)**.
And powered by voltage regulator (**U4100**)

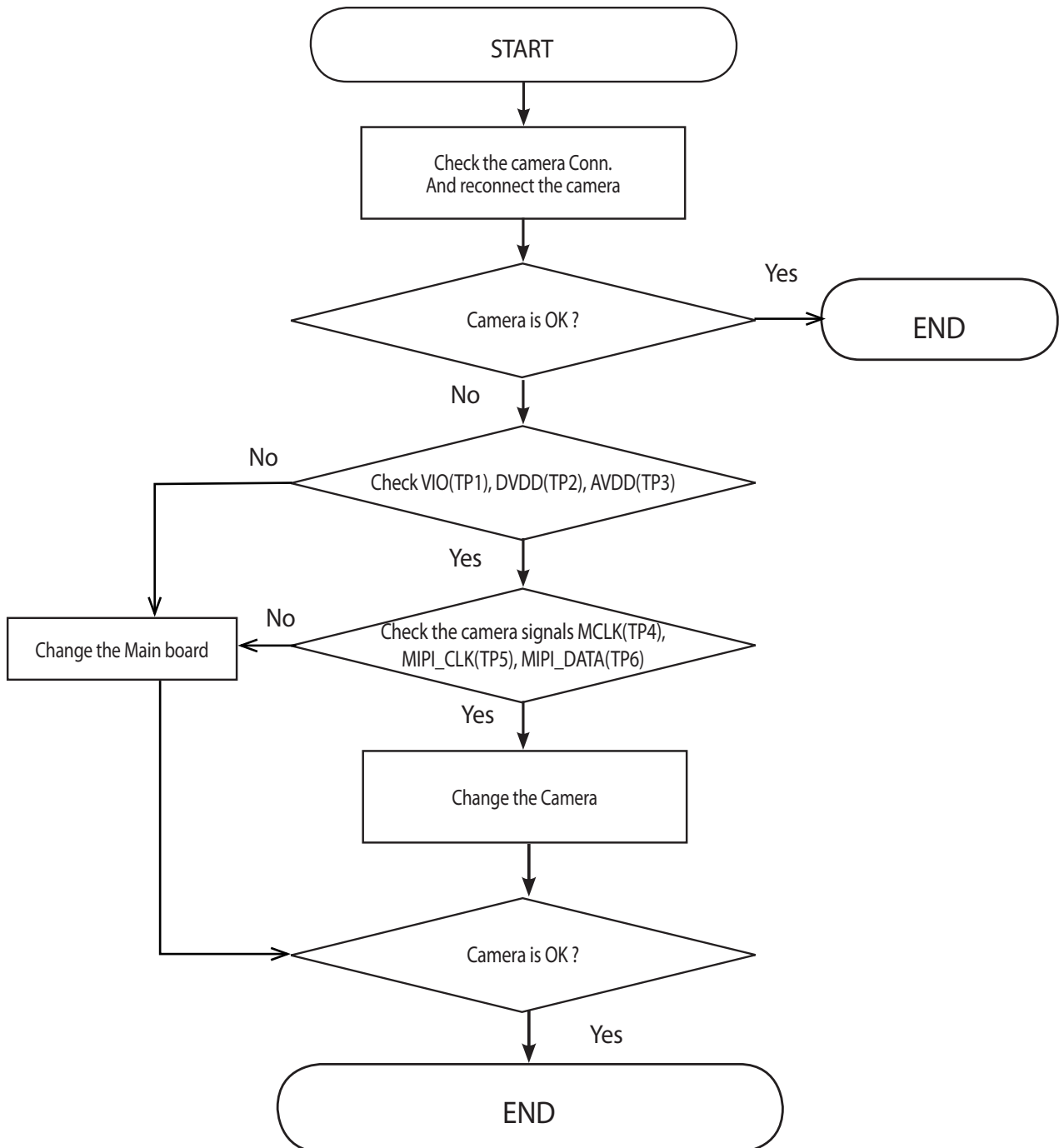


< Main_Camera_30pin_8M >

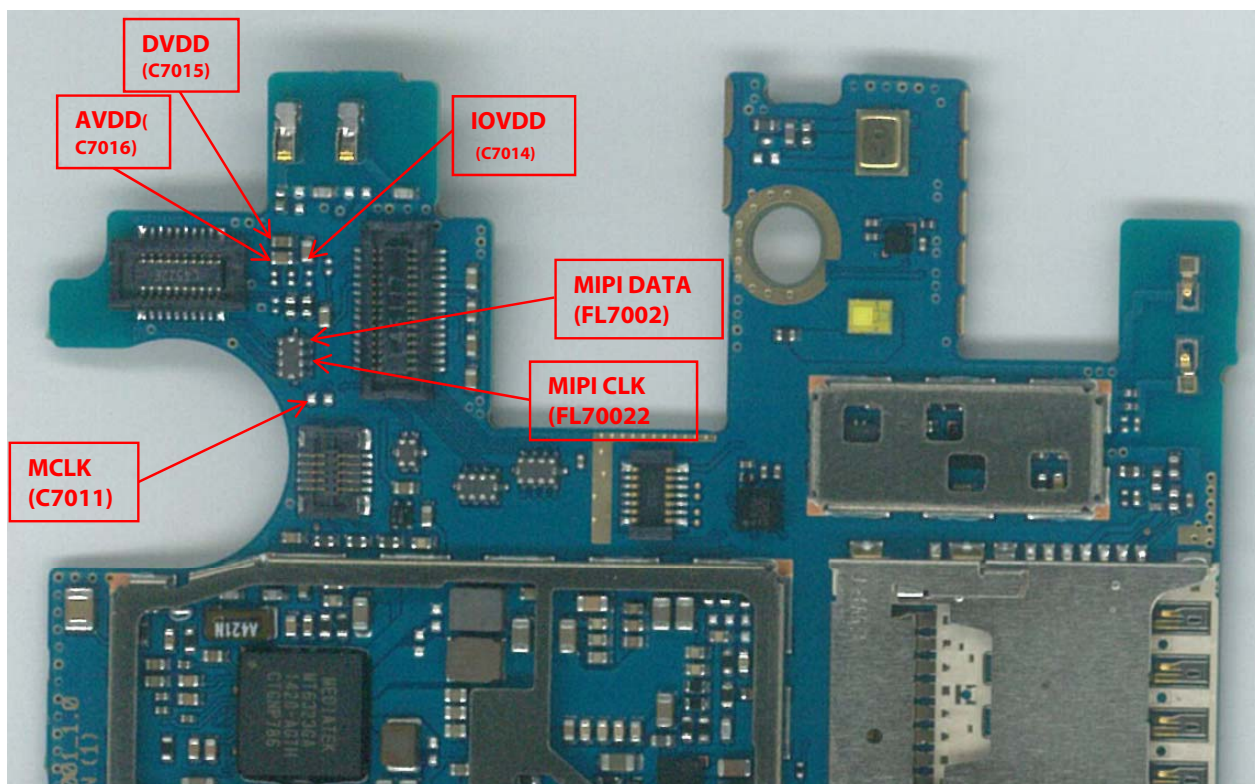
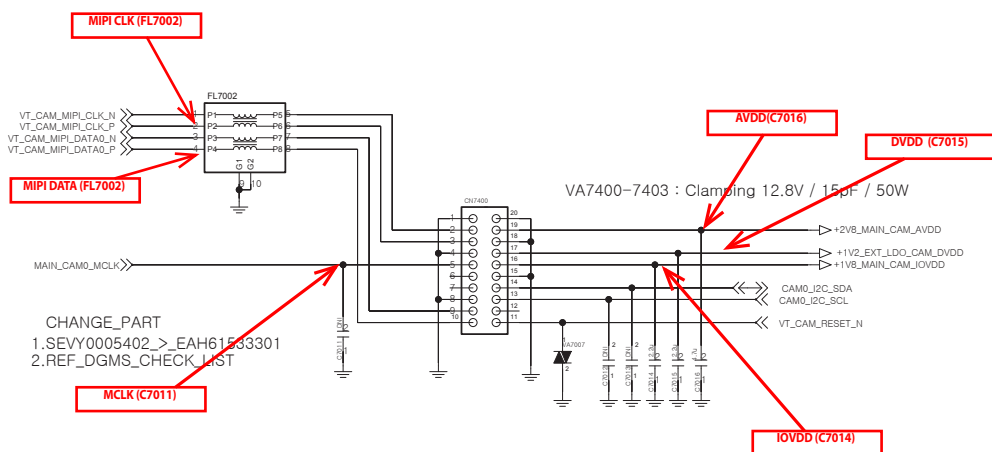


3.18. VT Camera TS

VGA camera control signals are generated by **MT6582 (U2100 : Main Chipset)**.
And powered by voltage regulator (**U4100**)

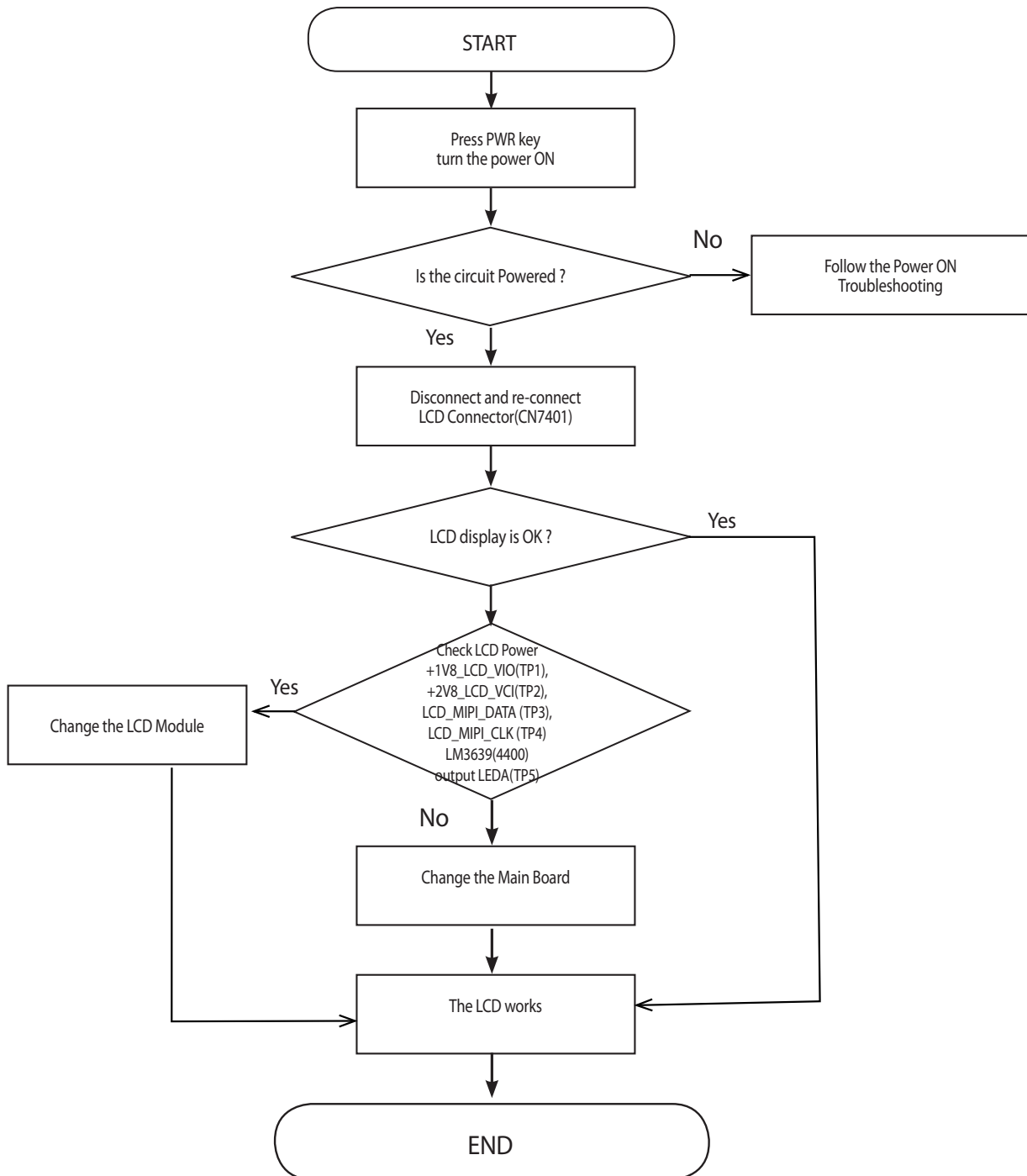


< 7-4-1_VT_Camera_20pin_VGA_1.26M_1.3M_2.4M>

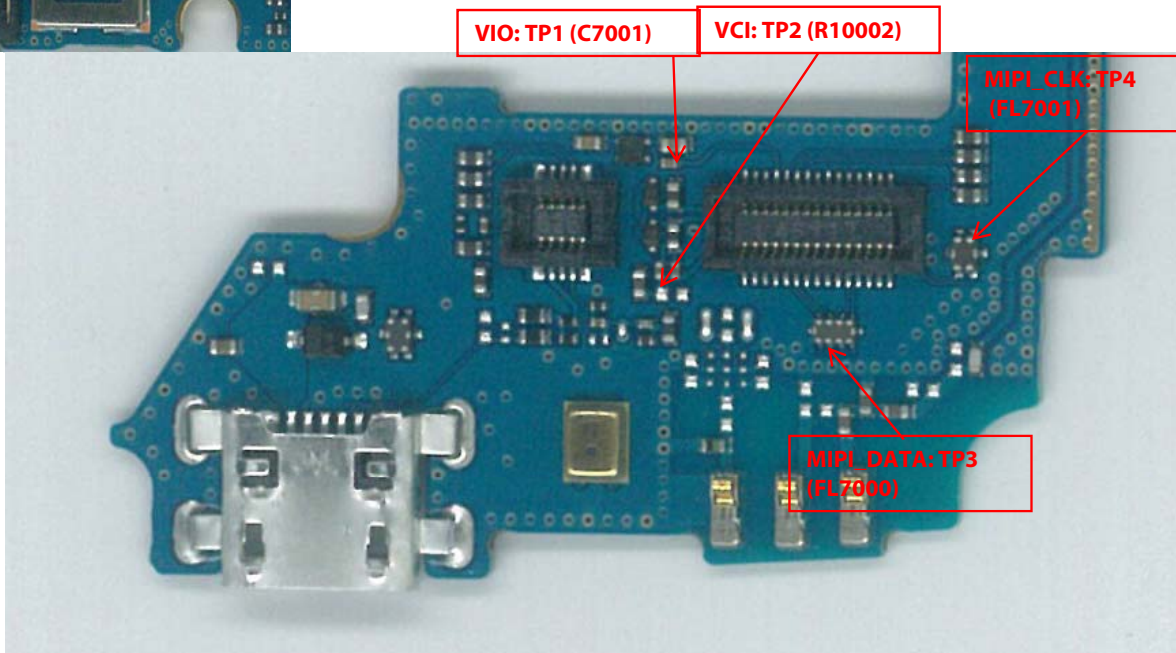
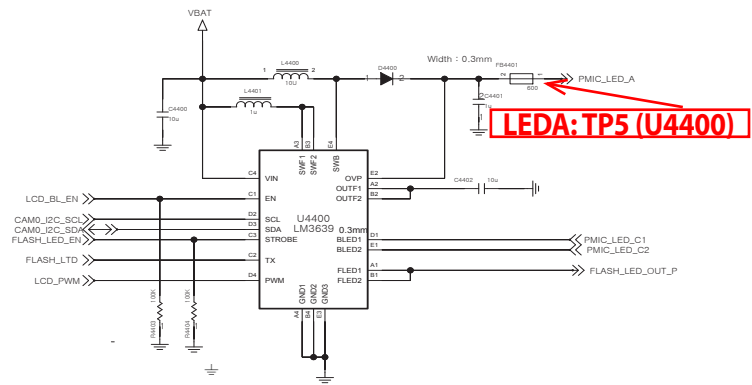
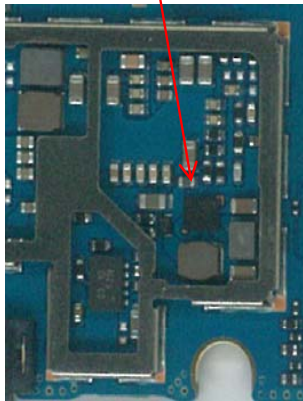
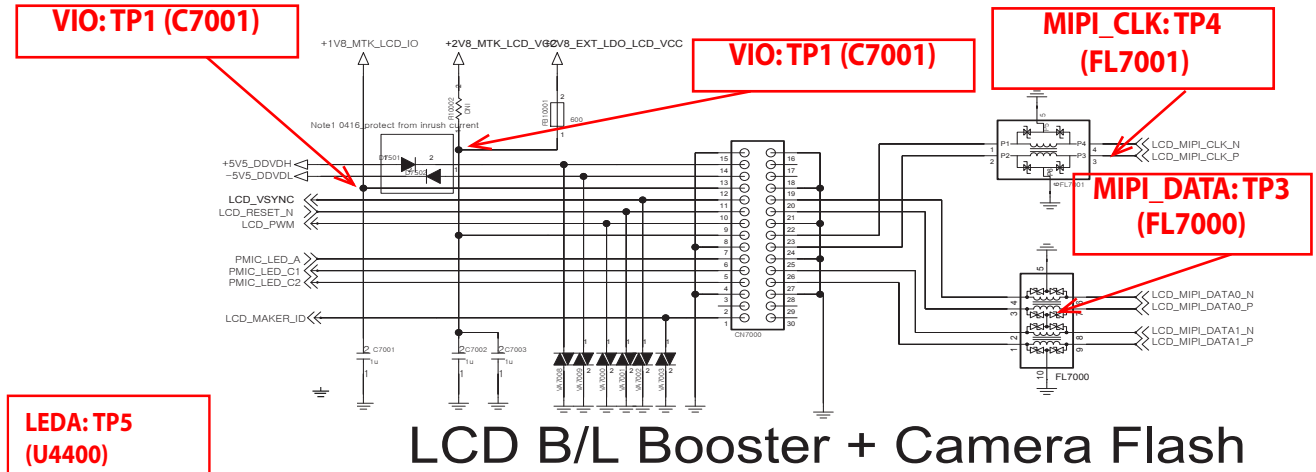


3.19. Main LCD TS

The LCD control signals are generated by **MT6582**. It needs three external powers like VIO, VCI. And its interface is MIPI having three data lanes and one clock lane.

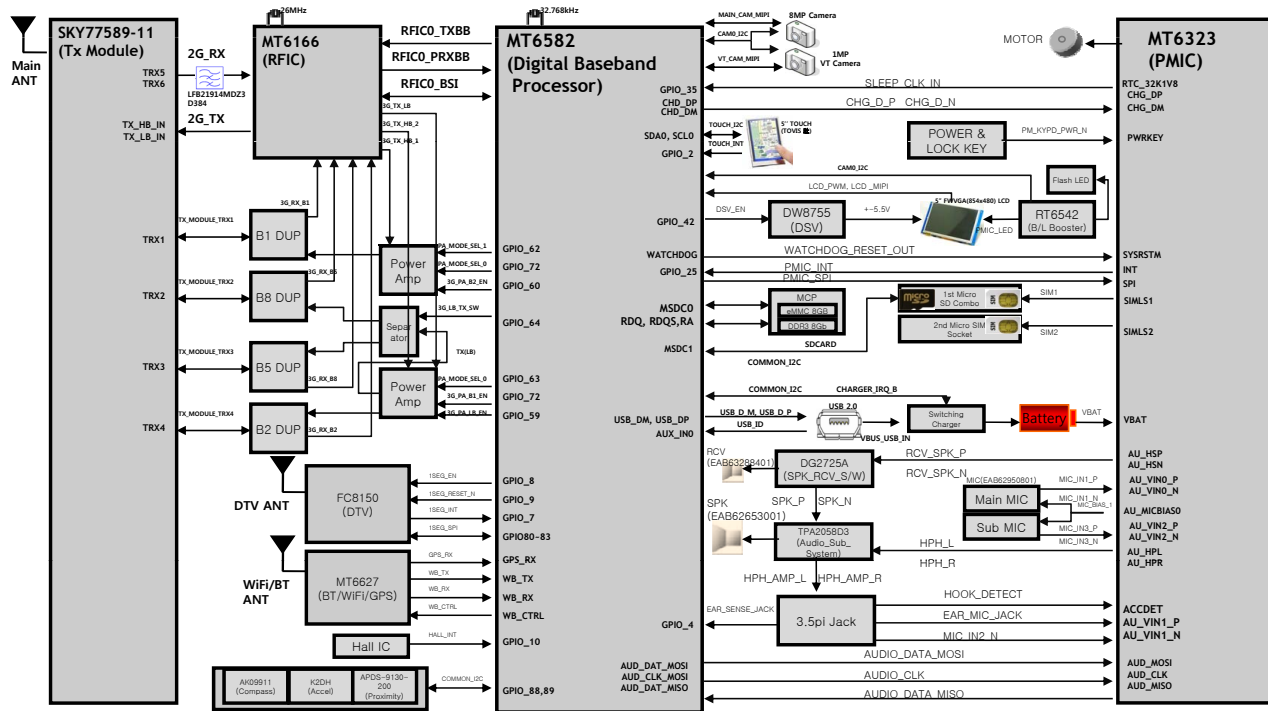


< 7-1-7-3-1_LCD_5_0inch HD >



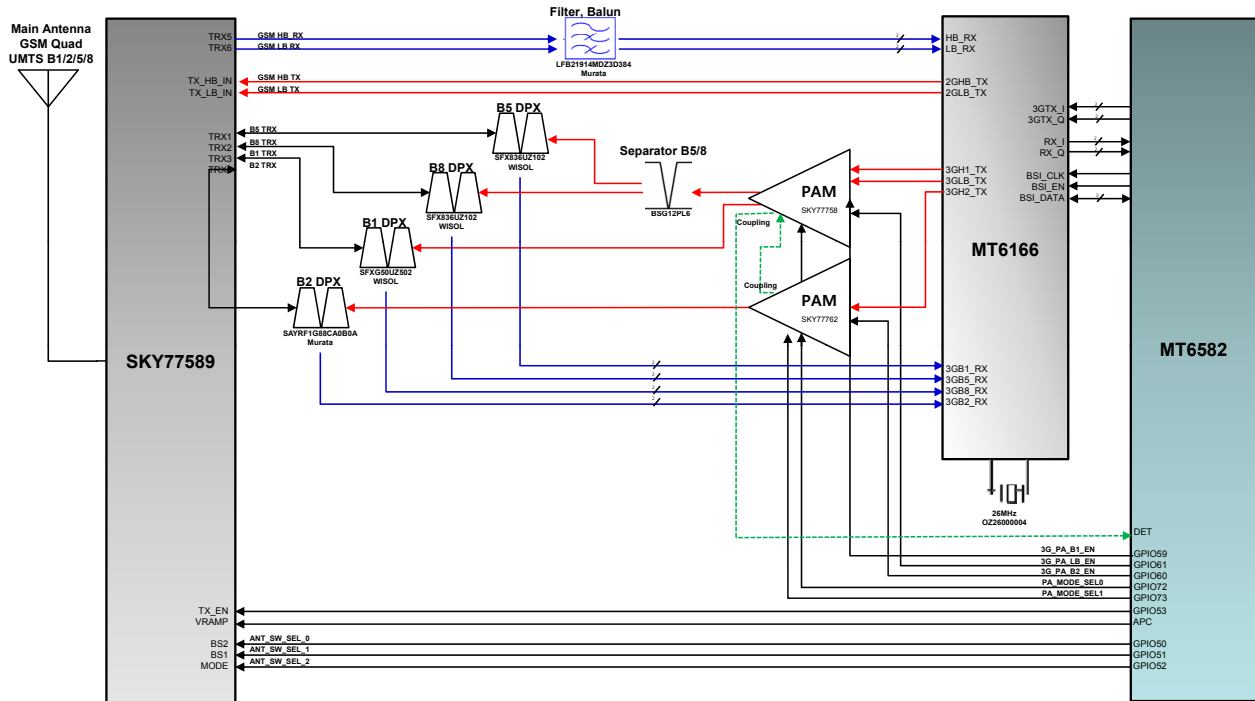
5. BLOCK DIAGRAM

Total Block Diagram

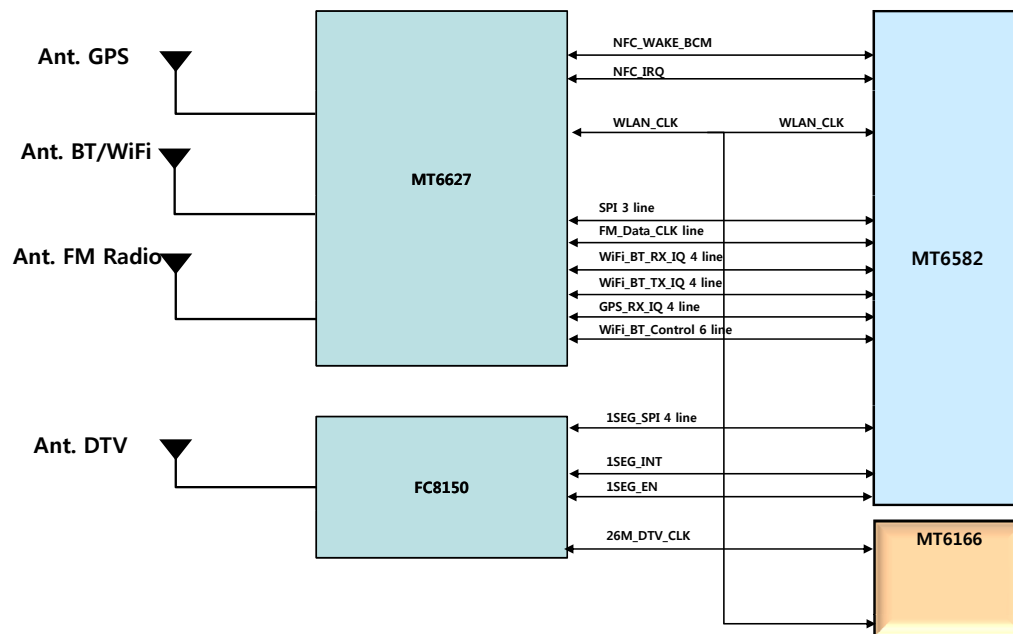


5. BLOCK DIAGRAM

RF Block Diagram

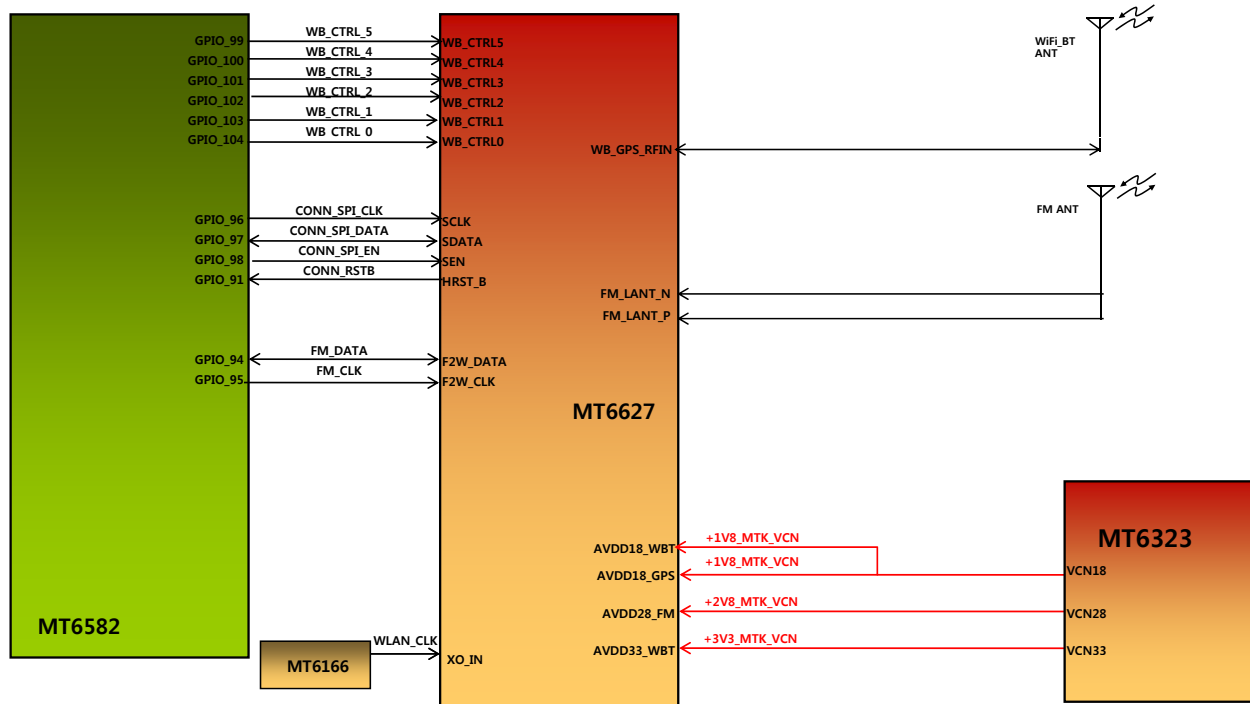


L80+ Connectivity Block-diagram



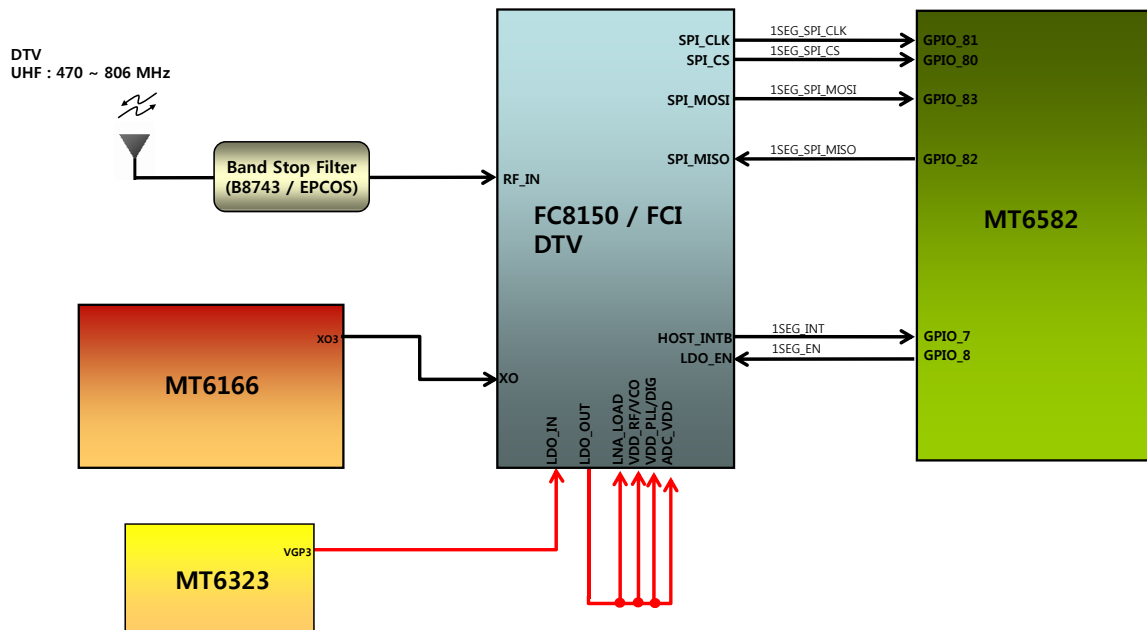
5. BLOCK DIAGRAM

L80+ Connectivity Block-diagram (BT/Wi-Fi/FM)

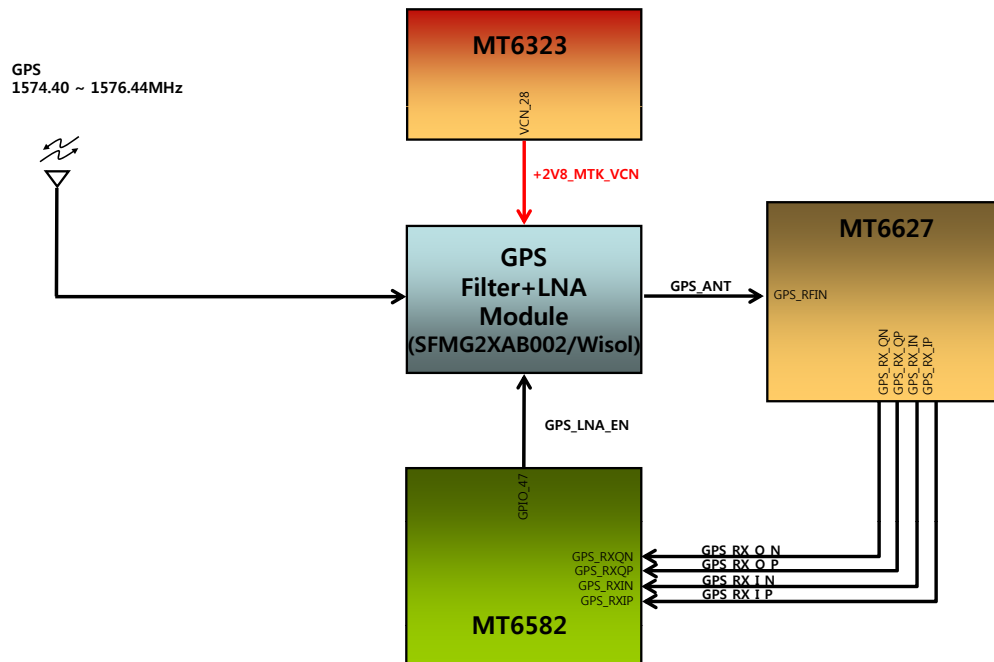


5. BLOCK DIAGRAM

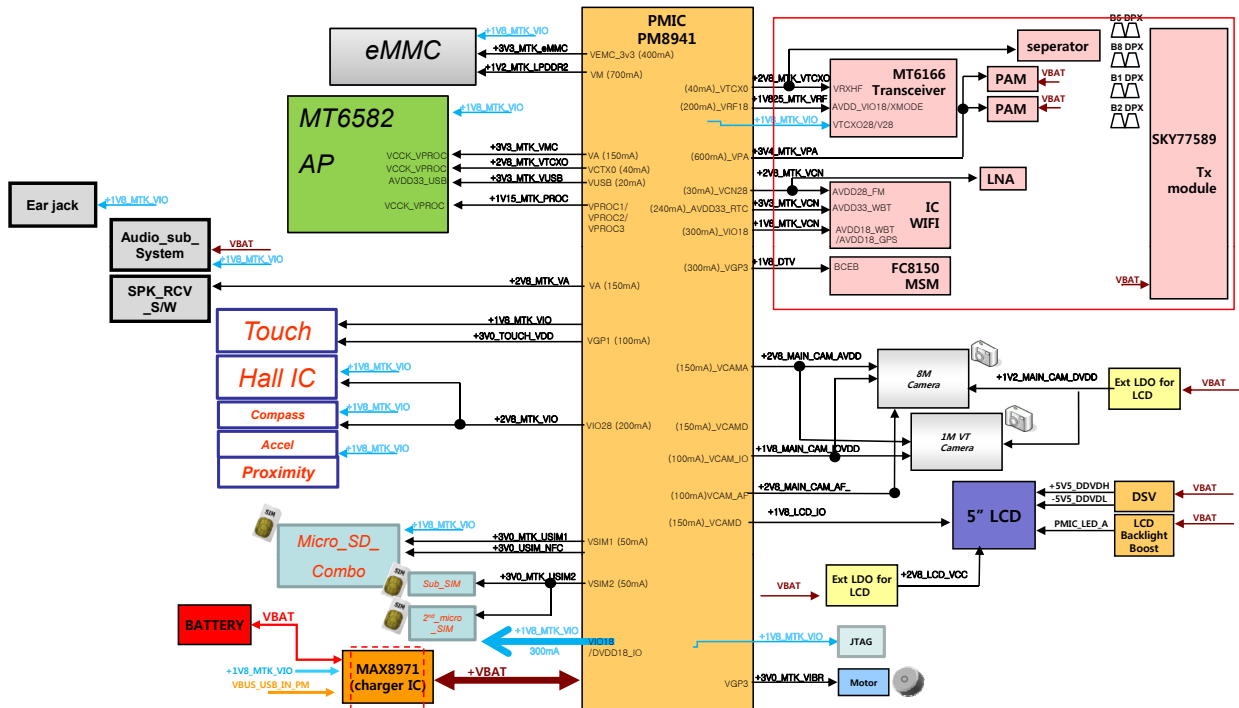
L80+ Connectivity Block-diagram (DTV)



L80+ Connectivity Block-diagram (GPS)

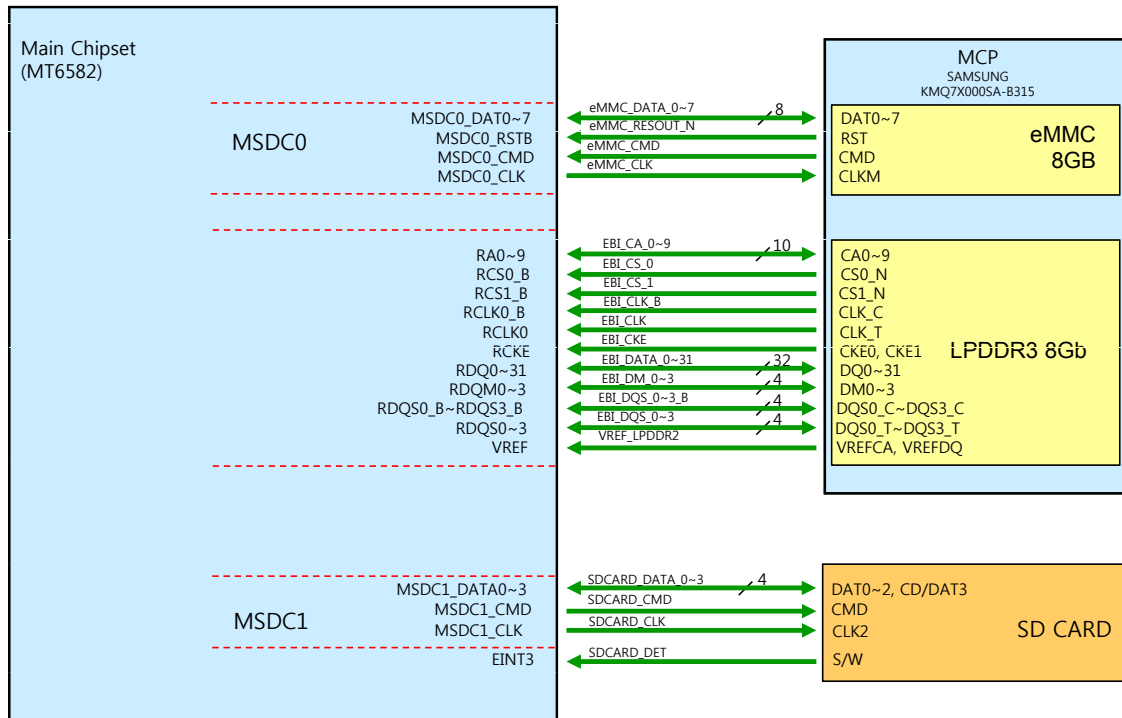


Output Power Management



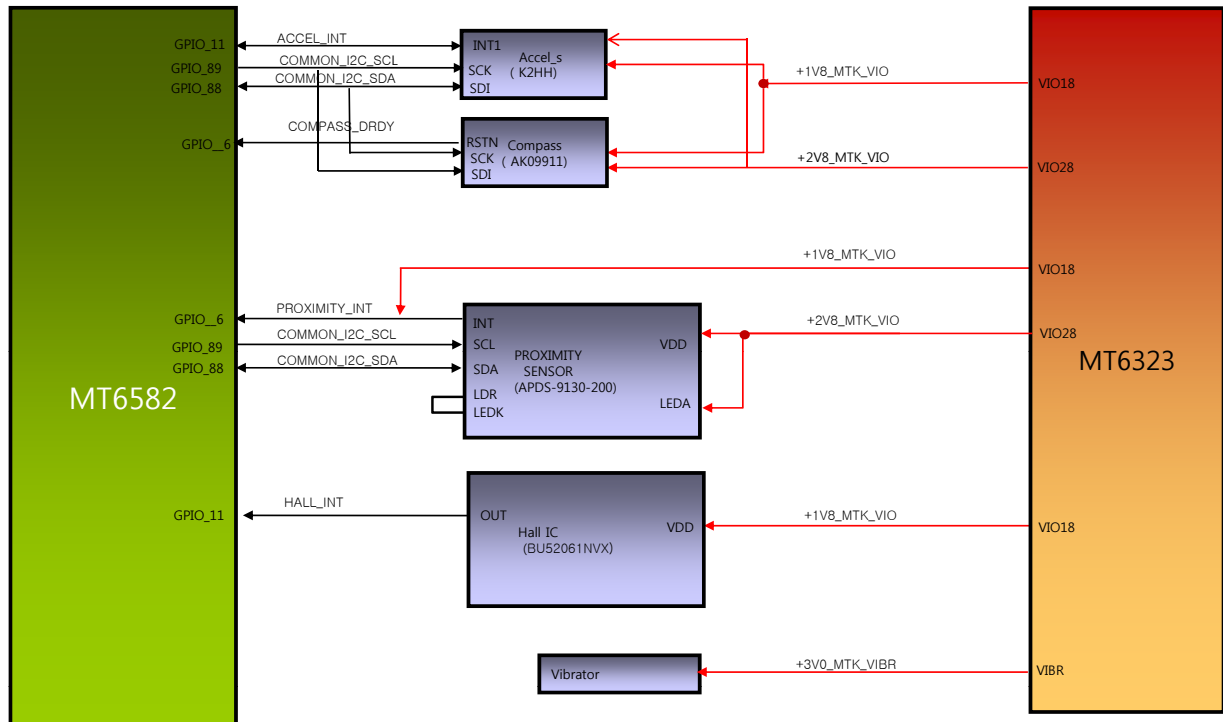
5. BLOCK DIAGRAM

Memory I/F

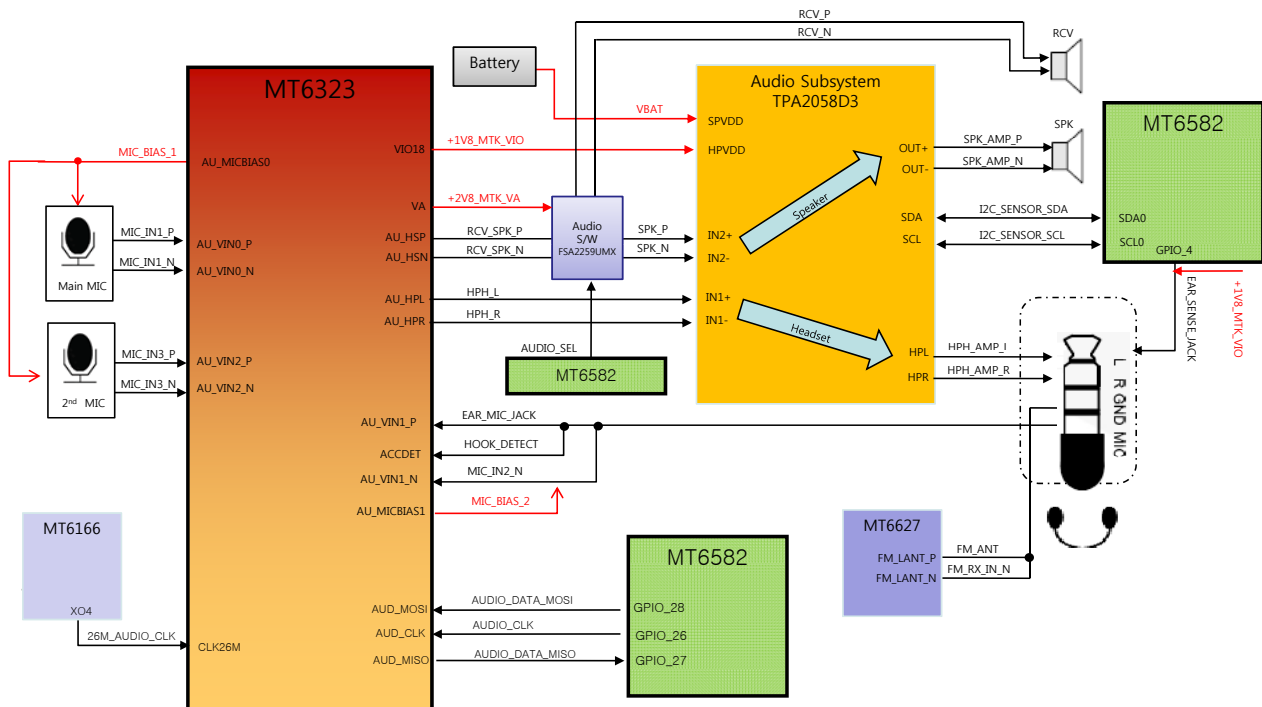


5. BLOCK DIAGRAM

L80+ Block Diagram - Sensor

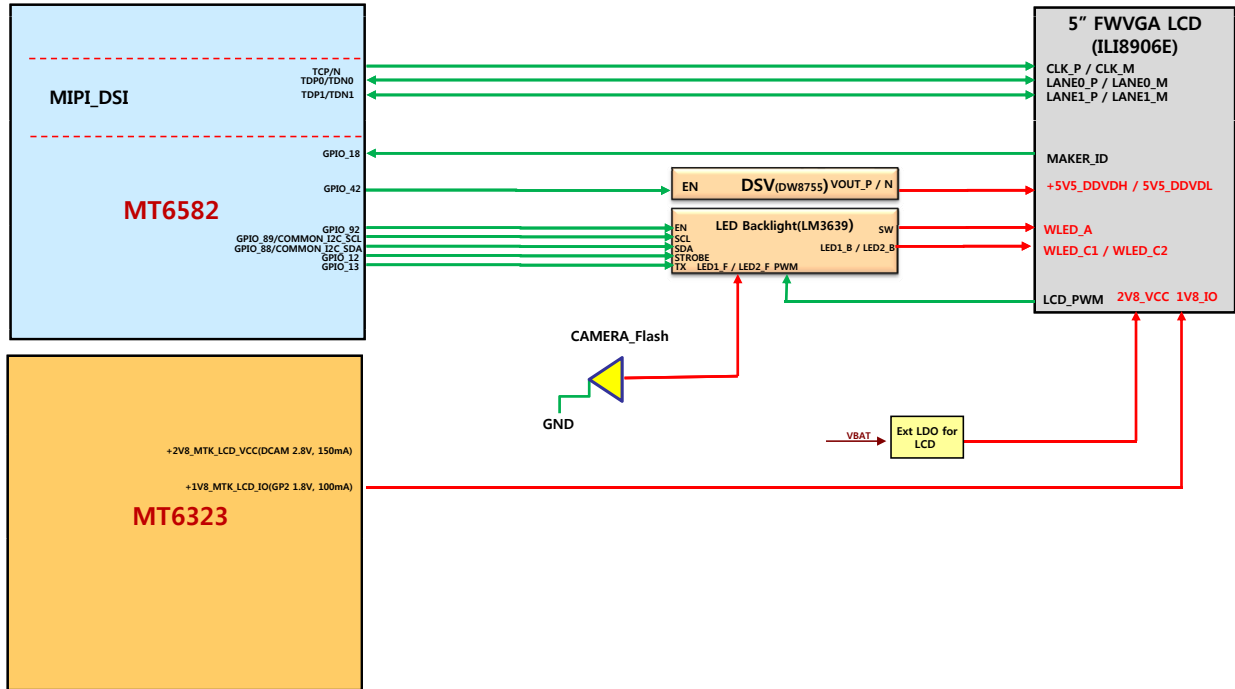


L80+ Block Diagram - Audio



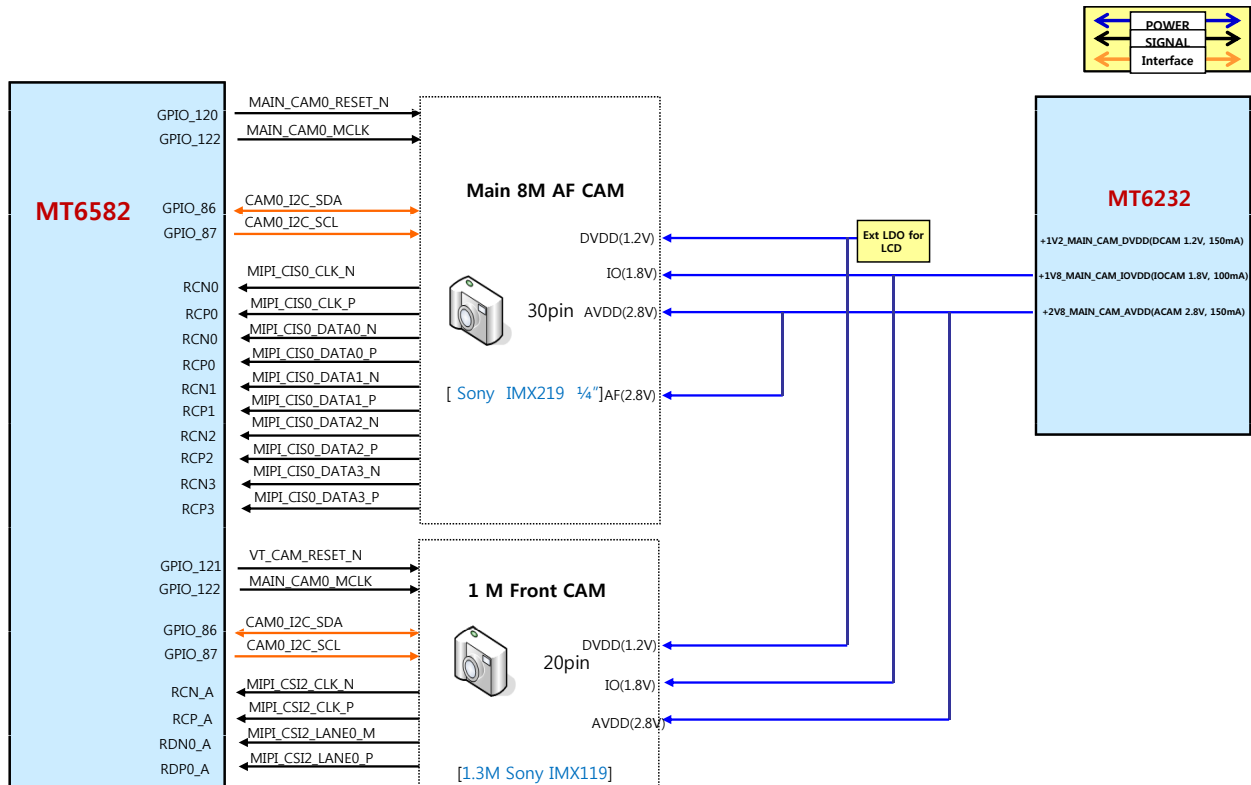
5. BLOCK DIAGRAM

Main Display_Camera_Flash_I/F



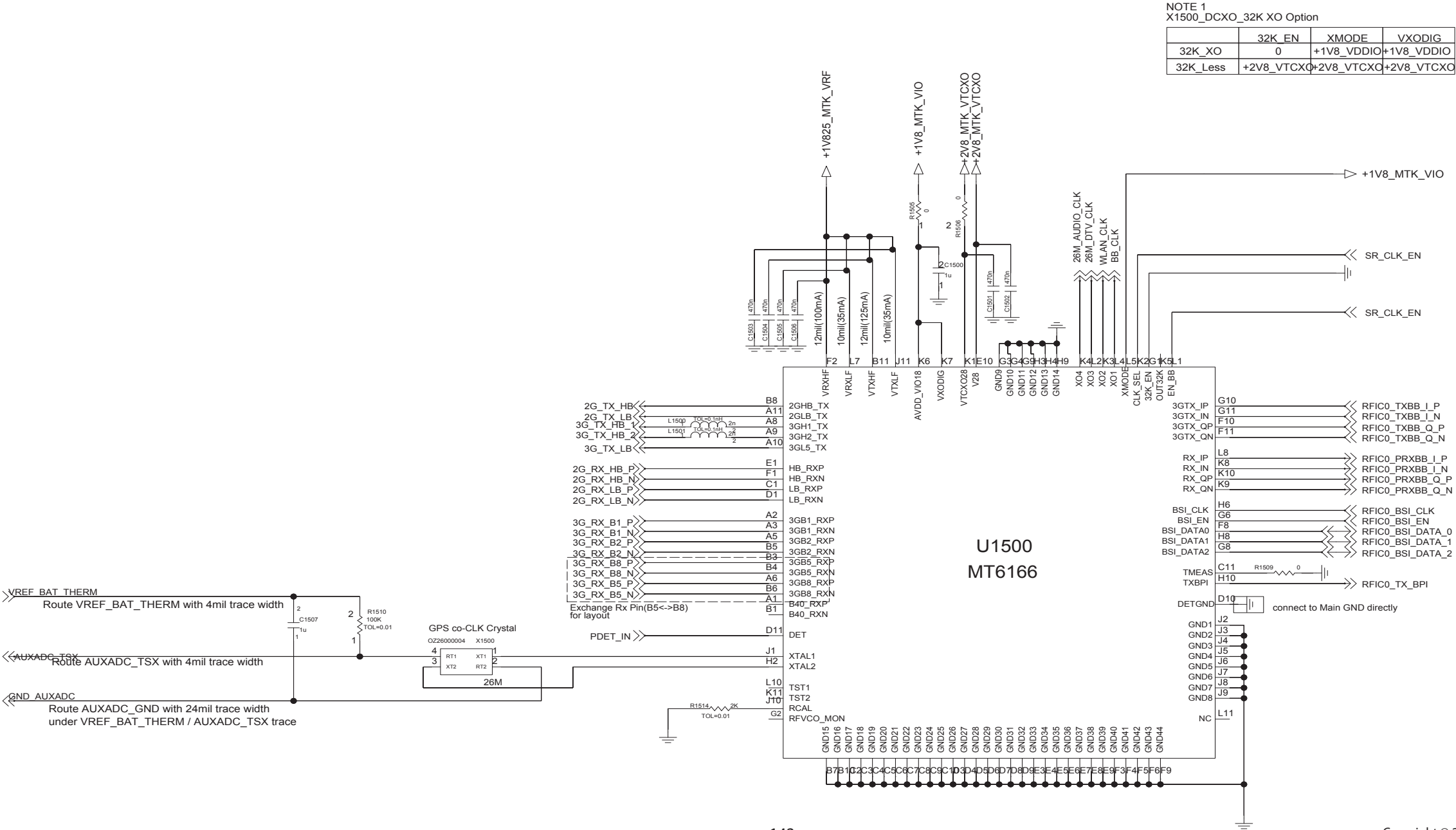
5. BLOCK DIAGRAM

8M AF Camera, 1.3M VT Camera



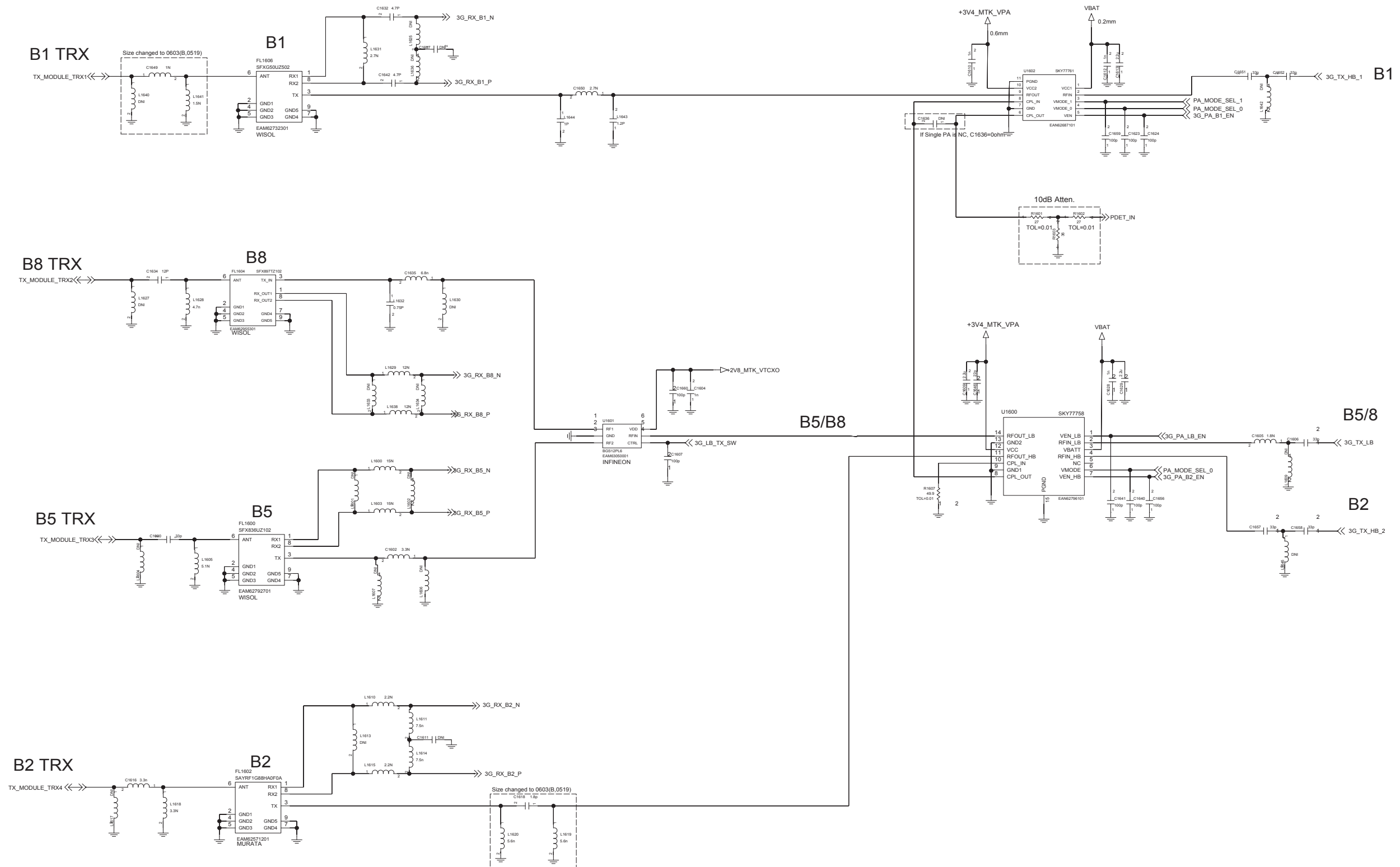
		TXEN	Mode	BS1	BS2
		CSM PA ENANT SEL2	ANT SEL1	ANT SEL0	
Stanby	Stanby	0	0	0	0
LB TX	G LB TX	1	0	0	1
HB TX	G HB TX	1	0	1	1
TRX1	B1	0	1	0	0
TRX2	B8	0	1	1	0
TRX3	B5	0	1	0	1
TRX4	B2	0	1	1	1
TRX5	G LB Rx	0	0	1	0
TRX6	G HB Rx	0	0	0	1

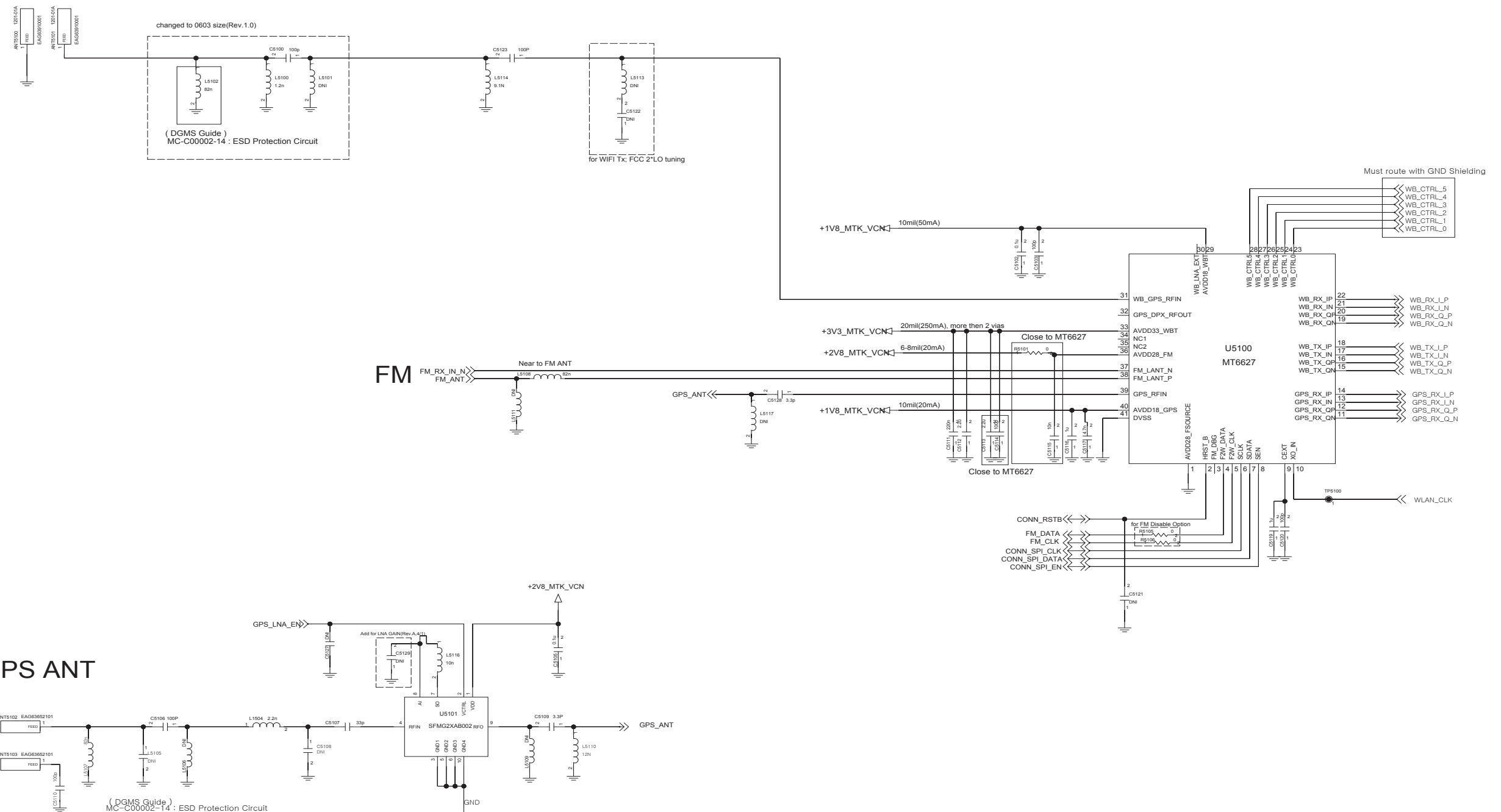
< 1-4-2-1_RFIC_MT6166 >



< 1-4-3-4_TRX_B1_B2_B5_B8_Quad band >

Rev_0.3

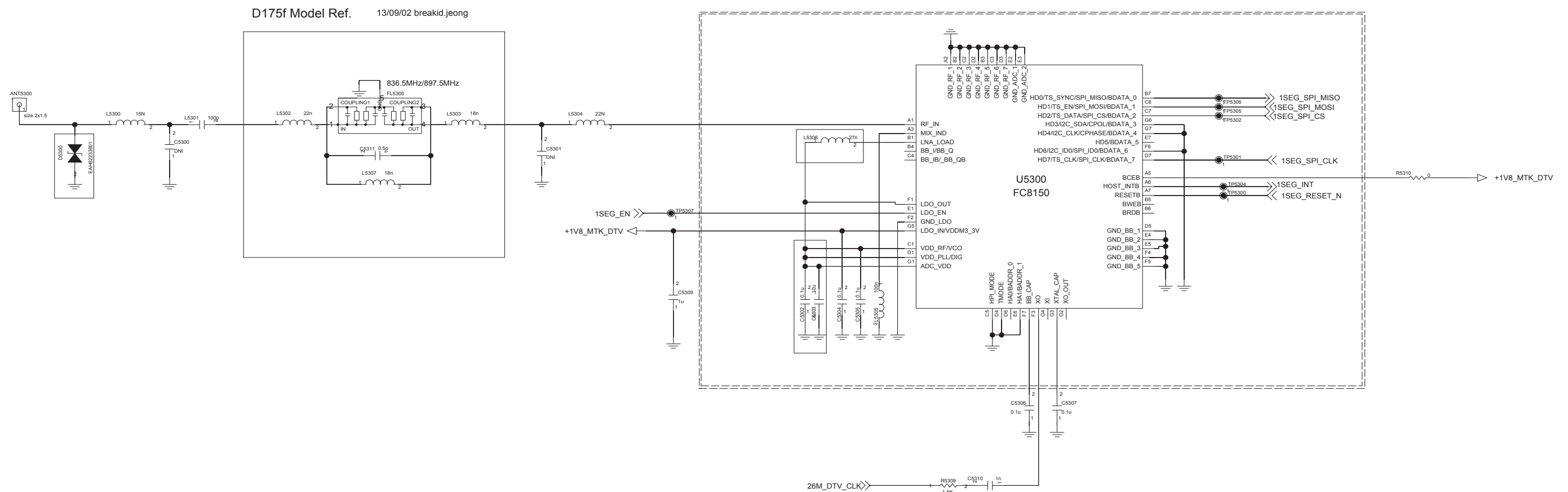




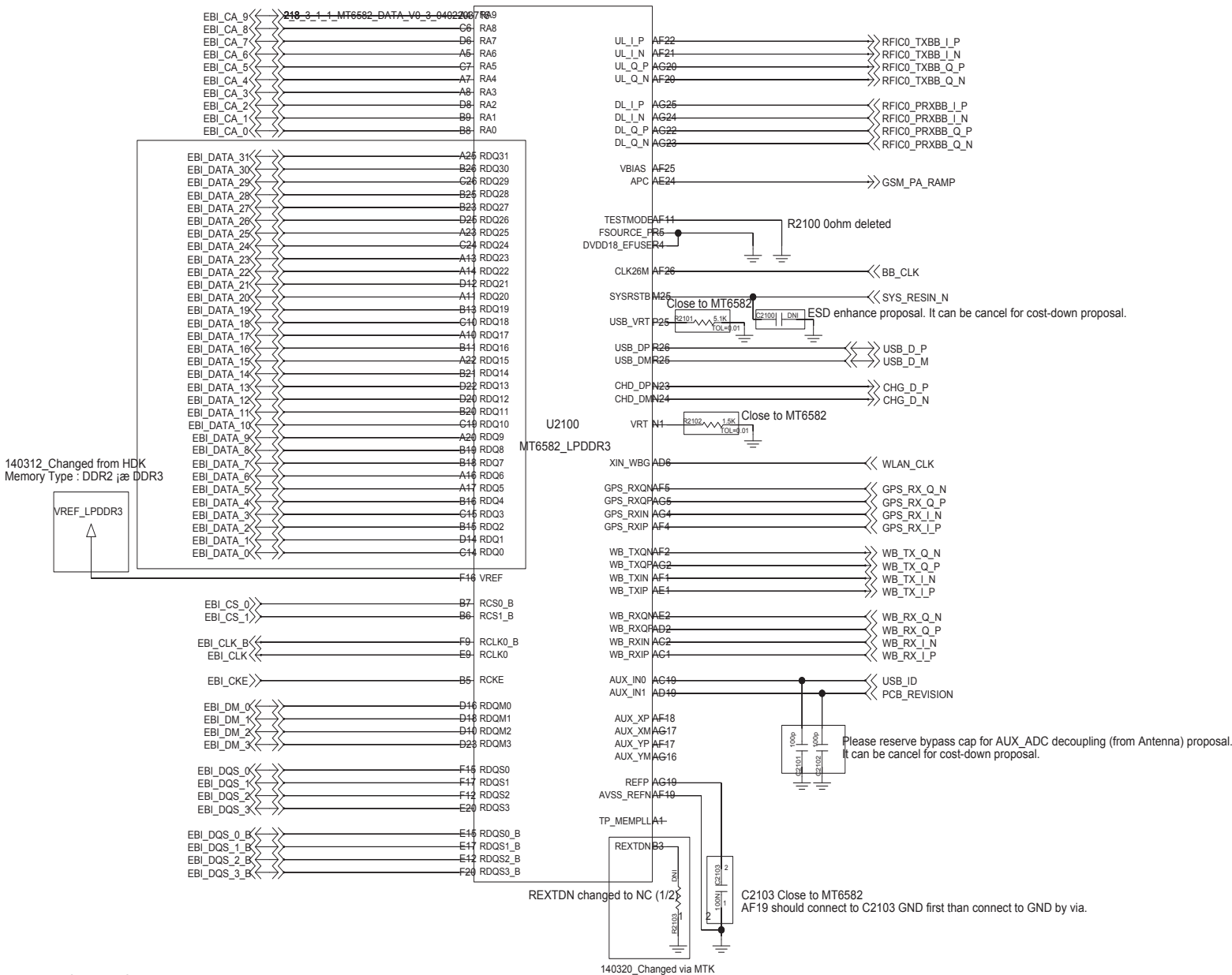
<5-3-1-3_Broadcasting_ONE-SEG_FC8150>

DTV Only

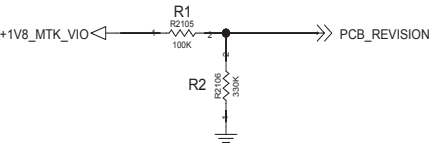
REV 1.0



<2-3-3-1-1_MT6582_DATA> Rev.0.3

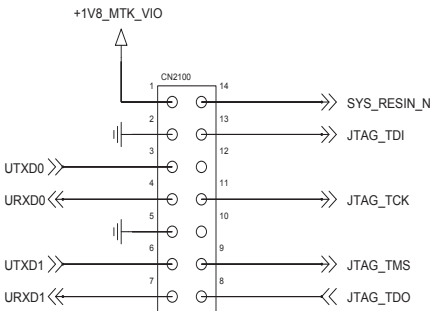


ADC



Revision	R1	R2	ADC Range(HEX.)
0.0	100K	9.1K	0.150
A	100K	30K	0.415
B	100K	51K	0.608
C	100K	82K	0.811
D	100K	130K	1.114
1.0	DNI	100K	0.000
1.1	100K	330K	1.438
1.2	100K	130K	1.114
1.3	100K	200K	1.301

< 9-3-1_JTAG > _DNI
Rev_0.4



Band_Indicator

SET_Indicator

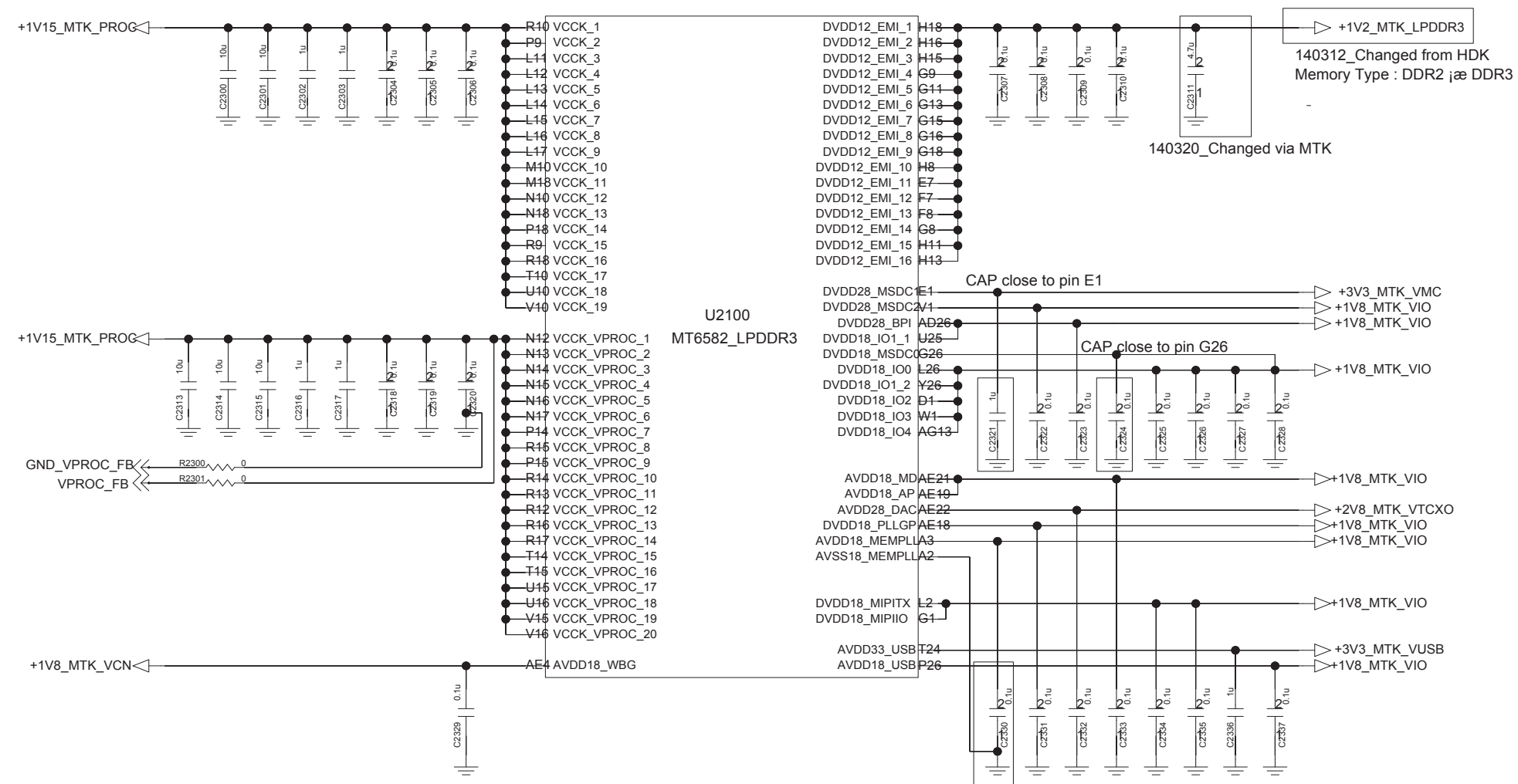
(for Telstra)

+1V8_MTK_VIO

LGE Internal Use Only

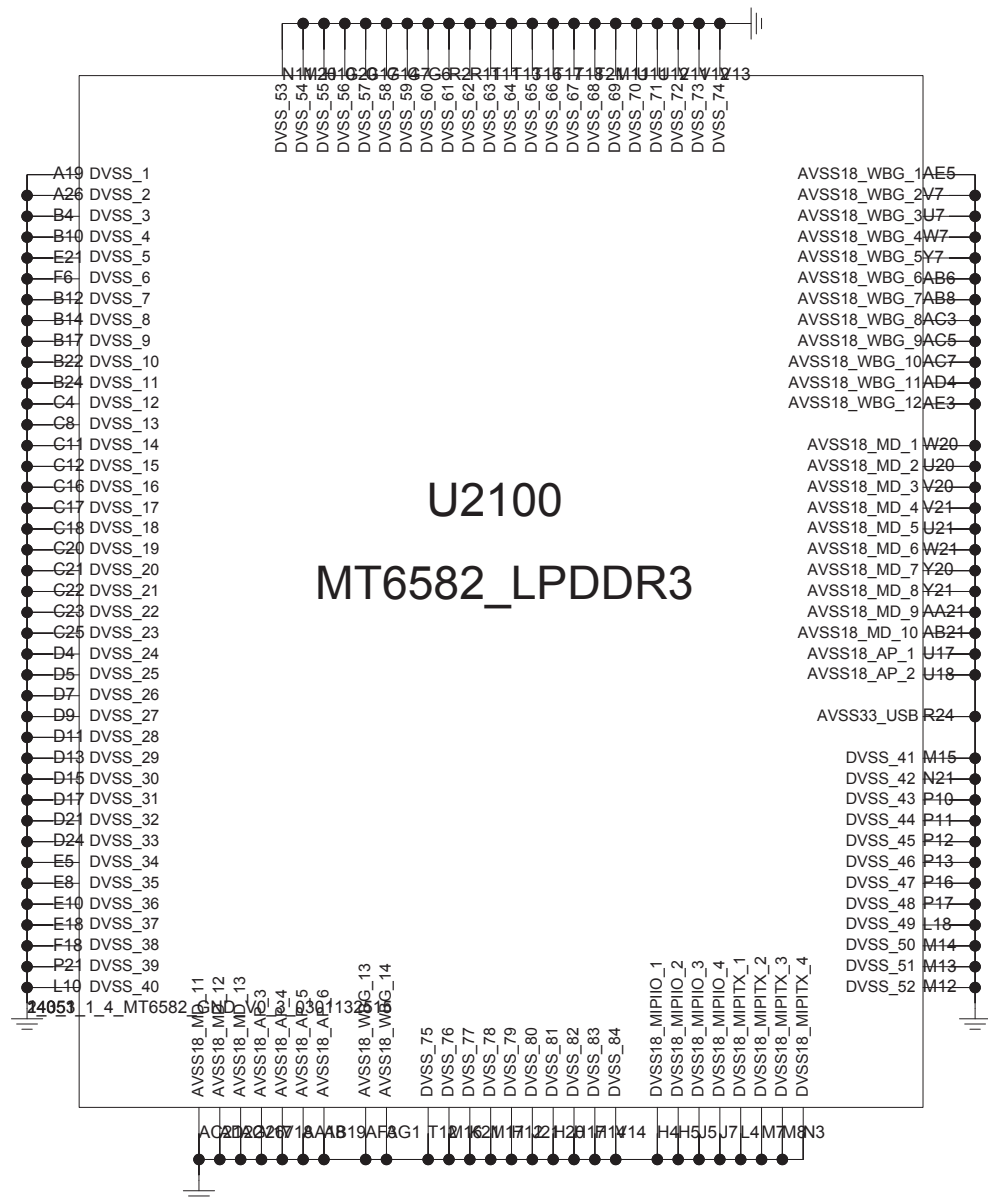
< MT6582_POWER >

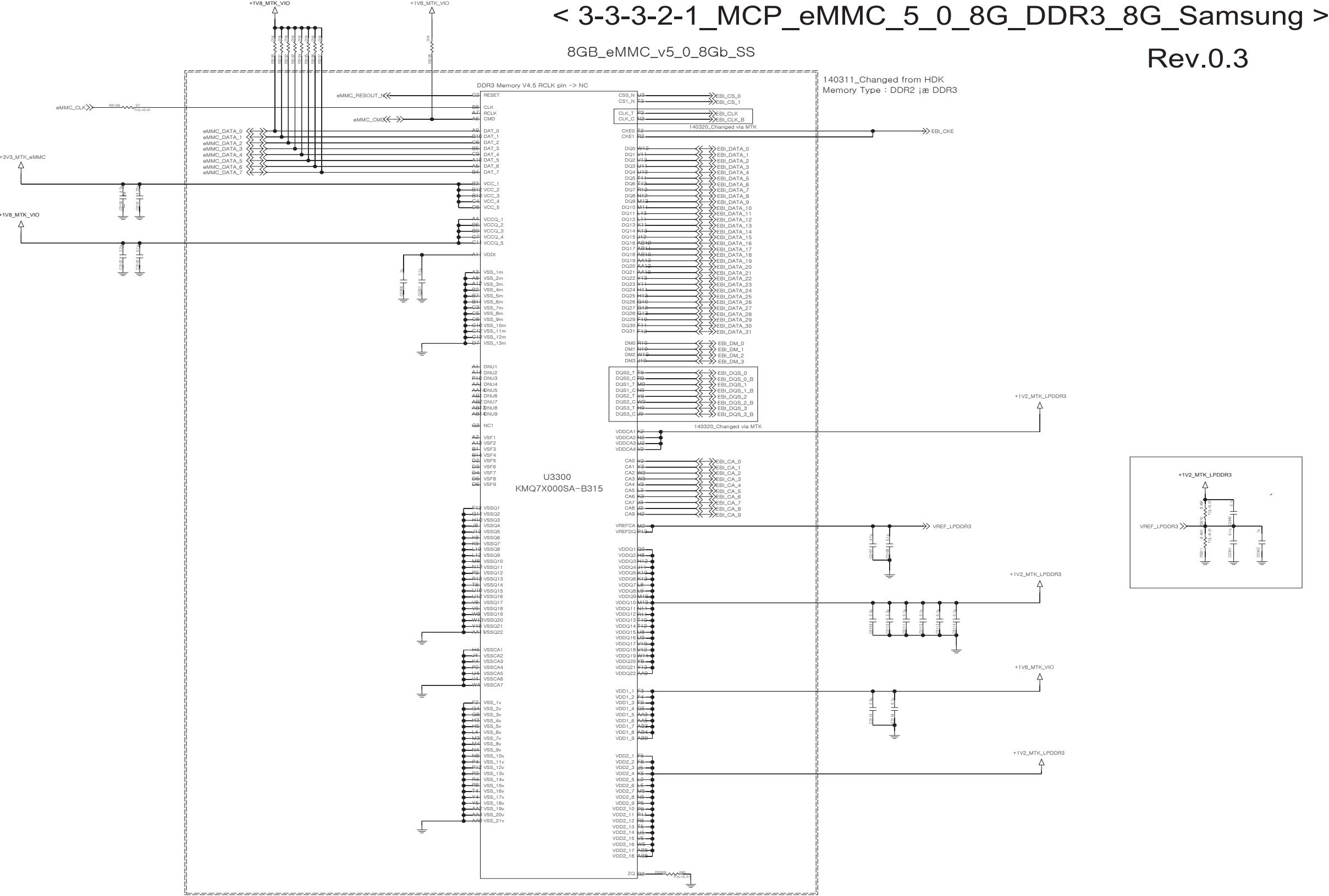
CCDS# : 2-3-3-1-3 Rev_0.3



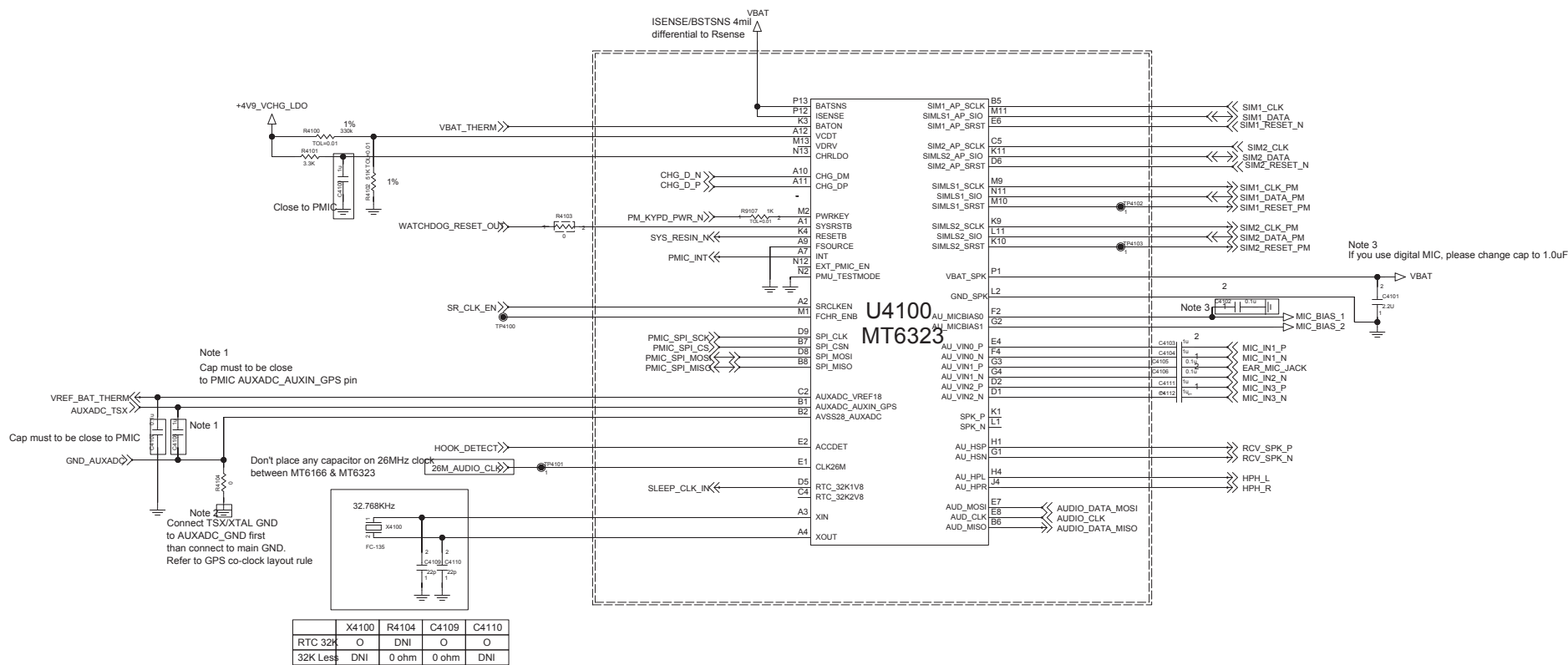
AVDD18_MEMPLL pin and AVSS18_MEMPLL pin should be connected with PCB CAP first, and then connected with PMU and PCB ground. CAP should be near MT6582 as possible as we can.

<2-3-3-1-4_MT6582_GND> Rev.0.3

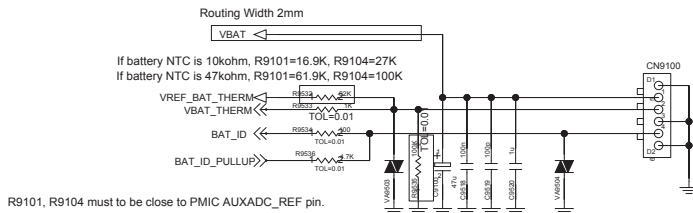




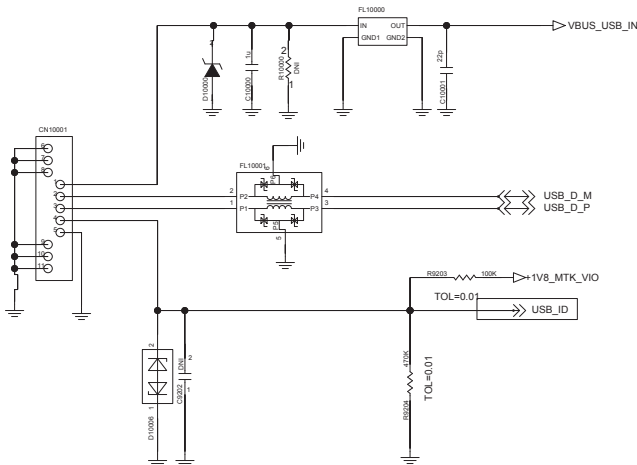
<4-1-7-1_MT6323_DATA> Rev_0.4



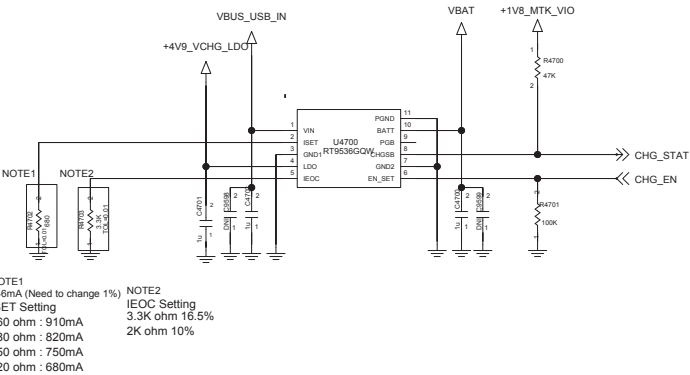
<9-1-2_Battery_CNT_4P> REV.0.4



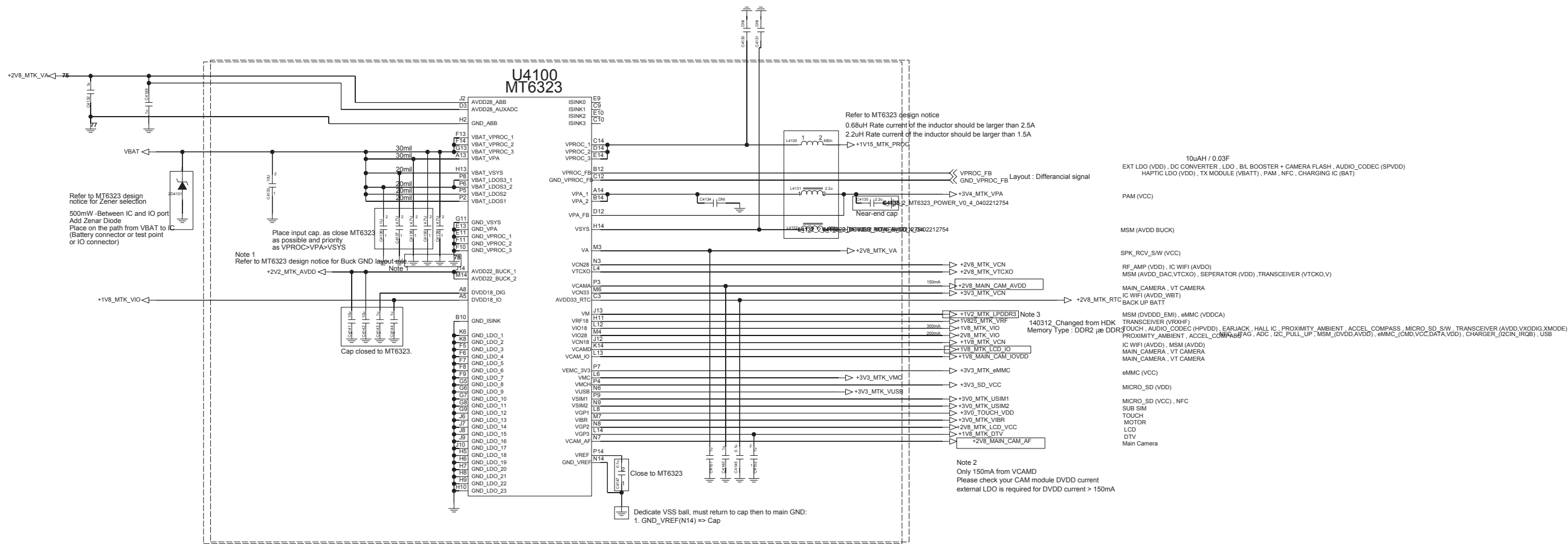
< 9-2-1_IO_USB2.0 > Rev_0.5



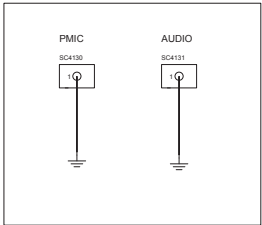
<4-7-1-1_Linear Charger_RT9536> Rev_0.3



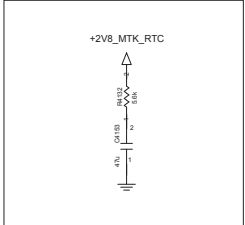
<4-1-7-2_MT6323_POWER> Rev_0.4



BOT CAN



BACKUP BATT



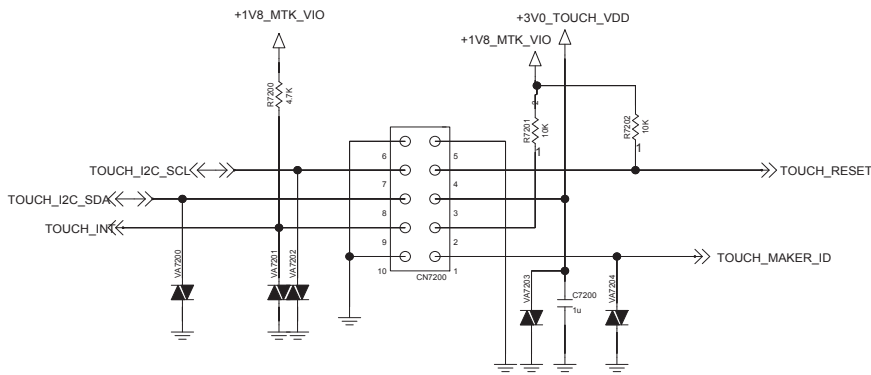
Buck converter brief specifications

VSYS	0.7-1.3V/1.35V (optional) (6.25mV/step)	2,800	Processor / Digital Core
VSYS	2.2	1,400	System LDO input
VPA	0.5V - 3.4V (0.1V/step)	600	3G power amplifier

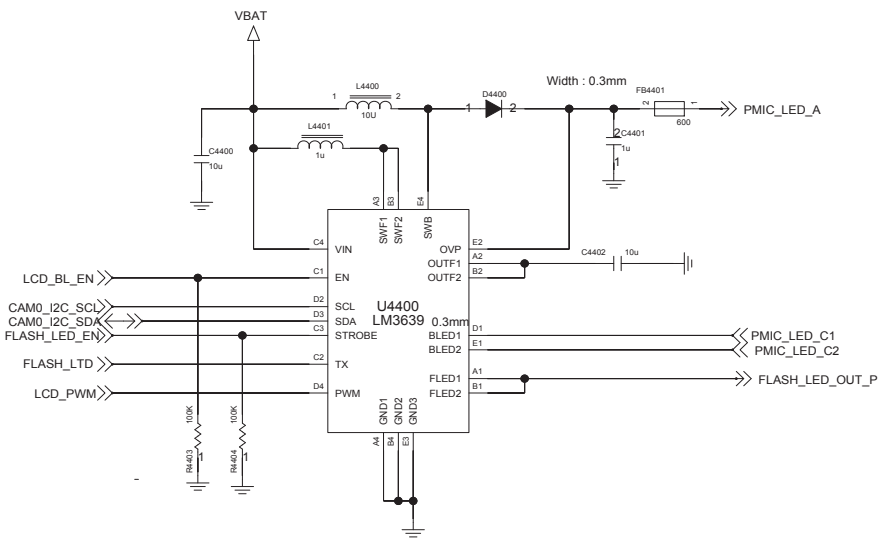
LDO types and brief specifications

Type	LDOname	Vout (Volt)	I _{max} (mA)	Application
ALDO	VCN28	2.8	30	RF chip
ALDO	VTCXO	2.8	40	13/26Mhz reference clock
ALDO	VA	2.8	150	Analog baseband
ALDO	VCAMA	2.8	150	Analog power for camera module
DLDO	VCN_3V3	3.3/3.4/3.5/3.6	240	WiFi
DLDO	VIO28	2.8	200	Digital IO
DLDO	VSIM1	1.8/3.0	50	1st Sim card
DLDO	VSIM2	1.8/3.0	50	2st SIM card
DLDO	VSUB	3.3	20	USB
DLDO	VGP1	1.2/1.3/1.5/1.8 2.0/2.8/3.0/3.3	100	General purpose LDO
DLDO	VGP2	1.2/1.3/1.5/1.8 2.0/2.5/2.8/3.0	100	General purpose LDO
DLDO	VEMC_3V3	3.0/3.3	400	3.3V EMMC
DLDO	VCAM_AF	1.2/1.3/1.5/1.8 2.0/2.8/3.0/3.3	100	AF application
ALDO	VMC	1.8/3.3	100	SD 2.0/3.0 memory card
ALDO	VMCH	3.0/3.3	400	SD 3.0 memory card
DLDO	VIBR	1.2/1.3/1.5/1.8 2.0/2.8/3.0/3.3	100	Vibrator
PTCLDO	VRTC	2.8	2	Real-time clock
VSYS LDO	VM	1.24/1.38/1.54/1.84	700	Memory power
VSYS LDO	VRF18	1.825	200	RF application
VSYS LDO	VIO18	1.8	300	IO pad power
VSYS LDO	VCAMD	1.2/1.3/1.5/1.8	150	Camera application
VSYS LDO	VCAM_IO	1.8	100	Camera IO application
VSYS LDO	VGP3	1.2/1.3/1.5/1.8	200	General purpose LDO
VSYS LDO	VCN18	1.8	120	General purpose LDO

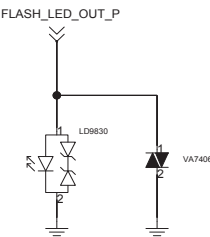
<5.0" TOUCH>



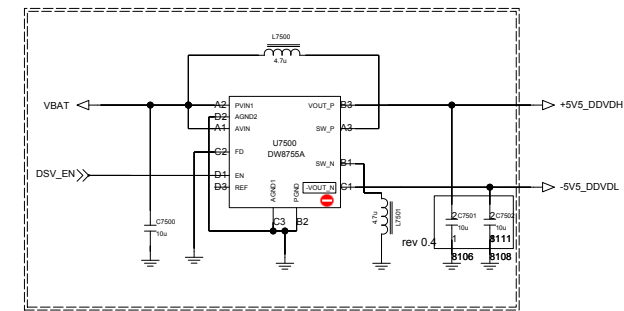
<4-4-3_LCD_Backlight_IC_RT8542>
LCD B/L Booster + Camera Flash Rev_1.0



<9-8-3_Flash_LED>

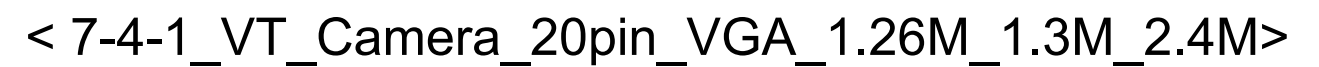


< 7-5-4_DSV_DW8755A > REV_1.1

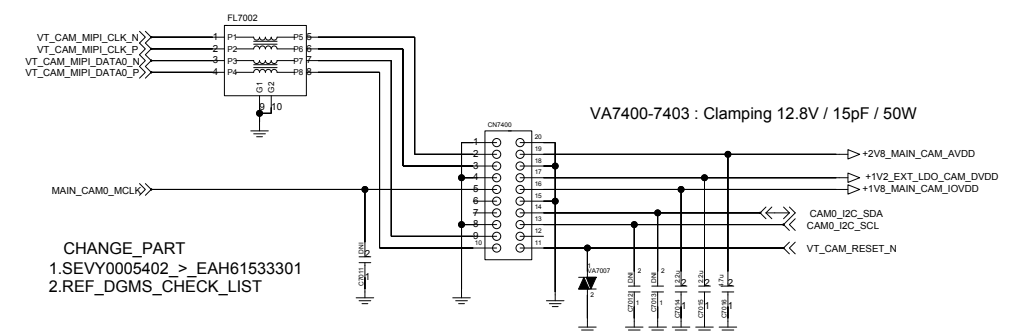


[FD operation Table]

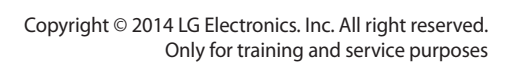
Ext. VCC LDO for LCD (2.8V)



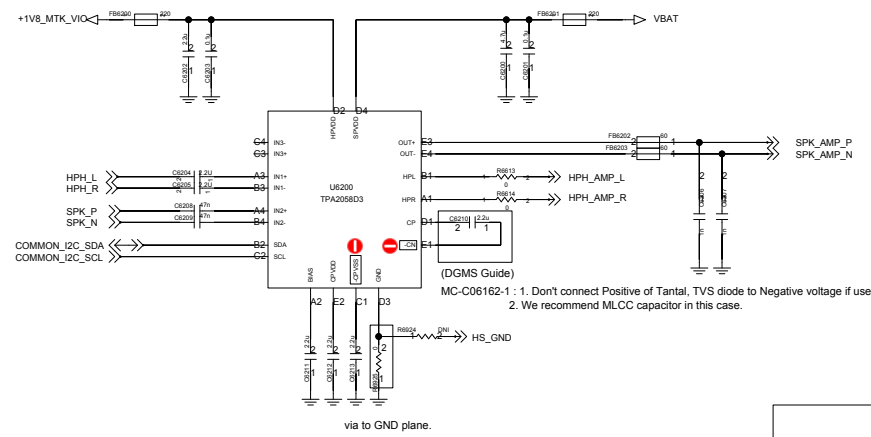
< 7-4-1_VT_Camera_20pin_VGA_1.26M_1.3M_2.4M>



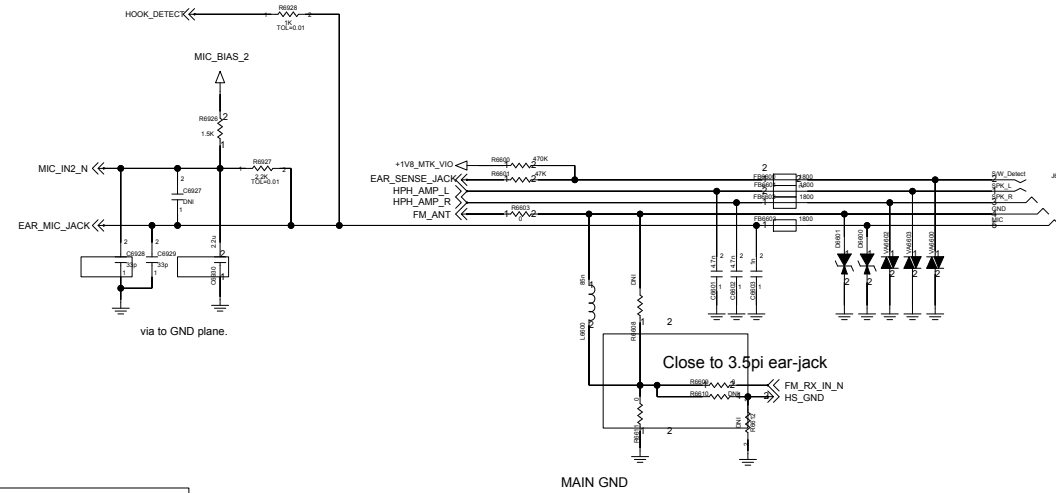
Ext. DVDD LDO for CAMERA (1.2V)



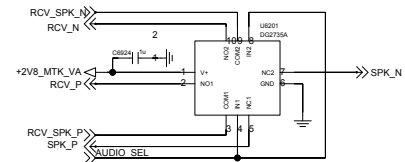
< 6-2-2-2_Audio_Sub_System_TPA2058D3 >
Rev_0.4



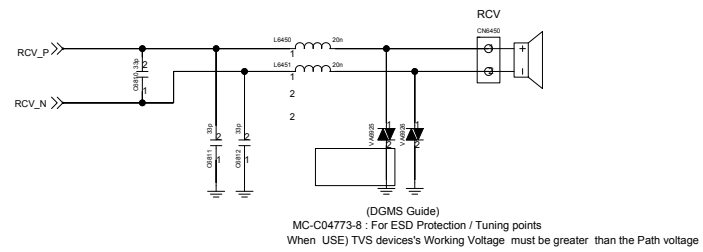
< 6-6-1_Earjack >



< SPK_RCV_S/W >

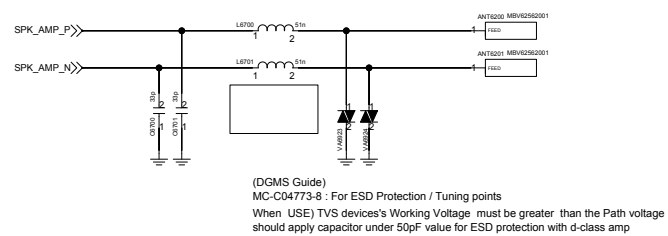


< 6-8-2_ReceiverL Rev_0.4

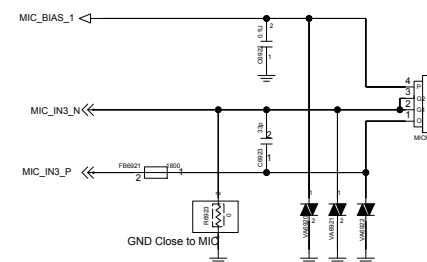


< 6-9-2_2MIC > Rev_0.4

< 6-7-1_Speaker > Rev_0.4

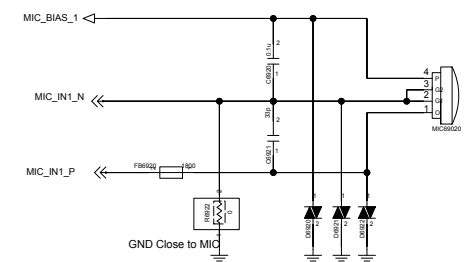


SUB_MIC



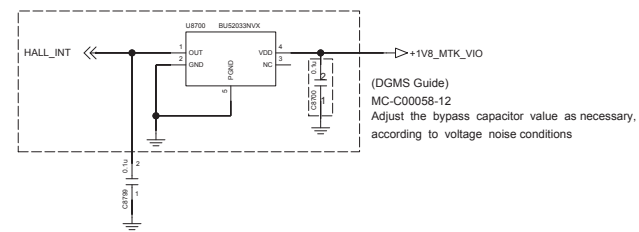
(DGMS Guide)
MC-C00074-11 : For ESD Protection / Tuning points
When USE) TVS devices's Working Voltage must be greater than the Path voltage

MAIN_MIC

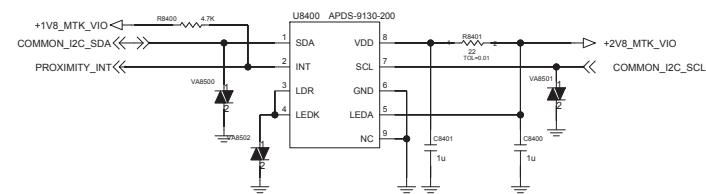


(DGMS Guide)
MC-C00074-11 : For ESD Protection / Tuning points
When USE) TVS devices's Working Voltage must be greater than the Path voltage

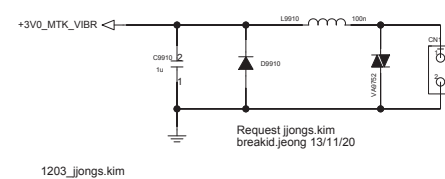
<8-7-1-6_Hall_IC_BU52033NVX> REV.1.0



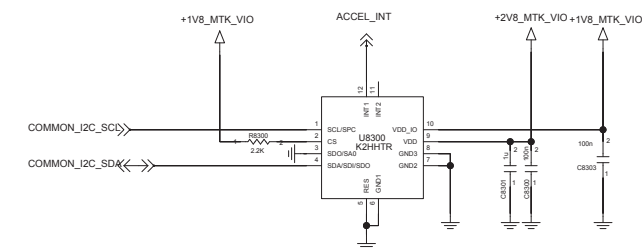
<8-4-1-1_Proximity_APDS-9130> REV.1.0



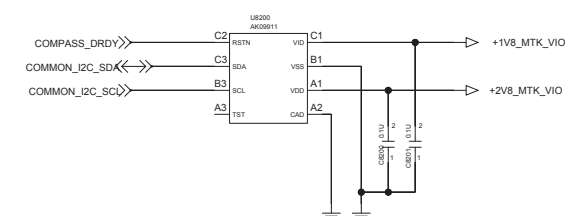
< 9-9-1-1_Motor > Rev_0.3



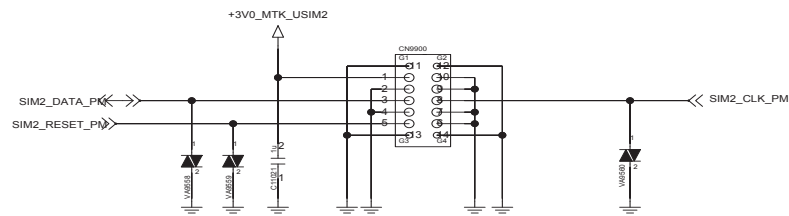
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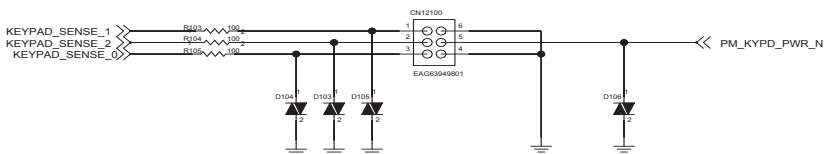
<8-2-1-3_Compass_AK09911> Rev.0.3



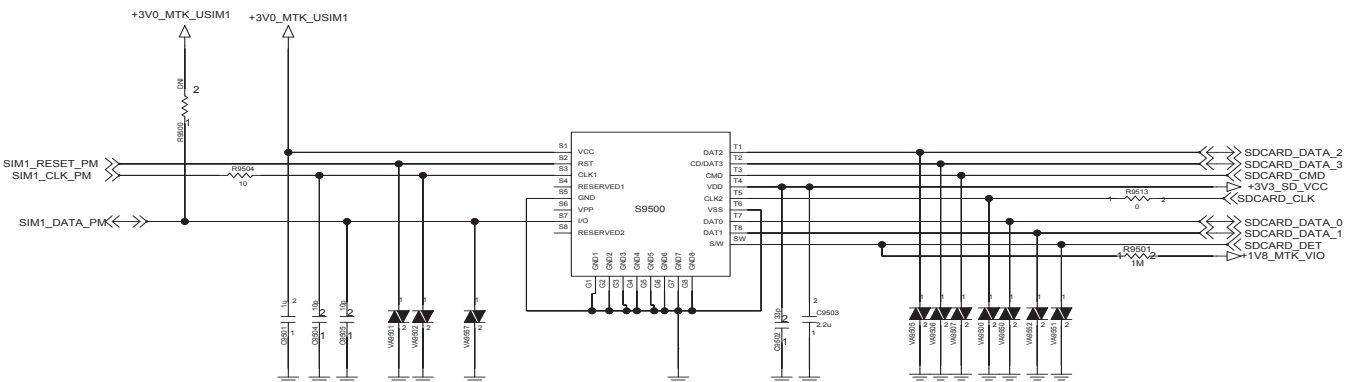
<Sub SIM Connector>



<Back Volume Key>

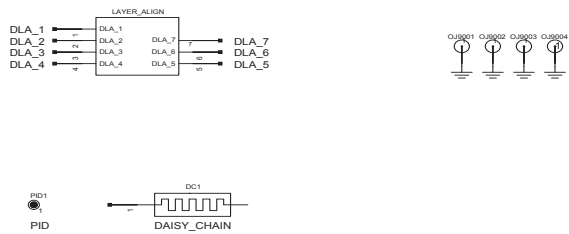


< 9-5-3_Micro_SD_Combo >



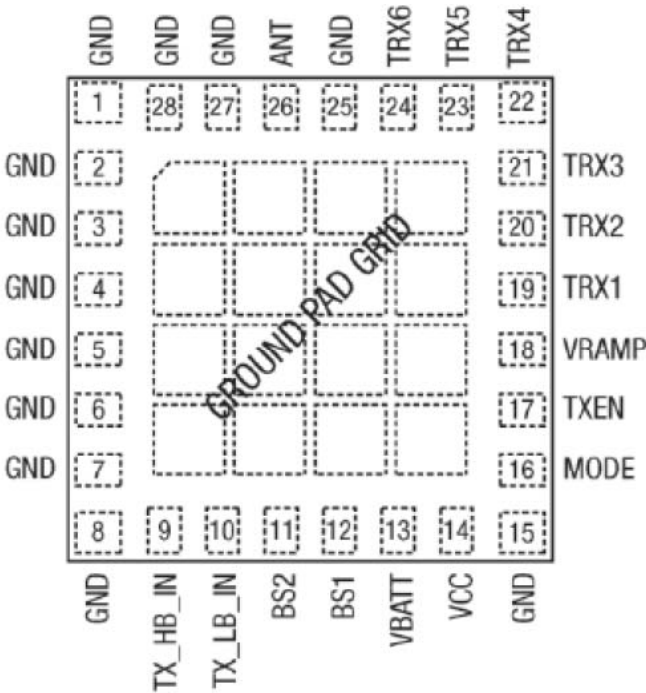
< PID >

Item for Manufacturing



7. BGA PIN MAP

U1100 SKY77589



- Not Used
- Used

7. BGA PIN MAP

U1500 MT6166 RF Transceiver

	1	2	3	4	5	6	7	8	9	10	11	
A	B40_RXP	3GB1_RXP	3GB1_RXN		3GB2_RXP	3GB8_RXP		3GH1_TX	3GH2_TX	3GL5_TX	2GLB_TX	A
B	B40_RXN		3GB5_RXP	3GB5_RXN	3GB2_RXN	3GB8_RXN	GND	2GHB_TX		GND	VTXHF	B
C	LB_RXP	GND	GND	GND	GND	GND	GND	GND	GND	GND	TMEAS	C
D	LB_RXN		GND	GND	GND	GND	GND	GND	GND	DETGND	DET	D
E	HB_RXP		GND	GND	GND	GND	GND	GND	GND	V28		E
F	HB_RXN	VRXHF	GND	GND	GND	GND		BSI_DATA0	GND	3GTX_QP	3GTX_QN	F
G	32K_EN	RFVCO_MG N	GND	GND		BSI_EN		BSI_DATA2	GND	3GTX_IP	3GTX_IN	G
H		XTAL2	GND	GND		BSI_CLK		BSI_DATA1	GND	TXBPI		H
J	XTAL1	GND	GND	GND	GND	GND	GND	GND	GND	RCAL	VTXLF	J
K	VTCXO28	CLK_SEL	XO2	XO4	OUT32K	AVDD_VIO 18	VXODIG	RX_IN	RX_QN	RX_QP	TST2	K
L	EN_BB	XO3		XO1	XMODE		VRXLF	RX_IP		TST1	NC	L
	1	2	3	4	5	6	7	8	9	10	11	

● Not Used

○ Used

U1600 BU52033NVX Hall IC

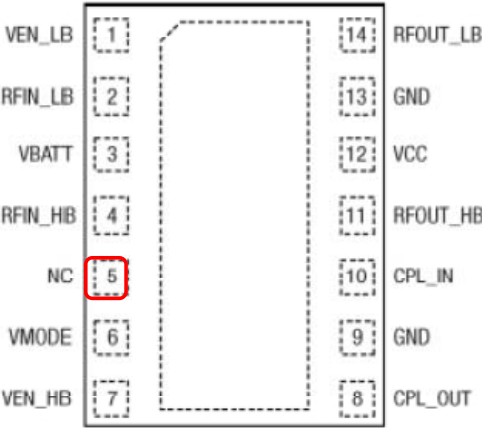
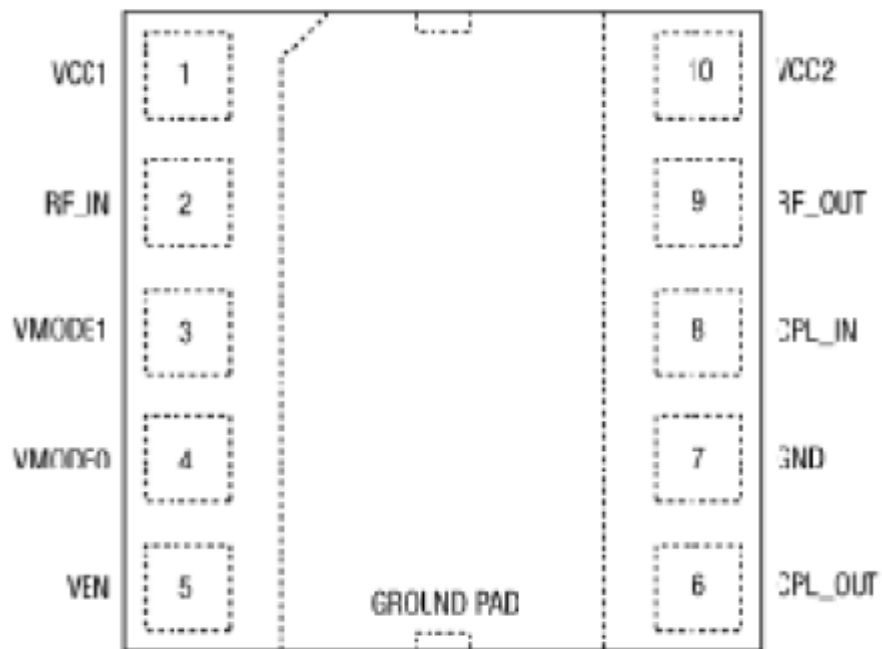


Figure 7.2 Pad Configurations (Top View)

- Not Used
- Used

U1602 SKY77761 B1 TRX



Pad layout as seen from Top View looking through the package.
GROUND PAD is package underside.

281717.006

● Not Used

○ Used

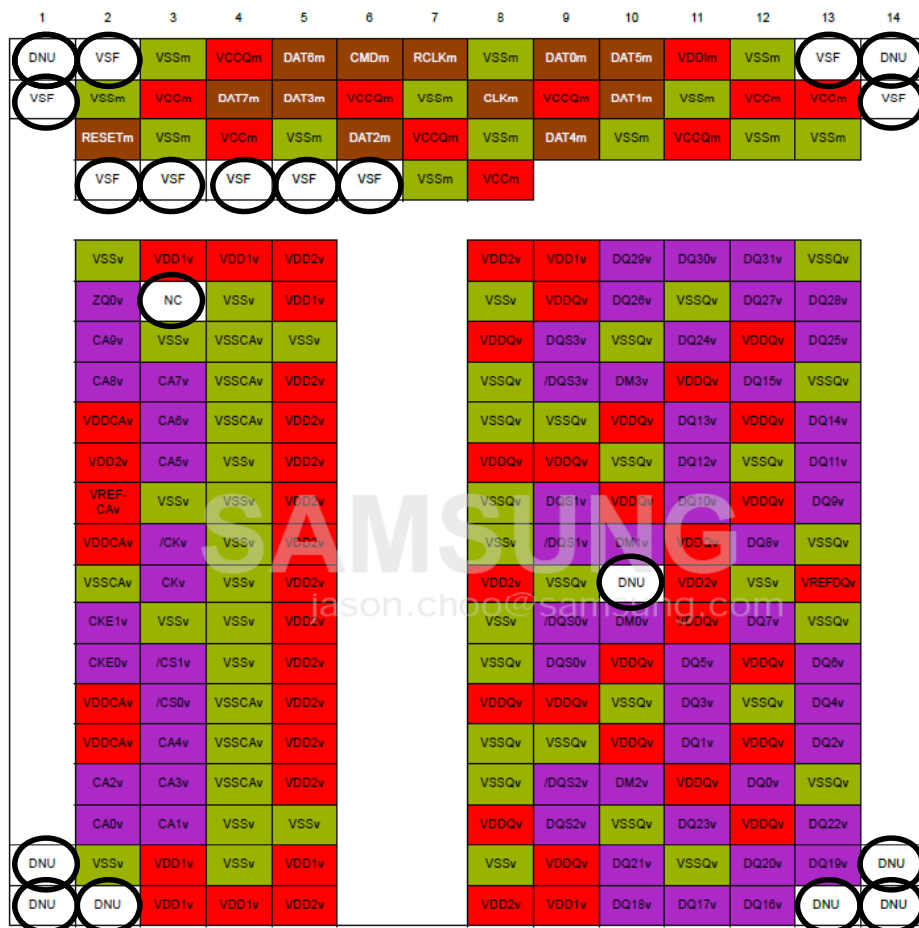
U2100 MT6582

	479	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	
A	TP1M VREF	AVSS18 MEM	AVDD18 MEM	RA0	RA6		RA4	RA3		RDQ19	RDQ23		RDQ25	RDQ22		RDQ4	RDQ6		DVSS	RDQ9		RDQ12	RDQ25		RDQ26	DVSS	A	
B	CMPC1 K	CMAC LK	EXTD N	DVSS	RCE	RC5L B	RC5H B	RA0	RA1	DVSS	RDQ16	DVSS	RDQ14	DVSS	RDQ3	RDQ7	DVSS	RDQ5	RDQ10	RDQ14	RDQ8	DVSS	RDQ28	DVSS	RDQ28	RDQ27	B	
C		MSDC1 DAT2	MSDC1 CLK	DVSS		RA8	RA5	DVSS		RDQ17	DVSS	DVSS		RDQ0	RDQ2	DVSS	DVSS	DVSS	RDQ11	DVSS	DVSS	DVSS	DVSS	RDQ30	DVSS	RDQ29	C	
D	DVDD18 IOI	MSDC1 DAT3	MSDC1 DAT3	DVSS	DVSS	RA7	DVSS	RA2	DVSS	RDQ15	DVSS	RDQ20	DVSS	RDQ1	DVSS	RDQ18		RDQ13	DVSS	RDQ15	RDQ16	DVSS	RDQ24	MSDC0 CLK	D			
E	DVDD18 MSD	CMOAT G	MSDC1 CMD	MSDC1 DAT0	DVSS		DVDD18 2. EM	DVSS	RCLD	DVSS		RDQ52			RDQ50		RDQ51	DVSS		RDQ53	DVSS		MSDC0 DAT6	MSDC0 CMD	MSDC0 DAT2	MSDC0 DAT3	E	
F		CMOAT S			RCP_A	DVSS	DVDD18 2. EM	DVDD18 2. EM	RCLD	DVSS			RDQ52			RDQ50		RDQ51	DVSS		RDQ53	DVSS		MSDC0 DAT7	MSDC0 DAT5			F
G	DVDD18 MIP	RD03	RD04	RD05	RD06	RD07	RD08	RD09	RD10	RD11	RD12	RD13	RD14	RD15	RD16	RD17	RD18	RD19	RD20	RD21	RD22	RD23	RD24	RD25	RD26	RD27	G	
H		RD03	RD04	RD05	RD06	RD07	RD08	RD09	RD10	RD11	RD12	RD13	RD14	RD15	RD16	RD17	RD18	RD19	RD20	RD21	RD22	RD23	RD24	RD25	RD26	RD27	H	
I		RD03	RD04	RD05	RD06	RD07	RD08	RD09	RD10	RD11	RD12	RD13	RD14	RD15	RD16	RD17	RD18	RD19	RD20	RD21	RD22	RD23	RD24	RD25	RD26	RD27	I	
J	RD02	RD02	RD01		DVSS18 MIP1	DVSS18 MIP1			DVDD18 2. EM	DVSS	DVDD18 2. EM	DVSS	DVDD18 2. EM	DVSS	DVDD18 2. EM	DVSS	DVDD18 2. EM	DVSS	DVDD18 2. EM	DVSS		DVSS						J
K	RD01	RD01	RD08		RCN	RCP																					K	
L	DVDD18 MIP	RD03	DVSS18 MIP1																								L	
M	TD03	TD03	TD03	TD03																							M	
N	VRT	TD01	DVSS18 MIP1																								N	
P	TD02	TD01																									P	
R	TD02	DVSS		DVDD18 2. EM	RDQ16 CLK		MSDC2 CLK		VCCCK	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	DVSS	R	
T	MSDC2 DAT2	MSDC2 DAT3	MSDC2 CMD		MSDC2 DAT3																						T	
U		EXTD N		125_LK CK				AVSS18 _WB0																			U	
V	DVDD18 MSD	EXTD N		125_BC K				AVSS18 _WB0																			V	
W	DVDD18 IOI	EXTD N		125_DA TA_IN				AVSS18 _WB0																			W	
Y	UTX00	DSI_TE						AVSS18 _WB0																			Y	
AA	SCL2	SDA2	SCL0					AVSS18 _WB0																			AA	
AB		SDA0						AVSS18 _WB0																			AB	
AC	WB_XN			AVSS18 _WB0			AVSS18 _WB0		AVSS18 _WB0		SPI_CS		SPI_M														AC	
AD	WB_XN			AVSS18 _WB0			AVSS18 _WB0		AVSS18 _WB0		SPI_CS		SPI_M														AD	
AE	WB_TX			AVSS18 _WB0			AVSS18 _WB0		AVSS18 _WB0		SPI_CS		SPI_M														AE	
AF	WB_TX			AVSS18 _WB0			AVSS18 _WB0		AVSS18 _WB0		SPI_CS		SPI_M														AF	
AG	AVSS18 _WB0	WB_TX		GPS_R XIN		GPS_R XIN		WB_SC LK		ENT55		GPS12	ITD1		DVDD18 IOI												AG	

Figure 2-1. LPDDR2 ball map view of MT6582



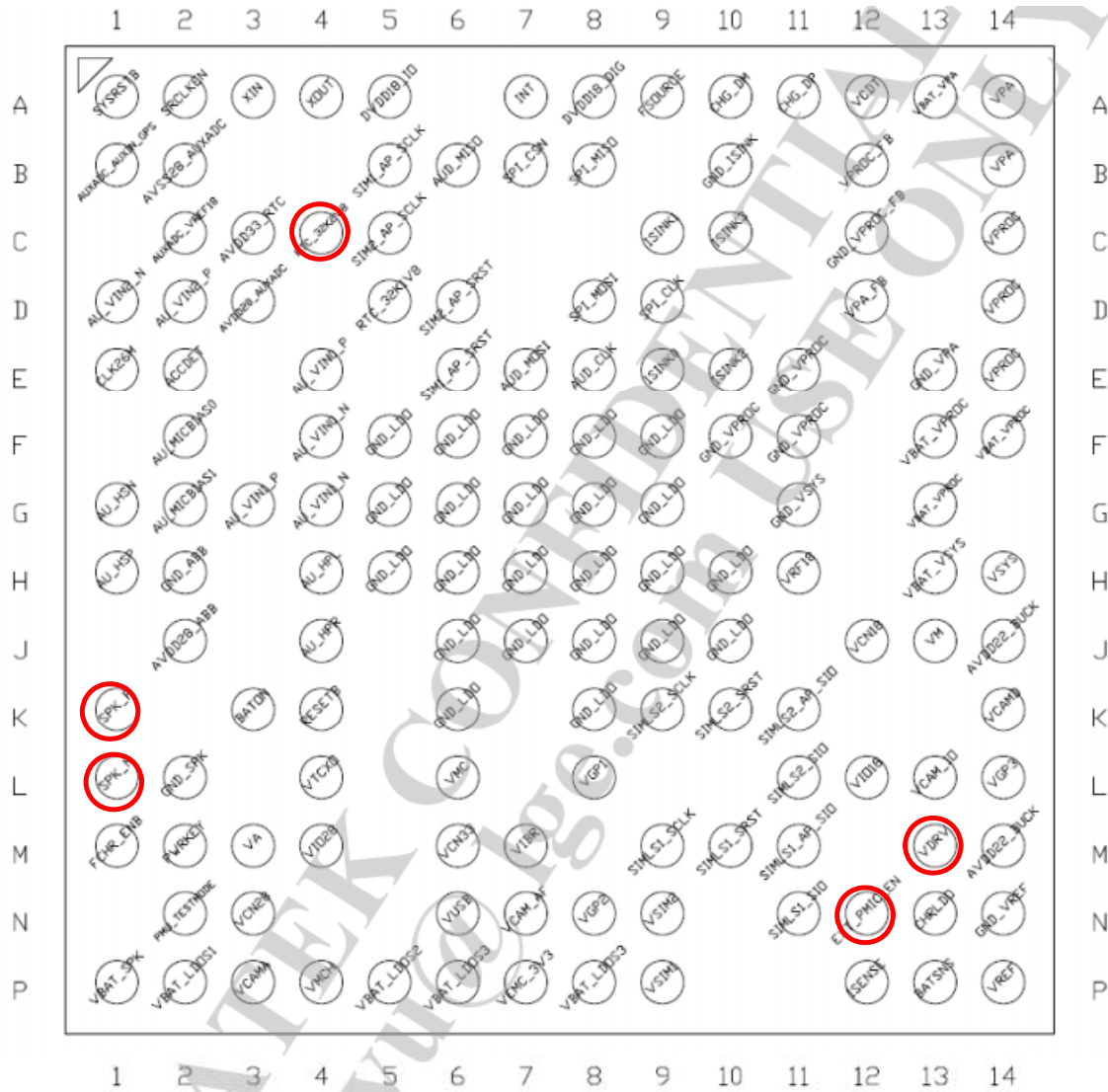
U3300 eMMC



○ Used

7. BGA PIN MAP

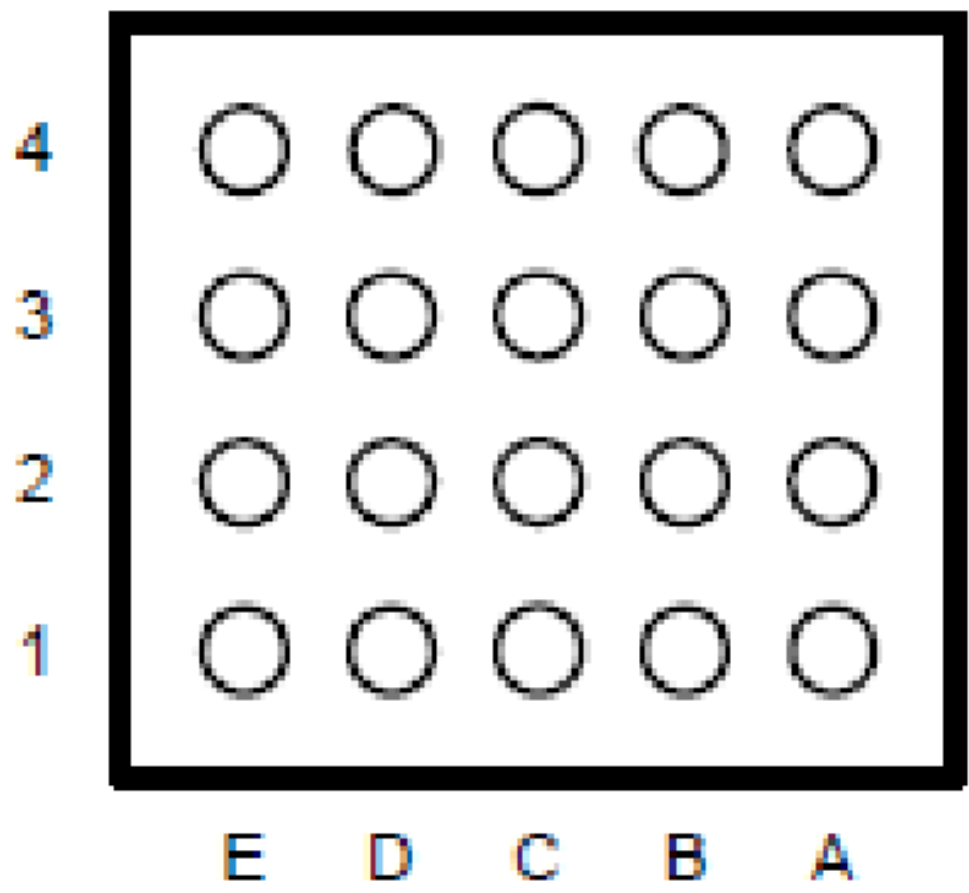
U4100 MT6323 PMIC



● Not Used

○ Used

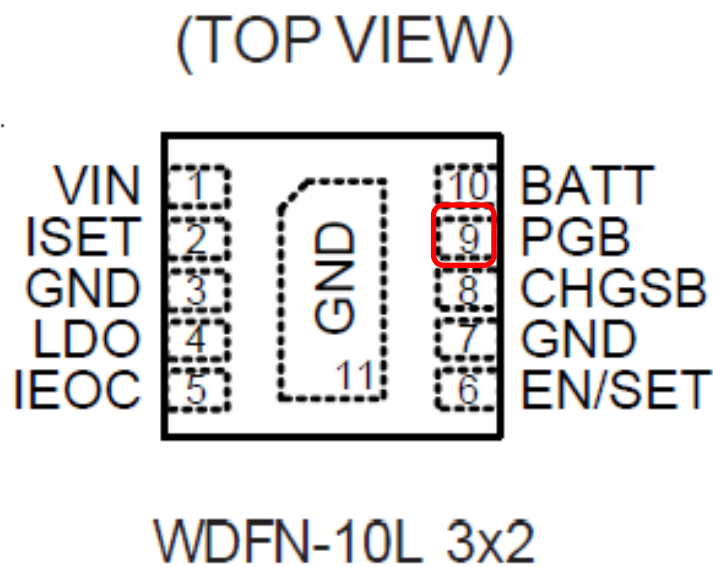
U4400 LM3639 LCD Backlight Booster



- Not Used
- Used

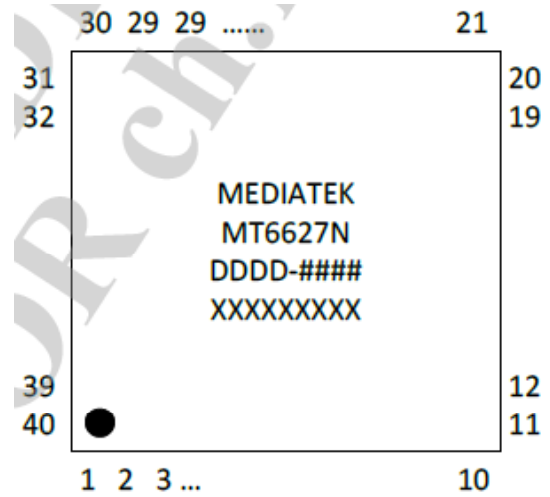
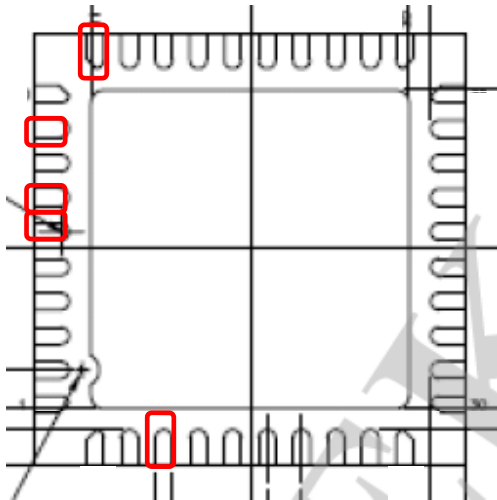
Bottom View

U4700 RT9536 Linear Charger



● Not Used
○ Used

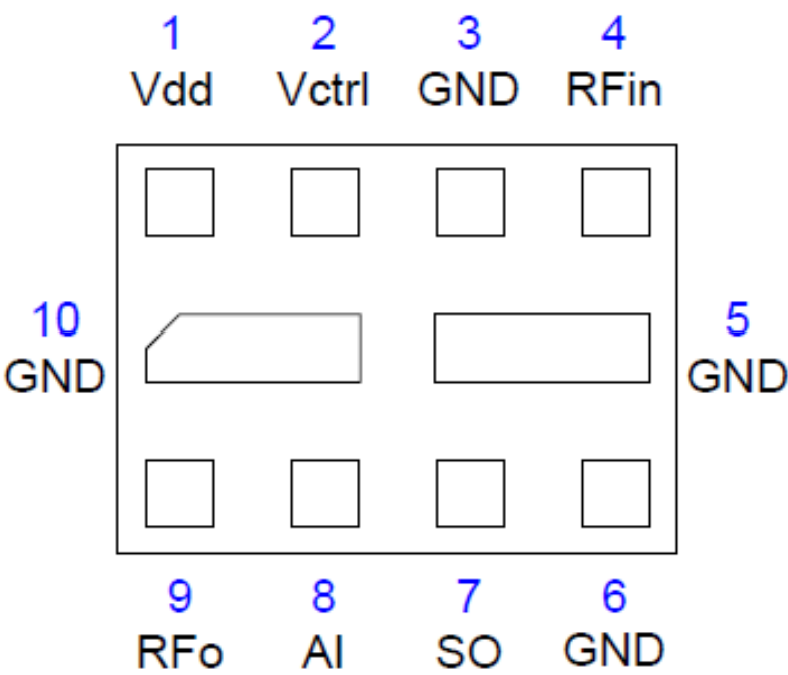
U5100 MT6627 IC,Wifi



● Not Used

○ Used

U5101 SFMG2XAB002 GPS LNA module



- Not Used
- Used

U5300 FC8150 DTV Chip

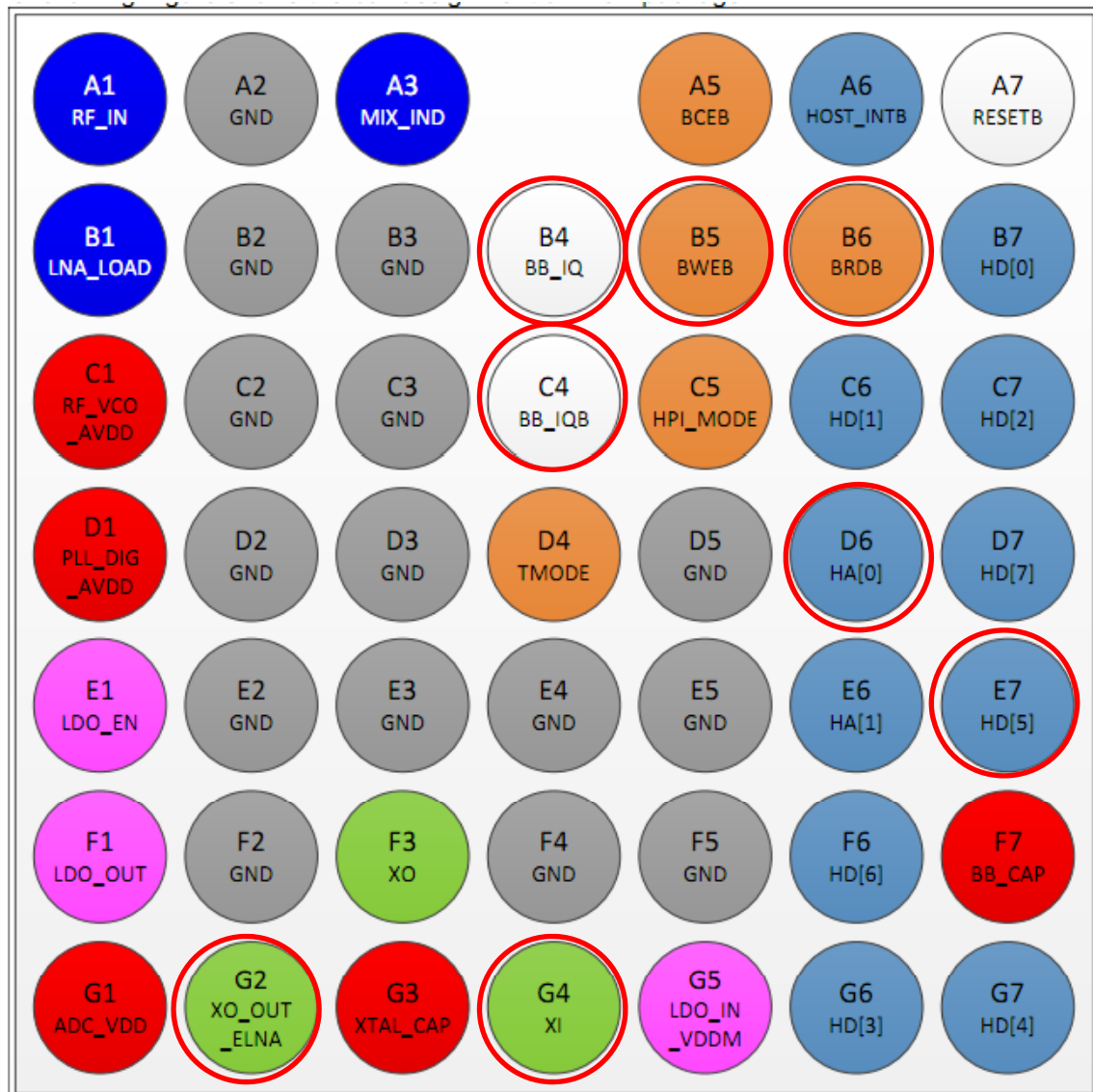
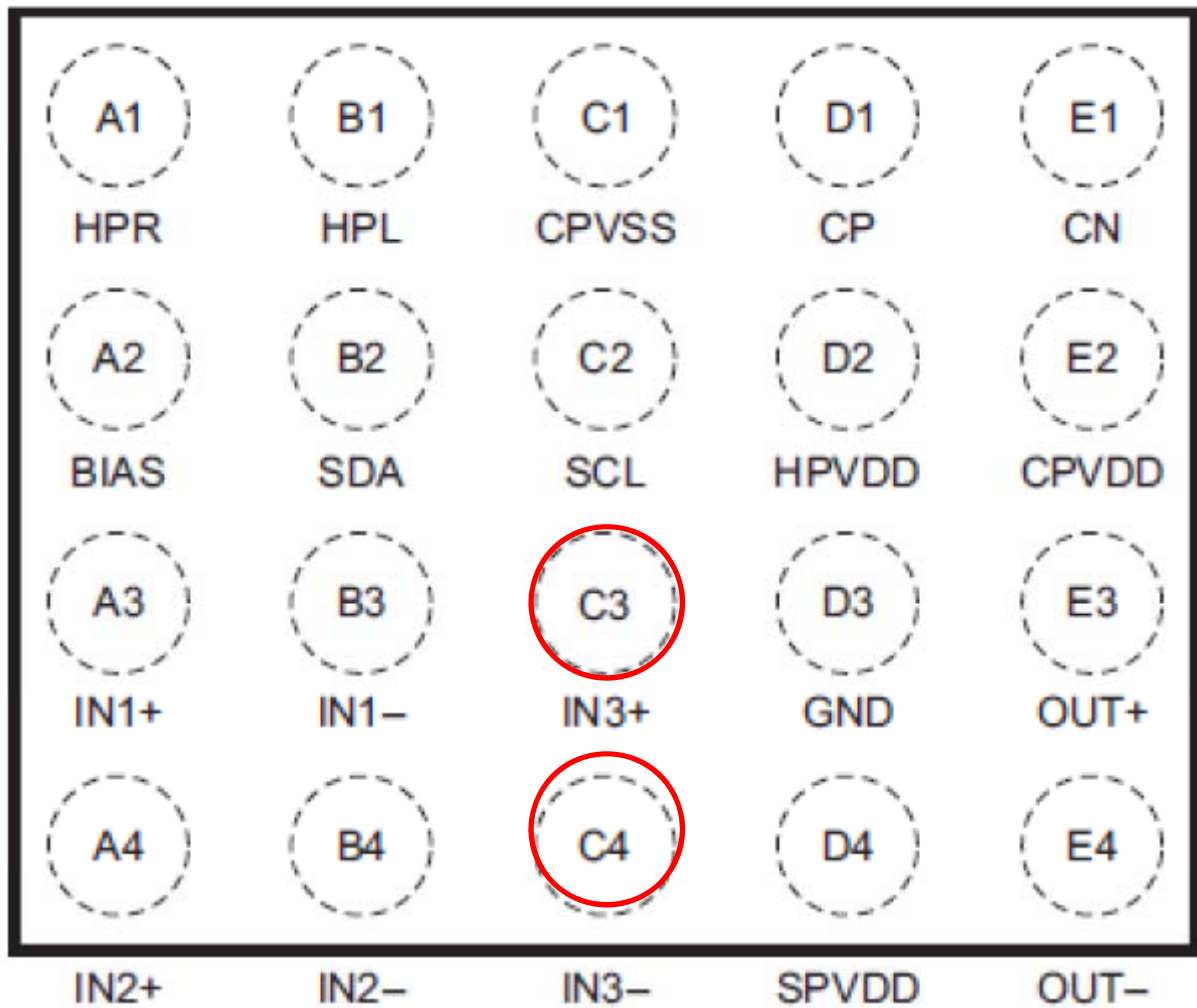


Figure 3.4.1 FBGA-48 pin assignment (Top view)

● Not Used

○ Used

U6200 TPA2058D3 Audio Sub System



● Not Used

○ Used

U6201 DG2735A SPK MIC

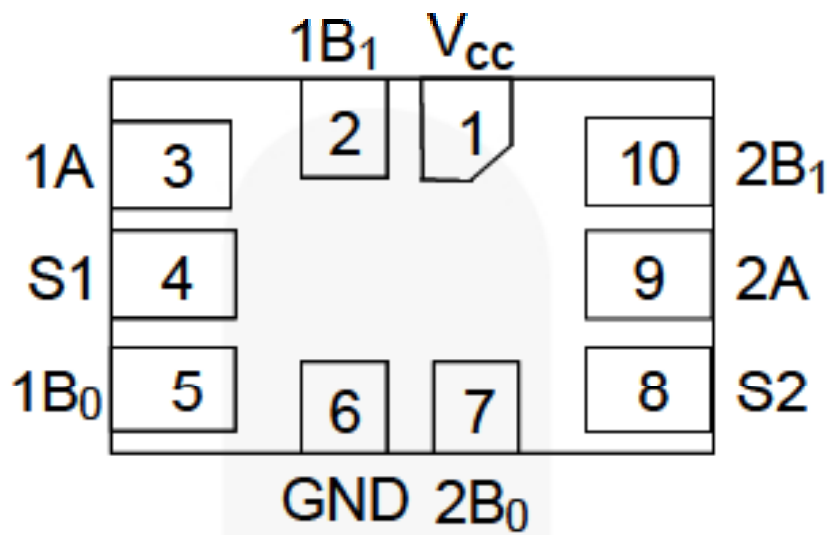
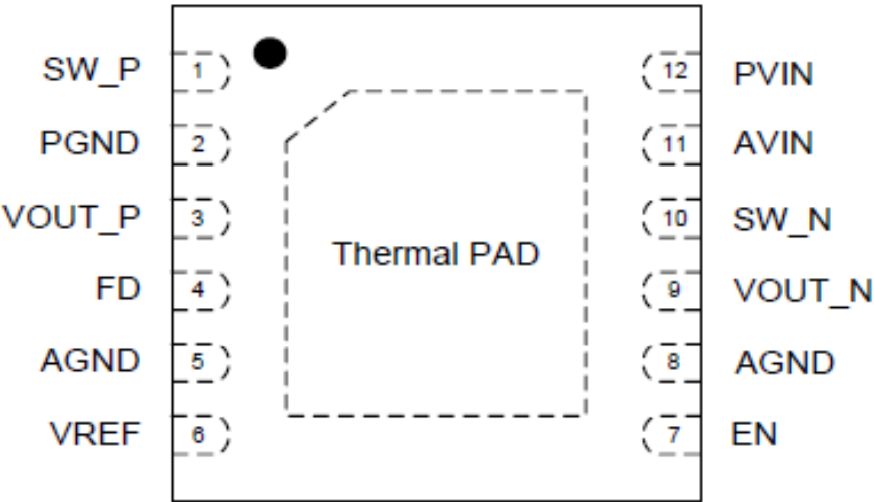


Figure 2. 10-Pin UMLP (Top Through View)

- Not Used
- Used

U7500 DW8755A DSV



[Top View]

- Not Used
- Used

U8200 AK09911 Compass

	3	2	1
C	SDA	RSTN	VID
B	SCL		VSS
A	TST	CAD	VDD

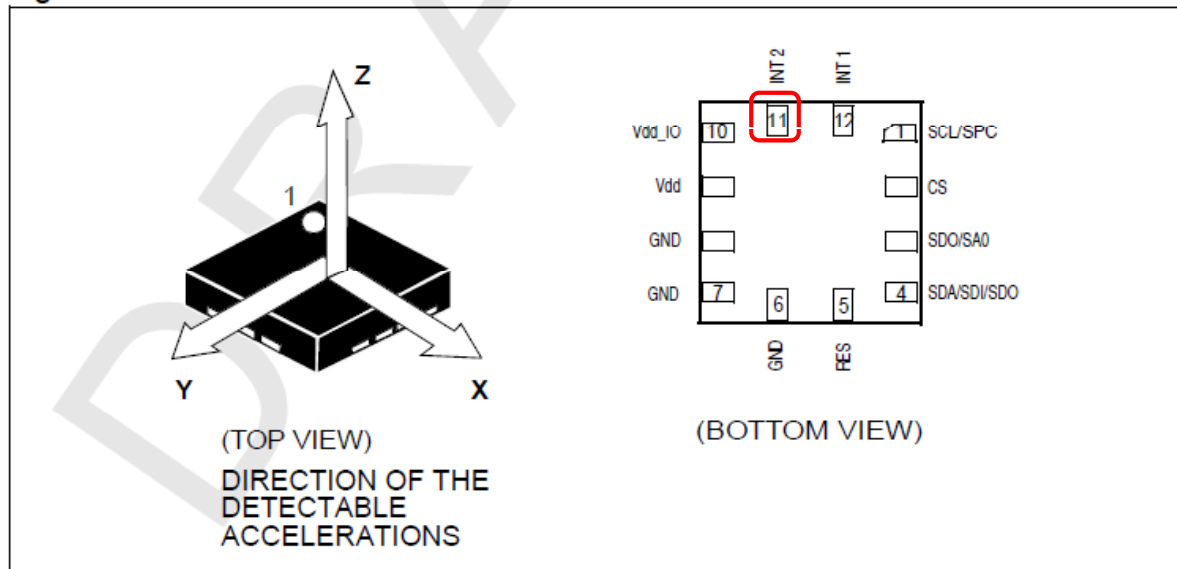
<Top view>

● Not Used

○ Used

U8300 K2HHTR Accel

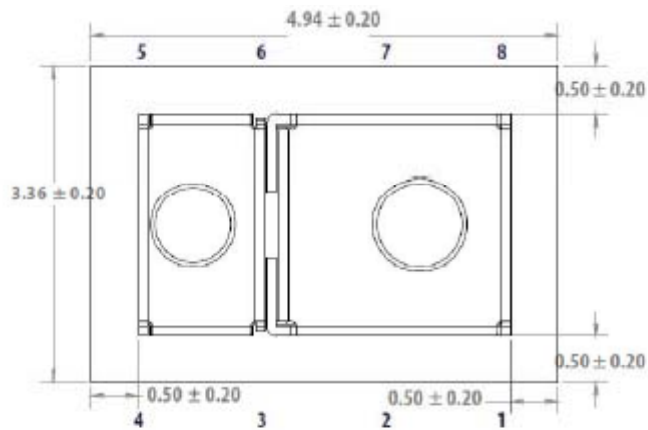
Figure 2. Pin connections



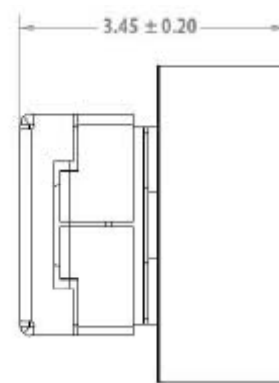
● Not Used

○ Used

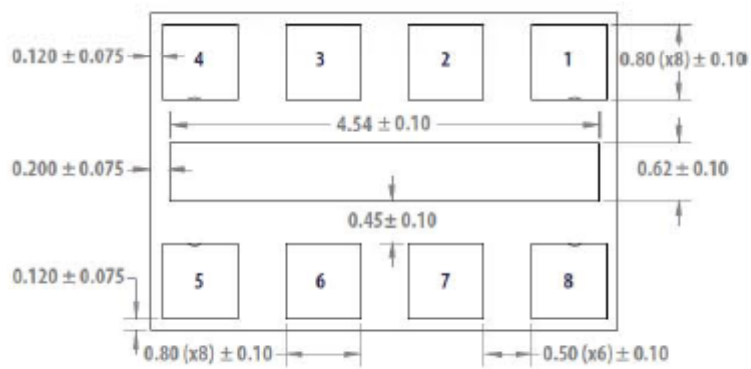
U8400 APDS-9130 Proximity Sensor



Top View



Side View

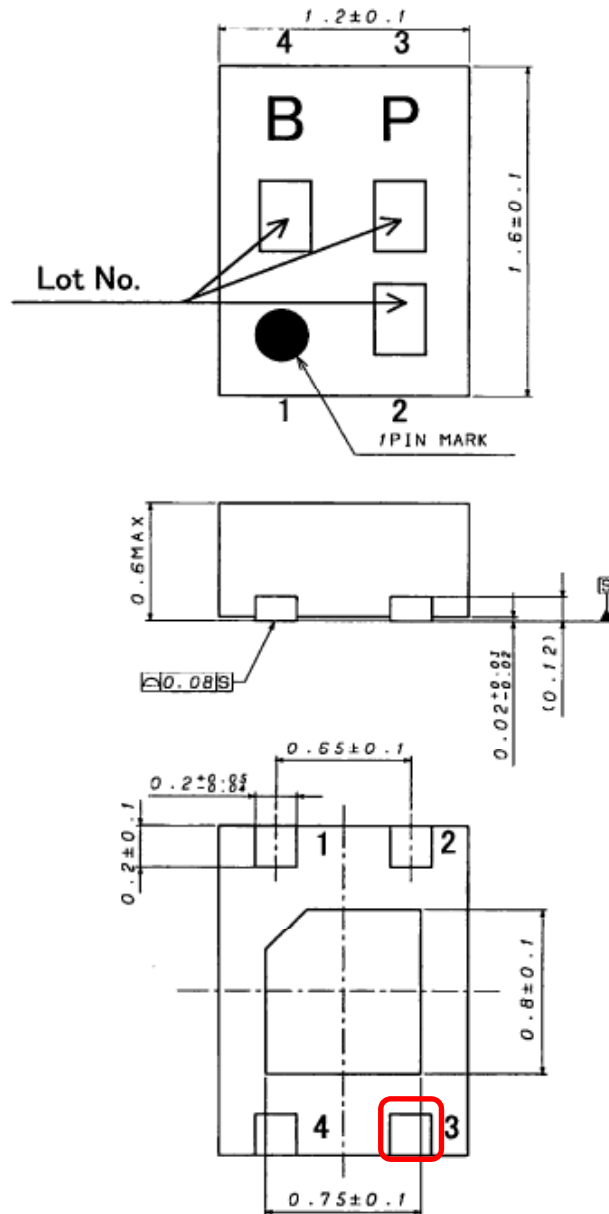


Bottom View

● Not Used

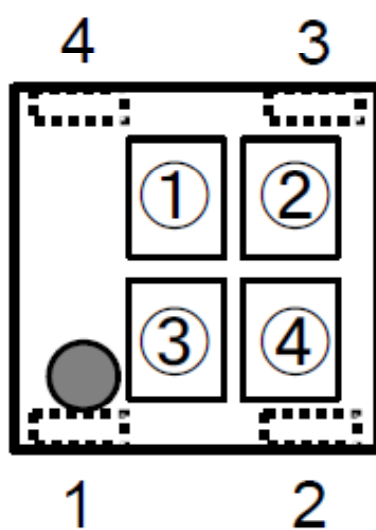
○ Used

U8700 BU52033NVX Hall IC



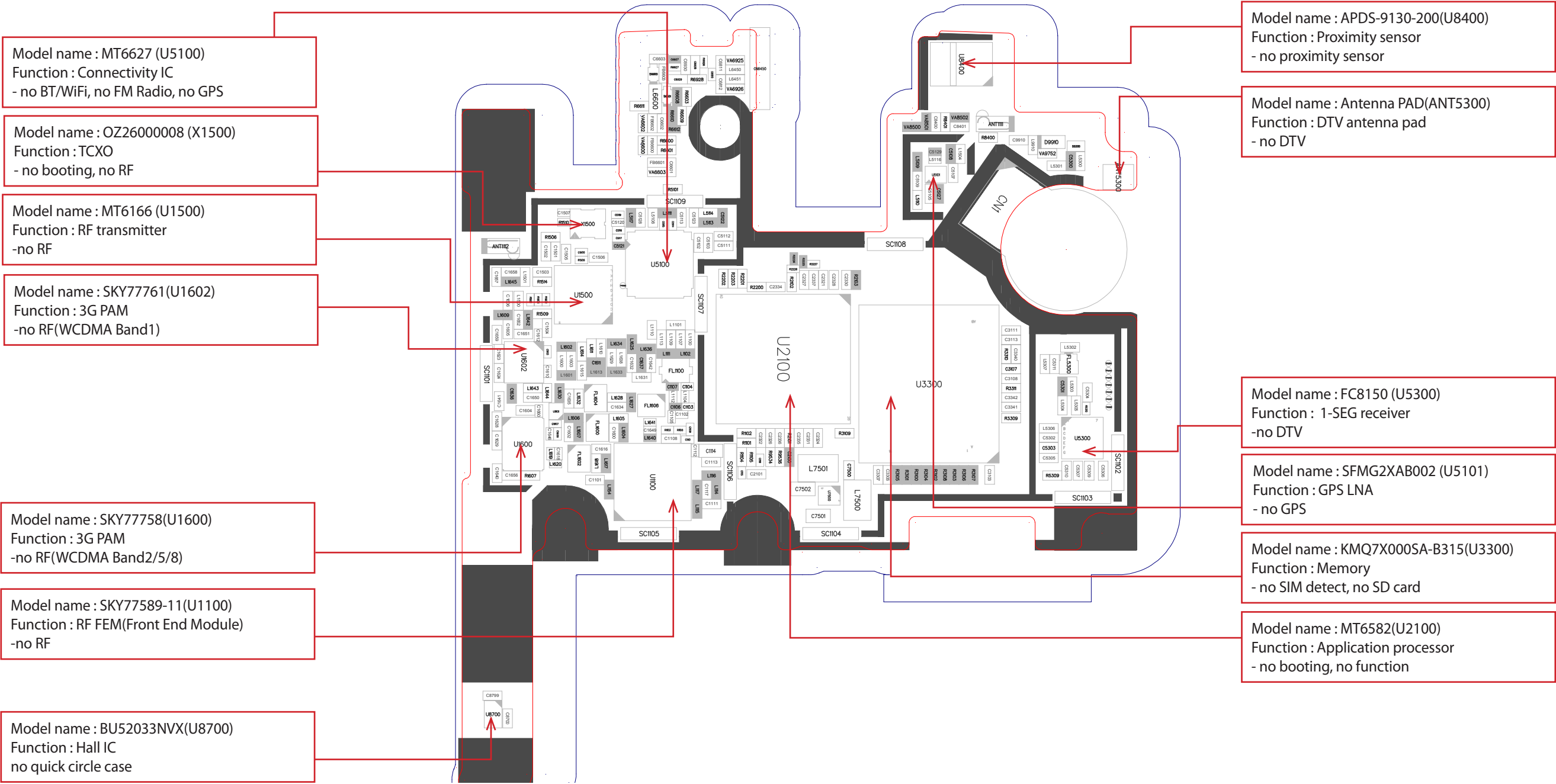
● Not Used
○ Used

U71001 RP103K121D-TR Ext. DVDD LDO for Camera

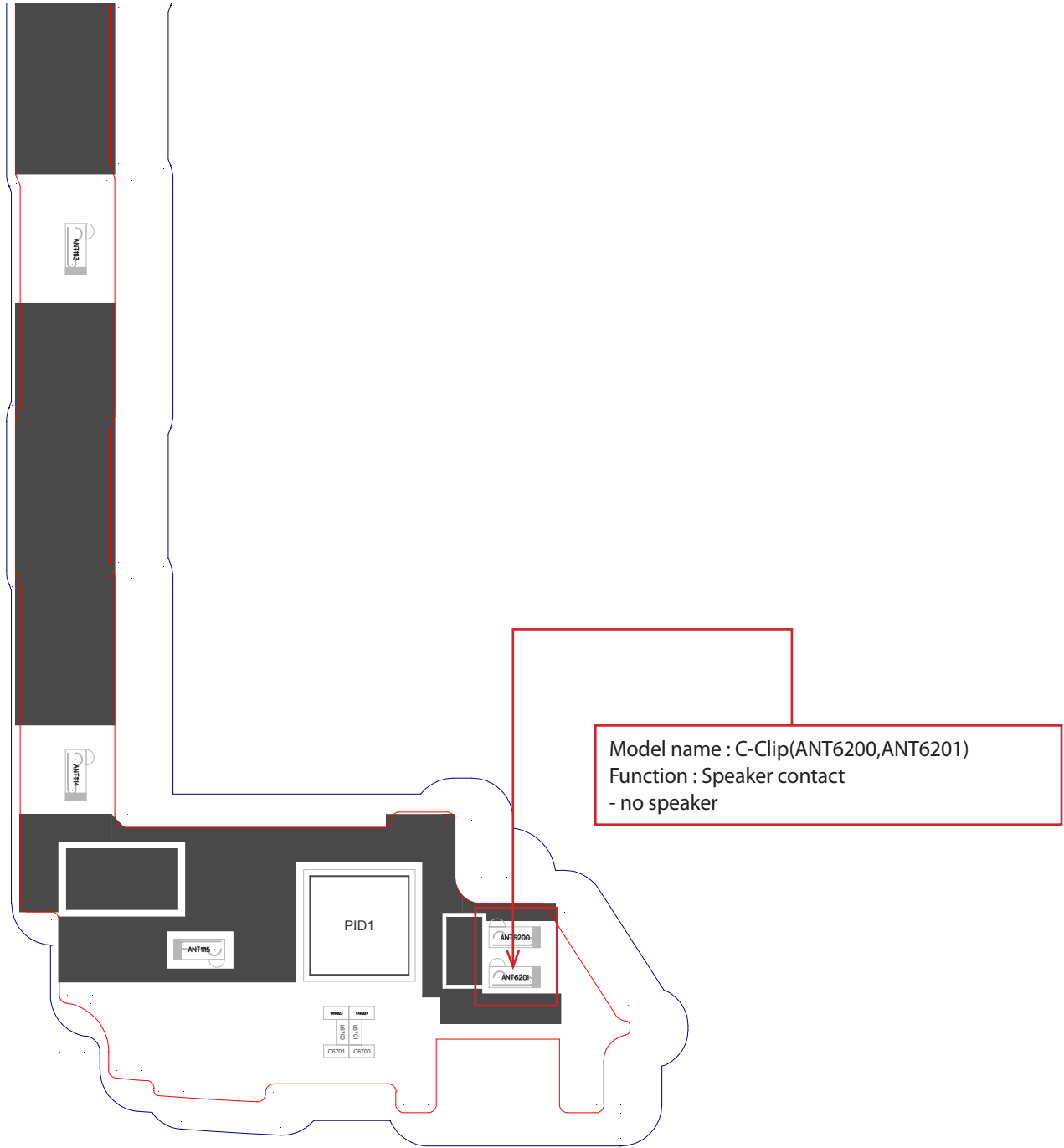


● Not Used
○ Used

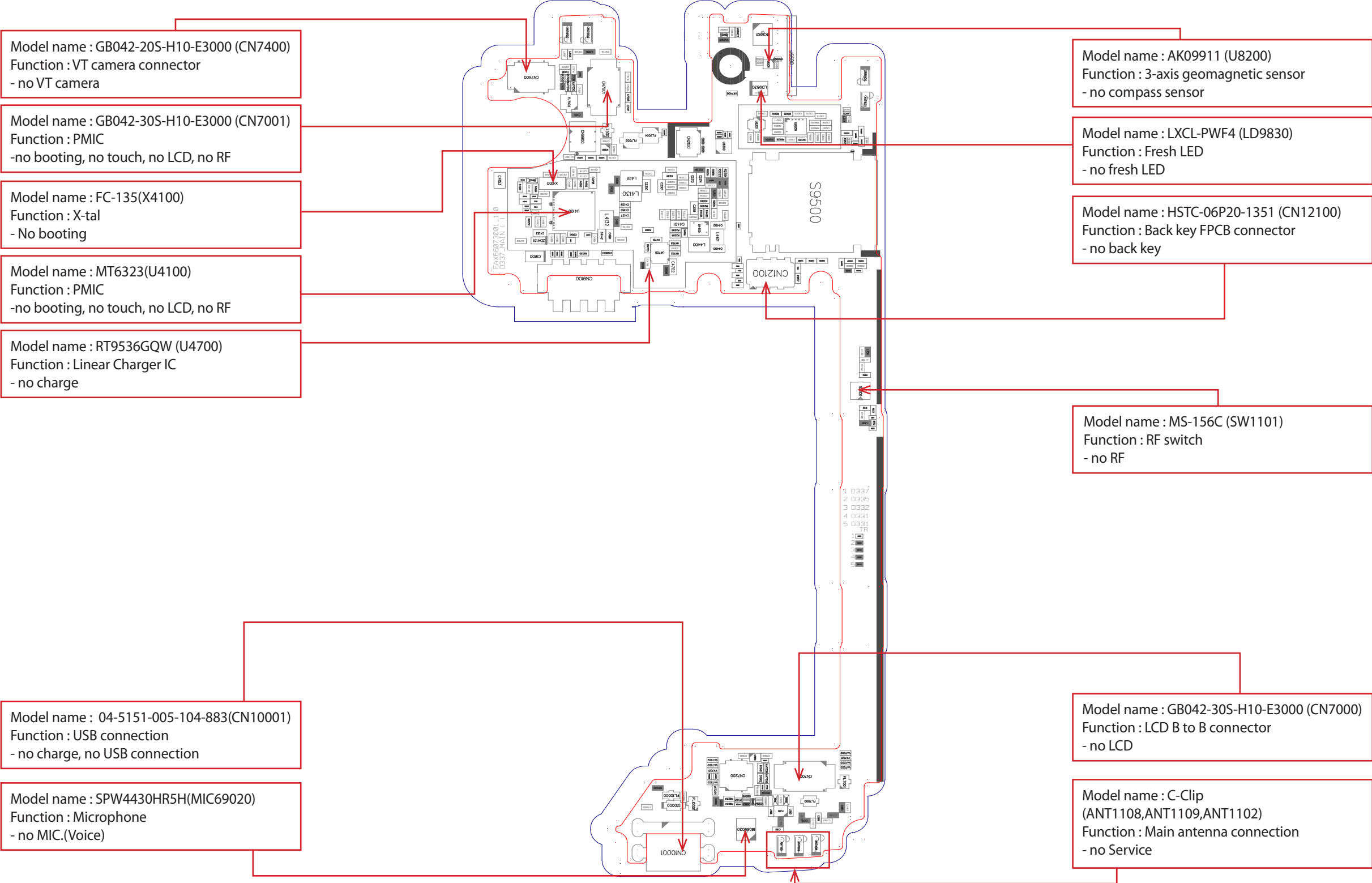
8. PCB LAYOUT



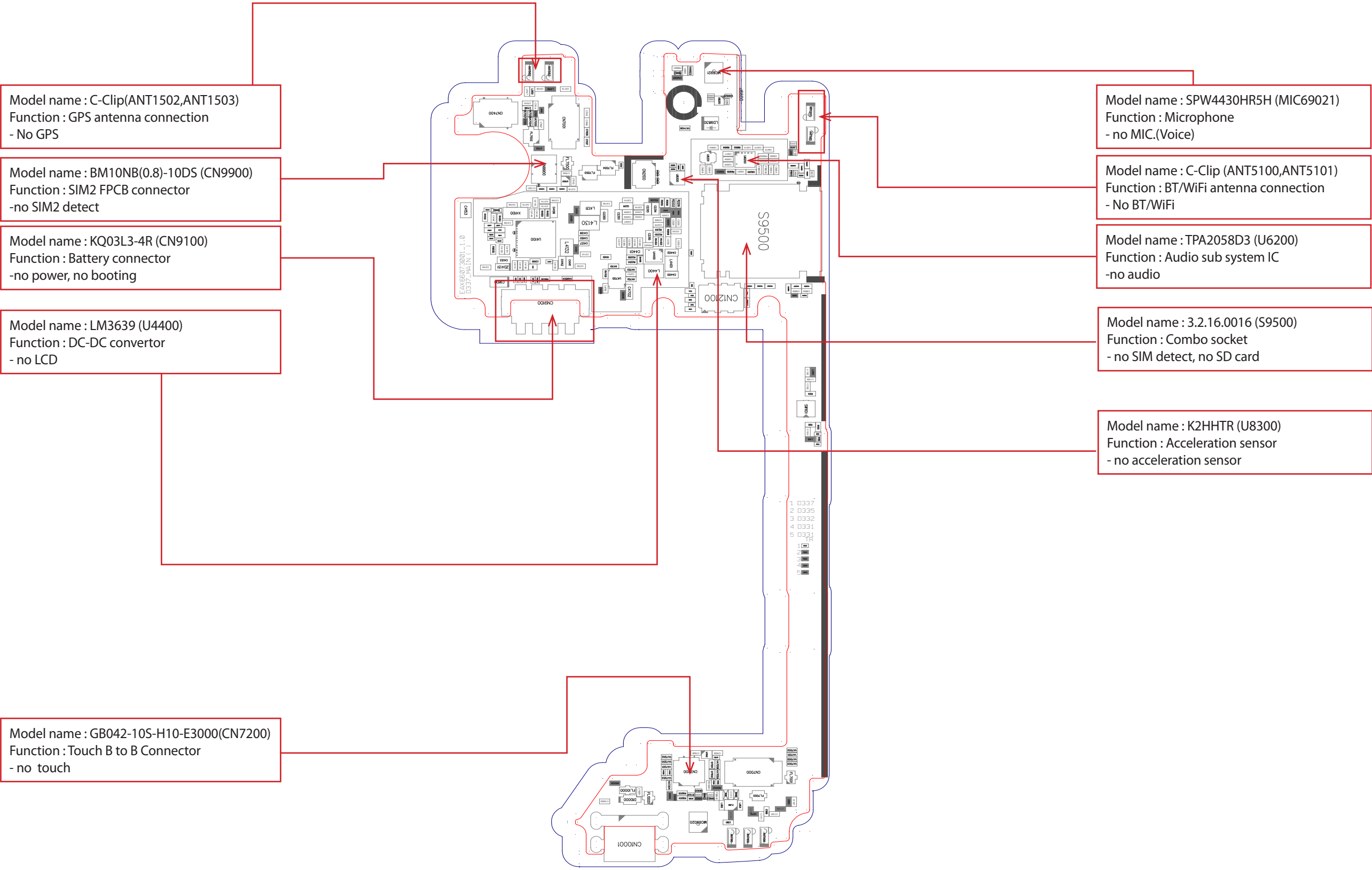
LG-D337_Main_EAX65632401_1.0_TOP-1



LG-L337-MAIN_EAX65632401_1.0_TOP-2

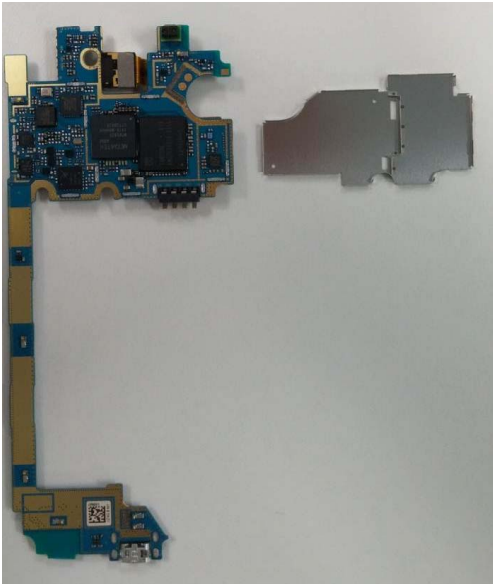


LG-D337_Main_EAX65632401_1.0_BOT-1



LG-D858_Main_EAX65632401_1.0_BOT-2

7. Disassemble PCB Ass'y top



Take apart the Shield can top from PCB

